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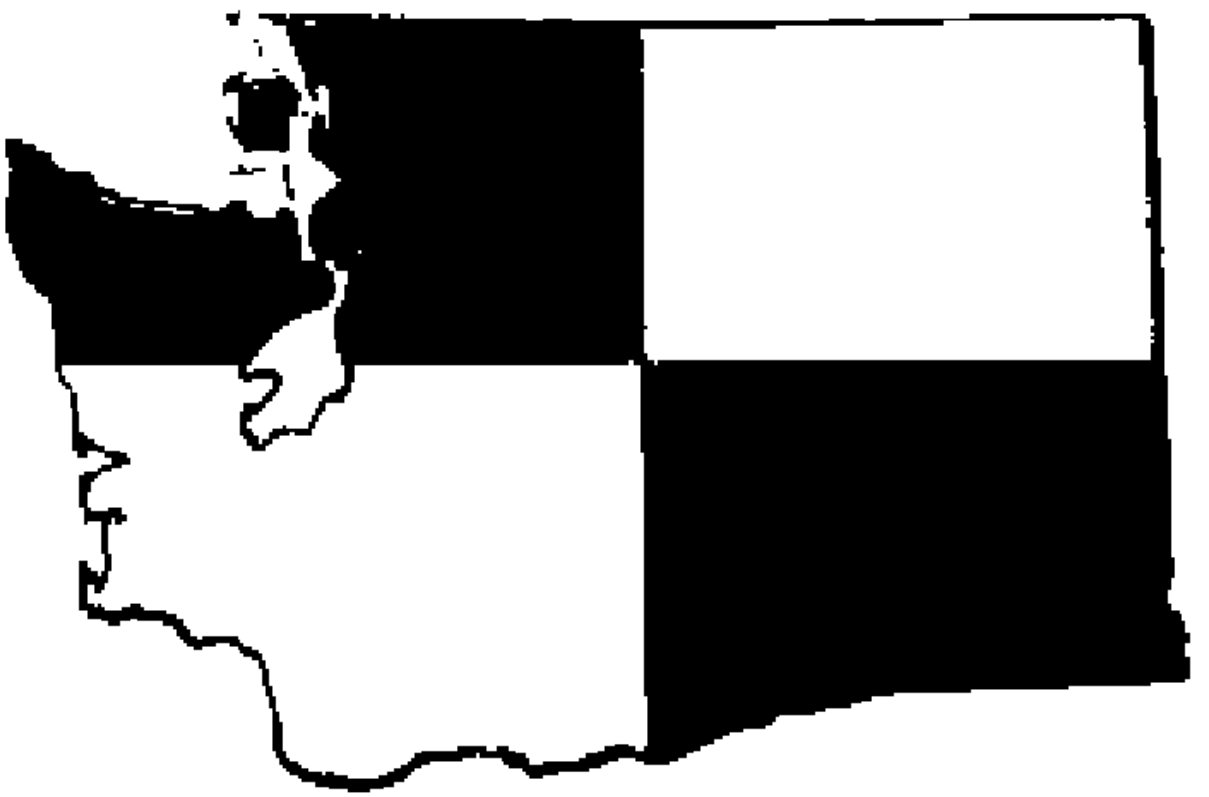
March 13-17, 1999


AMERICAN CONGRESS ON SURVEYING AND MAPPING
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Test Driving the FGDC National Standard for Spatial Data Accuracy

Donald F. Cooke, Geographic Data Technology, Inc (GDT) December 7, 1998
11 Lafayette St, Lebanon, NH 03766 800.331.7881 x1124 (donc@gdt1.com)

Mary M. Welsh, Statistics Consultant
198 Old Pine Tree Cemetery Rd, W. Lebanon, NH 03784 (mmwelsh@valley.net)

Abstract

The FGDC's NSSDA (National Standard for Spatial Data Accuracy), released in July 1998, was tested as a way to describe the spatial accuracy in the Census Bureau TIGER files. Practical and economic considerations for drawing a representative sample from a dataset containing 175 million coordinate measurements led us to choose nine areas across the country and develop a suite of independent check points in the field using a commercial DGPS (differentially-corrected global positioning system) unit.

A wide variance in accuracy was observed from site to site, which the authors explain by examining the pedigree of coordinate readings -- and of the TIGER database itself. The authors conclude that while accuracy for a continually updated database such as TIGER can be measured by the NSSDA, to be truly useful, individual accuracy measurements of every node and complete chain shape record should be carried as metadata in the TIGER database.

Background:

Since the World War II era, cartographic agencies in the United States have expressed accuracy of their products by a "National Map Accuracy Standard" promulgated by the Bureau of the Budget in the 1940s. To use the standard, one asserts that a map "complies with National Map Accuracy Standards" (NMAS). This means that the plotted location of no more than ten percent of well-defined points shall be in error by more than some small amount, either 1/30 or 1/50 of an inch, depending on the scale of the map.

At the end of the century, we find that this standard needs replacement for a number of reasons. First of all, NMAS is a binary standard, not a measure of accuracy. A map either complies with NMAS or not. There isn't a provision for saying "my map's twice as good as the standard" or "it's really very close to meeting the standard".

Second -- and more important -- the information we have been accustomed to storing in map form for centuries is rapidly migrating to computer storage. This migration has broader consequences than many realize, which will be addressed later in this paper.

Computerizing or digitizing can be achieved in two fundamentally different ways: One is by simply scanning paper maps and storing the scanned images. However, it turns out to be far more useful to "deconstruct" the map image and encode the knowledge portrayed on the map in computer storage so that information can be searched and retrieved in various ways. If this is done thoughtfully, functionality of the encoded map greatly exceeds that of the paper original or its scanned image. The original graphical map image can be reconstructed, and computer storage is minimized.

In July 1998 the Federal Geographic Data Committee (FGDC) responded to the new requirements imposed by map computerization with the National Standard for Spatial Data Accuracy (NSSDA). This report describes an effort to "test drive" the NSSDA by attempting to come up with an accuracy measure for the nationwide TIGER database.

TIGER

TIGER (Topologically Integrated Geographic Encoding and Referencing) is a digital map of the United States, developed and maintained by the US Census Bureau where it is used to geocode Census and survey data. TIGER has evolved into a crucial resource for the mechanics of administering the representative

government prescribed in the Constitution. In brief, equal representation in the House of Representatives and in state legislatures is achieved by drawing district boundaries on maps. By 1990, virtually all redistricting processes had "gone digital". The TIGER digital map has become the basis of constitutionally mandated redistricting.

TIGER evolved from a Census mapping project carried out by the New Haven Census Use Study in 1967. Researchers encoded the New Haven street network on IBM punch cards. Each card represented one "street segment", which was a line connecting two "nodes". Nodes were defined at street intersections, ends of cul-de-sacs, and at bends in the road. Each card carried the street name, address ranges on each side, Census statistical zone codes on each side, and coordinate measurements of the two nodes. In order to represent the boundaries of Census zones, streams, railroads, shorelines and political boundaries were treated as pseudo-streets. All in all, two boxes of punch cards (4000 cards) contained a complete encoding of the Census map, including the detailed street address information.

The Census Bureau made these "DIME" (Dual Independent Map Encoding) files for over 200 cities in the early 1970s. “GBF/DIME” coverage expanded to 345 metropolitan areas for the 1980 Census. By 1990 the Bureau achieved nationwide “TIGER” (Topologically Integrated Geographic Encoding and Referencing) coverage by using USGS 1:100,000 DLGs (Digital Line Graphs) to fill in rural areas not covered by GBF/DIME. As the Census Bureau approaches the 2000 Census, it is certain that annual updates of TIGER are a reality for the near future and a continually updated WWW-based TIGER resource will emerge soon.

Why is TIGER important? Because, as a public domain dataset, it supports a myriad of applications beyond the wildest dreams of the originators of DIME. The whole field of Business Geographics depends on TIGER and its commercial offshoots. Each year, thousands of companies save millions of dollars and make better decisions applying TIGER to vehicle routing, site selection, insurance risk evaluation and other business challenges. Offshoots of TIGER power the now ubiquitous WWW-based locator services and commercial CD/ROM atlases. Any systematic action you can take with a paper street map can be done better, faster, cheaper and more reliably with TIGER.

It's perhaps ironic that the most used and useful digital map database in the world was created and is maintained by an agency not usually recognized as a mapping organization. But such logical discontinuities are commonplace around paradigm shifts as major as the computerizing of a major body of knowledge like spatial information.

Measuring Spatial Accuracy of TIGER:

TIGER unquestionably contains all of the information previously encoded as lines and annotation on eleven acres of mylar at the Geography Division of the Census Bureau. The spatial component of TIGER is embodied in about 30 million latitude/longitude measurements of "nodes" (street intersections, etc), and another 140 million coordinate readings which describe the path of curvilinear roads, streams and other features as they go from node to node.

How do we describe the spatial accuracy of TIGER? Though it's a map -- in digital form -- it no longer has a scale. The location of objects is denoted by coordinate readings, not the position of a patch of ink on a sheet of paper drafted at a controlled scale. The old NMAS was wedded to scale. Because a digital map is scaleless, it is not meaningful to say that TIGER does or doesn't comply with National Map Accuracy Standards.

The FGDC's NSSDA takes a step in the right direction by prescribing a methodology for measuring and stating the accuracy of a "map" such as TIGER. The NSSDA prescribes selecting a set of well-defined test points, and statistically comparing the coordinates of these points in TIGER with an independently derived set of more accurate coordinate readings.

The NSSDA process results in an accuracy measurement, N, which has a statistical meaning: there is a 95% probability that a sample from this population of well-defined points has coordinates in error by less than N

meters. Statistically savvy users will recognize and relate to the form of this statement and will be able to employ it to determine the suitability of TIGER to a particular task.

Testing TIGER

The NSSDA accuracy test guidelines leave considerable discretion to the tester. Here are some key concepts relevant to testing TIGER abstracted from the Standard:

"...accuracy testing by an independent source of higher accuracy is the preferred test..."

For our study we compiled our own test dataset using differentially corrected GPS (DGPS).

"...the independent source... shall be the highest accuracy feasible"

We went through three iterations of GPS equipment, ending up with a Satloc 2001 unit, which has an integrated proprietary differential capability. We spent an inordinate amount of time trying to use more modest equipment and found through testing various units against first order control points, that they either were not sufficiently accurate or in the case of using CORS corrections for carrier-phase processing, were severely limited in usability. Our tests of the Satloc unit yielded an accuracy of 3.5 meters, which was sufficiently better than the best database coordinate readings we expected to encounter. *

"...the data producer shall determine the geographic extent of testing..."

We were attempting to measure the accuracy of a huge nationwide database. The geographic extent is the whole country. We decided to study a number of different areas believed to be representative of TIGER as a whole.

"...accuracy shall be tested by comparing...coordinates of well-defined points in the dataset with coordinates of the same points from an independent source of higher accuracy...."

The procedure was clear: measure coordinates of well-defined points in the field, and extract the corresponding coordinates from TIGER for comparison.

"...suitable well-defined points represent right-angle intersections of roads, railroads, or other linear mapped features...."

We used "T-" and cross- street intersections, which comprise most of the nodes in TIGER. In selecting these intersections we avoided major streets with lots of traffic and intersections with tree cover which disrupted GPS signals.

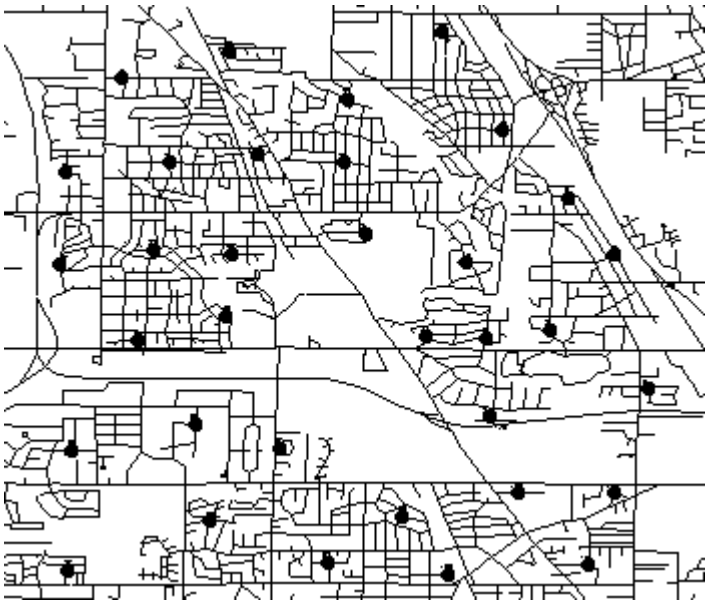
'...a minimum of 20 check points shall be tested, distributed to reflect the geographic area of interest and the distribution of error in the dataset..."

We captured 25 or more points in each field location.

"....data ...producers must determine check point location..."
"...check points may be distributed more densely in the vicinity of important features..."
"...when the distribution of error is likely to be nonrandom, it may be desirable to locate check points to correspond to the error distribution..."

* An early draft of the NSSDA standard urged that accuracy of check points be at most one-third of the accuracy of the dataset being evaluated. This was modified to "*The independent source of higher accuracy shall be of the highest accuracy feasible and practicable…*". According to the draft standard, we could use our Satloc gear and the checkpoints it generates to evaluate datasets with accuracy in the realm of 10 meters. This is important to Geographic Data Technology because this is the accuracy one achieves by aligning TIGER streets over USGS Digital Orthophoto Quads (DOQ).

Insofar as possible, we followed report guidelines in selecting checkpoints within a field location: no fewer than 20% of the points from each quadrant of the test area and an interval between test points of at least 10% of the diagonal distance across the test area. We expected error distribution in TIGER to be nonrandom, so we attempted to include points from the various sources of coordinate information in TIGER.



Check point locations in Deerfield, IL vicinity

We chose test areas in Greenfield, MA; Manchester, NH; Danvers, MA; Morristown, NJ; Ann Arbor, MI; Deerfield, IL; Denver, CO; Raleigh, NC; and San Mateo, CA. Criteria for these choices included coincidence with travel plans, sampling different kinds of TIGER coverage, and sampling various resources (Digital Orthophoto Quads, for example) that might be used as a better coordinate base for a commercially-improved TIGER file.

At publication time (December 7, 1998), partial results of the TIGER accuracy project were available:

	RMSE(meters)	Accuracy(meters)	Approximate NMAS scale
Ann Arbor, MI	47	81	1:140,000
Deerfield, IL	27	47	1:80,000
Greenfield, MA	21	36	1:62,000
Morristown, NJ	48	84	1:145,000
All four areas (weighted):	37	**64**	1:110,000

"RMSE" stands for "root mean square error". Error simply is the distance, in meters, from TIGER's coordinates for an intersection to the "true" value we measured in the field. To compute RMSE, square each error measurement, add up the squares, divide by the number of points (actually by one less than the number of points, say the statisticians), then take the square root of this quotient. The answer is the RMSE in meters.

"Accuracy" is the NSSDA statistic, which is simply RMSE times 1.7308. This factor translates the RMSE into a distance greater than the error of 95% of the points in TIGER. Another way to think of this is to use the complement to say – for example – that there's only a 5% probability that a sample of points in Greenfield would have accuracy greater (worse) than 36 meters.

For more detail, consult the two references listed at the end of the paper. Both are available on the WWW. Appendix 3-D to the FGDC standard suggests a way to relate the new NSSDA accuracy measure to the old NMAS map scale, which is applied in the above table.

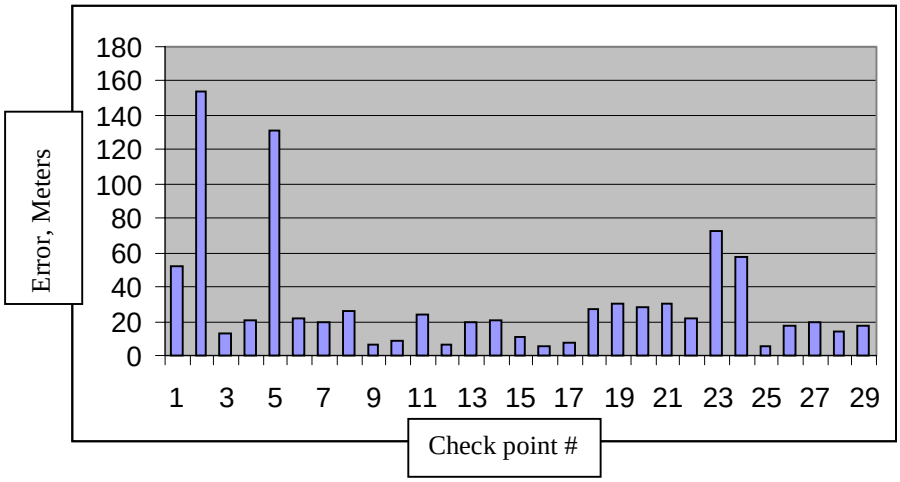
At this point, the NSSDA urges us to report TIGER accuracy in the following form:

"TIGER tested 64 meters horizontal accuracy at 95% confidence level".

Is this good or bad? How do we use this information? Why is Greenfield so much better than Ann Arbor? Does the NSSDA statement say everything we need to know about TIGER accuracy?

Analysis:

The first thing to observe is that the Greenfield coordinates originated from USGS DLG-E 1:100,000 coverage. Coordinates for the other eight areas came primarily from GBF/DIME digitizing done in the early 1980s.* One might jump to the conclusion that DLG coordinates are considerably better than those that came from the old GBF/DIME process. However, look at the distance between GPS coordinates and TIGER coordinates for the 29 check points in Ann Arbor:



Only five points (#1,2,5,23 and 24) are in error by more than 31 meters. *In every case, these were intersections of streets added to TIGER since the original GBF/DIME digitizing.*

Accuracy by sub-population:

	RMSE	Accuracy	Equivalent NMAS scale
5 "sketched-in" points	114 meters	197 meters	1:340,000
24 original GBF/DIME points	20 meters	34 meters	1:58,000

Knowing the TIGER compilation procedure, it is reasonable to conclude that coordinate error in TIGER – even taking into account differences between the GBF/DIME and DLG sources -- is far from randomly distributed. We really are dealing with several different populations of coordinate measurements when we evaluate TIGER, as a consequence of how TIGER was compiled and its ongoing update and maintenance processes.

Rather than label TIGER with a uniform 64-meter accuracy, we might say that the vast majority of points in TIGER, derived from DLGs, have a horizontal accuracy of 36 meters. Then in some metropolitan areas with coverage derived from DIME, accuracy may be even better than this. However, for some points, on streets added by free hand sketching in either GBF/DIME or DLG areas, the accuracy can be worse than 150 meters. Unfortunately, ***we don't know which points are which***.

* As a rule, TIGER coverage around the 345 largest metropolitan areas came from DIME; the rest came from DLGs.

Further – as you read this, new streets are being added to TIGER for the 2000 Census, and it's a near-certainty that TIGER updating by the federal government will become a continual process, as it already is at private data firms such as Geographic Data Technology. For the mission of both the Census Bureau and virtually all commercial TIGER usage, continual updating takes precedence over accuracy.

Much of the value and utility of databases like TIGER lies in this continual updating of the data. It is far more useful to have a new street represented in the database as soon as possible, even if it cuts in 200 meters down the road from where it should, instead of waiting months or years for the street to be added with a more accurate spatial representation. You can route deliveries, target direct mail, dispatch taxis, gerrymander legislative districts, evaluate insurance risk and perform many other applications even if a substantial number of streets are plotted one hundred or more meters away from their true location. Think about the inverse: if we don't have the new street in the database at all (because we're waiting for it's accurate spatial representation), users can't dispatch emergency vehicles to it, evaluate the insurance rating class of houses on it, or perform many other useful processes.

Accuracy is also important, because accurate coordinates yield good-looking graphical map renditions, and some application areas such as vehicle navigation require better than TIGER accuracy. Companies like Geographic Data Technology improve accuracy by realigning streets over Digital Orthophoto Quads. The Census Bureau and the TIGER user community would like to see spatial accuracy of TIGER improved, but no funding to do this has been appropriated.

Making TIGER a useful database requires adding new streets as soon as their existence is verified. But the prospect or reality of accuracy improvement projects makes it mandatory to set accuracy flags for each new coordinate reading entered into the database. The flags do not have to be precise accuracy measures; it would be adequate to have a few categories such as "original coordinate from TIGER" or "sketched in by hand", or "measurement determined by alignment of TIGER over USGS DOQ". From our test-drive of the NSSDA, we know the accuracy measures of each of these categories well enough to judge fitness for use for all known applications.

Point-by-point recordation of accuracy provides users the benefits of representation of the street, and allows them to take the street's location with a grain of salt if the accuracy flag shows that it had been sketched in roughly. Later the flags can serve as beacons for database technicians to follow in improving coordinate measurements of roughed-in streets when a new DOQ or accurate planimetric map becomes available.

The old paper map paradigm requires the map's graphical linework to provide the foundation on which to hang annotation and attributes. The new digital paradigm calls for a topological structure onto which attributes are attached. Some of these attributes are coordinate measures for topological objects such as nodes and complete chains. There's nothing particularly special about coordinate attributes; they can be updated just as one updates a street name or zipcode. The value of attributing each coordinate measure with an accuracy flag is apparent to practitioners dealing with daily updates to a digital nationwide street map.

Conclusion

The new NSSDA improves on the NMAS by providing a continuous accuracy measure rather than a binary "pass/fail" one. The NSSDA also has the virtue of independence from the concept of map scale, a cornerstone of the NMAS, which has lost all meaning and usefulness in the new paradigm of "map-as-database". The NSSDA will be especially useful for evaluating and describing paper maps and spatial databases compiled and maintained with fixed and uniform procedures.

Our experiments with test driving the NSSDA succeeded in yielding accuracy measures for TIGER, but also revealed the variance in accuracy endemic to dynamic and continually maintained spatial databases

We conclude that for dynamic spatial databases like TIGER, a single accuracy measure like the NSSDA leaves a lot unsaid. Practical realities of TIGER usage require that accuracy be compromised for the sake of being current, and that all coordinate attributes must be individually flagged with accuracy metadata.

Acknowledgements:

The authors acknowledge the contributions of the Vermont School-to-Work program and the Upper Valley Partnership. These organizations identified teacher "externs", Al Zielenski (River Bend Career and Technical Center, Bradford, VT) and John Kitzmuller (Richmond School, Hanover, NH), who did most of the work of evaluating GPS equipment, defining data collection procedures, and collecting and analyzing data. Without the diligence and intelligence of Al and John, this project would still be on the back burner.

Many people at GDT helped make this project work, in particular the "POSACC" (Positional Accuracy) team of Steve Zuckerman, Jay Clark, Peter and Mary Moeykens, Heather Thompson, and Mark Haugen. Special thanks go to Mike Gerling for years of ongoing support for this accuracy crusade.

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USE OF RANDOM ERROR THEORY IN SHOWING BOUNDARY CORNERS OF PARCELS OF LAND IN AN LIS

CARLTON A. BROWN
Graduate student, M.S. Geo-Information Science, Dept. of Geography
Salem State College, Salem MA 01970

ABSTRACT

In performing retracement surveys, land surveyors "follow the footsteps" of the original surveyor. Boundary corners not monumented by the original surveyor must be determined by the retracing surveyor, who gives a professional opinion of location by applying proper boundary law to the legal research and field investigation performed on the parcel of land.

Many equally competent retracing surveyors, each independently determining the location (by State Plane Coordinate (SPC)) of a specific unmonumented boundary corner, will result in many different locations (SPC) of that corner. Since each is equally competent, it is impossible to pick one retracing surveyor's opinion of boundary corner location as better than any other.

This paper will investigate the use of random error theory on these many locations of a specific boundary corner, which will result in a "most probable" location (SPC) and precision value for a given certainty. A boundary line drawn between two such boundary corners will not be a one-dimensional line but will instead be a "fuzzy" area of uncertainty.

The ramifications of this concept will be investigated for the creator of a cadastre or LIS that is to have a proposed life of more than 500 years.

INTRODUCTION

The success of a cadastre or parcel-based LIS requires that the true and actual boundary lines of each individual parcel be shown for three reasons:

1. If the boundary lines shown on the cadastre cannot be relied upon, then the user may not use the cadastre and the cadastre will fail.
2. If a user suffers a loss because of relying on boundary lines incorrectly shown on a cadastre, then the creator of the cadastre may be liable for damages.
3. Throughout its useful life, a cadastre will continually change as it tracks the dynamic nature of parcels of land being subdivided, re-subdivided, combined and re-combined. Since every new parcel is made up of parts of old parcels, the new parcel boundaries can only be as accurate as the old parcel boundaries from which they are made. Thus to show the true and actual future parcel boundaries requires that the true and actual boundaries of all the parcels that exist at the cadastre's inception be known.

The creator of a cadastre is therefore responsible for showing true and actual boundary lines of all the cadastre parcels.

A premise of this paper is that in the United States of America, for colonial or metes and bounds states (as opposed to states using the Public Land Survey System) the true and actual boundary lines cannot be definitively established. The role of land surveyors in making professional opinions of boundary locations will be discussed. Equally competent land surveyors who independently gather field and legal record evidence, and analyze it using boundary law will often determine a different boundary line determination. This paper will investigate the use of random error theory in describing boundary line locations. The end result will be a method that a cadastre creator can use to give an accuracy statement on boundary lines shown on the cadastre. Rather than showing definitive boundary lines with a one-dimensional line, a most probable boundary line with a fuzzy area of uncertainty can show a precision value for a given certainty.

OWNERSHIP OF ESTATES IN LAND

In colonial America, the concept of land ownership is that which was brought with the first settlers from England in the 1600's. The King of England through the right of force and discovery claimed absolute ownership over all the land. To show good title to a parcel of land requires that all the prior owners in the chain of title from the present owner back to the sovereign had good title for that parcel. The English tradition of land ownership dates back to the time of the Norman Conquest of England by William the Conqueror.

In 1066 William, Duke of Normandy, conquered England and was crowned King of England and supreme lord of the land. William rewarded members of his army, both leaders and rank and file, with English land. Those English landowners who fought against William or who had died during the war simply had their land taken over. How the transfer of the rest of the land happened is less clear. The remaining English landowners probably tried to hold onto their land, but much of it was taken by brute force by individuals of the conquering army in the years following the war. It took the next twenty years for the Normans to expropriate the lands formerly owned by the conquered English aristocracy. Thus by the year 1086, of the 180 major landowners only two were English, and of the 1400 or so smaller landowners, only about 100 were English (Wood 1987). Some Normans received land that was very valuable, others land that was barren. There was certainly squabbling going on between Normans about who had a legitimate or better claim to a certain parcel of land. The ownership of land at this point was in chaos. While William had given some land to specific Norman supporters, various members of the Norman occupying forces had simply violently and unjustly seized more. There were many disputes over ownership of land, especially between Normans who had rival claims. Many Normans attempted to misappropriate land and then try to have the courts of that day legitimize their claims. There were so many disputes brought to the courts and so much confusion over the ownership that William was forced to act. One purpose of the Domesday Book was to settle the question of ownership of land in England. William had his clerks investigate and determine three things: who owned the various parcels of land at the end of the English reign in January 1066 just before the Norman conquest; when did the present owner take possession of the land; and who possessed the land as of 1086 when the Domesday Book was written. Normans who had claims to land

were able to plead their case, and the outcome was an "official" judgement on their legalities. The Domesday Book listed all the Norman owners and described the land. Thus possession was approved only where William's authority had authorized it, whether directly or indirectly. The importance of the Domesday Book was that it legitimized all the land holdings in England. All the courts thereafter upheld the ownership claims of those named in the Book. In other words, to prove title an owner simply had to show that he had an unbroken chain of title (i.e. he could list all the prior owners and show that they all had good title to the parcel) back to the Norman listed in the Domesday Book. This tradition is still used today: to prove title to a parcel of land means to show an unbroken and good chain of title back to the sovereign.

Thus to prove title or ownership to a parcel of land requires that there be an unbroken and good chain of title back to the sovereign. In New England, between 1620 and 1789 the sovereign was the King of England, and after 1789 has been the individual state in which the land is located. Proof of title thus requires a determination of whether there is a "good" chain of title back to the sovereign.

In the Commonwealth of Massachusetts, this determination of "good" title is made in two ways: by the professional opinion of a lawyer and of a land surveyor; or by decree of a judge or jury. The opinions of lawyers and land surveyors are used unless there is a conflict in which case the judge and/or jury determine the adequacy of title in two distinct ways: quiet title action and through the Massachusetts Land Court.

In making a professional opinion, evidence must be gathered and the laws of evidence applied. Evidence is defined as: "At law, evidence is that which is legally submitted to a competent tribunal as a means of ascertaining the truth or untruth of any alleged matter of fact under investigation before it" (Brown 1994). "Evidence is not proof. Evidence leads to proof. A consideration of all evidence and conclusions to be drawn from the evidence, in accordance with the law of evidence, may produce proof." (Brown 1994).

The best evidence of land ownership are the written conveyances of property, the deeds, which are recorded at registries of deeds. The best evidence of parcel locations are monuments found on the ground which are those called for in the deeds. However, these deeds are not proof, they are evidence. Thus registries of deeds are repositories of evidence, evidence contained in deeds which are the written conveyances of property.

Evidence is not proof. Evidence can be used by a "competent tribunal" (defined by Black's Law Dictionary as a judicial court (Black 1979)) to draw a conclusion that may be proof. Land surveyors are not judicial officers, but in determining boundary line shape and location their opinion often has the effect of a judicial decision. Thomas M. Cooley, Jr., Chief Justice of the Michigan Supreme Court from 1864 to 1885, wrote a paper entitled "The Judicial Function of Surveyors" in which he stated: "Surveyors are not and cannot be judicial officers, but in a great many cases they act in a quasi-judicial capacity with the acquiescence of parties concerned..." (Brown 1994). Thus, unless one of the parties involved disagrees, the land surveyor is the "competent tribunal" and can make an opinion (conclusion of proof) on the shape and location of a parcel of land. In

like manner, lawyers gather evidence and apply the laws of evidence and make an opinion on the adequacy of title held by the owner of the land.

If there is no acquiescence of all the parties involved of the correctness of the professional opinions regarding parcel size and location and adequacy of title, then the judge and jury process must be followed. Quiet title actions often do not determine the complete parcel shape or location, nor do they determine definitive adequacy of title against all the people of the world. Rather only those specific boundary lines in conflict may be determined and adequacy of title is only determined against those individuals actually named in the proceedings instead of against all people. However, in Massachusetts there is a second judicial method, registration of land by the Land Court, which does definitively determine parcel shape and location, and does determine definitive adequacy of title.

For purposes of this paper only the determination of boundary lines by opinions of professional land surveyors will be discussed.

ACCURACY OF PARCEL BOUNDARY CORNERS

The underlying assumption of a cadastre is that it is correct and true. The parcel boundaries shown on a cadastre must be the true boundary lines that exist on the ground. Webster's Dictionary defines accuracy as the "quality or state of being free from mistakes or errors" and thus such a cadastre that shows the true boundaries of parcels is totally accurate and free from mistakes and errors. However, if the true boundaries are not definitively known, then the quality of accuracy must be determined.

Accuracy of parcel boundaries: opinions of land surveyors

The determination of parcel boundaries by land surveyors is not definitive and final because the acquiescence required of all parties can be withdrawn at any time. Only a judge can definitively decree the location of parcel boundaries such that forever more they are fixed and unmovable. Thus only when <u>all</u> parcels are decreed by a judge is a cadastre totally accurate. As mentioned above, this paper only describes boundary lines determined by opinion of professional land surveyors. This section will investigate the accuracy that can be assigned to parcel boundaries determined by opinions of land surveyors. The method will be to show that a land surveyor's <u>parcel boundary determination</u> has error sources that can be described in the same terms as those describing <u>measurements</u> made by land surveyors. In other words, a statistical analysis can be used to determine the magnitude of errors in the determination of property boundaries in much the same way that it is used on measurements. Land surveyors use measuring devices, limited by the refinement of the construction of the device, by environmental factors when measuring, and by limitation of humans to use the equipment and make a measurement. Such measurements will therefore have errors.

So too, the opinion of the location of a boundary corner is limited by the availability, meaning, intent and interpretation of the evidence, as well as the ability of the land surveyor to measure monuments and other field information and model this information

by a plan or a deed by defining the location by a State Plane Coordinate (SPC). There are many reasons why evidence and measurements can be interpreted differently and therefore why there can be different opinions as to boundary corner location. Only a few of these will be discussed as follows.

Deeds are made up of written words. The meanings of those words must be interpreted to determine the intentions of the parties to the conveyance. Sometimes meanings or words may be vague or ambiguous and more evidence must be relied upon to determine the intent of the parties. Even such simple words as “by”, “to”, and “from” now have very specific legal meanings because of the many court cases that have had to interpret what these words mean. Monuments called for in deeds may or may not be found on the ground. If they are found, an interpretation must be made as to whether they are in fact those called for monuments or did someone else set them at a different time. Perhaps the monuments called for in the deeds are found, but their location on the ground does not exactly correspond with distances or a direction called for in the deed, so there is a discrepancy between deed distances and called monuments.

If a person conveys a portion of his land, he cannot later convey more than his remaining land. The first buyer has senior rights, and the second buyer has junior, or remainder, rights. The first buyer is entitled to all the land conveyed by the initial deed. The second buyer is entitled to all the land conveyed by the second deed unless it conflicts with the rights of the first buyer. If it is determined that there is a conflict between the land conveyed, then the junior deed loses.

The gathering and interpretation of evidence is further hindered by the fact that boundary corners could have originally been created more than 350 years ago. There may be dozens or hundreds of links in the chain of title, each link of which must be checked using gathered evidence. The likelihood of having differing opinions as to the location of the boundary corner increases with the length of time that the boundary corner has existed.

Random error theory applied to measurements

Errors in measurement can be minimized. The three types of errors are mistakes, systematic errors, and random errors. Errors caused by blunders or mistakes can be minimized by attention to the measuring process to eliminate those mistakes caused by confusion or carelessness. Systematic errors are those errors that follow a physical law and thus can be predicted and removed by proper calculation. Thus a steel tape will contract when cold and expand when hot in a certain prescribed way. The inaccuracy caused by a tape that is too short or too long because of the expansion and contraction can be eliminated by proper calculations. Random errors are those errors that are left after all the blunders and systematic errors have been removed. They are the result of instrument and human imperfections. These errors are generally small and are as likely to be positive as negative. They do not follow any physical laws, but can be predicted with laws of mathematical probability. Error theory uses statistical methods to determine accuracy of measurements.

In an analogous way, errors in the opinion of boundary corner location can be minimized by careful attention to measuring in the field, by finding all the available evidence both in the field and at the registry of deeds and elsewhere, and by applying proper laws of evidence to the information. Thus errors or differences between opinions would be only based on the individual interpretation made by equally competent land surveyors who have found all the available evidence.

Error theory in measurements deals with the certainty or uncertainty that the measurements have. After making one measurement we may realize that with the uncertainty involved in a human using a manufactured measuring device, that the measurement may be uncertain. Perhaps we are uncertain that the measurement was the best that we could do. Perhaps we would be more certain if we repeated the measurement. If we did repeat the measurement we probably would end up with two measurements that were different. We would have no reason to pick one of the two measurements as more correct because they both were performed with the same procedure and both measurements seem reliable. If we were to repeat the measurement 100 times we probably would get many different measurements. Again, since all the measurements are performed with the same procedure and care, it would be difficult to pick one measurement as more correct than any other. However, we probably would be able to detect a central tendency to the measurements, a point where most of the measurements appear to cluster. If we were to take thousands, tens of thousands or hundreds of thousands of measurements we would probably find that the measurements were following the normal distribution function (the familiar "bell" curve) of probability. It still would be impossible to know what the "true" value of the measurement is no matter how many measurements were taken. However, it would be possible to use error theory to determine that for a given confidence level of certainty, the true value lies within a certain range.

Random error theory applied to determination of boundary corners

In like manner, suppose that the creator of the cadastre has a land surveyor determine the State Plane Coordinate (SPC) marking his opinion of a boundary corner location. The cadastre creator, uncertain of the accuracy of that first boundary corner SPC, has a second land surveyor give an opinion of the location of the same corner by determining a second SPC. This second equally competent land surveyor makes an entirely independent survey based solely on the evidence that he gathers, and computes a second SPC marking the boundary corner based on his opinion (totally disregarding the SPC determined by the first). The location of the two SPC would probably be different. The cadastre creator would have no reason to pick either of the SPC as more correct because equally competent land surveyors determined both. Now assume that 100 equally competent land surveyors perform independent evidence gathering and monument their boundary corner location with a SPC, again totally disregarding any SPC set by the other ninety-nine land surveyors. Since all 100 SPC were set by equally competent land surveyors, it would be impossible to pick one as more correct than any other. However, one could probably detect a central tendency of the SPC, a location around which the SPC marking the professional opinions of boundary corner locations appear to cluster. (Figure 1 shows an idealized example where 100 land surveyors have independently determined the SPC

of a boundary corner. The "central tendency" or most probable location of these locations is marked with a dot.)

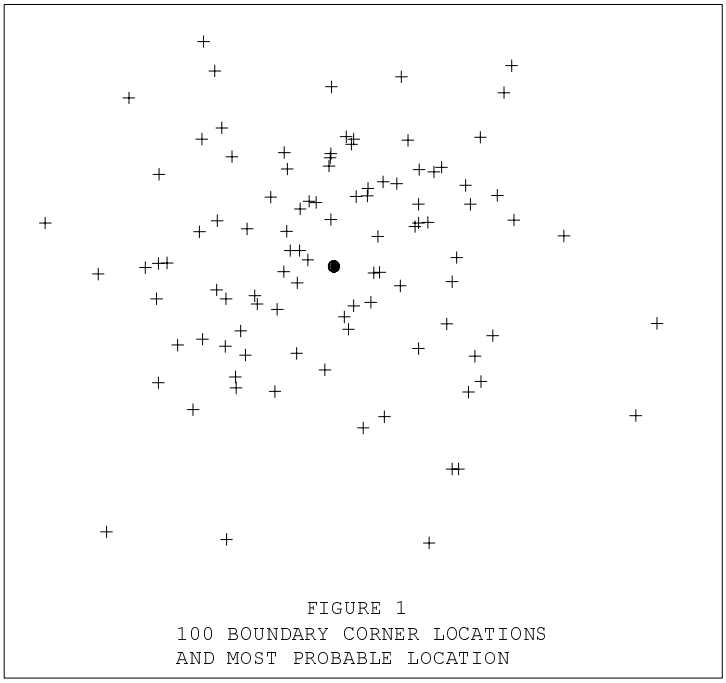


FIGURE 1
100 BOUNDARY CORNER LOCATIONS
AND MOST PROBABLE LOCATION

Thus we have described a procedure in which the independent interpretation and opinion on one land surveyor of the location of a boundary corner on the ground is treated as one "measurement" of that location. If many other land surveyors also form an independent opinion then there are many "measurements" of that boundary corner. Random error theory could then be used to rigorously analyze the discrepancies between land surveyors' opinions of location of boundary corners.

To understand how random error theory can be used in a cadastre, several terms must be defined in the context of boundary corner locations:

Population consists of all possible measurements that can be made of a particular quantity (Wolf 1997). Since the cadastre being studied will be usable for 500 years, then the population of the location of one particular boundary corner are all the locations that could be determined by every land surveyor who ever practices within that 500 year period.

A sample is a subset of data selected from the population (Wolf 1997). There are two reasons why only a sample of the population of all locations of a boundary corner are possible: 1) It would be logistically impossible for an individual land surveyor to determine the location of every boundary corner in a cadastre (for instance, a cadastre the size of the whole Commonwealth of Massachusetts); and 2) Since the population spans a time of 500 years, a sample can only be picked from those land surveyors actually practicing at that particular moment.

True value is a quantity's true or exact value. Since the whole population of boundary corner locations is determined only by opinions of land surveyors and since opinions of land surveyors are not definitive, it is not possible to determine the true value of a boundary location.

Most probable value is that value which, based on the observations of the sample set, has the highest probability of occurring. If all the observations have the same precision then the most probable value is the mean or average of the observations. The SPC defining the professional opinions of boundary corner locations have the same precision because of the assumption that all the land surveyors are equally competent

Degrees of freedom is defined as the number of observations that are in excess of those necessary to solve for the unknowns. Since only one determination of boundary location by one land surveyor is sufficient to define a corner, the number of degrees of freedom is one less than the total number of measurements or professional opinion locations.

Variance is defined as the value by which the precision for a set of data is given. The population variance (σ^2) applies to the entire population while the sample variance (S^2) applies to the sample set of data.

Standard error (σ) is the square root of the population variance. The standard error and variance of the population are indeterminate for two reasons: 1) the total population of observations is not able to be determined; and 2) the true value of a boundary corner location is indeterminate.

Standard deviation (S) is the square root of the sample variance and is an estimate of the standard error of the population. For a sample data set, 68.3% of all the sample data will theoretically lie between the most probable value plus or minus the standard deviation (S).

The total population of a specific boundary corner location consists of all the individual determinations of all land surveyors who ever practice within that 500-year period. Since each determination is merely an opinion, an exact true location is not possible. However, by analyzing this total population, the creator of the cadastre could make a definitive statement about the most probable value (the mean) of the boundary corner location and the precision or discrepancy between all the locations (For instance, the maximum distance within which 95% of all the population locations are located from the most probably value could be determined.)

For example, for a certain cadastre suppose that the location of a boundary corner can only be determined by 100 independent land surveyors. The total population of the location of this boundary corner is therefore the 100 SPC determined by the 100 land surveyors. Figure 1 shows the relative location of the total population of SPC. The most probable location of the "true" boundary corner is the average or mean location of the population, shown in Figure 1 with a dot. The dispersion or precision of the boundary corner can be calculated using random error theory and can be shown by a circle of

uncertainty having a radius centered at the point of most probable location. Figure 2a shows a circle of uncertainty within which 39.4% (1σ) of the total population of 100 SPC should be contained. The 39.4% uncertainty circle has a radius length of $R_{39.4\%}$.

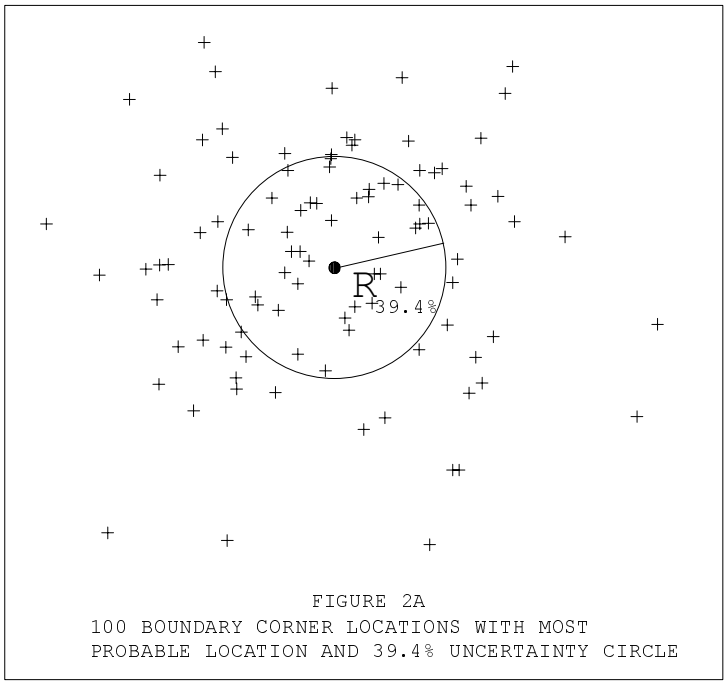


FIGURE 2A
100 BOUNDARY CORNER LOCATIONS WITH MOST
PROBABLE LOCATION AND 39.4% UNCERTAINTY CIRCLE

Figure 2b shows a circle of uncertainty within which 86.5% (2σ) of the SPC should be contained. The 86.5% uncertainty circle has a radius length of $R_{86.5\%}$.

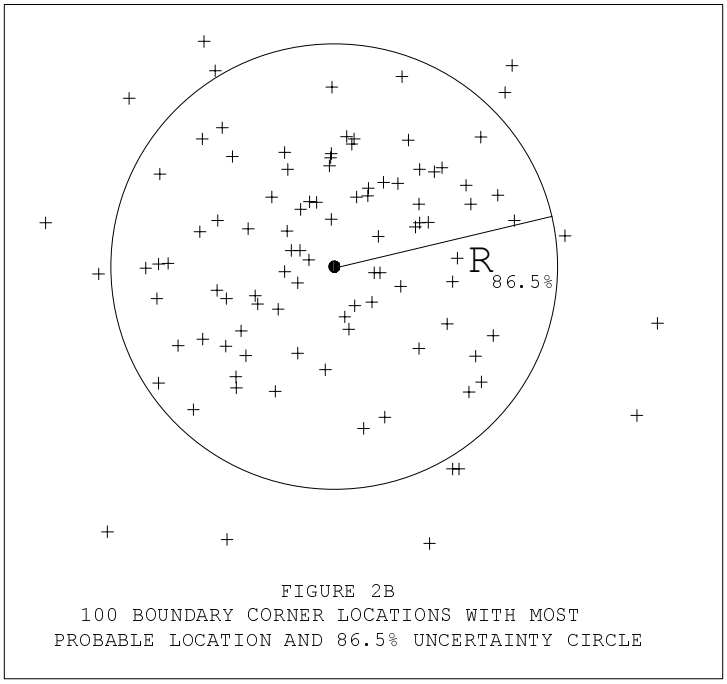


FIGURE 2B
100 BOUNDARY CORNER LOCATIONS WITH MOST
PROBABLE LOCATION AND 86.5% UNCERTAINTY CIRCLE

Figure 2c shows a circle of uncertainty within which 98.9% (3σ) of the SPC should be contained. The 98.9% uncertainty circle has a radius length of $R_{98.9\%}$.

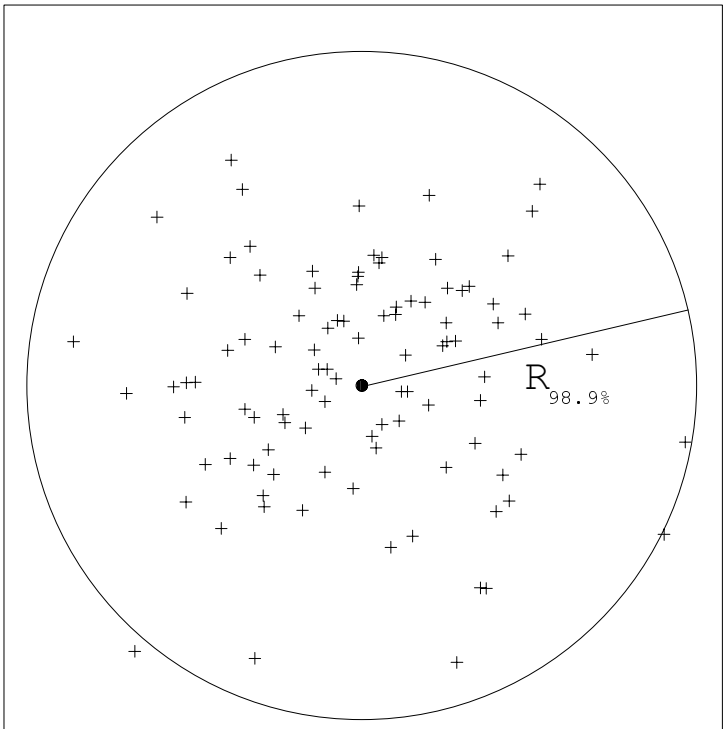


FIGURE 2C
100 BOUNDARY CORNER LOCATIONS WITH MOST
PROBABLE LOCATION AND 98.9% UNCERTAINTY CIRCLE

Using sample sets to estimate total population mean and variance

If the total population is known, then the most probable value and the precision of the boundary corner location can be definitively determined. However, additional analysis of the mean and variance are required if only a sample of the population is used. Suppose a sample of twenty land surveyors out of the total population of 100 are chosen to each independently determine the location of a particular boundary corner. Based on the resulting twenty locations, a sample mean (most probable value) and variance (measure of precision) of the location of the boundary corner could be calculated. Now suppose that a second, different sample of twenty land surveyors are chosen and a second mean and variance are calculated based on this second set of twenty locations. The mean location of each set would probably be different, as would the variance between the measurements of each set. Now suppose that a different sample of twenty land surveyors was chosen from the population a total of fifty times which would result in fifty determinations of the mean location of each group of twenty land surveyors. To determine which sample mean and variance of the fifty separate groups of twenty land surveyors most nearly matches the mean and variance of the total population (as shown in Figure 2) would not be possible. In statistics, the relationship between sample sets, their sizes, and the mean and variance calculated from the sample set values is part of the sampling distribution theory (Wolf 1997).

Example of estimating population statistics using sample sets

For the total population, Figure 2c shows the most probable value of the boundary corner location and the radius of precision for a confidence level of 98.9% (3σ). Now suppose that instead of using the SPC determined by the full population of 100 land surveyors, only the SPC determined by a sample (Sample 1) of twenty land surveyors randomly picked from the 100 could be used (Figure 3 shows all the land surveyors of the total population marked from 1 to 100. Figure 3a shows Sample 1 in which 20 of the 100 surveyors were picked at random.). Based only on the 20 SPC determined by the land surveyors in Sample 1, a most probable value and a 98.9% confidence circle could be calculated, but this would only be an estimate of the actual population's most probable value and the population's 98.9% confidence circle.

1	11	21	31	41	51	61	**71**	**81**	91
2	12	22	**32**	42	52	62	72	82	92
3	**13**	23	**33**	**43**	53	63	73	**83**	93
4	14	24	34	44	54	64	74	84	**94**
5	15	**25**	35	45	55	65	75	85	95
6	16	26	36	46	56	66	76	86	**96**
7	17	27	37	47	57	67	77	87	**97**
8	**18**	28	38	**48**	58	68	78	88	**98**
9	19	29	**39**	49	59	69	**79**	89	99
10	20	30	40	50	60	**70**	80	90	**100**

Sample 1: 20 land surveyors randomly picked from the total population of 100
FIGURE 3A

Suppose a second sample (Sample 2) of 20 land surveyors is randomly picked from the total population of 100 (Figure 3b shows Sample 2 in which a second random selection of 20 surveyors was picked from the total 100.).

1	11	21	**31**	41	51	61	71	81	91
2	12	22	32	42	52	**62**	**72**	82	92
3	**13**	23	33	43	53	63	73	83	**93**
4	**14**	24	**34**	44	**54**	64	74	84	94
5	15	**25**	**35**	45	55	**65**	75	85	95
6	**16**	26	36	46	56	66	76	**86**	96
7	17	27	37	47	57	67	77	87	97
8	18	28	**38**	48	**58**	68	78	88	**98**
9	19	29	39	**49**	59	**69**	79	89	99
10	**20**	30	40	50	60	70	**80**	90	100

Sample 2: 20 land surveyors randomly picked from the total population of 100
FIGURE 3B

Sample 2 would most probably consist of different land surveyors than Sample 1 and therefore the most probable value of the boundary corner location and 98.9% confidence circle calculated using Sample 2 would probably be different than those calculated in Sample 1. There is no reason to pick the results of either sample as more correct in estimating the population's most probable location or 98.9% confidence circle. Figure 3c shows Sample 3 in which a third random selection was made.

1	11	21	31	41	**51**	**61**	71	81	91
2	**12**	**22**	32	**42**	**52**	62	**72**	82	92
3	13	23	33	43	53	63	73	83	**93**
4	14	24	34	44	54	64	**74**	84	94
5	15	25	35	45	55	65	75	**85**	95
6	16	26	36	46	56	66	76	86	96
7	17	27	37	47	57	67	**77**	87	97
8	18	**28**	**38**	**48**	58	68	78	88	98
9	19	29	39	**49**	59	69	79	89	99
10	20	30	40	**50**	60	70	80	90	100

Sample 3: 20 land surveyors randomly picked from the total population of 100
FIGURE 3c

Now suppose that samples of 20 land surveyors randomly picked from the full population of 100 are picked for a total of 50 samples of 20 land surveyors. Each of these 50 samples could be used to estimate the population's most probable location and 98.9% confidence circle. There is again no reason to pick any of the 50 samples as more correct than another. However, there would be probably be a central tendency in the 50 locations of the 50 samples of the most probable locations of the boundary corners and of the length of radius of the 98.9% confidence circle. (Figure 4 shows the results of what could be expected if the results of Samples 1, 2 and 3 are superimposed on each other.)

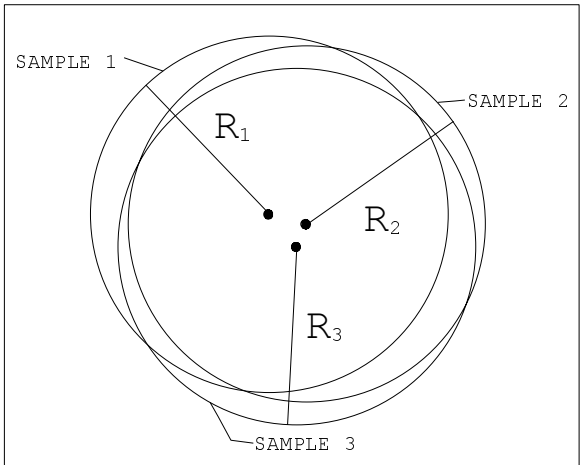


FIGURE 4
MOST PROBABLE LOCATION AND UNCERTAINTY
CIRCLE FOR SAMPLES 1, 2 AND 3

As we have shown, the mean and variance computed from a sample is an estimate of the mean and variance of the population. These estimates vary from sample to sample and thus the sample estimates of the mean, and sample estimates of the variance have their own population distributions. Sampling distribution theory is used to describe or qualify the reliability of the mean and variance estimates so that a statement can be made about how good the estimates are for a given level of confidence of the actual population mean and variance. In other words, for a selected level of probability (or level of confidence), a range or interval can be determined within which the population mean and variance can be expected to fall. This range interval is called the confidence interval. Sampling distribution theory is used to determine the confidence interval for the mean (t distribution) and the confidence interval for the variance (chi-square distribution).

The chi-square distribution compares the relationship between the population variance and the sample variance of the sample sets and is based on the number of observations in the sample set. This distribution is used to determine the range in which the variance of the population can be expected to occur based on 1) some specified probability; 2) the value of the sample variance of the sample set; and 3) the number of degrees of freedom in the sample. The chi-square distribution is therefore used to construct a probability statement about the population variance being within a range centered on the sample variance of the sample set. The larger the number of degrees of freedom (i.e. the larger the number of land surveyors used in determining the boundary corner location), the smaller the range in which the population variance can be expected to occur.

Statement of accuracy of boundary corner locations

The end result is therefore a State Plane Coordinate of the most probable location of the boundary corner with the associated estimated standard error or range or interval of uncertainty for a given confidence level (e.g., Massachusetts Coordinate System, NAD 1983, Mainland Zone, N915,362.361 meters, E86,399.933 meters, plus or minus 0.561 meters at a 98.9% confidence level). (An error ellipse in which the range of uncertainty is split between two dimensions can be used. However this paper will treat the dimension of the semimajor axis of the error ellipse as the one (and only) value of the range of uncertainty.) The range of uncertainty can be made smaller by increasing the size of the sample of the total population (i.e. having more land surveyors participate in independently determining the location of the boundary corner).

Fuzzy Boundary Lines or How to Show Boundary Lines Based on Indefinite Boundary Corners

Now that the cadastre creator has boundary corners defined by a most probable location with an associated standard error, how are boundary lines to be shown on the cadastre? Certainly, if the boundary corners at the end of the property line are definitively known (i.e. the radius of the 100% confidence circle is zero) then the boundary is the one-dimensional line connecting the two corners. However, for boundary corners that have an estimated standard error, the boundary line connecting the two will be a “fuzzy” two-

dimensional quadrilateral within which, for the specified confidence level (e.g. 98.9%), the “true” boundary line will lie. Figure 5 is an example of such a boundary line.

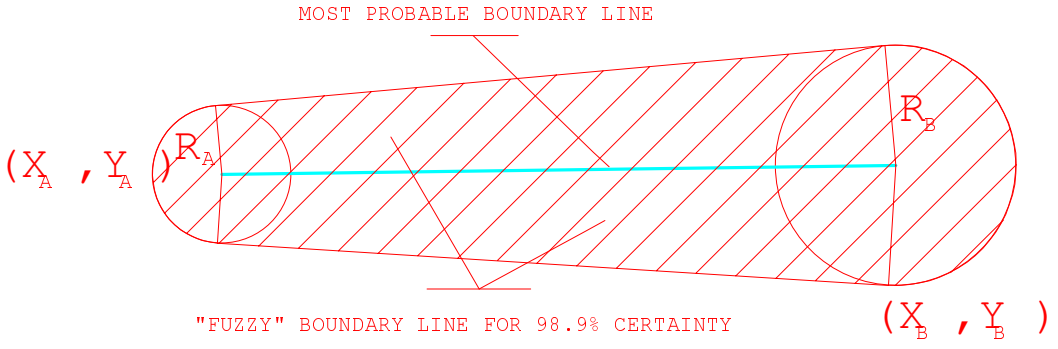


FIGURE 5
FUZZY BOUNDARY LINES CREATED BY UNCERTAIN BOUNDARY CORNERS

In Figure 5, the boundary line between boundary corners “A” and “B” is to be determined. For boundary corner “A” the cadastre creator has determined a most probable State Plane Coordinate location of (X_A , Y_A) and the 98.9% uncertainty circle has a radius length of R_A . For boundary corner “B” the most probable location was found to be (X_B , Y_B) and the 98.9% uncertainty circle has a radius length of R_B. The most probable location of boundary line A-B is the line connecting the most probable locations of corners “A” and “B” shown as the heavy solid line in Figure 5. However, with 98.9% confidence, the true boundary line is located somewhere within the shaded region of Figure 5.

In like manner, other boundary lines of a cadastral parcel of land can be shown. Figure 6 shows the procedure applied to a four-sided parcel of land (note that the scale of the “fuzzy” region is blown up to show detail).

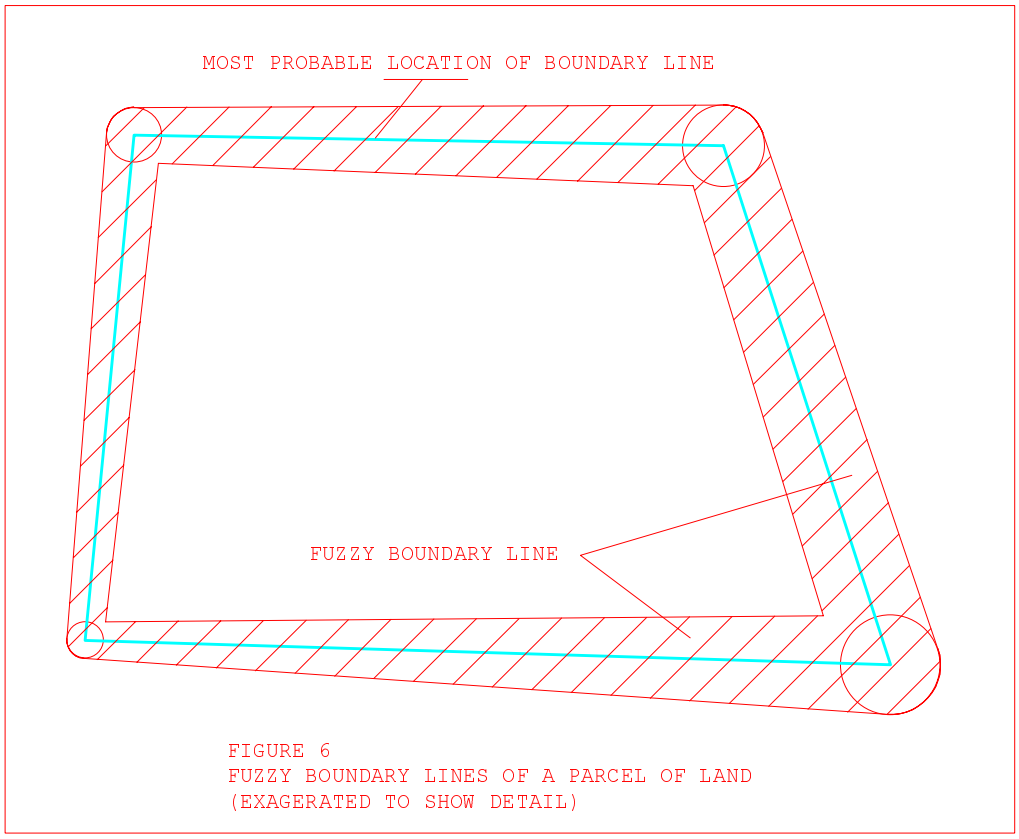


FIGURE 6
FUZZY BOUNDARY LINES OF A PARCEL OF LAND
(EXAGERATED TO SHOW DETAIL)

Conclusion

The success of a cadastre requires that the true and actual boundary lines of each individual parcel be shown. For colonial or metes and bounds states in the United States of America the true and actual boundaries cannot be definitively determined but rather are determined by professional opinion of land surveyors. Equally competent land surveyors working independently in determining the State Plane Coordinate (SPC) of a boundary corner location will probably result in many different locations for the boundary corner.

Random error theory can be used to analyze the total population of the separate locations of the one boundary corner in question (the separate locations determined by every land surveyor who was, is or ever will be in practice while the cadastre is in existence). Thus a cadastre creator can state to a user that based on all the possible locations of the boundary corner, the corner shown on the cadastre is the most probable location and the 98.9% confidence circle has a radius of some length (e.g., Massachusetts Coordinate System, NAD 1983, Mainland Zone, N915,362.361 meters, E86,399.933 meters, plus or minus 0.561 meters at a 98.9% confidence level (3σ)). The user would therefore know that 98.9% of all land surveyors who ever determine that particular boundary corner would locate it within that confidence circle (the radius of which would be 0.561 meters in the example).

Only a sample of the total population can be determined at one time and therefore the most probable location and confidence circle of the population can only be estimated by using the sample. Sampling distribution theory is used to determine the confidence interval for the most probable location or mean (t distribution) and the confidence interval (circle) of the variance (chi-square distribution). The larger the number of degrees of freedom (or the larger the sample or more land surveyors used to determine boundary corner locations), the smaller the range in which the population mean (most probable boundary corner location) and population variance (confidence circle) can be expected to occur.

Using this system, for each boundary corner of a parcel, a cadastre creator would have a sample of boundary corner locations determined by using the professional opinions of at least two (and optimally many more than two) land surveyors who each independently determine the boundary corner location. Using sample distribution theory (t distribution and chi-square distribution) the cadastre creator could give an accuracy statement of the location of the boundary corner as shown on the cadastre. The cadastre itself would show boundary lines connecting the boundary corners as "fuzzy" lines of uncertainty.

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PROGRESS TOWARD INTEGRATING SPATIAL TECHNOLOGIES

FOR URBAN INFRASTRUCTURE MANAGEMENT

Elizabeth K. Burns and Elizabeth A. Wentz
Department of Geography
Arizona State University
Tempe, AZ 85287-0104
Ekburns@asu.edu

ABSTRACT

Small isolated communities with few technical, managerial, and data resources face considerable obstacles in setting up a modern data management system. These cities and towns have the initial problem of inventorying their infrastructure as the basis for on-going operations and construction activities. Spatial technologies of global positioning systems (GPS) and geographic information systems (GIS) can provide high-precision, low-cost solutions for documenting infrastructure location. Benefits include the ability to plan and manage each portion of a utility network. In November 1997, the City of St. Johns, Arizona, took the opportunity to inventory its water and sewer facilities as part of the planned installation of natural gas pipelines. Global positioning satellite readings of pipe, valve, and manhole locations were saved as separate GIS layers. Apache County tax assessor parcels were attributed with occupant addresses after a utility customer-billing file was updated with parcel numbers. The general issue of integrating the precise City GPS facility inventory with the general accuracy of the County GIS parcel layer was resolved by adjusting parcel boundaries.

OVERVIEW

Infrastructure management is a core concern for large and small urban governments. Municipalities often provide water and sewer service to individual parcels also served by telephone and electric systems. Inventorying facilities, operating and maintaining these systems, and billing users are common tasks. Spatial technologies of GIS and GPS are in general use in large communities to inventory, maintain, and monitor extensive utility systems. GIS allows integration of related datasets such as parcel boundaries, pipe locations, and street center lines with addresses. GPS offers precision in location. Small communities using these technologies can reap similar benefits in allocating their scarce resources of staff time, construction funds, and maintenance activities.

Getting a GIS system in place is a first step for any community before the full potential of a system of data management is in place for querying, updating, and combining with GPS information and other municipal datasets such as land use and zoning. Literature on information systems provides a basis for

understanding how spatial technologies are being merged with managerial practices (Burns, 1994; Burns, Fry, and Scott, 1998). In this case study, sufficient conditions for a successful project existed:

- The partial availability of needed files,
- The resources to plan for new infrastructure installation,
- The opportunity to add a city GPS-GIS project to a county GIS project, and
- The availability of a qualified technical team trained in GPS and GIS.

BACKGROUND

The City of St. Johns, Arizona, has a distinctive regional situation and local site. The town, 1990 population of 3,400, lies in northeastern Arizona on the Colorado Plateau along the Little Colorado River. It serves as the County seat for rural Apache County, 1990 population of 43,000. Apache County extends 220 miles from the Navajo nation in the north to established towns and increasing numbers of retired urban immigrants living in isolated subdivisions in the south. St. Johns is a two-hour drive from Flagstaff and a five-hour drive from metropolitan Phoenix.

Colonizing Mormon pioneers moving south from Utah laid out its town plat in 1870s. Its grid plan has the streets 100 feet wide, blocks of four acres, and large original parcels designed for a nineteenth-century agricultural society (Reps, 1979). This original town plat, laid out by compass and chain, fits within the U.S. surveying grid of township and range lines. There are small adjustments in street right-of-way and block layout for the regional State Highway 191. The present population lives primarily within this original town framework. One-story houses on large wooded lots front on wide streets. Two paved lanes are in the center of each street. Sidewalks, water and sewer lines are located in the unpaved rights-of-way. Storm water ditches are next to property lines. Recent plat expansions to the north and west include stub streets to a mobile home park, high school complex, a local church, and the airport. Water and sewer lines extend outward from the original town plat along section lines to serve a few new houses.

In November 1997, an opportunity arose to create an accurate inventory of the locations of water and sewer facilities. A natural gas company planned to install pipelines in portions of the city. Municipal approval was dependent on resolving concerns that construction would not disturb existing services. The complete locations of current water and sewer facilities were not known, however. Outdated partial paper maps and the memories of City staff members provided the day-to-day basis for operating activities such as adding and replacing pipes and stopping leaks. Moreover, potential natural gas customers on specific parcels could not be identified. There is no home delivery mail service. City customer utility billings are sent to owners at post office box mailing addresses rather than to occupants at unique street addresses. Clearly,

information on accurate facility locations and corresponding occupant addresses was needed.

In March 1997, Arizona State University staff and students started updating existing Apache County tax assessor parcel files as part of an extensive GIS database development. This research group approached the St. Johns' city manager with a proposal to create an infrastructure inventory by combining the accuracy of global positioning satellite readings with the spatial data integration of geographic information systems. With the County staff's approval, this additional project made updating the St. Johns parcel maps a high priority.

APPROACH

An initial survey of current information about the City's infrastructure quickly revealed that existing, but scattered, resources could be assembled for portions of the inventory. Detailed information about water and sewer lines existed as incomplete, outdated paper maps and in the memory of experienced staff members nearing retirement. Apache County tax assessor files contained owner addresses for individual parcels and confirmed that there are substantial numbers of absentee owners. A City of St. Johns' utility customer billing file provided a partial list of parcel occupants and addresses that could be linked to parcels.

Infrastructure inventory using a global positioning satellite system

Arizona State University staff conducted GPS field surveys in winter 1997 and spring 1998 with City staff. These surveys located water pipes and valves and sewer lines and manholes to an accuracy of four inches. A RELIANCE GPS system was used consisting of a base station and a receiver. The two equipment cases and tripod fit in a station wagon trunk.

The following procedures were used. Scheduling of fieldwork required identifying the availability of a minimum of four satellites on workdays for City staff. Once these periods were known, a team of at least two people traveled to St. Johns with the equipment. Each day, this team first set up a base station at the known point of latitude, longitude, and elevation located at the St. Johns regional airport about two miles from the center of town. This point was established during a 1995 Federal Aviation Administration GPS airport survey. Acquisition of this signal took no longer than five minutes. The team then picked up the City staff members who were available that day to start the field work. Because the temperatures were often as low as 5 degrees Fahrenheit and shortened battery life, the base station was housed in a City pickup with the heater on. The GPS rover was kept within the vehicle as much as possible. Someone sat either at the folded-down station wagon trunk or at the open door so that the rover had access to acquire satellite readings. The vehicle then was

driven slowly along and stopped where there were water valves and sewer manholes. The water and sewer line readings were saved as a line file in the GPS software, while the valves and manholes were saved as a point file. These separate layers were then imported as ArcView files at the end of each day. Because of low temperatures, two sets of batteries were required each working day and batteries were recharged overnight. Emergency activities of the City staff and the short daylight hours of the winter season regularly interrupted the fieldwork. The fieldwork was completed in four trips.

Quality control issues emerged. The discussion between university and city staff in the field quickly became a dialogue about the history of local installation. Both GIS displays of the GPS locations, which were available the next day after a field survey, and later paper plots, provided once the initial survey was done, jogged staff memories of pipe locations, connections, and widths. Sewer pipes, for example, do connect throughout the town although the initial field survey missed some segments; these missing segments and extensions were readily identified and corrected. The staff refined the fieldwork by commenting on just where pipes were placed in relation to each other and to nearby property lines on the GIS tax assessor parcel data layer.

While the town plat appears to be a regular grid, the GPS readings show the variations on the ground, including streets that diverge from their true headings and blocks that are rhomboids, not rectangles. These results confirm an early 1980s City land survey that showed where the streets actually went. This mismatch between the original plat and actual layout illustrates a common problem that plagues efforts to integrate GPS with pre-GPS data.

Parcel occupant addresses using geographic information systems

A parallel task involved identifying the mailing address of each parcel's occupant and attaching this information to the GIS coverage of tax assessor parcels. Two existing data files were combined into a revised utility customer-billing file with a common field that could be added to the GIS parcel coverage. A data field for tax assessor number is present in both the revised billing file and the parcel coverage.

A first step merged the original utility customer-billing file with a data file provided by the Arizona Department of Revenue. Both the billing file and the revenue file had a field that contains the mailing address for the parcel record. There are several duplicate mailing addresses in both files, however, as many people are the owners of many separate parcels, each with a unique assessor number. In order to achieve a better match rate when the data files were joined, unique identifiers were added to the mailing address field for each duplicate record in both the utility and revenue files. The City of St. Johns provided the assessor numbers for the duplicated mailing address records in the billing file.

Each separate utility customer billing record could then be matched with the correct assessor number when the two files were joined.

A second step involved merging the data files. The utility customer billing file and revenue files, originally in Excel, were exported in Dbase formats and converted to INFO files for use in ArcInfo software. The JOINITEM command joined the two data files. This revised utility customer-billing file now contains the tax assessor number data in the “parcel” field.

A third step involved attributing the parcel coverage with the revised billing file. Using ArcInfo JOINITEM command again, the revised utility billing file was joined to the polygon coverage that represents the land parcels in the City of St. Johns. The tax assessor number, contained in the “parcel” field in the revised utility billing file, was used as the common item between the two datasets. The data contained in the utility billing file are now present as attributes for the parcel polygons in the coverage. Each parcel is attributed with the assessor number, ownership, and utility customer billing information.

This process of first joining two data files and then joining the resulting data file to the GIS parcel coverage was effective. There are 2,119 total parcels in the St. Johns GIS tax assessor parcel coverage. While 1,578 parcels had tax assessor addresses, the limiting factor was the number of service address records in the customer utility billing file. Some 1,018 (86%) of the 1,170 utility customer billing address records matched tax assessor parcel addresses in the Arizona Department of Revenue file and could be added as attributes to the GIS parcel coverage. Parcel attribute data questions remain. The City of St. Johns expects to identify the parcels of the 152 unmatched utility billing records through direct contact with customers. Apache County is evaluating the paper and electronic records for the tax assessor parcels (540) with no address information or tax assessor number (218).

INVENTORY RESULTS

The inventory of water and sewer infrastructure can now be viewed in combinations of GIS layers based on GPS locations. A comprehensive view of the study area appears in Figure 1. Two separate layers for water lines and water valves are displayed on the adjusted parcel base map. Water lines provide denser network service along original town plat streets and extend service along selected section lines. Gravity sewer lines (not shown) display a tree structure that drains to the north. This figure also shows the interruption in the eastern portion of the original town plat by the diagonal route of State Highway 191.

Figure 1. The location of water lines and valves in relation to tax assessor parcels for the City of St. Johns

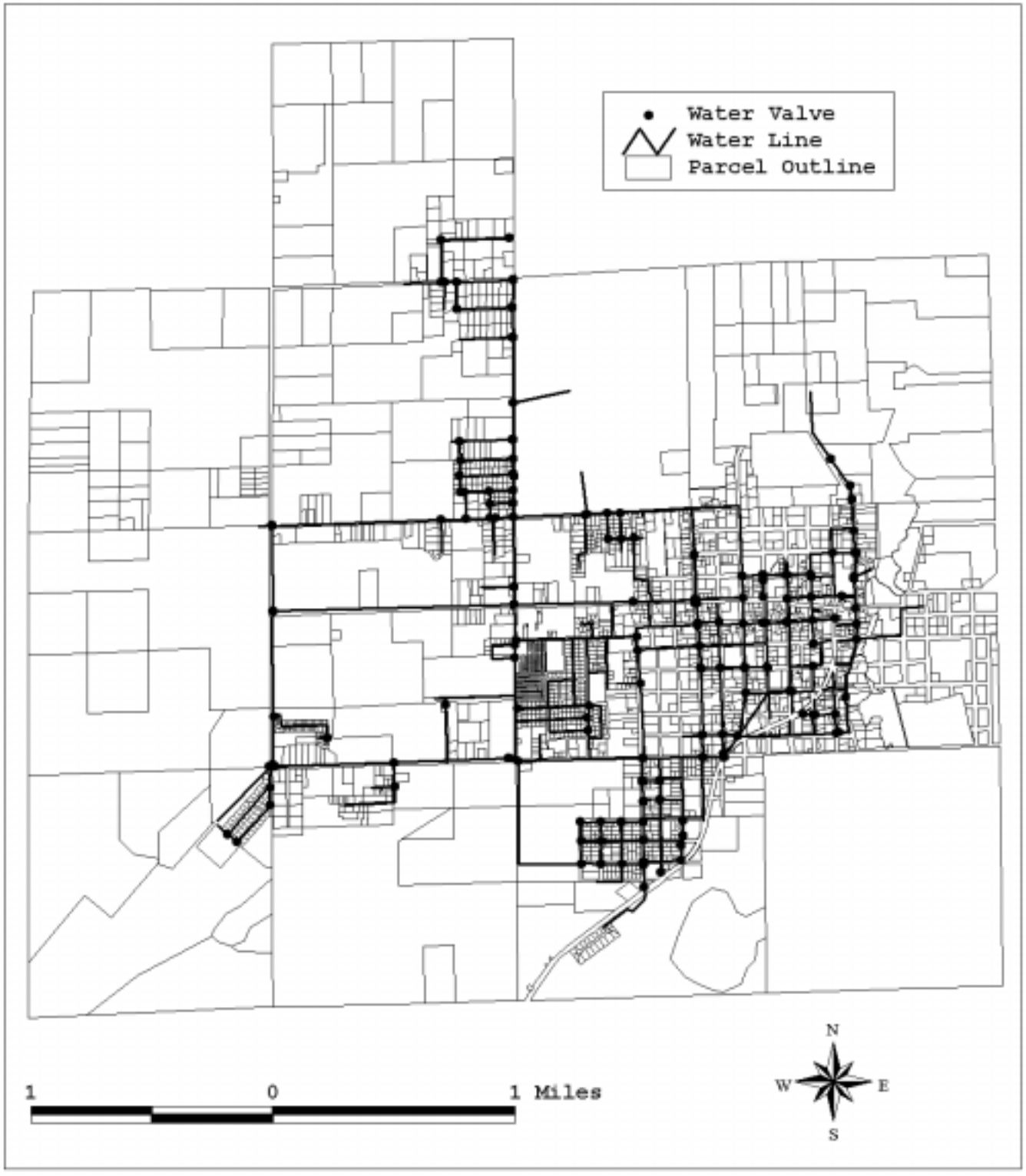

Water Valve
Water Line
Parcel Outline
1
0
1 Miles
N
W
E
S



Figure 2. A detail of water and sewer lines showing placement in street rights-of-way and adjacent parcel boundaries.

The City of St. Johns staff now uses the combined detail of all water and sewer facilities for planning, operations, and maintenance activities. Figure 2 shows an enlarged segment of the original town plat with water lines and valves as well as sewer lines and manholes. The placement of lines in the street rights-of-way provides critical information for installation of new natural gas lines. Water and sewer lines are usually installed on opposite sides of a street; water meters occur in clusters at intersections. Current parcel boundaries show the extent of lot fragmentation within the original town blocks.

Future utility management is expected to benefit from the revised utility customer-billing file, especially as the data problems identified during the file revision are resolved. This file's data fields now include: (1) customer name, (2) parcel – a blank field initially that is now filled with the tax assessor number, (3) service address, (4) mailing address, (5) city, (6) state, and (7) zip code.

DISCUSSION

This paper documents how small communities can move into more sophisticated management of their infrastructure through the use of spatial technologies. In this case study, sufficient conditions for a successful project existed. They included (1) the availability of existing data files, (2) the financial resources to plan for new infrastructure installation, (3) the opportunity to add a city GPS-GIS project onto an existing county GIS project, and (4) the availability of a university team trained in GPS and GIS.

One significant technology integration issue emerged that will be a recurrent problem for other spatial technology integration efforts. The precise global positioning satellite and generally accurate geographic information systems results represent different levels of spatial detail. Combined results may appear inaccurate (Huxhold, 1991). In this case, the precise locations of the water and sewer facilities appeared inaccurate when displayed in relation to the tax assessor parcel boundaries that were based on earlier land surveys and assumptions of regular street and blocks. This city made the decision to adjust the individual parcel outlines to appear accurate in relation to the water and sewer lines. More extensive adjustments would have required comprehensive GPS surveys throughout the entire county. The result is two GIS layers of City of St. Johns tax assessor parcels. The “adjusted” layer is used with the GPS results as displayed in Figures 1 and 2. Both Apache County and the City of St. Johns retain the “official” parcel layer.

The initial preparation of spatially referenced infrastructure files moves a city’s operations beyond the stage of a mixture of electronic files, paper records and personal memory for its infrastructure management. A new operations situation, however, emerges. Staff unfamiliar with spatial technologies and related software must take over the maintenance of files once the consultant team leaves. In this case, the inventory project increased staff technical awareness. Each time the staff viewed a GIS map, they recognized field errors more quickly. Actual software training has started. Continued commitment by the City administration ensures that the present inventory effort will become the basis for future operations.

ACKNOWLEDGMENTS

We wish to gratefully acknowledge the funding support of the City of St. Johns, Arizona, and particularly the assistance of Mr. Steven Anderson, City Manager. Further acknowledgments go to Joshua A. Burns and Max M. Wyman for their fieldwork and to Charlene Howard McDonald and Maria Asteriadou for their computer assistance.

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The Development of 2D and 3D Triangular Irregular Network Programs

Alias Abdul-Rahman and Jane Drummond
Department of Geography and Topographic Science
University of Glasgow
Glasgow G12 8QQ
United Kingdom

Email:
alias@geog.gla.ac.uk
jdrummond@geog.gla.ac.uk

Abstract

This paper concentrates on a development of two-dimensional (2D) and three-dimensional (3D) Triangular Irregular Network (TIN) in raster domain. Two programs for generating the 2D TIN and 3D TIN were presented. Fundamental concepts involved in the program development, i.e. the concept of Borgefors Distance Transformation (DT), and Voronoi tessellations were highlighted. We demonstrated the application of the algorithms using rasterized data sets generated by our 2D and 3D rasterizing programs. Examples of the 2D and 3D TIN were given. Finally, the paper discusses the possibility of using the generated TIN data structures for future development of 3D spatial information system.

1.0 Introduction

Data structuring for terrain surface data has been studied and investigated since the late seventies (Peucker *et al.*,1978), when the suitability and the adaptability of data structures for terrain surface representation were considered. Then a triangular irregular network (TIN) data structure was established. Several methods and techniques have emerged for TIN generation since then (McCullagh and Ross, 1980; Watson, 1981; Mirante and Weingarten, 1982). Most of the developments were in the vector-based domain and computationally complex (de Berg, 1997). An alternative to the vector-based algorithm, a raster approach, was proposed (Pilouk, 1992; Chen *et al,* 1994; Pilouk, 1996). Since the alternative approach provides for far less complex computation, we adapted this approach in our 2D and 3D TIN development with an intention of developing a 3D spatial information system. The 2D and 3D TIN are part of the ongoing research and they form the major discussion of this paper.

We will describe the development of the 2D TIN program in the second section, and the 3D TIN program follows in the third section. The programs involve three major tasks; distance transformation, Voronoi or Dirichlet tessellations, and triangulations. The end result of each development is presented graphically in the form of a wireframe visualization. The remainder of the paper is devoted to the discussion on

the possibility of further development of the TINs programs in object-oriented (OO) environment with a 3D GIS prototype in mind.

2.0 The 2D TIN Program

The program consists of three tasks. The tasks are:

$ Distance transformation,
$ Voronoi tessellation, and
$ Triangulation.

Distance transformation (DT)

The task of DT is to generate a distance-transformed image of object pixels. Object pixels of a raster image may be in the form of random points, digitized points, digitized lines, etc. In this algorithm, the DT works as follows: all object pixel=s are changed to zero (i.e. a value 0) and the rest of the pixels (i.e. the background pixels) to the highest possible value. In this case, the highest integer value of 32767 could be used. Second, scan the image in two passes (i.e. forward and backward passes) using a 3 x 3 mask of Chamfer 3-4 of the Borgefors DT (Borgefors, 1986), see Figure 1.

The forward pass (scans with upper-mask) begins from the first or top-left pixel and goes to the last pixel of the image. During the forward pass, all the pixels which were covered by the mask get a new value. Each pixel=s value was added either to 3 or 4 depending on the pixel location. Then, the minimum value is determined from the five possible candidates and assigned to the current pixel location. The mask is then moved to the next pixel location. At this next location, the minimum value for this pixel is again determined and assigned. This process continues to the last pixel location (i.e. the bottom-right pixel) in the image. The result of the forward operation is used for the second pass which operates in reverse (from the last pixel to the first pixel). It is a recursive operation. Finally, a DT image is created after these two passes were carried out. Thus, all pixels will contain the approximate distance to the nearest object pixel.

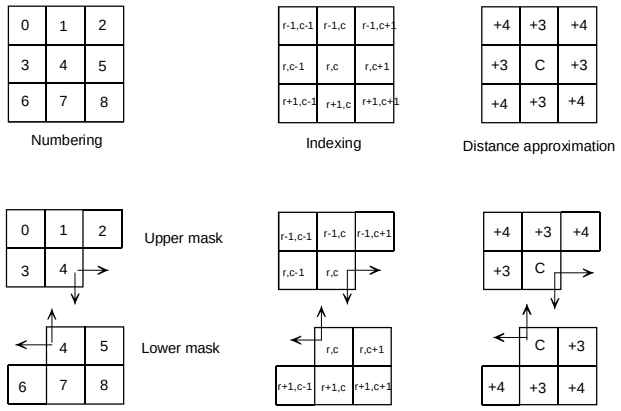


Figure 1 Masks for the DT

Voronoi tessellation

The Borgefors DT and Voronoi tessellation tasks could be performed in parallel or in

stages. In this algorithm, we carried out the tasks in parallel. If we re-examined the DT as described in the previous section, it involved three steps, first, change the object pixel value to zero (i.e. 0) and the background image to the highest possible value. Second, determine the minimum value of the current pixel location among five possible candidates of the upper mask. Third, assign the minimum value to the current pixel location. In other words, the pixel value represents a distance value of the pixel as calculated from the nearby object pixels. To obtain the Voronoi-tessellated image in parallel, the value of an original pixel needs to be assigned to the current pixel location and written to a different output file. Figure 2 shows the parallel process for the DT and Voronoi tessellation.

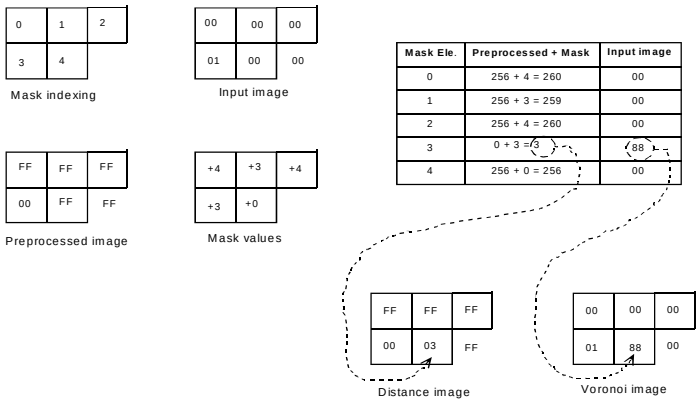


Figure 2 DT computation and Voronoi image generation during the forward pass

In the Voronoi tessellation task, the result of the forward pass is used as an input for the backward pass. We tested the algorithm by using digitized contours. Figure 3 and Figure 4 show the DT image and the corresponding Voronoi tessellation image.

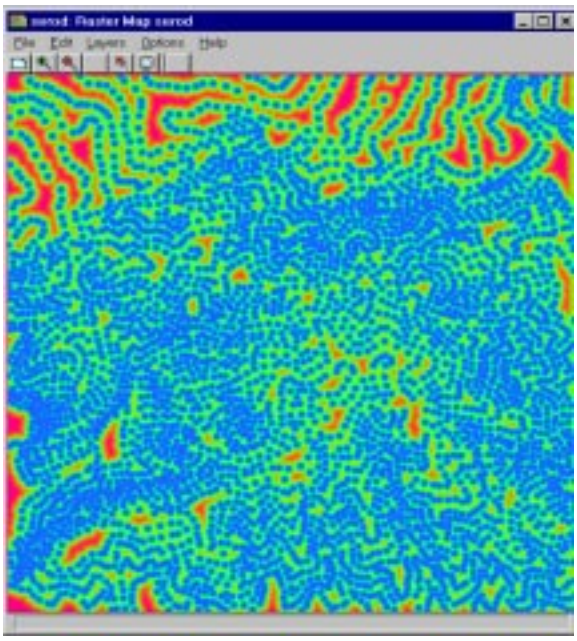


Figure 3 DT image of digitized points

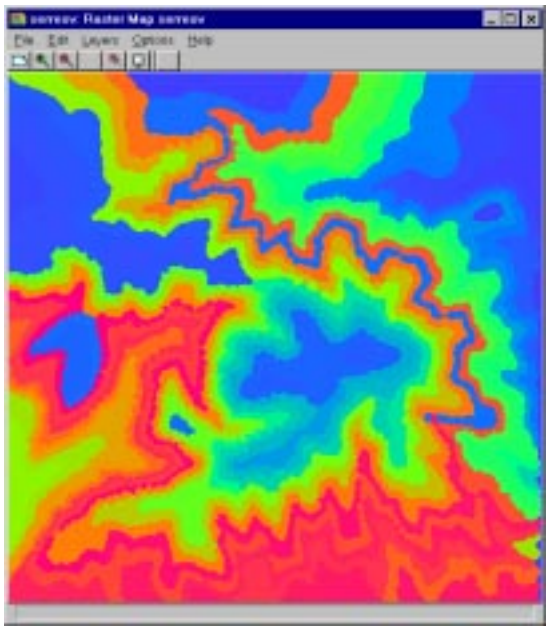


Figure 4 The corresponding Voronoi image of the digitized points

2D Triangulation

In this development, triangles were generated based on the Delaunay triangulation principle and attributed to Delaunay (1934). As shown in Figure 5, basically we need three kernel points of the neighbouring Voronoi polygons to form a triangle. If we have more than three neighbouring polygons, say 4 polygons, then there will be two possibilities for triangles formation.

A correct TIN topology must be established, and this is done by searching 3 Voronoi polygon neighbours. In order to find a unique set of 3 points from a Voronoi-tessellated image, a 2 x 2 mask is used. The mask is designed to detect only two specific situations where 3 or 4 different pixel values fall inside the mask at a time. These different pixels correspond to the neighbouring Voronoi polygons and the kernel points of these polygons were used to form the triangle. Figure 6 shows the mask for detecting the triangle topology.

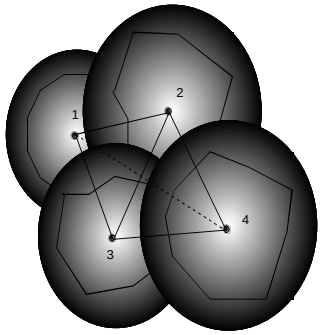


Figure 5 The two possible triangles formation

A triangle is found if the four different pixels match either of the imposed matching conditions. The mask was separated into two parts with the aim of avoiding overlapping (crossover) triangles, a situation not allowed in the Delaunay triangulation criteria.

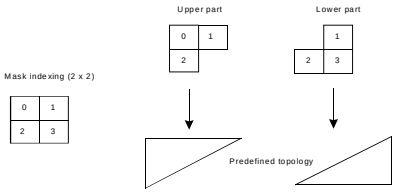


Figure 6 Mask (2 x 2) for TIN topology detection

The next task in the pipeline is to establish the triangle=s attributes so that one could make use of the generated triangles, e.g. for queries and visualization purposes. For query purposes, a more complicated structure needs to be established. Some of the issues in TIN data structuring were studied by Abdul-Rahman (1992). However, in this development, a simple structure is devised where

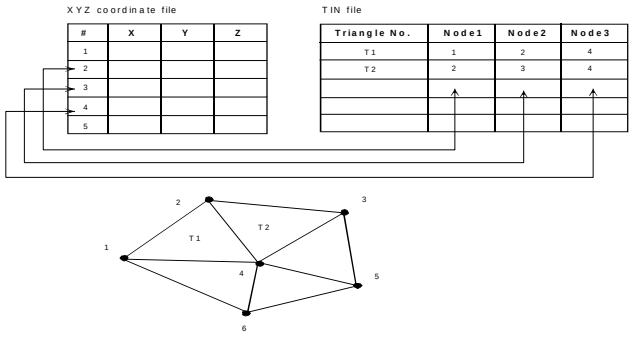


Figure 7 The link of XYZ coordinates and the TINs

the purpose is only for visualizing the TINs; see Figure 7 for the structure.

TINs visualization

It has been claimed in de Berg (1997) that visualization of TINs is one of the major issues in TIN development. In this paper we have only developed a simple display program for visualizing generated TINs. One of the fundamental tasks of any GIS or DTM package is to perform the visualization of data.

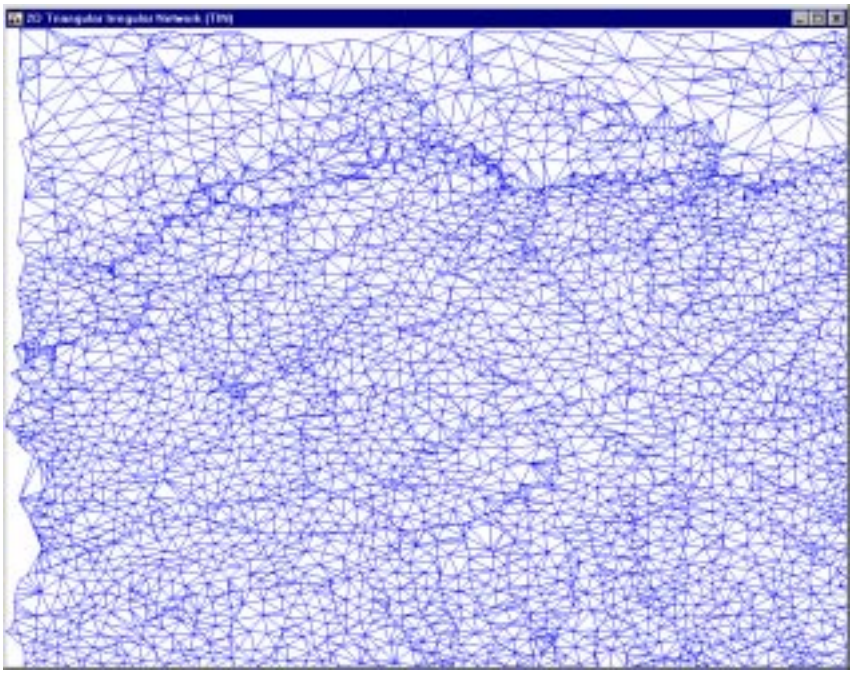


Figure 4

Figure 8 shows a simple TIN visualization output developed in this process. The program takes two input files, a XYZ coordinate file, and the TIN table file. The triangles three nodes (i.e. Node1, Node2, and Node3) can be linked to the corresponding XYZ coordinate table for the nodes with the appropriate pointers as shown in Figure 7. Based on values in the XYZ file triangles could be shaded according to slope, elevation etc., for further visualization.

3.0 The 3D TIN Program

3D Distance transformation

Digital distance transformations in 3D have been considered for more than a decade and the technique has been applied in a number of scientific fields such as image processing (Borgefors, 1996). In this paper, we utilised the DT technique to generate a DT image, a Voronoi image, and tetrahedrons. The previous 2D DT algorithm can be extended to the third dimension relatively straightforwardly by due to the nature of the raster data structure. Thus, the same DT principle is utilised for the 3D TIN development. We used a 3D mask of dimension 3 x 3 x 3 as proposed by Borgefors (1996) known as Chamfer 3-4-5, see Figure 9. Other types of mask are also applicable such as the Chessboard mask, and the City-block mask (Borgefors, 1996).

The Chamfer mask is used due to its computational simplicity and is capable of generating quite accurate distance images. Each voxel in the mask is assigned a local

distance either with a value 3, 4 or 5, depending on the voxel location, see Figure 9. The centre voxel is surrounded by 26 other voxels in x, y, z directions, where each voxel has three types of neighbours. They are called face neighbours, edge neighbours, and node or vertex neighbours. The face neighbour voxels are assigned the value 3, the edge voxels the value 4, and the vertex voxels the value 5.

To generate a distance image of a 3D raster image, the first step is to set the voxel

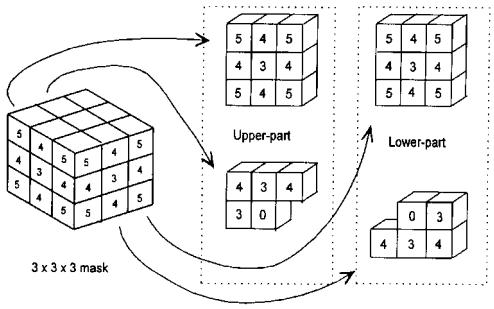


Figure 9 The 3-4-5 mask for the 3D DT

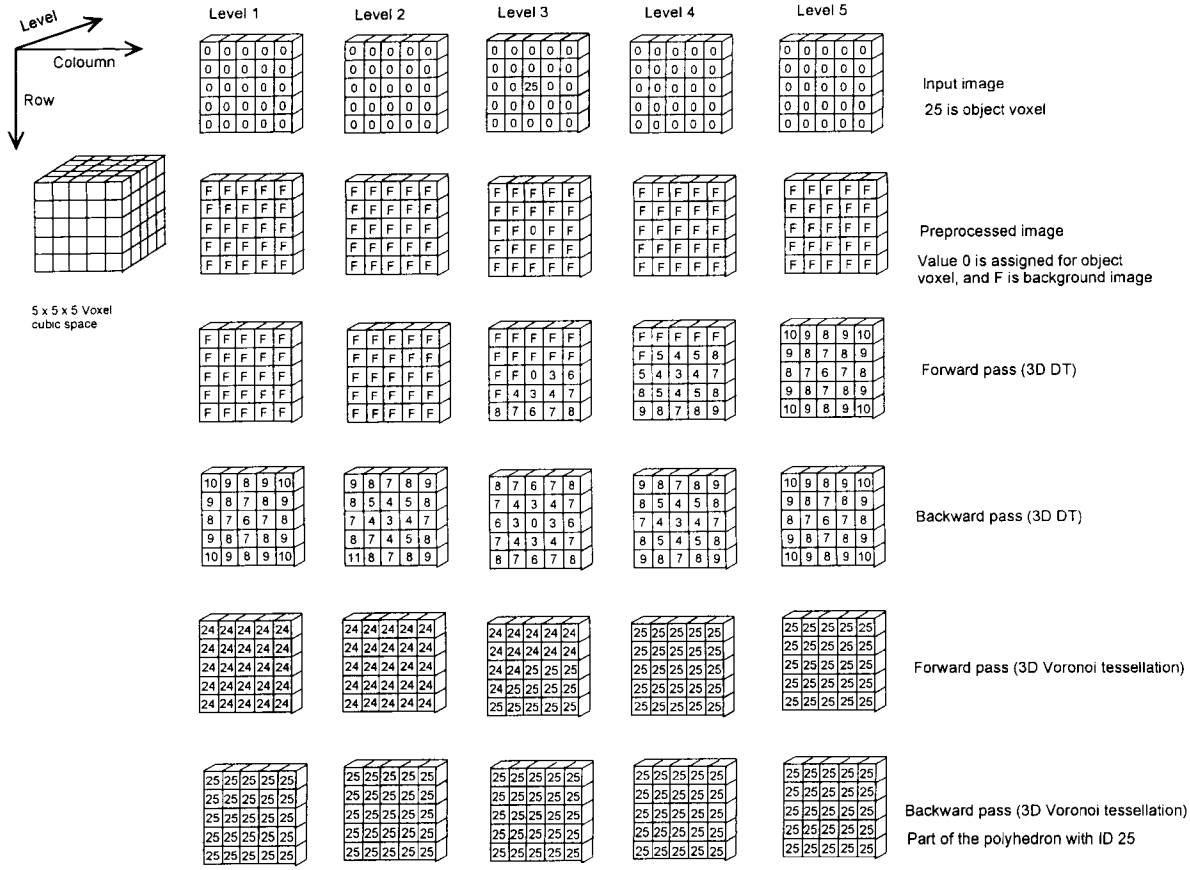


Figure 10 Slice of images (along the Z or level direction) for the 3D DT and 3D Voronoi tessellation

background image with highest integer value and the object voxels with zeros (i.e. 0). The image is then scanned in two passes, i.e. forward and backward passes. The forward pass (using the upper-part mask) begins from the first voxel to the last voxel. At this stage, the voxels surrounding the object voxels will get new values. The new value is the minimum distance from the 14 possible voxel candidates. The result of the first pass is taken into account for the second pass. This time, the image is scanned with the lower-part mask (i.e. the backward pass) beginning from the last voxel and moving to the first voxel, see Figure 10 for the accumulated distance of the

5 x 5 x 5 cubic space. Finally, a 3D distance-transformed image is formed after the two passes were carried out.

3D Voronoi tessellation

A Voronoi image is generated from the DT image. Again, we generated these two images in parallel. The task also involves three steps: First, cover the image with the mask. Second, the values of the mask are added to the value of the voxels being covered by the mask. Third, a minimum value from the 14 voxel candidates is determined and assigned to the current voxel location. The original voxel value of the current voxel location is taken, assigned, and written to the 3D Voronoi file. This is done prior to the mask being moved to the next voxel location. The process continues until the last voxel of the image is reached. Again, the result of this forward pass is taken into account in the backward pass which begins from the last voxel proceeds to the first voxel of the image. Figure 10 shows how the 3D Voronoi polygons (i.e. polyhedrons) were delineated from one object voxel with ID = 25. In other words, a polyhedron of the voxels with ID 25. Visualization of the 3D DT and 3D Voronoi images or polyhedrons can be by a true 3D viewing package.

Tetrahedron Network (TEN) Generation

Using the same principle as for the 2D TIN, the algorithm for the 3D TIN utilised a mask of 2 x 2 x 2, see Figure 11. It has 8 voxel elements. It provides a unique way of establishing a tetrahedron. In order to obtain non-overlapping tetrahedrons, several predefined conditions have to be imposed during voxel scanning. There are 6 possible non-overlapping tetrahedrons that we can get from the mask. The mask is then used to scan the voxel=s Voronoi tessellated-image once.

Once the tetrahedron was detected (based on the imposed conditions), it is then written to a file. The file contains a record of tetrahedrons where each record has 4 nodes, it is an ASCII file and structured as in Figure 12. Thus, it is one way of establishing a simple tetrahedron data structure. This data structure together with a table of point coordinates provide a means for further manipulation of the data, e.g.

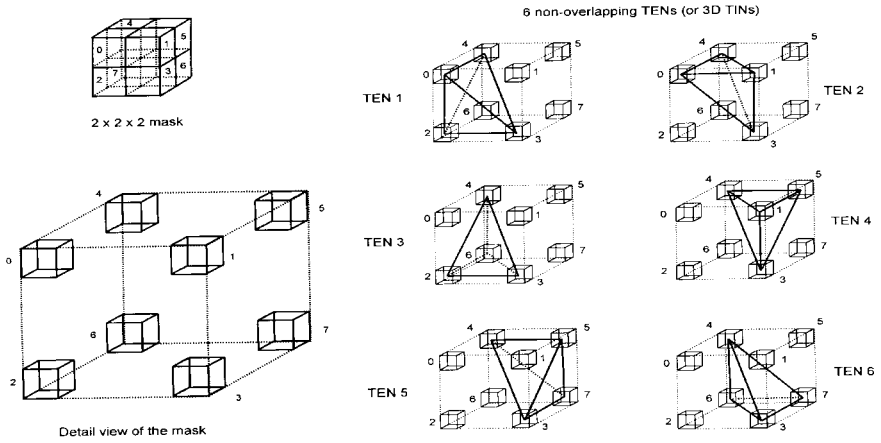


Figure 11 The six non-overlapping TENs

visualization.

We implemented the algorithm and tested it by using simulated 3D raster data sets. This data set was generated by our 3D point-to-raster program. We also developed a wireframe display program for visualizing the TENs, see Figure 13 for the output display.

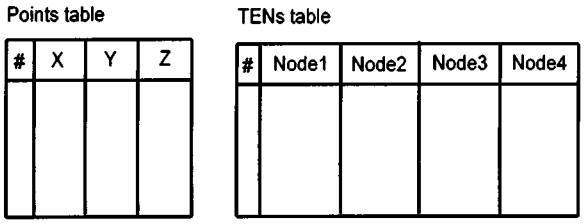


Figure 12 TEN data structure

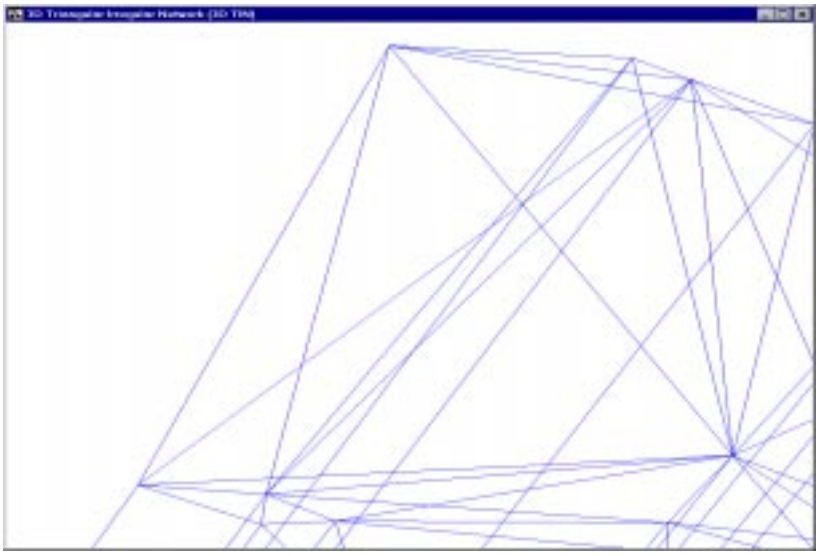


Figure 13 The TENs visualization

4.0 Object-Oriented Design of TIN-based Modelling

A data model in an information system is a collection of conceptual tools for describing data, data relationships, data semantics, data constraints and operations on data. In this section, a data model based on an object-oriented technique is described. The model realised the object-oriented concepts such as objects, encapsulation, classification, generalisation and inheritance, and aggregation. These OO terms are not described here, but the reader can consult them from literatures and various OO textbooks, e.g. Khoshafian and Abnous (1995), Isadale and Lee (1996), and Stroustrup (1997). Furthermore, OO data models have been considered to satisfy many aspects of the data modelling requirements of GIS (Worboys *et al.*,1990; Webster, 1990). If we consider our previous 2D and 3D TIN modelling programs as object classes then we may relate them with other spatial domain as

shown in Figure 14. Basically, a spatial object may be described by its geometric and thematic data. Then, these two data (i.e. geometric and thematic) types could be subdivided into several spatial primitives, e.g. node, edge, surface, and solid. See Figure 14 for the derivation of objects. We say a node represents point feature, an edge for line (contains at least two points), a surface for area feature (i.e. polygon), and a solid for volumetric feature. In this figure, the ASpatial object@ is the base class, and two were derived classes, namely, AGeometric@ and AThematic@ class. Several other classes were derived from the geometric class. Four classes were derived from the thematic class, they are AForest@, ARiver@, AFaultlines@, and ABoreholes@. As for an example, the following are some definitions of the prominent classes:

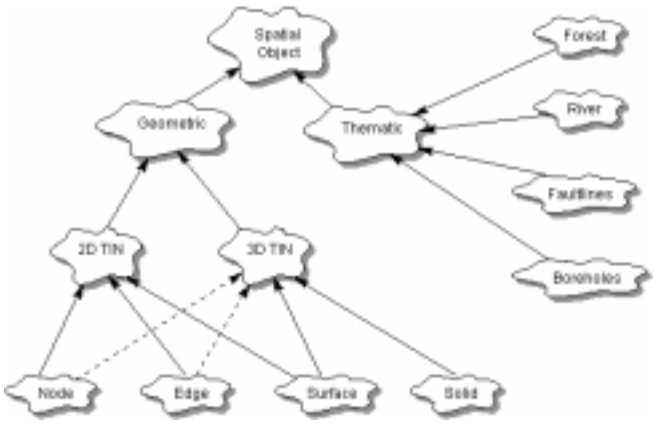


Figure 14 TINs within spatial object hierarchy

class 2D TIN:
parent class: Geometric.
methods: e.g. 2D DT, 2D Voronoi, TIN table generation, visualization.

class Node:
parent: 2D TIN, 3D TIN
methods: e.g. create, delete, show identifier, GetCoordinates, etc.

class Edge:
parent: 2D TIN, 3D TIN
methods: e.g. create, delete, show edge, etc.

The implementation of this OO modelling including the development of the database and spatial operations become our next research task of prototyping a 3D GIS system.

The 3D GIS System

Traditionally, two-dimensional spatial data can be efficiently handled by existing GIS systems. However, the need for 3D systems is acknowledged by the GIS research community as reported in Raper (1990) and Houlding (1994). Indeed the fields of photogrammetry, simulation and modelling, mining, geology, hydrology, geo-engineering, urban and rural planning are likely to benefit from such system. The call for better geo spatial data management also provides an initiative for this so called 3D GIS. Undoubtedly, the need for such a system has been discussed in various geo-related research communities. Concepts contributing to the development of this system have been addressed in Weibel and Heller (1990), Fritsch (1996), Bruenig (1996), and Pilouk (1996). We propose the following scheme for the 3D GIS system (Figure 15):

1. a module supporting the user-interface
2. a module supporting 2D and 3D TIN modelling
3. a module supporting the object database, and
4. a module supporting the spatial operators.

The user interface provides a means of interaction between user and system. The 3D modelling component contains two sub-components, 2D TIN and 3D TIN. TINs have been considered as a useful data structure in 3D GIS by many researchers (Raper, 1990). Others such as Qingquan and Deren (1996) indicated that the structure was considered a powerful vector structure for 3D GIS. The structure of TEN could be organized in various ways. Figure 12 shows one of the possible structures for TEN which was developed in this research. The proposed TEN structure should be able to define the topological relationships among the primitives, i.e. point, line, surface, and solid. These TIN programs are used to generate 3D data structure from the given input of 3D datasets. The spatial operator module provides a means for spatial manipulations and analysis, e.g. 3D visualization, geometric transformation, etc. The object database is for the management of TINs data together with their attribute and topological data.

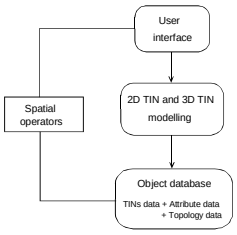


Figure 15 the possible components for the 3D GIS system

5.0 Discussion and Outlook

We have implemented and demonstrated the two programs for generating 2D and 3D triangular irregular networks. The graphics output of the TINs suggests that the algorithms work for random points in two and three-dimensional space. We also described the possible development of the TINs in object-oriented environment. The later development in OO stimulates us for further research on establishing a prototype of 3D GIS system.

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ISO 15046-13
A FRAMEWORK FOR QUALITY INFORMATION

Leslie Godwin
Geography Division, GRaSS
Bureau of the Census
U.S. Department of Commerce
Washington, DC 20233-7400
lgodwin@census.gov

ABSTRACT

The International Organization for Standardization Technical Committee 211 *Geographic information/Geomatics* (ISO/TC 211) is developing a structured set of standards in the field of digital geographic information. The set of standards is numbered 15046. 15046 Part 13 *Quality Principles* (15046-13) establishes the principles for describing the quality of geographic data, specifies the components for reporting quality information, and provides an approach to organizing knowledge about data quality. Although it has a theoretical basis (a conceptual model, or schema, is provided), 15046-13 is surprisingly concrete in that it describes the components of quality that must be provided for a conforming dataset and should be considered when evaluating a dataset. This part of the International Standard provides the framework for identifying, collecting, and reporting quality information for a geographic dataset.

The development of 15046-13 is reviewed with a look at the standards development process, the major areas of concern during the development process, their resolution, and possible impacts of their resolution on other parts of the International Standard. A status report on progress is provided.

THE BIG PICTURE: ISO/TC 211

What and Who is ISO/TC 211?

The International Organization for Standardization (ISO) is a worldwide federation of national standards bodies (national bodies) whose work of preparing international standards is carried out through technical committees. ISO Technical Committee 211 *Geographic information/Geomatics* (ISO/TC 211) was established as a technical committee in 1994 to develop a structured set of standards for information concerning objects or phenomena that are directly or indirectly associated with a location relative to the Earth. ISO/TC 211 consists of Active members from more than twenty-five countries (alphabetically ranging from Australia to Yugoslavia and geographically representing five continents) and Observing members* from more than fifteen countries. ISO/TC 211 is working closely with more than fifteen liaison groups, including the Open GIS Consortium, Incorporated, to develop standardization in the field of digital geographic information.

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* Observing members may participate in the process but cannot vote.

The Organizational Structure of ISO/TC 211

ISO/TC 211's internal structure reflects the primary areas of interest within which the nineteen parts of the standard are being developed; the parts are alternately referred to as "work items," "projects," or more formally "this part of the International Standard." Taken together the parts form the suite of standards known as 15046. ISO/TC 211 is organized into five Working Groups (WG), each responsible for developing specific parts of 15046. Each WG has an elected convenor and each part an assigned project leader. Additionally, some parts have acknowledged editors. The parts were determined by vote, and a first order of business was agreement on their scopes. The number of parts is not necessarily static; ISO/TC 211 is continually contemplating the addition of new parts to ensure that all areas of geographic information are adequately covered. As the work evolved, there also have been recommendations that some parts be combined. With nineteen current parts, the work of ISO/TC 211 can be compared to a giant jigsaw puzzle with some uncertainty still existing over how the pieces will fit together. Figure 1 illustrates the five WGs and the parts of 15046 each is responsible for completing. The convenor's country (shown in italics) and ultimately the convenor is responsible for completing the parts assigned their WG.

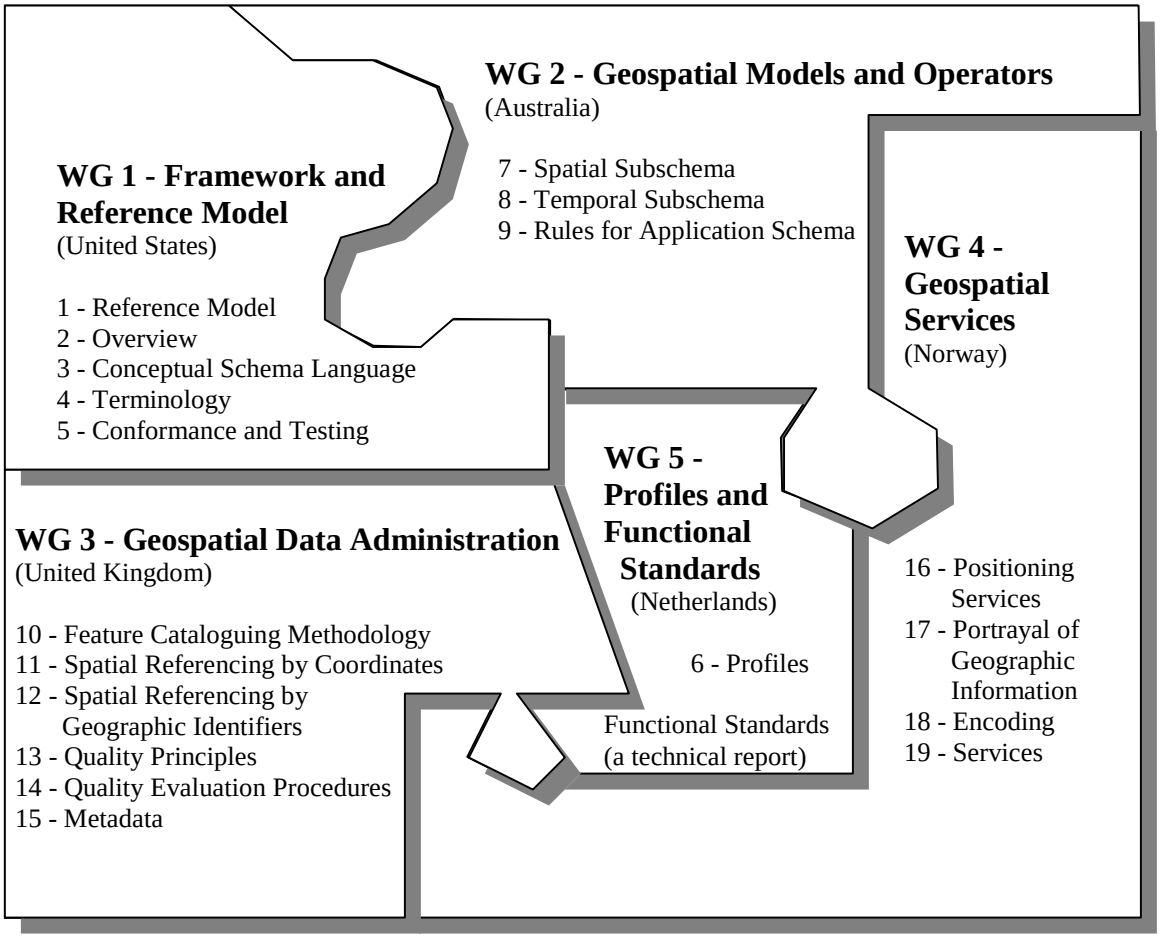


Figure 1. ISO/TC 211 Working Groups and the parts of 15046 each WG is developing.

US Involvement

The American National Standards Institute (ANSI) is the United States national body participating on all ISO TCs. ANSI is represented on TCs by Technical Advisory Groups (TAGs). ANSI has delegated the TAG administration function for ISO/TC 211 to its Accredited Subcommittee, the National Committee for Information Technology Standards (NCITS), who in turn has delegated the function to its subgroup, the L1 *Geographic Information Systems* Subcommittee. L1 elects subject matter experts to become TAG members. The U.S. TAG is very active in ISO/TC 211. The United States has the convenorship of WG 1 and is ultimately responsible for the five work items assigned the WG. Additionally, United States experts are the project team leaders of six parts of 15046, specifically Part 1 Reference Model, Part 8 Temporal Schema, Part 10 Feature Cataloguing Methodology, Part 13 Quality Principles, Part 15 Metadata, and Part 16 Positioning Services.

A BRIEF HISTORY OF 15046-13*

The Participants

ISO 15046 Part 13 Quality Principles is lead by Dr. Joel Morrison (United States); this paper's author is editor. More than twenty countries expressed interest in participating in developing Part 13 and an active core of approximately ten countries has worked on this part of the International Standard.

Evolution of a Name

Part 13 Quality Principles began life simply as Part 13 Quality. Its metamorphosis from Quality to Quality Principles resulted from a need to distinguish it from and establish boundaries between 15046-13 and 15046-14 Quality Evaluation Procedures. The logic for developing two distinct parts is that one aspect of quality (its principles) is fairly mature while another aspect of quality (the quantitative methodology for determining and recording it) is still in its infancy. By separating the topic into two parts, ISO/TC 211 has been able to provide a stable framework in one area while acknowledging on-going work in another area.

As editor of 15046-13, the author often emphasizes the importance of quality to geographic information by pointing out ISO/TC 211 chose to devote two of its nineteen parts (or one tenth of the International Standard) to the topic of quality.

Those members of ISO/TC 211 interested and active in 10546-13 Quality Principles are interested and active in 15046-14 Quality Evaluation Procedures. The two project teams are almost one and the same. Partially because of this, the developers of 15046-13 considered the practicalities of measuring quality when devising the framework within which to measure quality. ISO/TC 211 as a whole still has difficulty distinguishing

* Specific parts of an International Standard, such as 15046-13, are correctly referred to as ISO/*xx* 15046-##, where *xx* is the stage of current development and ## is the part of the International Standard. Stages of development are Working Draft (WD), Committee Draft (CD), Draft International Standard (DIS), and Final Draft International Standard (FDIS). For example, ISO/CD 15046-13 is the correct reference for 15046-13. For simplicity, ISO/xx has not been included when referring to a specific part of the International Standard in this paper.

between the two parts, and most members acknowledge both parts of the International Standard are highly dependent upon one another. The continuing debate over the need for distinct parts surfaced again in comments to the Editorial Committee that suggested the two be combined. The comments were ruled beyond the scope of the Editorial Committee and referred to the ISO/TC 211 Secretariat.

Evolution of a Scope

ISO/TC 211 provided an original scope for 15046-13:

Develop a "*definition of the schema for quality applicable to geographic data.*" *

While this scope technically remains the scope statement under which the project team works, the team developed the following extended scope that, in their opinion, better stated their purpose:

Develop a "*definition of the schema for quality applicable to geographic datasets. A quality schema establishes the principles for describing the quality of geographic data and offers a data quality model for organizing knowledge about data quality. The components of data quality are defined and the requirements for their use specified.*"

Reference Materials

Two bodies of existing work served as the basis for 15046-13. The International Cartographic Association, Commission on Spatial Data Quality published "*Elements of Spatial Data Quality*" in 1995. The book's chapters, highlighting proposed components of quality, were the starting point for developing 15043-13's data quality elements. The European Committee for Standardization (CEN), Technical Committee 287 Geographic Information's proposed standard: "*Geographic Information – Data Description – Quality*" was valuable in developing an approach to standardizing data quality. Many of the 15046-13 team members either participated in or were familiar with the CEN effort.

Schedule

The 10546-13 project team held its first official meeting at a joint Working Group meeting in Norway in October 1995. The first working draft of this part of the International Standard was completed by May 1996. The project team continued meeting twice yearly in conjunction with ISO/TC 211 plenaries, completed and submitted a working draft for review and comment by all national bodies in April 1998. More than one hundred comments were received. An Editorial Committee was convened (consisting of the Convenor of WG 3, the editor of 15046-13, and three additional volunteers from the WG 3 membership). Meeting prior to the ISO/TC 211 Plenary in China in September 1998, the Editorial Committee adjudicated the relevant comments. The editor subsequently revised the document. The next steps in the process are (1) a

* As a counterpoint, the scope of 15046-14 is to develop "*guidelines for the methods of specifying/ evaluating data quality.*"

Committee Draft Standard (CD) vote[**] (November 1998), (2) a Draft International Standard (DIS) vote (May 1999), and (3) a Final Draft International Standard (FDIS) vote (November 1999).

The current schedule calls for 15046-13 to become an International Standard (IS) in May 2000. The Editorial Committee, however, has recommended that the schedule be delayed because of harmonization concerns. The Editorial Committee feels 15046-13 in its revised form should be recirculated with 15046-14 for review and comment by national bodies (the work of 15046-14 was designed to be six months behind that of 15046-13). Since they are closely related, the joint review would allow national bodies to properly review 15046-14 within the context of 15046-13 Quality Principles; national bodies could evaluate and submit comments pertinent to either or both parts. The Editorial Committee would like to see these two parts of the International Standard proceed in tandem through the process, although this does mean delaying 15046-13's completion date by six months. The Editorial Committee believes the delay is justified and would aid in the ultimate success of the Standard.

THE THEORY BEHIND 15046-13

The Quality Model

The quality model, or schema, is fundamental to describing the components and their interrelationships. It was the project team's first area of discussion and has remained an ongoing subject of controversy throughout the development process. The fact so many team members had strong and conflicting conceptual views was evidenced in the many national body comments specifically addressing the model. This was partially a result of the many compromises made by team members during work on the model.

Although fine-tuning of the model was constant throughout the development process, the model's conceptual core remained fairly stable and was central to reaching agreement on the principles of quality. Discussions about the theoretical model strengthened the practical nature of 15046-13 and the importance of the ability to measure and quantify quality information. The model was based on a majority of prior quality models that emphasized a two-step process of (1) abstracting reality to a "universe of discourse[*]", then (2) creating a dataset quality of which is a measurement of the degree of success in approximating the abstracted reality. Figure 2 illustrates the initial and final quality models.

[**] A Committee Draft vote is a vote by all national bodies for approval to move the document from the Committee Draft stage to the Draft International Standard stage. Similarly, a Draft International Standard vote is a vote for approval to move the document from the Draft International Standard stage to the Final Draft International Standard stage and a Final Draft International Standard vote is a vote for approval to move the document from the Final Draft International Standard stage to become an International Standard.

[*] The "universe of discourse" is alternately referred to as the "abstract universe" or "nominal ground" in publications. The current and still contested ISO/TC 211 definition of universe of discourse is "view of the real or hypothetical world that includes everything of interest".

Both the initial and final models emphasize two points: (1) the importance of a product specification against which to measure quality and (2) the fact that data users commonly have differing needs or requirements than the requirements driving the data producer's creation of the dataset.

Although central to developing the principles of quality, the quality model actually appears in an informative Annex and is provided only for information. There are no specific requirements attached to the quality model or its use.

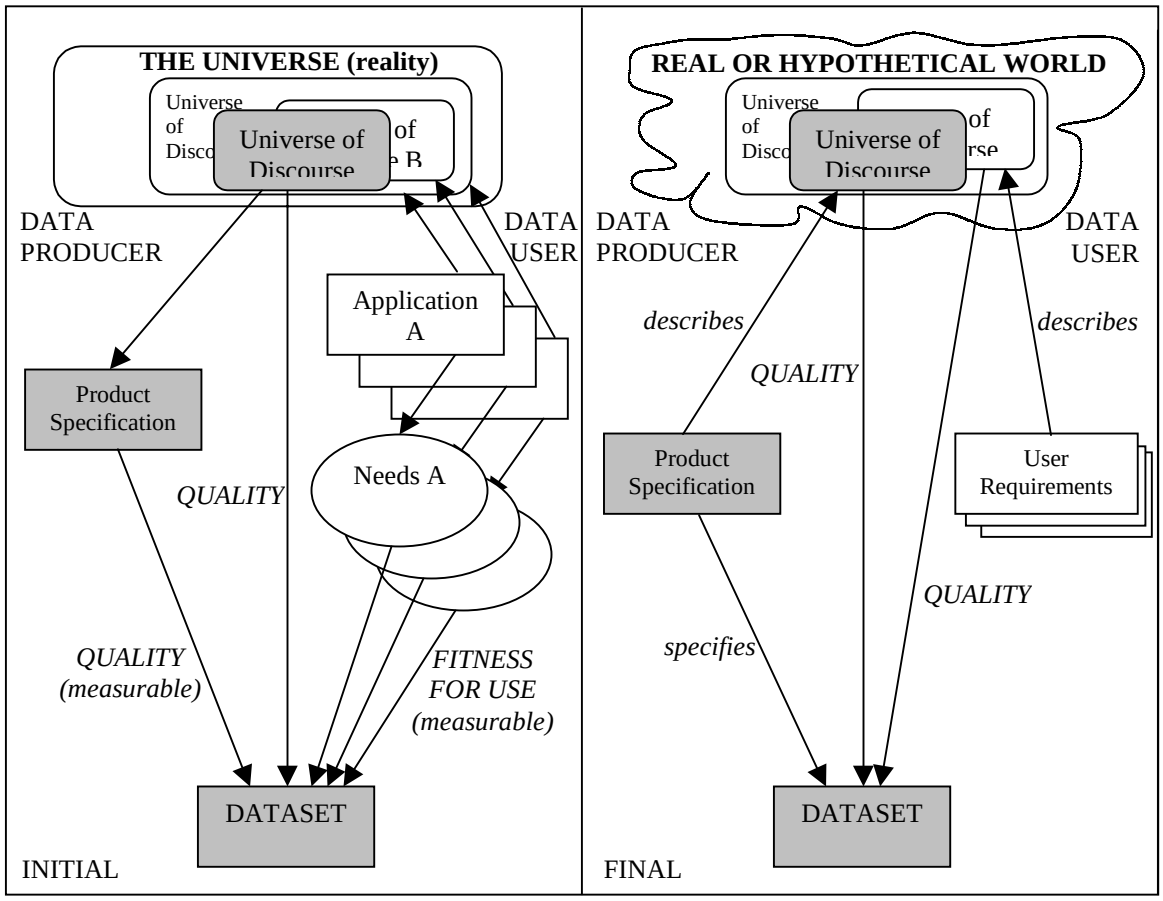


Figure 2. Initial and final quality models.

THE REQUIREMENTS OF 15046-13

Thou Shall

Requirements of a standard are expressed using the verb "shall." The whole of 15046-13 can be summarized as: the following shall be accomplished for compliance with this part of the International Standard -

- The quality of a dataset shall be described using two components: data quality elements and data quality overview elements.

Data quality elements, together with data quality subelements and the descriptors of a data quality subelement, report how well a dataset meets the criteria set forth in its product specification and provide quantitative quality information.
Data quality overview elements provide general, non-quantitative information.

- The identified data quality elements and data quality subelements shall be used to describe quantitative quality.
  Figure 3 provides a complete listing of data quality elements and their associated data quality subelements.
- The product specification shall be used to determine the applicable data quality subelements.
- The six mandatory descriptors of a data quality subelement shall be used to report information for an applicable data quality subelement.
  The six mandatory descriptors of a data quality subelement and their definitions as given in 15046-13 are:
  Data quality scope – extent or characteristic(s) of the data for which quality information is reported.
  Data quality measure – type of test applied to a data quality scope.
  Data quality evaluation procedure – operation(s) used in applying and reporting a data quality measure.
  Data quality result – value or set of values resulting from applying a data quality measure or the outcome of evaluating the obtained value or set of values against a specified acceptable quality level.
  Data quality value type – value type or unit for reporting a data quality result.
  Data quality date – date or range of dates on which a data quality measure is applied.
- The data quality overview elements "Purpose," "Lineage," "Usage," and "User-defined" shall be used to describe general, non-quantitative quality.
- All quality information shall be reported as metadata in conformance with the requirements of 15046-15 Metadata.
- Quantitative quality information shall additionally be reported as an Additional Quality Report in conformance with the requirements of 15046-14 Quality Evaluation Procedures.

EDITORIAL COMMITTEE RESOLUTIONS – THE MOST RECENT MODIFICATIONS

The Comments

The Editorial Committee reviewed more than one hundred comments submitted by national bodies on the proposed Committee Draft. A majority of the comments resulted in minor sentence structure and document organization modifications to improve the readability of this part of the International Standard. Several comments brought to light technical problems that had not been resolved through the consensus building process in developing 15046-13. Major areas of conflict and their resolution are described below.

Quality and Distinct Views: The Data Producer and the Data User

The project team felt there were two views of quality based on an important distinction: whether one was a data producer or a data user (data producers actually are a unique subclass of data users). Data producers need to evaluate and report how well a dataset met the criteria set forth in its product specification. Data users need to evaluate how well a dataset meets the needs of their specific application(s). A data user's application(s) often differ from the application for which the dataset was created.

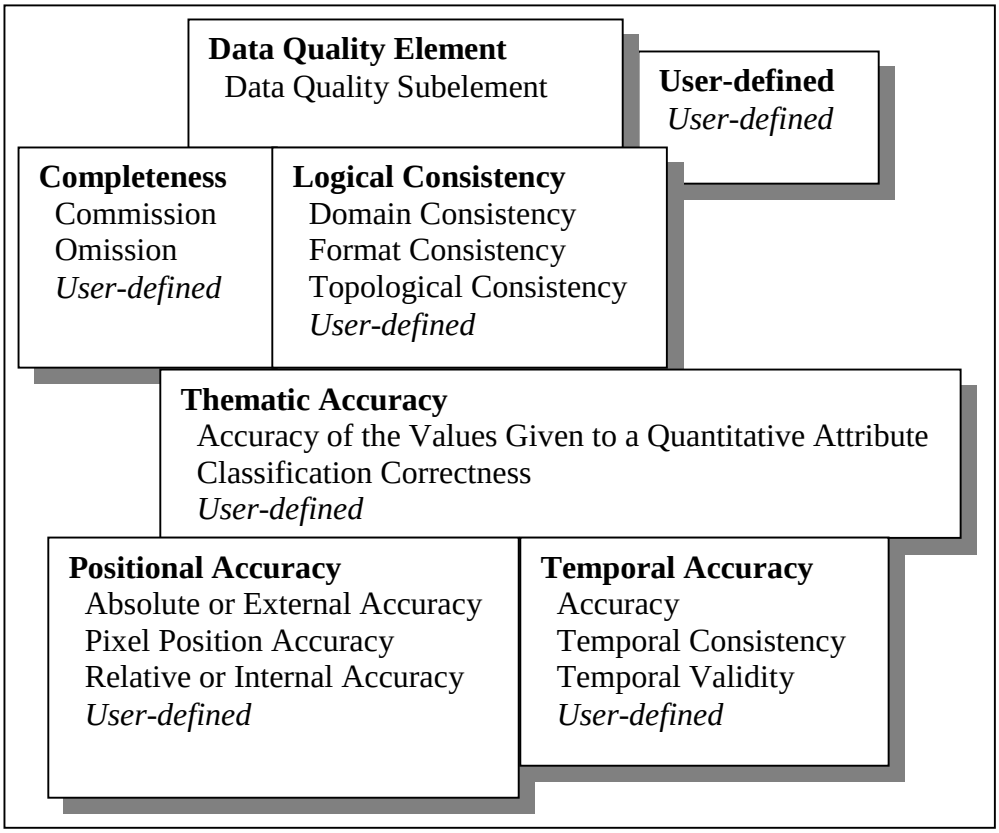


Figure 3. Listing of data quality elements and their associated data quality subelements.

The two views of quality are clearly reflected in the informative quality model, both initial and final. It was decided during the Editorial Committee review, however, that the emphasis 15046-13 placed on the data producer was extreme. 15046-13 placed all the burden of quality upon the data producer. Requirement sentences began "The data producer shall …" while the scope indicated "This part of the International Standard is applicable to data producers … This part of the International standard may additionally be of use to data users for describing their quality requirements." The comments suggested the requirements of 15046-13 are not restricted to data producers and clarification was required. Clarification was achieved by removing data producer from requirement statements.

De-emphasizing the role of the data producer may have an effect on Quality Evaluation procedures. 15046-14 provides two distinct quality evaluation models, one for the data

producer (for determining and reporting a dataset's quality information) and one for the data user (for evaluating a dataset's quality information against user requirements.) In the latest draft of 15046-15, (October 1998) the two models were normative (required).

The Distinction between Quality and Fitness for Use

In reviewing the available literature, the project team determined just as with two distinct points of view on quality, there were two distinct "types" of quality: quality and fitness for use. 15046-13 defined quality as:

Totality of characteristics of a product that bear on its ability to satisfy stated and implied needs.

NOTE: Within the quality model, quality information reports how well a dataset meets the criteria set forth in its product specification. This information is a measure of how well a dataset meets the purpose for which it was created. Quality information may be used to assess fitness for use.

15046-13 defined fitness for use as:

Ability of a product to satisfy the requirements of an application.

NOTE: Within the quality model, fitness for use is an evaluation made by a data user using the quality information reported by a data producer of the appropriateness of a dataset for a specific need or application (which often differs from the purpose for which the dataset was created.)

Comments suggested the distinction between quality and fitness for use was a fine one, advanced in notes rather than in the definitions that were, in fact, synonymous. The Editorial Committee agreed and adopted the definition of quality that appeared in ISO 8402. All references to fitness for use were deleted from 15046-13. The ISO 8402 definition adopted by 15046-13 is:

Totality of characteristics of a product that bear on its ability to satisfy stated and implied needs.

The concept of "fitness for use" is found in other parts of the International Standard, though not to the extent it was present in 15046-13. 15046-14 has incorporated the concept into its two models of quality evaluation procedures in emphasizing the unique roles of data producers and data users.

Finalizing Data Quality Elements and Data Quality Subelements

15046-13 was never intended to provide an exhaustive listing of the components and aspects of quantitative quality. A "user-defined" data quality element and data quality subelement were included in the respective listings to ensure the areas of quantitative quality not specifically covered could be addressed. The "user-defined" concept provides an extensible standard, one that can mature with the field of quality evaluation.

The project team attempted to provide data quality elements and data quality subelements that generally have been accepted by the geographic profession as important indicators of the quality of a dataset.*

Many comments questioned either the inclusion of or description (definition) of data quality elements and data quality subelements. Resolution was on a case by case basis.

Positional Accuracy Data Quality Subelements. The existing volume of work in the area of positional accuracy contributed to the on-going discussions about its data quality subelements. At one point in the development process, more than fifteen data quality subelements were proposed for positional accuracy. Many were requested by the Part 16 Positioning Services project team. The Editorial Committee determined such data quality subelements as Positional Stability and Relative Positional Stability were used by and specific to a small community and should be identified as "user-defined" data quality subelements in a profile of 15046-13 provided by 15046-16 rather than as a part of 15046-13. Another data quality subelement removed by the Editorial Committee was Shape Fidelity. Shape Fidelity initially was included in 15046-13 as it is a recognized concept within the geographic community. However, it was argued that a suitable measure of shape fidelity has not been found, and this precluded it being a usable data quality subelement. This concentration on plausible measures is an indicator of the practicality expected of the standard.

Thematic Accuracy Data Quality Subelements. Three data quality subelements were identified for thematic accuracy: (1) Accuracy of a Feature Attribute Defined by a Value, (2) Classification Accuracy, and (3) Misclassification. One comment suggested Accuracy of a Feature Attribute Defined by a Value could be more clearly identified as Accuracy of a Value Given to a Quantitative Attribute. This suggestion was accepted. Another comment suggested there was not really a worthwhile distinction between Classification Accuracy and Misclassification. The Editorial Committee determined there was no significance difference, removed Misclassification, and renamed Classification Accuracy as Classification Correctness to more accurately reflect its measurement.

Modifications to the data quality elements and data quality subelements have a major effect on 15046-14. 15046-14 describes the data quality evaluation procedures that can be used to apply data quality measures for data quality subelements. Data quality subelements additionally are carried as accepted domain values for quality-related metadata elements in 15046-15 and referenced in 15046-9 Rules for Application Schema; minor modifications are required in these parts of the International Standard.

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* The applicability of a data quality element and its data quality subelements to a dataset depends on both the dataset's content and its product specification; all data quality elements and/or their data quality subelements may not be applicable to all datasets.

OTHER RELEVANT PIECES OF THE PUZZLE

The nineteen parts of 15046 are being developed in parallel and somewhat in isolation. Harmonization was difficult during the initial phases of the development stage as each part struggled to solidify its individual schema and hadn't the time to worry about the big picture. Communication within a working group tended to be better than communication between working groups. As a result, parts within working groups have to date achieved a greater harmonization. The focus of ISO/TC 211 is shifting to the harmonization effort as more parts progress to the Committee Draft vote.

It was evident early in the process that 15046-13, 15046-14, and 15046-15 must present a single, harmonized view of quality. Consequently, the three project teams have worked together during the last three years with each team acknowledging that changes to the one affect the other. 15046-13 provides the quality framework and theory for 15046, while relying on 15046-14 to provide quality evaluation procedures and relying on both 15046-14 and 15046-15 to provide the mechanisms for reporting quality information. Because all three projects are located within Working Group 3 *Geospatial Data Administration*, the harmonization has been simplified by a shared, common approach to geographic data.

CONCLUSION

In conclusion, the author invites the reader to additionally read the two papers prepared and presented by a U.S. expert who is a member of the project team for 15046-14 Quality Evaluation Procedures (Frederick R. Broome) and the team leader of 15046-15 Metadata (David Danko). The three selections together provide the totality of Quality as it is being approached by ISO/TC 211.

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MULTIPATH EFFECTS OF GPS CONTINUOUS TRACKING STATIONS IN TAIWAN

Jenn-Taur Lee
Department of Surveying and Mapping Engineering
Chung Cheng Institute of Technology
Tahsi, Taoyuan, Taiwan 33509

ABSTRACT

Multipath is one of the crucial effects in GPS precise positioning for engineering applications and scientific researches, since multipath may constitute of obstructions from station environment, ground antenna, satellite antenna, atmospheric disturbance and others. Once multipath effects could be resolved completely, the outcomes of GPS positioning may be very close to the true positions based on the center of earth. In this study, a approach which is derived from GPS phase residuals was developed to point out the possible place and period of multipath effects occurred and investigate the quality of GPS observations.

A weekly data set of five GPS continuous tracking stations in Taiwan was chosen to examine this new approach. These five GPS stations are located around Taiwan island and each one has installed with Turbo Rouge SNR-8100 GPS receiver. The preliminary results do indicate that this approach is effective and efficient for detecting multipath effect. The resolution in this case study is below 5.0 mm in root mean square of phase residuals on average. In addition, the quality of GPS observations were also evaluated with seven IGS GPS stations located in the State of California, which is several thousand kilometers away. The repeatability of positions is 1.0 mm of northern, 2.0 mm of eastern, and 3.0 mm of upper, on average.

Key words: Global Positioning System (GPS), Antenna, Multipath Effect

INTRODUCTION

Global positioning system (GPS) is one of the most accurate positioning techniques to establish the geodetic control with efforts of investigation, derivation, and verification over ten years. In addition, GPS relative positioning technology is also adopted to perform the conventional surveying tasks, such as short baseline control surveying of urban planning, land administration, and others. The reliability of GPS relative positioning is about 5 to 10 parts per million-ppm (Leick, 1995) and the main error resources of GPS are satellite orbital coordinates, atmospheric effects, internal noise of receiver, and environment of station. The latter two may pass the major uncertainty of positioning to the final results, if the shorter observation session and the obstructions of station are occurred. In other words, the cycle slips of GPS raw observations may not be detected and repaired, when the amount of multipath were large (Breivik, 1997; Weill, 1997).

For short baselines of relative positioning, the multipath of satellite signal between stations may be similar, so the differencing techniques may mitigate this type of error (Young, 1985). The multipath of receiver's antenna is that the interference of received signal happened repeatedly after one sidereal day of GPS observation. This may come

from the reflected signal caused by obstructions around the GPS receiver. Evans (1986) did analyze the pseudorange and carrier phase measurements from multiple days of observation session and indicated that the correlation coefficient of pseudorange multipath and station location is very high. Bishop (1985) also pointed out that the multipath of pseudorange measurements may be estimated from ionospheric delay of GPS signal. Greenspan (1982) evaluated the residual of phase measurements to predict the amount and timing of multipath.

The object of this study is to investigate the multipath of GPS measurements and its effect in the relative positioning. The algorithm of detecting multipath was developed and based on the geometrical relationship of GPS satellite signal and antenna. This new approach may point out the timing and amount of multipath to verify the cycle slips and atmospheric noise in the measurements. The experiments of short baseline and five GPS continuous tracking stations were done to test this approach.

CHARACTERISTCS OF GPS MULTIPATH

The multipath of GPS measurements is a type of GPS signal noise obtained by GPS receivers through indirect path. The indirect GPS signal may be caused during satellite signal transmission or obstructions around GPS receivers. It is assumed that the multipath of GPS signal reflected from ground is displayed in Figure 1 and the mathematical relationship is listed as the following (Tranquilla, 1986):

$$ep = 2h\sin\theta \tag{1}$$

$$\beta = \frac{ep}{\lambda} \times 2\pi + \pi = \frac{4\pi h \sin\theta}{\lambda} + \pi \tag{2}$$

Where:
ep = Excess path length
h = Antenna height
θ = Satellite elevation angle
β = Phase variation of reflected signal
λ = Wavelength of signal

Figure 1. Geometry of multipath

The combination of direct and reflected signal is as the following (Raquet, 1996):

$$V_{measured} = V_{direct} + V_{mp} \tag{3}$$

Where:
$V_{direct} = A0\ell^{j\phi}$
$V_{mp} = \alpha A0\ \ell^{j(\phi+\beta)}$
ϕ = Phase of incoming GPS signal

ϕ_m = Phase measurement
$\delta\phi$ = Phase error caused by multipath
V_{direct} = Direct signal
V_{mp} = Multipath signal
$V_{measured}$ = Measured signal
A0 = Amplitude of direct signal
α = Decay factor of reflected signal

In Figure 2, if β is changed, the multipath rotated around the incoming signal may produce the positive and negative phase errors because the true phase error does have the relation of multipath phase and its amplitude. As phase error $\delta\phi$ is the core of multipath, V_{direct} rotated about real axis and A0 = 1, then amplitude of measured signal may be observed. The amplitude of measured signal $V_{measured}$ and its phase error could be displayed as the following mathematical formula:

$$\mathrm{Re}(V_{measured}) = 1 + \alpha\cos\beta \tag{4}$$

$$\mathrm{Im}(V_{measured}) = \alpha\sin\beta \tag{5}$$

$$\left|V_{measured}\right| = \sqrt{\mathrm{Re}(V_{measured})^2 + \mathrm{Im}(V_{measured})^2} \tag{6}$$

$$\delta\phi = \mathrm{atan}\left[\frac{\mathrm{Im}(V_{measured})}{\mathrm{Re}(V_{measured})}\right] \tag{7}$$

Figure 2. Amplitude and phase error of measured signal

DESCRIPTION OF EXPERIMENTS

The multipath detecting algorithm in this study does focus on both satellite and receiver multipath, but not including phase center variation of GPS antenna and its imaging effect. Two experiments, short baseline and continuous tracking network, were designed for this research. The short baseline experiment was located on campus of Chung Cheng Institute of Technology (CCIT) which did consist of four stations aligned at the south-north and east-west directions. The length between any two stations is about 5 meters. The second experiment including five GPS continuous tracking stations, YMSM, FLNM, TMAM, PKGM, and KDNM, which are located around Taiwan island is shown in Figure 3 (Lee, 1998). The pseudorange and phase measurements obtained from dual frequency receivers may be used to examine the previous algorithm. All observation session were 24 hours long and 30 seconds of recording interval. The developed multipath detecting program and the existed GPS packages of AOSS and GAMIT were used to compute the multipath noise and baseline vectors to evaluate the new detecting approach and the multipath effect on

positioning.

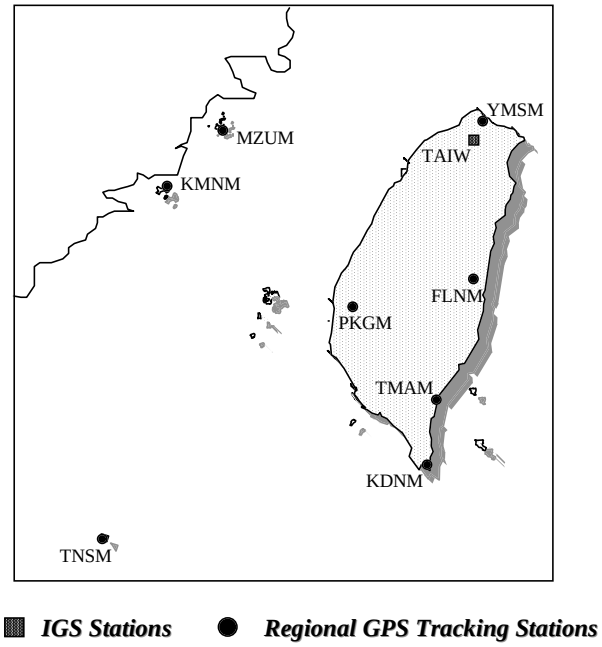


Figure 3. Taiwan tracking network

PRELIMINARY RESULTS AND ANALYSIS

The preliminary results do indicate that this approach is effective and efficient for detecting multipath effect. The outcome from short baseline experiment is about 40 to 70 cm of multipath in the GPS raw data displayed in Table 1. The multipath variations from same satellite and different time period are shown in Figures 2 and 3. About the Taiwan tacking network experiment, the resolution in this case study is below 5.0 mm in root mean square of phase residuals on average. The largest one appeared at FLNM station in Figure 6. In addition, the quality of GPS observations were also evaluated with seven IGS GPS stations located in the State of California, which is several thousand kilometers away. The repeatability of positions is 1.0 mm of northern, 2.0 mm of eastern, and 3.0 mm of upper, on average.

Table 1. Average multipath values of L1 and l2

Station	Period	Ant Height	Multipath of L1	Multipath of L2
N001	Day290-Day291	1.5 m	0.402512 m	0.412157 m
	Day291-Day292	0.14 m	0.718322 m	0.556603 m
S001	Day290-Day291	1.5 m	0.392499 m	0.443800 m
	Day291-Day292	0.14 m	0.460446 m	0.544685 m
E001	Day290-Day291	1.5 m	0.509404 m	0.408777 m
	Day291-Day292	0.14 m	0.577119 m	0.607208 m
W001	Day290-Day291	1.5 m	0.359885 m	0.386843 m
	Day291-Day292	0.14 m	0.592130 m	0.652890 m

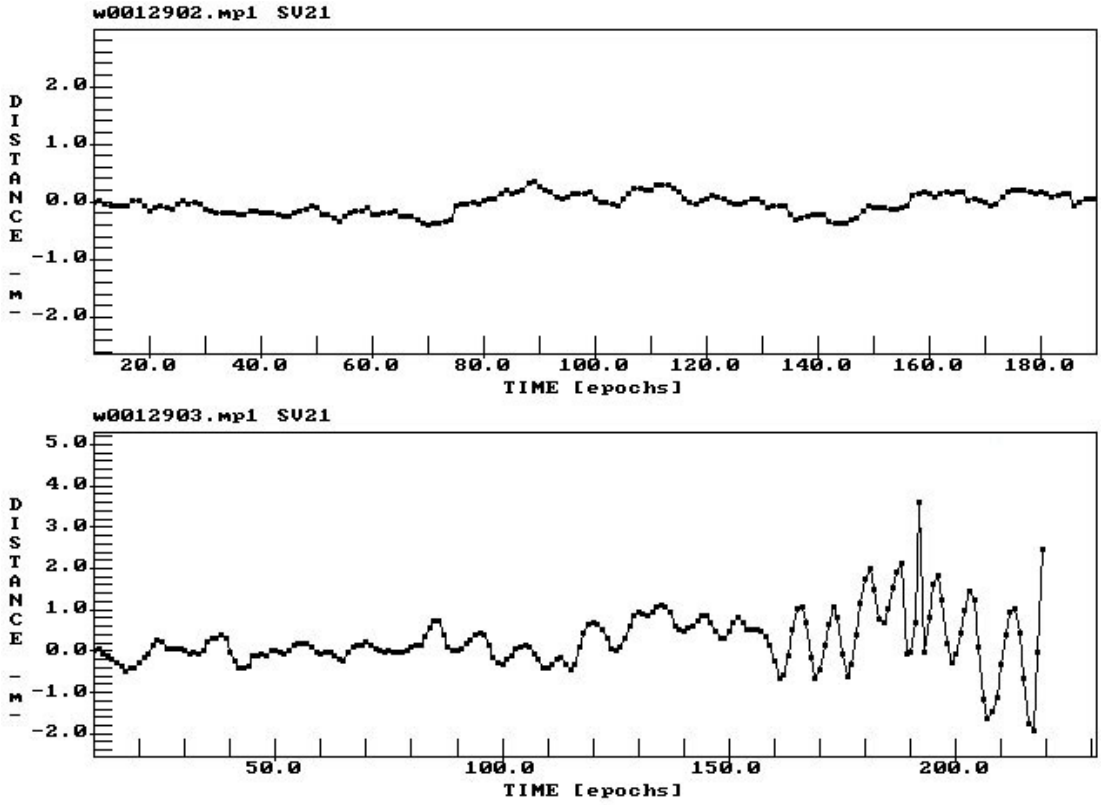


Figure 4. Multipath effect from same satellite but different time period

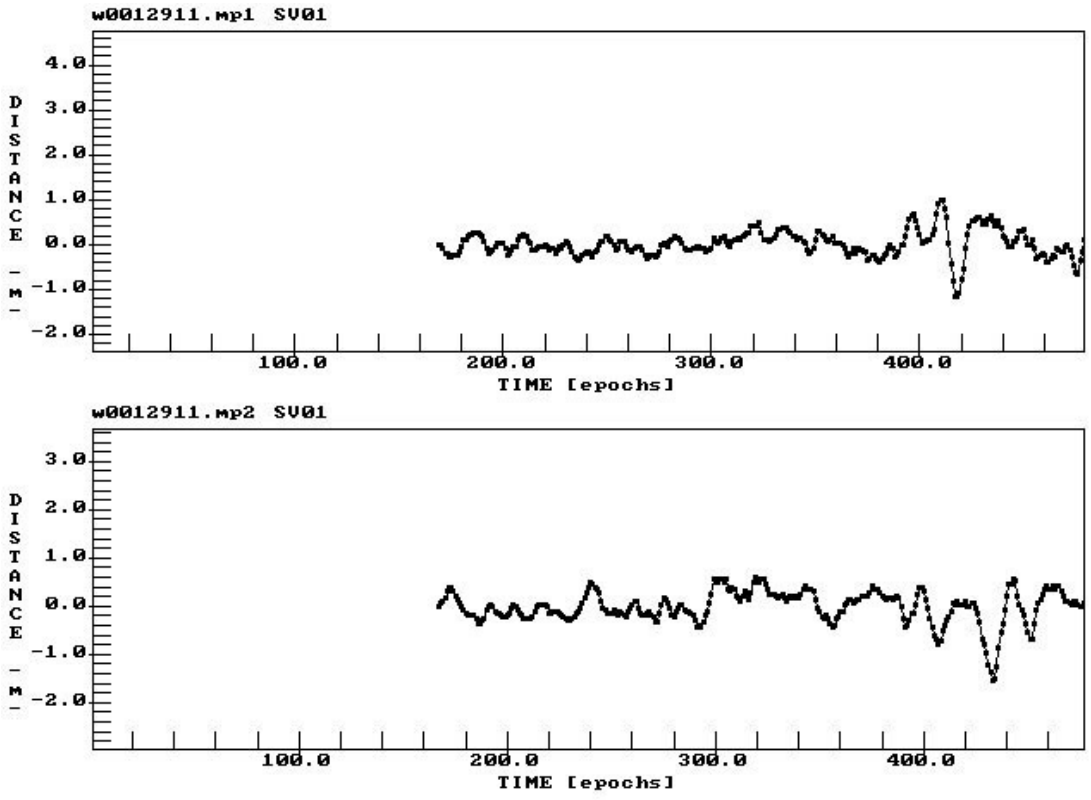


Figure 5. Multipath effects of L1 and L2 from same satellite

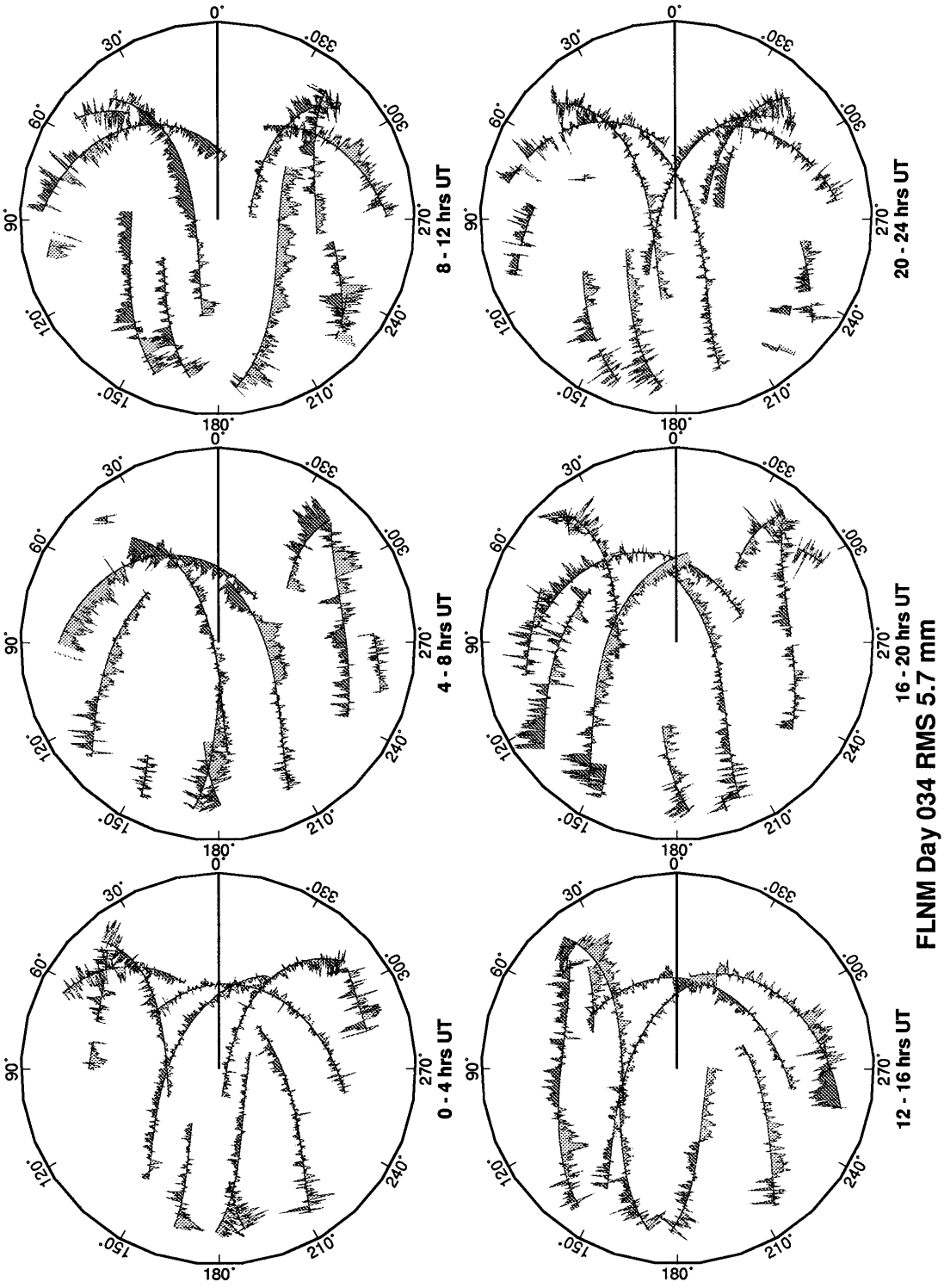


Figure 6. Multipath effect of Taiwan tracking station-FLNM

CONCLUSIONS

The preliminary results do indicate that this approach is effective and efficient for detecting multipath effect. The multipath and antenna height is highly correlated on GPS raw measurements, but it may not affect the results of baseline vectors from relative positioning since the differencing techniques are adopted. The amount of phase residual obtained from long baseline may indicate that not only multipath but also atmosphere noise which may perform further research for forecasting the climate changes. Of course, more investigations are requested to verify the detecting algorithm, such as different receiver and antenna pair, different length of baselines, etc.. In addition, the possibility of real time detecting multipath is another interesting topic.

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TIN-BASED GEOMORPHIC FEATURE EXTRACTION PROCESSES FOR HYDROLOGICAL MODELING

Honglei Zhu (hzhu@clarku.edu) and Kristin Schneider
Clark Labs, Clark University
950 Main Street, Worcester, MA 01610

ABSTRACT

Geomorphic features, such as channel networks and delineated watersheds, are basic features required for hydrologically related GIS modeling. In the past, extracting such features from a raster grid digital elevation model (DEM) has drawn more attention than from triangulated irregular network (TIN) models. Based on TIN models, this paper presents several geomorphic feature extraction algorithms designed and implemented in a popular GIS software. New algorithms for pit removal and watershed boundary delineation are proposed. 1) Pit removal. First, features like pits and outlets are identified. Then the preliminary channel networks are extracted. Based on the channel network, pit-affected watershed boundaries are delineated, then, pits are filled. 2) Watershed boundary delineation. Three procedures, preprocessing, watershed delineation and refinement, are proposed for watershed boundary delineation. Starting from a junction point between two neighboring channels, triangles are split if necessary along flow direction traced upwardly. After delineating watershed boundaries, a procedure for refining the boundaries is discussed.

Keyword: Triangulated irregular network, channel network, watershed.

1.0 INTRODUCTION

To perform hydrologic modeling, a significant amount of information must be gathered before the actual analysis can be done. Much of the information is related to geomorphic features of a watershed. Several researchers have investigated techniques for extracting geomorphic features from a digital terrain model. The terrain models are mainly raster grid digital elevation models (DEM) and triangulated irregular network models (TIN).

In the past, extracting such features from the raster DEM has drawn more attention because of the easy access of the data sets. Researchers have presented methods and algorithms based on DEMs. Peucker and Gouglas (1975) were among the first to develop and present such algorithms. O'Callaghan and Mark (1984), Douglas (1986), Jenson and Domingue (1988), among others, contributed to the refinement of the process for extracting drainage networks, watershed, and other hydrologic features from DEM. These techniques generally were based on neighborhood operations where calculations and decisions are made for each pixel on the basis of values in its spatially adjacent neighboring pixels.

Several researchers have studied the extraction of geomorphic features for hydrologic modeling based on a TIN model. Silfer *et al.* (1987) developed a TINFLOW system for flow hydrographs based on partitioning the flow across triangles. Algorithms for delineating watershed boundaries based on a TIN model have been investigated (Jones *et al,* 1990; Nelson *et al,* 1994; Tachikawa *et al,* 1996; Guercio and Soccodato, 1996)

Researchers have pursued geomorphic feature extraction based on other models too. More et al. (1988) showed that data organized in the form of contour strings could be useful in defining drainage because flow direction is always orthogonal to the contours.

This paper presents a study for pit removal and watershed delineation using the TIN models. New algorithms are proposed and validated.

2.0 CHANNEL NETWORK DELINEATION

First, some concepts and assumptions are introduced. A *pit* is a point where none of its neighbors have lower elevations. An *overflow point of a pit* is a point on the boundary of a basin that is characterized by having the minimum elevation difference relative to a pit. A *watershed outlet* is a point on the boundary of a basin that has the lowest elevation in the whole watershed. A *ridge point* is a point which has no drainage inputs. A *drainage channel* represents points at which runoff sufficiently concentrated such that fluvial processes dominate over slope processes. Detailed definition and description can be found in O'Callaghan and Mark's (1984) work.

Before delineating a channel network from a TIN model, flat triangles, flat ridges and channels must be processed and removed from the TIN model. We assume that such kinds of flat features in a TIN model have been processed when describing the following procedures for delineating a channel network.

2.1 LABEL PITS AND OUTLETS

If a point forms an edge with another point, this point is processed as a visible point by that point. In the process of labeling pits and outlets, for each vertex, only its visible points are examined. After checking all of the vertices in a TIN, pits can be identified. At this stage, we are only concerned with the outlets located at boundaries. Therefore, if a point occurs at the boundary of the data set, and it is a local minimum, it is identified as an outlet. Pit and outlet information is stored in a particular data file. We can read this file to get the information when it is needed.

2.2 LABEL TRIANGLE EDGE'S FLOW ATTRIBUTE

The azimuth of the steepest slope can be calculated from a triangle. According to the azimuth relation of a triangle edge and the triangle's steepest slope azimuth, an edge's flow attribute is assigned. The edge is classified by one of the three attributes: *flow in, flow out* and *flow along.* When there is surface flow, a flow in edge means surface water flows into the triangle from the edge. A flow out edge means water flows out from this edge, and a flow along edge means that water flows along the edge. These three values account for the three possible flow situations for a TIN edge.

Because a triangle facet is used to describe a surface unit, in an individual triangle, there is at least a flow in edge and a flow out edge as well as at most one flow along edge, two flow out edges, and two flow in edges. Figure 1 illustrates the azimuth of the steepest slope of a triangle and its three edges' attributes. Once the three vertex (X, Y, Z) values are input, the triangle's steepest slope can be calculated, and its three edges' attributes can be labeled. Figure 2 illustrates all of the possible attributes of a triangle's edges.

2.3 LABEL TRIANGLE EDGE'S FUNCTIONALITY

The following criteria are used for determining a TIN edge's functionality. If both neighboring triangles flow into a shared edge, this edge is a channel edge. If a shared edge is labeled flow out for both neighboring triangles, this edge is a ridge edge. If a shared edge has a flow along attribute for both neighboring triangles, this edge is a flow path edge. If a shared edge has a flow in attribute in one neighboring triangle and has a flow out attribute in the other triangle, this edge is a flow across edge. If an edge is only used by one triangle, then, the edge is a boundary edge.

According the above criteria, each edge of a TIN is classified with a functionality label from the five values: a channel edge, a ridge edge, a flow path edge, a flow across edge, and a boundary edge.

2.4 DELINEATE A CHANNEL NETWORK

In this study, a temporary channel network has to be extracted before processing for pit removal. The pit removal procedure requires a channel network to provide certain information. There are many discontinuous channel edges in current TIN models, therefore, extending the discontinuous channel edges is necessary before extracting the channel networks. Following along the steepest gradient, a discontinuous channel is extended starting at its lowest end and tracing until it reaches either a channel network or the boundary of the TIN model (Tachikawa *et al.*, 1996). Then, a stream network for a TIN is automatically delineated by starting at terminus points and recursively tracing uphill along channel edges until all branches of the stream are reached (Jones *et al.*, 1990). The channel network delineation process is illustrated in Figure 3.

The delineated channel network data are saved into a file to be read when the channel network information is needed.

3.0 PIT REMOVAL PROCESS

In this study, pits are removed in two steps. The first step is to detect the pit-affected watershed boundaries. The second step is to fill the pits. Since a watershed identifier has to be specified before

beginning a pit removal process, the following algorithm is used to assign each triangle a watershed identifier.

(1) Starting from the centroid point of each triangle in a TIN, trace downwardly along the steepest slope.

(2) Check the functionality of an intersected edge.

a) If an edge is part of a channel network, then the trace is stopped. The channel branch's identifier is assigned to the starting triangle.

b) If the edge is part of the boundary of a TIN, the starting triangle is assigned with a boundary identifier which identifies that the starting triangle contributes to a watershed which is outside of the study area.

c) If the edge's functionality is a low across, the tracing is continued till it reaches an edge with the functionality of a) or b).

3.1 DELINEATE PIT-AFFECTED WATERSHEDS BOUNDARIES

Previous research has almost universally recognized that pits are the nemesis of determining hydrologic flow directions because the pits must fill before the flow can continue. Some pits are data errors introduced in a TIN generation process, while others represent real topographic features. Several researchers have attempted to remove pits by smoothing the DEM (O'Collaghan and Mark, 1984). A second approach is to "fill" pits by increasing the values of cells in each pit to the values of the cell with the lowest value on the pit's boundary (Jenson and Trantwein, 1987; Jenson and Domingue, 1988). The perimeter of a pit feature is the highest-valued closed contour surrounding a minimum level (Jenson and Trantwein, 1987). In this study, we introduced another concept, i.e., *pit-affected watershed*. A pit-affected watershed is defined by all of the watersheds which contribute to a pit when there is a surface flow. All of the upward channels' watersheds are among the pit-affected watersheds. The algorithm for delineating a pit-affected watershed boundary is as follows:

1) For each pit in the data set, search all of the upward connected channels. All of the channel branches are used to determine whether a triangle contributes to the pit-affected watershed or not.
2) Search for a pit-affected watershed boundary edge. The edge is a portion of the pit-affected watershed boundary.
3) Delineate the pit-affected watershed boundary. Start from the above pit-affected watershed boundary edge, delineate this pit-affected watershed boundary, until a complete polygon is formed.

3.2 PIT FILLING PROCESS

Once all of the pit-affected watershed boundaries are delineated, the pits filling process is launched. The algorithm is presented as follows:

(1) Along the boundary of pit-affected watersheds, find the lowest point. This point is the overflow point of the pit-affected watersheds.

(2) Start searching and tracing from the overflow point, check all of the connected points within this pit-affected watershed boundary. Put all of the points into a stack if they are lower than the overflow point, take the lowest point as the next point.

(3) Continue searching and tracing until finding a lowest point which is higher than the overflow point, or reaching a boundary edge of pit-affected watersheds, which ever comes first.

(4) Using distance from the first point to the last point as a weight, modify all of the middle points' elevation values.

(5) If the stack is not empty, take one out and go to step (3).

The algorithm is illustrated in Figure 4.

4.0 WATERSHED BOUNDARY DELINEATION

The relationship between a triangle and a watershed falls into three categories: 1) the triangle contributes all of its surface flow into the watershed; 2) the triangle contributes a portion of its surface flow into the watershed; 3) the triangle does not contribute to the watershed. There are three steps in a watershed boundary delineation process related to such phenomena.

4.1 PREPROCESSING OF WATERSHED BOUNDARY

Preprocessing of a watershed boundary is designed for delineating boundary edges between two neighboring watersheds. The following algorithm is proposed:

1) Starting from each channel junction point, search all of the upward connected channels.
2) Check if there is a ridge or a flow path edge between two neighboring channels. If there no such edges are found, start splitting from the junction point, tracing upwardly along the steepest slope. The tracing is continued till a ridge edge is reached.

4.2 WATERSHED BOUNDARY DELINEATION

The preprocessing procedure delineates most of the boundary edges.

(1) For each channel branch, start tracing from the left and right side of lower ending point respectively.
(2) Find the connected points, select the point that has the highest elevation. Check the condition of the edge:
   a) If it is used as a ridge, a flow along edge, or a boundary edge, it is taken as a piece of sub watershed boundary. If the edge is not a ridge edge, one triangle is chosen from its two neighboring triangles which includes the steepest slope of going upwardly from the current point.
   b) Then, the triangle is split. The intersection point is calculated. X, Y and Z values of the intersection point are interpolated, and the TIN data is updated.
(3) Continue trace until a boundary point, or a ridge edge is met.

After preprocessing, watershed boundaries are delineated. The following algorithm is used for delineating a watershed boundary.

1) For each channel branch, take the lower ending point of current piece of channel, start tracing from the lower left side.
2) Trace along the watershed boundary until the starting point is met, and a complete polygon is formed.

The above algorithm is illustrated in Figure 5.

4.3 REFINEMENT OF WATERSHED BOUNDARY

If a boundary triangle contributes surface flow to more than one watershed, this triangle needs to be further processed until all of the boundary triangles contribute only to one watershed. The following algorithm is proposed.

For all of TIN edges, if a TIN edge is a watershed boundary edge:

If the current edge satisfies one of the following conditions, it is accepted as a final watershed boundary edge.

a) It is a boundary edge of the TIN data set.
b) It is a ridge edge.
c) It is a flow along edge.

If this edge does not satisfy the above conditions, a triangle is split along its steepest slope. And the TIN data is updated.

5.0 CASE STUDY

A contour data set of a small valley, ranging 3.21 km from East to West and 3.35 km from South to North, is used to validate the proposed methodology and algorithms. A TIN data set is generated based on the contour data set supported by the IDRISI for Windows GIS and Image Processing software. The interval of the contour data is 20 meters, and elevation ranges from 1120 to 1680 meters. Figure 6 shows the contour data set, and Figure 7 shows the optimized TIN generated based on the contour data set. The detailed TIN optimization process can be found in Zhu, Eastman and Schneider (1999).

5.1 CHANNEL NETWORK DELINEATION

Figures 8 and 9 are a TIN data set and channel network delineated based on the TIN. A channel edge extending process was not applied before delineating the channel network. In Figure 9, the channel networks are scattered along the valleys. Because we start delineating a channel network from a pit or an outlet, some channel edges in Figure 8 are not traced. Figures 10 through 13 illustrate better results after applying the channel edge extension process.

Considering the extending process results in a significant increase in number of triangles, and that is very time consuming, a parameter for limiting the process was specified. Five splitting processes were given in Figure 10. Thirty splitting processes were used as a limit in Figure 12, and Figure 13 is the result.

5.2 PIT REMOVAL PROCESS

After finishing the channel network delineation process, pit-affected boundaries were delineated. Figure 14 presents pits and pit-affected boundaries. Pits were overlaid on to pit-affected watershed boundaries. If two neighboring pits shared a common overflow point, the overflow point became a new pit after removing these two neighboring pits. Therefore, the pit removal process became iterative. The process was repeated until all of the pits in the TIN were removed. Figure 15 illustrates the updated information. The number of pits was decreased after processing. Four iterations were made before removing all of the pits from the TIN.

5.3 WATERSHED DELINEATION

After the pits removal process, watersheds were delineated based on the TIN and channel networks. Figure 16 illustrates the result. The channel network data set and contour data were overlaid on to the watershed data set. Detailed channel networks were used to delineate watersheds. From Figure 16, we can see that flow path edges, though they were used as a portion of channel networks, did not have accumulation areas.

6.0 CONCLUSIONS

The preliminary case study results have validated the algorithms and procedures proposed in the paper. Based on the primary information in a TIN model, together with its channel networks and watershed information, more information such as surface flow accumulation and flow length can be extracted.

This study has been done as part of a research and development project at the Clark Labs. Further research and development is being undertaken as well as the continued testing of currently developed algorithms in different geographic regions.

7.0 ACKNOWLEDGEMENT

The authors would like to give thanks to J. Ronald Eastman for his directives and valuable discussions. The authors would also like to give thanks to James Toledano, John Auble, Michel Fulk, and Stefenpho Crema for their valuable discussions and suggestions.

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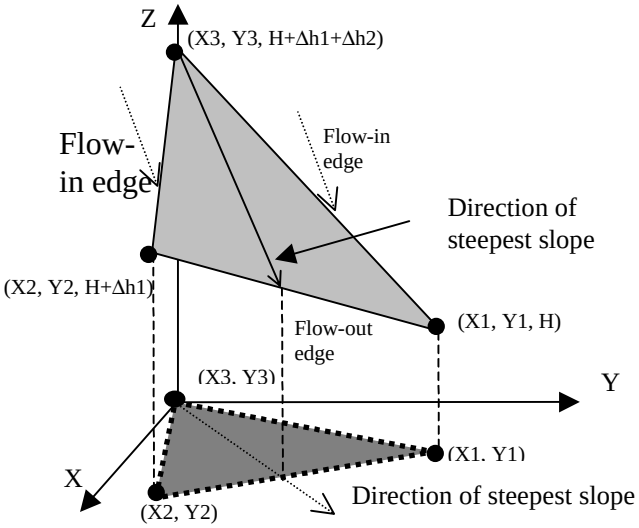


Figure 1. A spatial triangle and it surface water flow directions

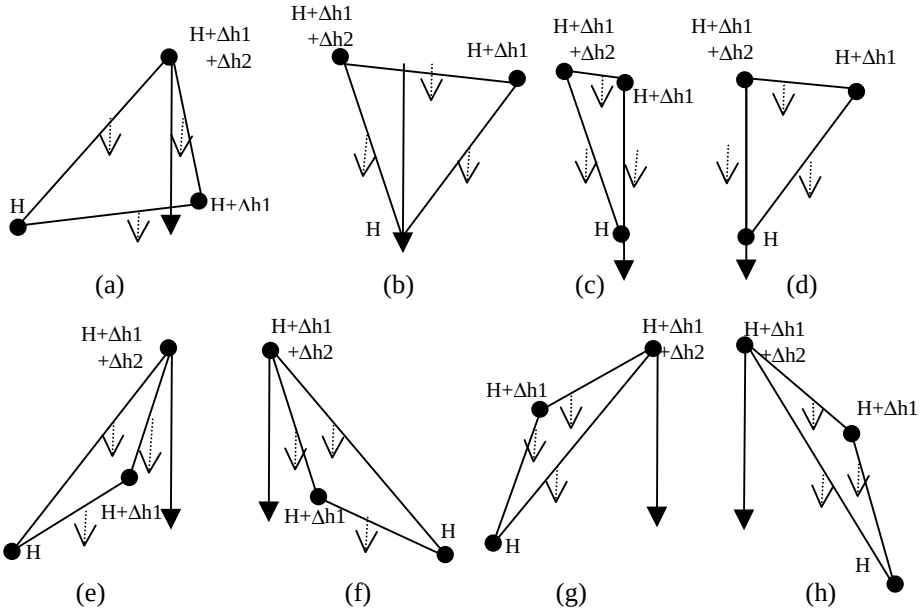


Figure 2. Label edge's attributes with *flow in, flow out,* and *flow along.* Arrows indicate the steepest slope.

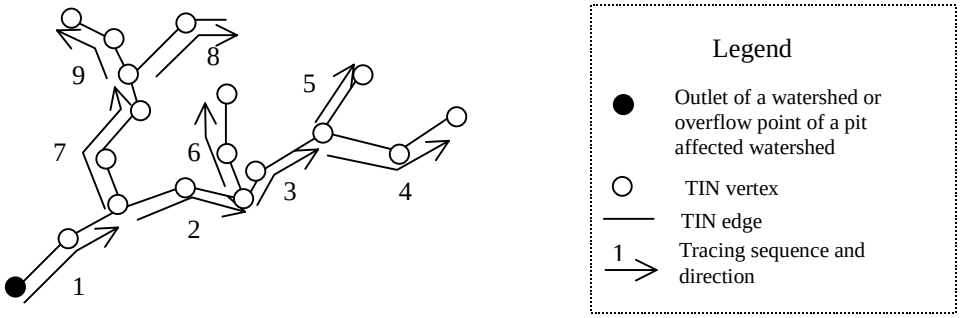


Figure 3. Channel Network delineation

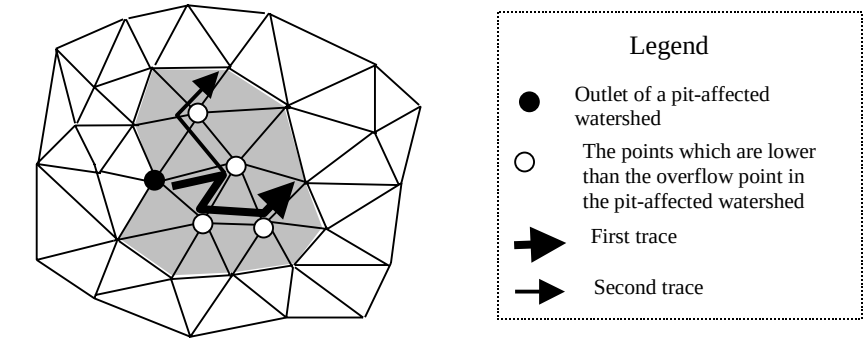


Figure 4. Pit Filling Process.

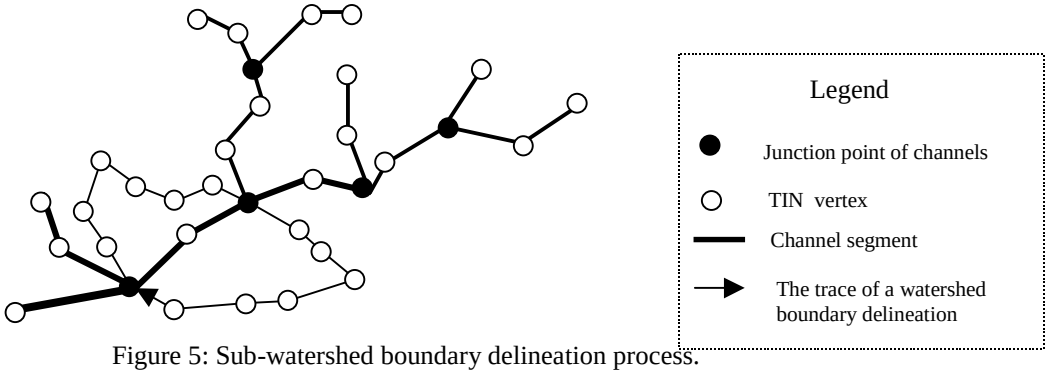


Figure 5: Sub-watershed boundary delineation process.

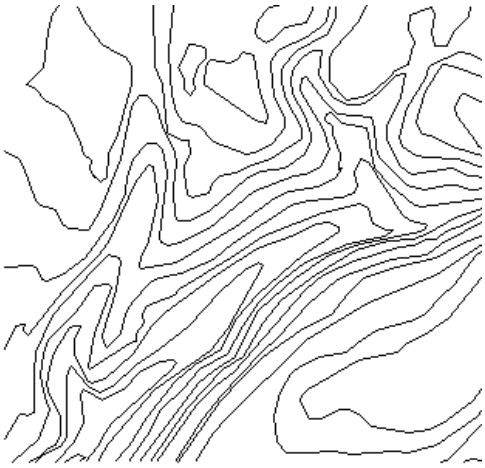


Figure 6. Contour data set of a study area.

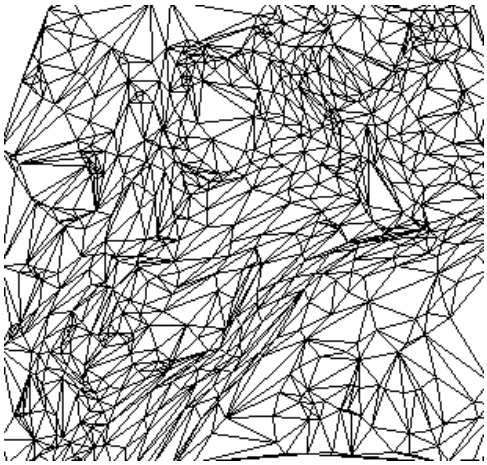


Figure 7. TIN data generated from a contour Data set.

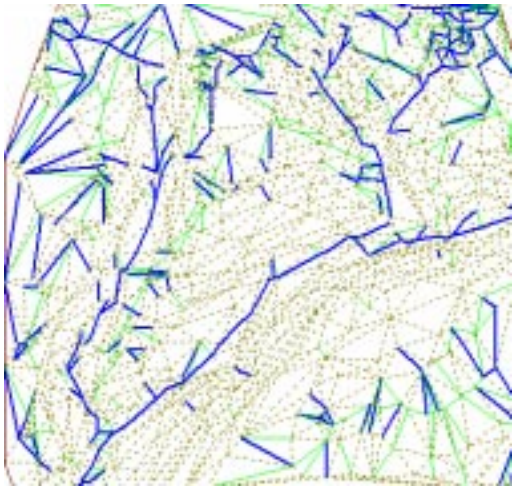


Figure 8. Edge attributes of the TIN. Thick blue lines represent channel edges, green lines represent ridge edges, and the dotted brown lines represent flow across lines.

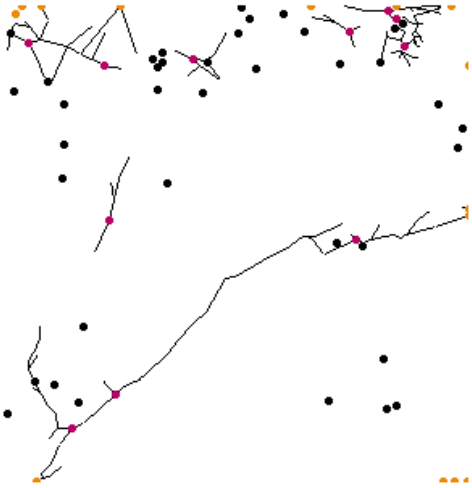


Figure 9. The extracted channel network. Outlets are in orange, pits are in purple, and peaks are in black.

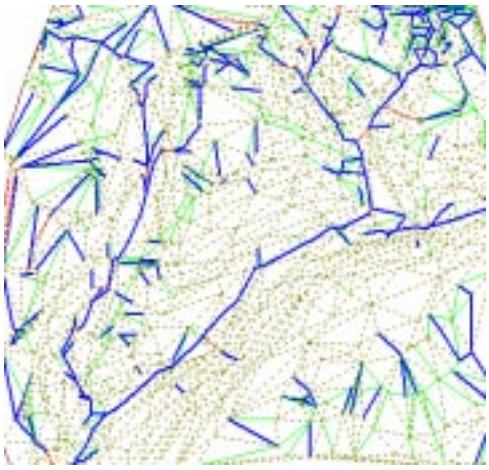


Figure 10. Edge attributes of the TIN. Discontinuous channel edges have been extended along the steepest slope. Five edges are given as a limit of the extension process. Thick blue lines represent channel edges, green lines represent ridge edges, red lines represent flow path edges, and the dotted brown lines represent flow across lines.



Figure 11. The extracted channel network. Outlets are in orange, pits are in purple, and peaks are in black.

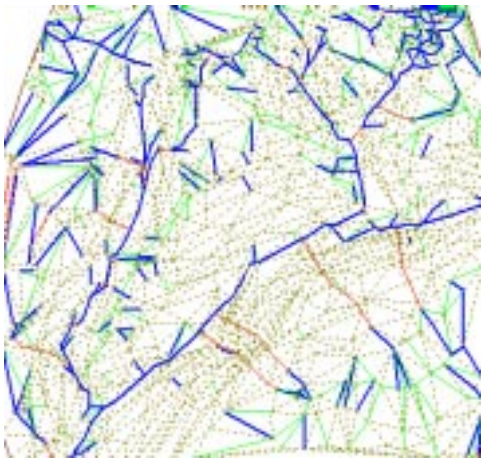


Figure 12. Edge attributes of the TIN. Discontinuous channel edges have been extended along the steepest slope. Thirty edges are given as a limit of the extension process. Thick blue lines represent channel edges, green lines represent ridge edges, red lines represent flow path edges, and the dotted brown lines represent flow across lines.

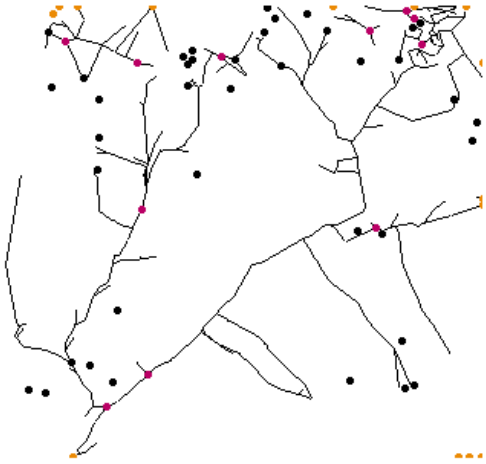


Figure 13. The extracted channel network. Outlets are in orange, pits are in purple, and peaks are in black.

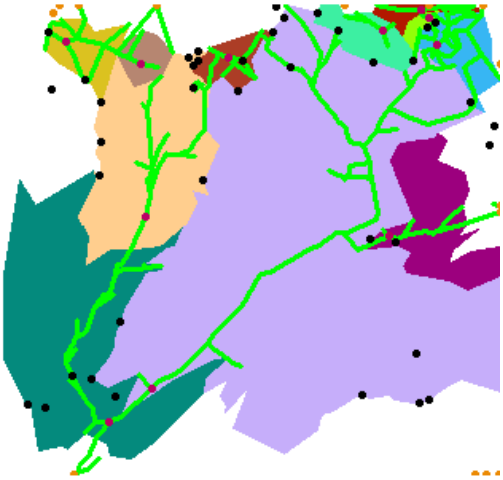


Figure 14. First pit-affected watershed boundary delineation result. Outlets are in orange, pits are in purple, and peaks are in black.

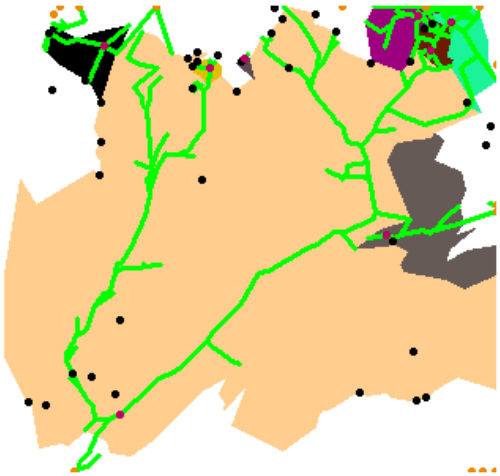


Figure 15. After the previous pit removal process the number of pits have been decreased. Several pit-affected watersheds are merged. Outlets are in orange, pits are in purple, and peaks are in black.

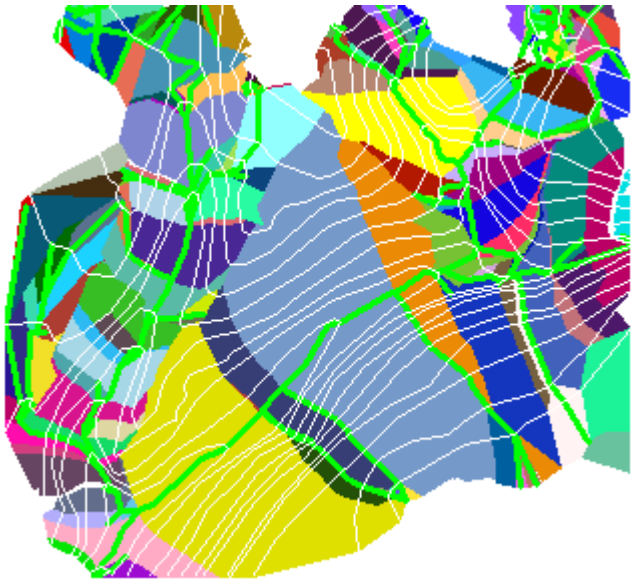


Figure 16. Watersheds extracted based on the TIN. The contour data set and channel networks are overlaid.

GIS and the Artist:
A Collaboration of Traditional and Computerized VisualizationTechniques in Participatory Design

Kheir Al-Kodmany,
University of Illinois at Chicago

Abstract

Public participation is a key component in environmental planning anddesign. Yet too often, planners and designers are not equipped witheffective tools and visualization techniques to generate meaningful publicinput. This paper explains how a Geographic Information System and an artist played critical roles in aparticipatory planning process in Chicago's Pilsen neighborhood. The GIS provided community leaders, planners,architects and designers with an interactive visualization of theneighborhood context. It also provided examples of design prototypes in relation to theirgeographic context. The artist, on the other hand, translated neighborhoodresidents' ideas into quick sketches, merging their ideas and thoughts into a sharedneighborhood vision. Both of these elements – the GIS and theartist – provided a means for residents to visualize past, present and futureneighborhood conditions, enabling them to have a greater voice in thedesign of their neighborhood. Our findings reinforce the viewthat visualization is a key component in public participation (King, 1998),and that effective visualization may be achieved through combiningtraditional and computerized visualization tools.

Key Words

GIS, Artist, Visualization, Interactivity

Introduction

This paper describes how design professionals at the University ofIllinois at Chicago combined traditional freehand sketching withcomputerized visualization techniques to facilitate public participation ina neighborhood planning process in Chicago's Pilsen community. Frustrated by the limitations of traditional planningand presentation methods, such as 35 mm slides, paper maps, and large foamboards, UIC planners developed an interactive Geographic Information Systemimage database. Used in tandem with an artist drawing "real time" sketches thatreflected community residents' emerging ideas, a powerful planning and visualization technique wascreated. This technique helped to open up the design process and buildtrust between the University and the neighboring Pilsen community, at thesame time increasing residents' design awareness, enhancing communication, and creating a sharedneighborhood vision. As a result, residents were able to create a designtemplate for their neighborhood to guide future redevelopment plans by theUniversity and private developers.

COPC at University of Illinois Chicago

The Community Outreach Partnership Center program of the U.S.Department of Housing and Urban Development is designed to assistuniversities indeveloping programs that involve faculty and students in applied projectsthat benefit neighborhoods and communities. The University of Illinois atChicago Neighborhood Initiative is one of 42 funded COPC programsnationwide. This initiative sought to alleviate the fear and distrust of neighborhood residents by buildingproductive partnerships. For the University of Illinois at Chicago (UIC),the COPC program and funding reinforced institutional priorities tostrengthen the quality of life in these neighborhoods through collaborative programs involving university faculty,community organizations, local government, and corporations (Wiewel, 1998).UIC's Great Cities program and the City Design Center express theuniversity's commitment to direct its teaching, research, and service programs to address urban issues in the Chicagometropolitan area. Both programs created partnerships between UIC andorganizations in neighborhoods surrounding the university, especiallyPilsen and the Near West Side.

Brief History of Pilsen/Little Village

Stretching east to west from the Chicago River to the city limits andnorth to south from roughly 16th Street to the Stevenson Expressway, thePilsen/Little Village community has a long history of community activism.Long used as a water passage for Native Americans traveling between the GreatLakes and the Mississippi
River, the Pilsen/Little Village area continued to be of vitalimportance after European settlement, becoming an economic and culturalcrossroads. As a port of entry for recent immigrants, the Pilsen area has welcomed countlessimmigrants to begin a new life in America. In the 19th century, workersarrived to build the Southwestern Plank Road; the Illinois and MichiganCanal; and the Chicago, Burlington, and Quincy Railroad. Many stayed to live in the neighborhood (Marzullo, 1998).

By the late 19th century, rapid industrialization and urbanization hadtransformed the largely Czech and German working-class neighborhood into anational center of labor activism. Poles, Croatians, Lithuanians, Italians, and members of other ethnicgroups settled in the area, and all have left their imprint. Now one of thelargest Mexican communities in the United States, Pilsen/Little Village andits residents have, for more than three decades, engaged in a struggle for politicalrepresentation, educational reform, social justice, and workers' rights(Marzullo, 1998).

The Project

The University of Illinois at Chicago is located in the heart of thePilsen community. The shared history of UIC and its neighbors includes notonly the displacement of homes and businesses to accommodate the University

rquotes need for expansion, but also large, well-publicized, and eventuallydiscontinued community service programs. These issues have created a nearly universal distrust of the University in Pilsen/Little Villageand the Near West Side. It was within this context that Universityplanners and community leaders spearheaded a planning and design processfor the neighborhood.

Pilsen, which this research focuses on, is a largely Mexican-Americanand Mexican immigrant community of nearly 50,000 people. Leaders in thePilsen community expressed an interest in a participatory approach to theplanning and design of the future of their neighborhood. Like many other low-income neighborhoods, Pilsen neededto address a host of urban issues, including heavy traffic, run-downbuildings, vacancies and crime. The community was anxious to harness thecreative energies of its members as a way of fostering the enthusiasm required to take serious actions towardimproving the neighborhood. The community felt that the meaningfulinvolvement of the entire community would strengthen members' sense of community, and a cooperative effort would help present a"unified front" when funding opportunities arose. Aplanning team was formed that included 25 community members, twoarchitects, two planners, and one artist.

The University team's objectives, on the other hand, wentbeyond the actual neighborhood planning and design process. UIC's objectives included creating a mutually respectful partnership withneighborhood residents, preserving neighborhood history, providing abroader understanding and context of urban issues, and exploring effectivevisual communication methods. Building trust was the highest priority inthe planning process. Trust arises from consistently meeting expectationsand creating outcomes that all partners perceive as beneficial. One of thefirst lessons that the University team would learn was that effective visualization was a key to buildingtrust throughout out the process.

The Problem

After a short period of involvement, University design professionalsrealized that the presentation techniques at their disposal were not promoting trust and meaningful public participation. Slide images ofthe neighborhood were presented at the beginning of the session to displaycurrent site conditions. Slides were set and presented in a fixedsequence. As the discussion moved from the project introduction to the design development stage, there was nointeractivity between the present conditions of the neighborhood and thedeveloped future images. As a slide projector lacks navigationcapabilities, the images were not readily available during the design discussion. When participants requested viewing aspecific image, it was impractical to search for it in the slide tray.Similarly, when the artist and the planners asked for specific images tohelp them remember contextual details of the site, the images were inaccessible. Consequently, the processlacked a means of placing what was being proposed within the context ofwhat currently existed. Residents and community representativesexperienced disorientation, as they had to relyon their memories of the site. Planners, architects and artists also grewfrustrated with the limitations of the design process (Al-Kodmany, 1998).

The UIC team began searching for a visualization environment thatcould effectively connect the two traditional stages of a project: orientation and design development. There is anintimate relationship between proposed design alternatives and theirphysical context. A visual connection between the two had to beestablished.

Method

The GIS

A system was needed that would geographically illustrate theneighborhood's cultural and architectural history; its present conditions, including itsstrengths, weaknesses, opportunities and threats; and provide some designprototypes to foster discussion about how the neighborhood might look in five, ten and twenty years. First, the databaseneeded to illustrate the neighborhood context, including its history,geography, architecture, and cultural values. An interactive GIS imagedatabase was developed. It consistedof maps, images, tabular data, and textual information about the Pilsenneighborhood and its surroundings. Thematic layers were created for platmaps, land use maps, zoning maps, base maps, historic maps (fire insurancemap), and current aerial photographs. Historic images showing the neighborhood characteristics in varioushistoric periods were collected. These photographs were hot-linkedrespectively to historic maps of various periods. The images werecoordinated with a key plan showing direction and the position of each image. This plan was overlaid on the historic maps.This arrangement was meant to aid orientation for future reference in thedesign process.

Second, the database had to show existing conditions of theneighborhood. A digital camera systematically documented the present condition of the neighborhood, andimages were hot linked to their geographic location. Members of theuniversity design team photographed the long-distance implications of theneighborhood from vantagepoints, i.e. tall buildings. Conversely, on-neighborhood shots recorded spatial linksfrom inside out: routes, and shuttered, filtered, sneak, and panoramicviews into the surrounding space. The team also shot all access routes andvisual links—sneak glimpse, views, etc.—from around the immediate neighborhood periphery looking in.On-neighborhood photographs also documented the mass and details of theimpinging forms: materials, openings, textures, colors, and the like.Photographs were hot-linked to their geographic location. The above photographs were also coordinated with a key plan showingdirection and the position of each shot. As with the historic photos, thisarrangement was meant to aid orientation during future reference in design.

Finally, a GIS library of environmental design prototypes was incorporated in the database. Thislibrary was created by the City Design Center. It consisted of photographsof key developments in Chicago's neighborhoods, particularly those adjacentto the campus. The artist on the University design team annotated these photographs regarding quality,historical significance, architectural style, and building materials, andthese photographs were hot linked to their geographic locations. Thisarrangement was intended to visually represent types, architectural styles, and locations of buildings and designsthat could be incorporated into the neighborhood plan.

Implementation

Workshops were planned for four consecutive Saturdays, from 9 a.m. to5 p.m. at a church in the Pilsen neighborhood. Equipment included a computer, an electronic whiteboard, twoprojectors, and two large screens. One screen was used to display theartist's sketches, and the other was used to display the GIS images. Thepresident of the Pilsen community organization served as the host for the event, welcoming and introducing communityand University participants, describing the goals and objectives of theworkshops, and describing the contribution of participants in the long-termplanning process. Several ground rules were set for the workshop discussions, including 1) Speak only foryourself; let others speak for themselves; 2) Don't criticize anidea; instead, suggest alternatives; and 3) Don't focus on solutions; rather, brainstorm for alternatives.

The design process ran counter to conventional practice. It is common thatarchitects and planners prepare a master plan and then proceed to addressdetails (King, 1998). The workshop reversed the process by addressing theparts of the design first and then progressing to the whole. The design process began with the life style of theinhabitants, continued to the design of individualbuildings and then tothe entire neighborhood. This is the opposite of the traditional process,in which the first communication with the public is presenting an entire plan.

The artist wrote the name of an activity at the top of the electronicsketch board. Then, she drew a figure engaged in the activity, and addedmore figures as directed by the participants. Through questionand response, the participants suggested furniture, planting, cars,buildings, and other environmental features. At each suggestion, theartist glanced at the eyes of the participants to check their agreement onthat the image was developed as it should. Soon, rough images began to appear of Pilsen streets atdifferent seasons (e.g. winter, summer), of people, roller skating,jogging, cycling, strolling, and sitting at cafe tables or open-air restaurants, or dancing at an outdoor concert. A myriad of vividimages developed depicting a rich display of urban activities. Theseimages would later be incorporated into an overall plan for the area.

The GIS image database tremendously assisted the planning team invisualizing the past and present conditions of the neighborhood. It engagedthe community members in developing alternative solutions. It also helpedin visualizing current urban development examples in the city. The artist, with traditional felt tips, markers,and sketch paper, translated and incorporated the audiences' ideas ontopaper. The constant reference to the image database – includingmaps, existing buildings and lots– made the discussion contextual and more realistic. The imaged database helped theaudience articulate their ideas in relation to neighborhood context.Together, the artist and the GIS image database reinforced each other increating a common visual language. Thecombination of the high-tech GIS and the low-tech sketches facilitated adynamic communication among planners, architects, designers, and thecommunity. The project demonstrated how a combination of

basic sketchingand visual techniques along with advanced GIS visualization power allowed participants to reach an optimalvisualization environment.

Once participants agreed on an idea, the artist sketched it on paper.Sketches were immediately projected on a large screen, side by side withthe screen containing computer images. This positioning of screens allowed participants toconstantly observe how the artist modified images and incorporated them inthe design. Everyone was potentially able to voice an opinion or concern.Such a setting reinforced the reliance on visual cues and minimized reliance on jargon. Consequently, itincreased participation in the design process. Incrementally, when theaudience discussed new ideas, the artist further developed these freehandsketches. This time, instead of two distinct phases -- orientation (through use of a slide show) and designdevelopment (artist with a sketch pad), the GIS database and the artistworked interactively, resulting in a more fluid, contextual design process.This system helped participants view the gradual design process and witness the impact that their input hadin shaping the design and ultimately, their neighborhood.

Following the four-week series of workshops, sketches were put onpublic display in the church hall for two weeks. The display represented a logical extension of the public design process, allowingpeople who did not attend the workshop an opportunity to view the sketchesand plans. In order to personalize the design process, the signatures ofthe community leaders and theUniversity designers were imprinted on the drawings. Following the publicdisplay period, the drawings and comments were gathered into a report. Thereport represented the community's vision for Pilsen and served as a reference guide for futuredecision-makers, developers, planners, urban designers and architects.

Results

The combination of the GIS and the artist had many positive outcomes. First, the use of this technique promoted strong community involvement in the planning process, which was an objective of the community and the University. The GIS and the artistworking in tandem helped community residents articulate their ideas inrelation to neighborhood context. Together, the artist and the GIS imagedatabase reinforced each other in creating a common visual language. People not trained in the design professionssometimes have a difficult time communicating ideas about architecture andurban design, but most people have definite design preferences. Tofacilitate this, the GIS image databasecontained examples of numerous developments near the neighborhood andthroughout the city. Images that represented design examples andprototypes were used as anchor points for discussing developmentalternatives. As participants suggested solutions, the planner would display images on the large screen that most closelymatched the participants' ideas. Design examples were used to probeand support the audience's ideas.

Second, the GIS helped highlight the importance of cultural values andhistory in the future design of the neighborhood. One of the major concerns of thePilsen community is to preserve their cultural heritage represented in thephysical form. The GIS images reminded the artist, the planners, and thecommunity residents of the cultural artifacts and

environmental elements in Pilsen. These images supporteddiscussion of cultural issues in the neighborhood. Images helped theartist to incorporate some of the cultural symbolic features and artifactsof the community in the new designs. Also, the GIS showed the geographic distribution and clusters of buildingsof significant cultural and historic value. Consequently, it was clear toeveryone the areas of great cultural and historic significance and theareas of less importance.

Finally, and most importantly, the workshops and the visualization tools helpedbuild a relationship of trust between the University and the community.The GIS and the artist helped empower residents to plan and design for thefuture of their community. The designs that were created by the planners and designers reflected their wishesand input and respected their cultural heritage. At the end of theprocess, the community felt that the purpose of the University was not todestroy their lifestyle but to revitalize their community. This helped overcome some of the distrust problemsthat were experienced in the past.

Building the GIS database was a tremendous undertaking, requiring morethan three months. University planners and designers exceeded the budgetfor this project due to the labor-intensive activities required in gatheringand assembling the images, maps and historical data. However, the benefitsof this system for the University and the neighborhood far outweighed thecost. The visual context provided by the GIS image database was critical to the success of the project.People’s perceptions about the built environment vary tremendously by theirgeographic location, cultural background, length of stay, age, gender,socioeconomic status and the like. The GIS helped bridge those perceptions, helped level the playing fieldbetween the technical experts (the planners and designers) and theneighborhood. Everyone on the planning team had access to the samecontextual information and could formulate their ideas and designs accordingly.

The cost of this project is also justified because the City DesignCenter, the Great Cities Institute and the Urban Data Visualization Programat the University of Illinois have longer-term projects involvingvisualizationof Metropolitan Chicago. This project was viewed as a "test" to theeffectiveness of visualization tools in public settings. The findings ofthis research reinforce views about the importance of visualization inparticipatory design. The most important finding is that each of the artists' visualization capabilities and theGIS visualization capabilities were irreplaceable. Collaboration betweenthe GIS and the artist provided a unique visualization environment thatimproved participatory design.

Conclusion

For participatory design, this project suggested that a collaborationof traditional and GIS tools could be effective. It showed howcollaboration of basic sketching and visual techniques along with advancedGIS visualization power created an optimal visualization environment. Visualization is a key in publicparticipation because it is the only common language that all participantscan relate to. Visualization provides a focus for a

community's discussionof their design ideas; it guides them through the design process; it raises their design awareness, and facilitatesbetter communication. Consequently, exploring alternative visualizationtechniques could be a necessity for ensuring successful communication.Planning processes vary in nature and different situations, ends and goals, may require differentvisualization tools.

In describing the workshop experience, this paper aims at reinforcingthe work of others in the field of public participation. By sharing ourexperience, we may arrive atbetter methods of cooperative community design. Clearly, the developmentof methods and skills in community design are in their infancy, still atthe exploratory and discovery stage. This paper is a step in thedevelopment of such skills – progressing toward the art of designing with people.

Author's Note:
The author wishes to thank several individual from the University ofIllinois for their help in this project including: Robert Brugmann andPeter Hales (Art History); Charles Hoch, Wim Wiewel, and Tinwei Zhang,(Urban Planning), Roberta Feldman (City Design Center and Architecture); George Hemmens (City Design Center and UrbanPlanning); James Hudson (Arc/Info Technology Lab); Yequao Wang (GeographyDepartment); Kate Pravera (Great Cities Institute). The author also wishesto deeply thank Angie Mark for editing the paper. Finally, the author wishes to thank Mike Schwartz andIsmail Sumairah for useful comments.

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HAPs, Micromaps and GPL - Visualization of Geographically Referenced Statistical Summaries on the World Wide Web

J. Symanzik*, D. A. Axelrad, D. B. Carr, J. Wang, D. Wong, T. J. Woodruff

* Jürgen Symanzik
George Mason University
Center for Computational Statistics 4A7
Fairfax, VA 22030
e-mail: symanzik@galaxy.gmu.edu

Abstract

In this paper we report on an approach to visualize the U.S. Environmental Protection Agency (EPA) hazardous air pollutant (HAP) data on the World Wide Web (WWW). Long-term cumulative concentrations for 148 HAPs have been estimated as a part of EPA's Cumulative Exposure Project (URL:CumulativeExposureProject) for each of the 60,803 census tracts in the 48 contiguous states. Confidence bounds are available as well for this data set.

Our WWW approach is based on the Graphics Production Library (GPL) (Carr, Valliant, and Rope, 1996) and extends it with micromaps (Carr and Pierson, 1996; Carr, Olsen, Courbois, Pierson, and Carr, 1998). The GPL is a JAVA-based application, developed and maintained within the Bureau of Labor Statistics (BLS), that allows to display statistical data as row-labeled plots and time series plots on the WWW. This data visualization approach closely follows recent recommendations on statistical graphics by Carr (1994) and Cleveland (1993, 1994). Micromaps are a series of small generalized maps that highlight the geographical region associated with statistical features in accompanying plots.

When added to the GPL, micromaps serve two purposes. In addition to providing the geographical component of a linked statistical summary, they serve as a navigation tool for drilling down through a hierarchy of maps. A WWW user of our application can click on a state and advance to the underlying county-within-state micromap display. Selection of a county leads to information at the census tract level. This hierarchy of clickable maps and statistical displays allows the WWW user to look at estimates and distributional summaries at the desired geographic resolution. Atypical values in the statistical summaries may lead the investigator to points of interest at a higher spatial resolution level.

Additional interaction through the Web browser is possible. The user can easily toggle between individual HAPs, look at formatted tabular data, and display and download the raw unformatted data.

1. Introduction

This paper reports on an approach to access and visualize the U.S. Environmental Protection Agency (EPA) hazardous air pollutant (HAP - also called air toxics) data on the World Wide Web (WWW). The Cumulative Exposure Project (URL:CumulativeExposureProject) Web page (Figure 1) has been designed and implemented to provide insight into the scope and the underlying modeling process of the project. A major goal of the Web page is to provide access to the modeled 1990 HAP data at different spatial resolutions.

The Cumulative Exposure Project Web site has scheduled updates. A first version of the Cumulative Exposure Project Web page went online in March 1998 to provide background information on the project. An updated version that provides access to the modeled 1990 HAP data at different spatial resolutions is scheduled for release in December 1998. Additional graphical components will be added to the page in January 1999.

In the Cumulative Exposure Project, EPA has developed modeled concentration estimates of 148 air toxics for every census tract in the continental United States. Air toxics are pollutants known or suspected to cause cancer and other serious human health effects. The modeled concentrations are annual averages for the year 1990, and uncertainty bounds have been developed for each estimate. Development of the modeled concentrations for the 60,803 census tracts in the continental U.S. is described in Rosenbaum, Axelrad, Woodruff, Wei, Ligocki, and Cohen (1999), and analysis of the modeled concentrations is presented in Woodruff, Axelrad, Caldwell, Morello-Frosch, and Rosenbaum (1998) and Caldwell, Woodruff, Morello-Frosch, and Axelrad (1998).

The purpose of the web page is to provide easy, fast, and understandable access to the model results. Since the results are numerous, involving 148 x 60,803 x 3 numeric values, we have taken a hierarchical spatial approach in their display. The WWW user starts at the top (US) level with data at the state level being displayed. Selecting a state causes its county level estimates to be shown. Selecting a county causes its underlying census tract level to be displayed. It is possible to toggle between individual HAPs at any stage. Mouse-clickable maps, menus, and geographical names in the Web document allow the user to easily move up and down in this hierarchy of maps and displays. Thus, the user can look at data and summary statistics at any desired spatial resolution.

In Section 2 of this paper we introduce micromaps and the Graphics Production Library (GPL) and explain their joint use in the Cumulative Exposure Project Web page. Section 3 focuses on the tabular displays of the Web page. Section 4 concludes with a discussion of achievements so far and mentions possible extensions. Additional details on the Cumulative Exposure Project and the related Web page can be found in Symanzik, Wong, Wang, Carr, Woodruff, and Axelrad (1999).

2. Linked Micromap Plots and the GPL

Linked micromap (LM) plots, often simply called micromaps, provide a new way of viewing spatially indexed estimates and summaries statistics. Carr and Pierson (1996) and Carr, Olsen, Courbois, Pierson, and Carr (1998) provide the basic descriptions, several examples, and cite the connections of LM plots to other statistical graphics. A LM plot consists of parallel sequences of micromap panels, label panels, and statistical summary panels. The micromap panels are typically map generalizations or caricatures. The caricatures preserve region neighbors and enlarge very small regions so their color is visible. The label panels provide regions names. The statistical summary panels represent estimates, confidence bounds, and related information using the most perceptually accurate representation, i.e., the position along a scale. Individual panels take familiar forms such as dot plots, bar plots, and box plots. The representation typically uses the encoding that has the highest perceptual accuracy of extraction, i.e., the position along a scale extraction (Cleveland and McGill, 1984). The sequence of panels results from sorting and logical or perceptual grouping. Color and position link corresponding elements within the parallel sequences.

The design of LM plots contrasts with the design of classed choropleth maps. In LM plots, the maps are caricatures but sufficiently convey the location or regions associated with the statistical summary. The statistical summaries have high perceptual accuracy of extraction. Classed choropleth maps use the best representation, i.e., the position along a scale, and most of the space to represent political boundaries. The classed choropleth map design typically discards confidence bounds on statistical estimates, degrades the estimates into a few classes, and represents the order classes with a poor encoding, usually color. Thus the LM plots representation places more emphasis on the estimates and their quality. Knowing the census tract, the county, or state name and general location is usually good enough for an interpretation of the statistical data.

The Graphics Production Library (GPL) is a set of JAVA class libraries for interactive statistical graphics (Carr, Valliant, and Rope, 1996). The GPL was initially intended to add interactivity, such as drag and drop comparisons, panel reordering and rescaling; and pan and zoom to the row-labeled plots of Carr (1994). It also follows recent recommendations on statistical graphics as given in Cleveland (1993, 1994). The design of the GPL also addressed the display of times series and provided for incorporation of metadata, such as warning flags on the time series and links to articles on the time series adjustments. The library was developed to facilitate web distribution of statistical summaries from the Bureau of Labor Statistics (BLS). The GPL, as maintained by the BLS, provided a reasonable starting point for incorporating the micromap capabilities and producing LM plots on the WWW.

Even though LM plots will not be fully available on the Cumulative Exposure Web Page before January 1999, we can provide a first idea how these LM plots will look like. Figure 2 shows a micromap at the top (US) level with states sorted in alphabetical order. The total modeled 1990 HAP concentration of all 148 HAPs, aggregated over the census tracts in each state, is displayed. In our final version on the Web, this display will allow

the user to Mouse-click on a state and move to next underlying level. As an example, Figure 3 shows a micromap for Michigan. Again, the total modeled 1990 concentration of all 148 HAPs, aggregated over the census tracts in each county, is displayed. However, although we used the total modeled concentration as a summary for all HAPs in these two figures, this information will not be provided on the Web page. Obviously, the total is quite meaningless with respect to the toxicity of a particular HAP. A relatively small quantity of one air toxic may be far more poisonous than a much larger quantity of another air toxic. The total might easily lead to wrong conclusions when publicly being displayed on the Cumulative Exposure Project Web page.

3. Tabular Displays

Similar to the graphical display based on LM plots and the GPL, we can access our data through formatted data tables. These tables are available at the same spatial resolutions as the LM plots. We start at the top (US) level where data is being displayed for one selected HAP, here benzene, for all 48 states and the District of Columbia (Figure 4). Summary statistics with respect to the selected HAP are being displayed.

We can now select a new state either by changing the menu option for States on top of the table or by Mouse-clicking on the desired state. In our case, we decided to look at data for Rhode Island. The resulting table is displayed in Figure 5. Obviously, the selection menu on top reflects this state selection, allowing us to select one of Rhode Island's five states. The same summary statistics as in the previous figure with respect to the selected HAP are being displayed, however this time at the state level.

Finally, we can now select a single county either by changing the menu option for Counties on top of the table or by Mouse-clicking on the desired county. In our case, we decided to look at data for Rhode Island's Bristol County. The resulting table is displayed in Figure 6. Obviously, the selection menu on top reflects this state/county selection. Now the modeled 1990 concentration of benzene and a corresponding 90% confidence interval are being displayed in the table.

At any time, the user can toggle to a different HAP, a different representation (micromaps or raw data), or move up and down in the spatial hierarchy.

The final display type, the raw data representation, gives an unrestricted insight into the raw data as it comes out of the modeling process. In particular, no rounding takes place. This representation can best be used to download data for one or multiple regions and perform additional calculations or display the data in a different format. The data is comma-delimited and can be transferred by copy/paste from the Web browser window into any application window. At the next stage of this Web page, it is planned to provide access to entire data files at a higher spatial resolution to ease further usage of the data.

4. Discussion and Outlook

When finalized in January 1999, the Cumulative Exposure Project Web page will provide an easy-to-use and up-to-date interface for access and display of a large environmental data set at different spatial resolutions. The combination of LM plot design and the GPL provides the basis of a hierarchical clickable approach to the display of spatially-indexed estimates and summaries. The hierarchical spatial approach is intuitive and has been easily understood by WWW users with little statistical experience. The approach provides a visual query language that can provide access to summary tables and raw data. Future work will add more user preferences and allow for an additional analysis of the data.

This work promotes the display of confidence (or uncertainty) bounds. Such bounds are regularly reported with professional political polls. The presence of such bounds suggests the use of appropriate statistical methodology and the absence of bounds raises the question, do the reported estimates have any statistical validity? The modeling of HAPs results in wide uncertainty bounds. The Web page provides background on these modeled bounds. While methods for calculating bounds are subject to debates and refinements, publishing wide bounds reduces the chances the reader will be badly lead astray by poor estimates.

There are several possibilities for additions to this Web page after the completion of the micromaps display in January 1999. Currently, it is not possible to access the modeled 1990 concentration for all 148 HAPs for a particular location (either state, county, or census tract) at the same time. Instead, one has to toggle through 148 different Web pages to learn about all the HAPs at one location. One goal is to make this data available at one glance, which requires to display data of very different dimensionalities.

Also, currently there is no easily accessible site on the Web that allows the mapping of a particular ZIP code to a census tract number which are used throughout the Cumulative Exposure Project. We plan to add such a mechanism to our Web page.

At this stage, only the modeling of the 1990 air toxics is finished. Once available, we plan to dissiminate the modeled data for exposure levels for chemical contaminants found in public and private drinking water supplies and exposures to contaminants in foods in a similar way through this Web page.

Acknowledgments

EPA funded the majority of the work behind this paper under contract No. 098272 and cooperative agreement No. CR825564-01-0. Additional federal agencies, BLS and NCHS, supported some facets of this work. The article has not been subject to review by any of these agencies so does not necessarily reflect the view of the agencies, and no official endorsement should be inferred. The conclusions and opinions are solely those of the authors and are not necessarily the views of the agencies.

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URL:CumulativeExposureProject http://www.epa.gov/CumulativeExposure/

Figures

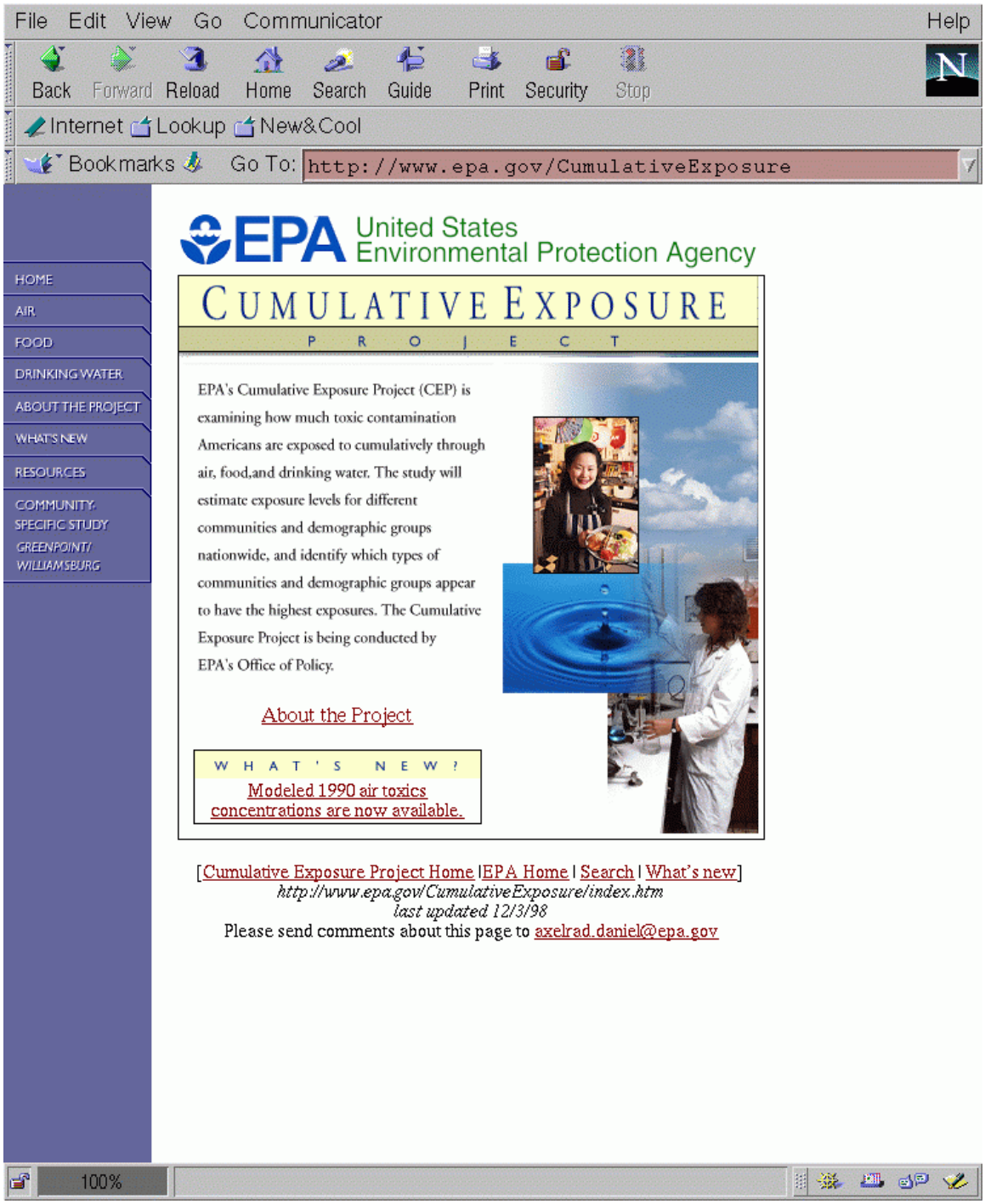


Figure 1: The starting page of the Cumulative Exposure Project Web page, accessible at http://www.epa.gov/CumulativeExposure/.

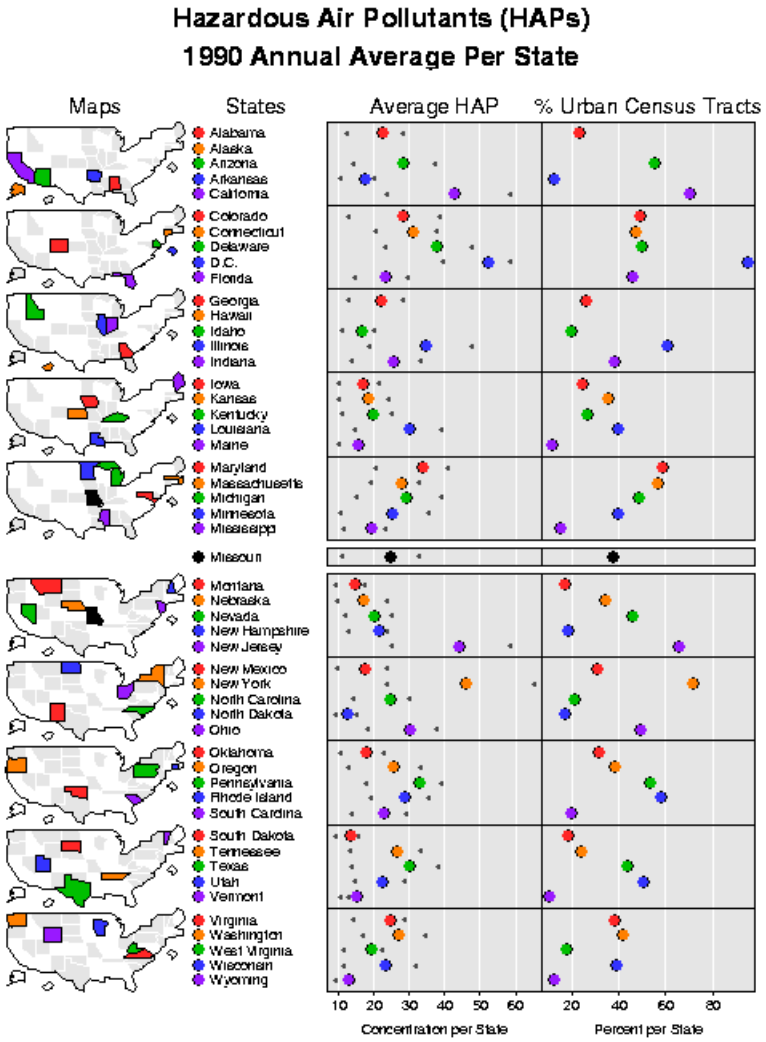


Figure 2: Micromap display at the top (US) level for all 50 states and the District of Columbia. The total modeled 1990 HAP concentration in micrograms per cubic meter for all 148 HAPs has been averaged over the number of census tracts in each state. The 25th and 75th percentiles have been displayed as well. For some states, e.g., Vermont, the distribution of HAP concentrations is extremely skewed which becomes obvious when the displayed arithmetic mean falls above (or below) the 75th (or 25th) percentile. The accompanying second data column (which will not be available on the Web page) indicates the percentage of urban census tracts. A strong correlation between the two displayed variables can be visually detected.

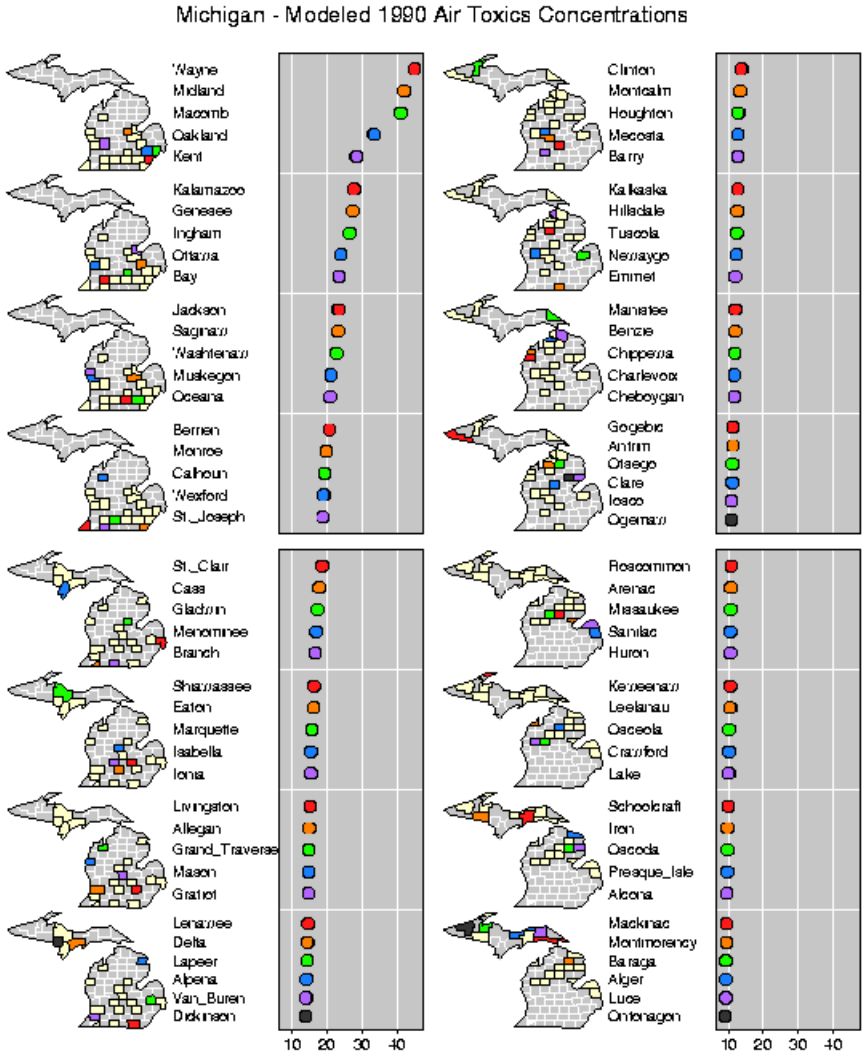
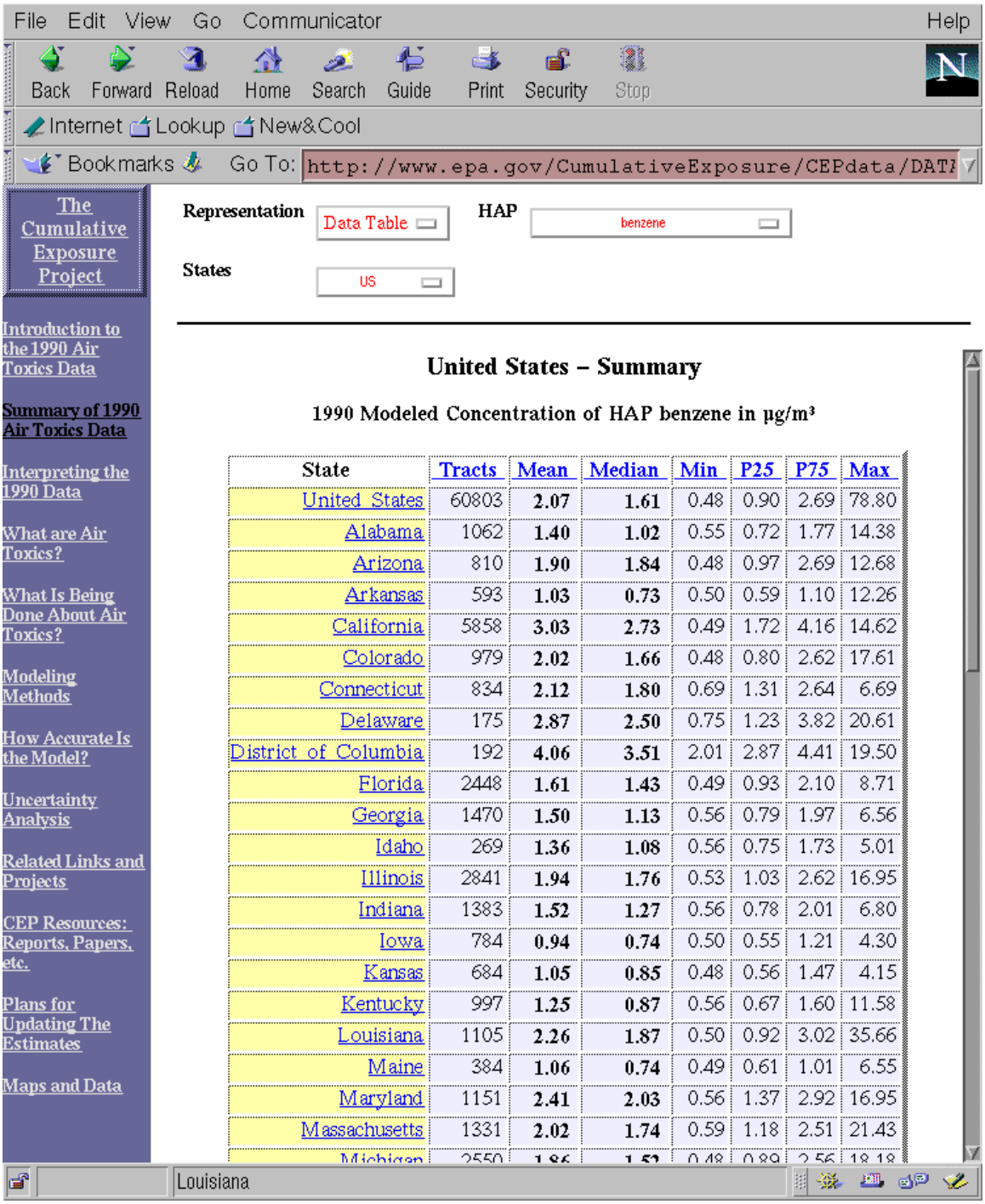


Figure 3: Micromap display at the state level for Michigan for all of its 83 counties. The total modeled 1990 HAP concentration in micrograms per cubic meter for all 148 HAPs has been averaged over the number of census tracts in each county. Counties are ordered from highest to lowest with respect to the total modeled 1990 HAP concentration.



State	Tracts	Mean	Median	Min	P25	P75	Max
United States	60803	**2.07**	**1.61**	0.48	0.90	2.69	78.80
Alabama	1062	**1.40**	**1.02**	0.55	0.72	1.77	14.38
Arizona	810	**1.90**	**1.84**	0.48	0.97	2.69	12.68
Arkansas	593	**1.03**	**0.73**	0.50	0.59	1.10	12.26
California	5858	**3.03**	**2.73**	0.49	1.72	4.16	14.62
Colorado	979	**2.02**	**1.66**	0.48	0.80	2.62	17.61
Connecticut	834	**2.12**	**1.80**	0.69	1.31	2.64	6.69
Delaware	175	**2.87**	**2.50**	0.75	1.23	3.82	20.61
District of Columbia	192	**4.06**	**3.51**	2.01	2.87	4.41	19.50
Florida	2448	**1.61**	**1.43**	0.49	0.93	2.10	8.71
Georgia	1470	**1.50**	**1.13**	0.56	0.79	1.97	6.56
Idaho	269	**1.36**	**1.08**	0.56	0.75	1.73	5.01
Illinois	2841	**1.94**	**1.76**	0.53	1.03	2.62	16.95
Indiana	1383	**1.52**	**1.27**	0.56	0.78	2.01	6.80
Iowa	784	**0.94**	**0.74**	0.50	0.55	1.21	4.30
Kansas	684	**1.05**	**0.85**	0.48	0.56	1.47	4.15
Kentucky	997	**1.25**	**0.87**	0.56	0.67	1.60	11.58
Louisiana	1105	**2.26**	**1.87**	0.50	0.92	3.02	35.66
Maine	384	**1.06**	**0.74**	0.49	0.61	1.01	6.55
Maryland	1151	**2.41**	**2.03**	0.56	1.37	2.92	16.95
Massachusetts	1331	**2.02**	**1.74**	0.59	1.18	2.51	21.43
Michigan	2550	**1.86**	**1.52**	0.48	0.89	2.56	18.18

Figure 4: Tabular display at the op (US) level for all 48 states and the District of Columbia. The currently invisible states can be displayed by scrolling down the slider on the right hand side of the table. Displayed is the modeled 1990 concentration for benzene in micrograms per cubic meter. Summary statistics include number of census tracts, mean, median, minimum, 25th percentile, 75h percentile, and maximum with respect to the underlying census tracts for each state.

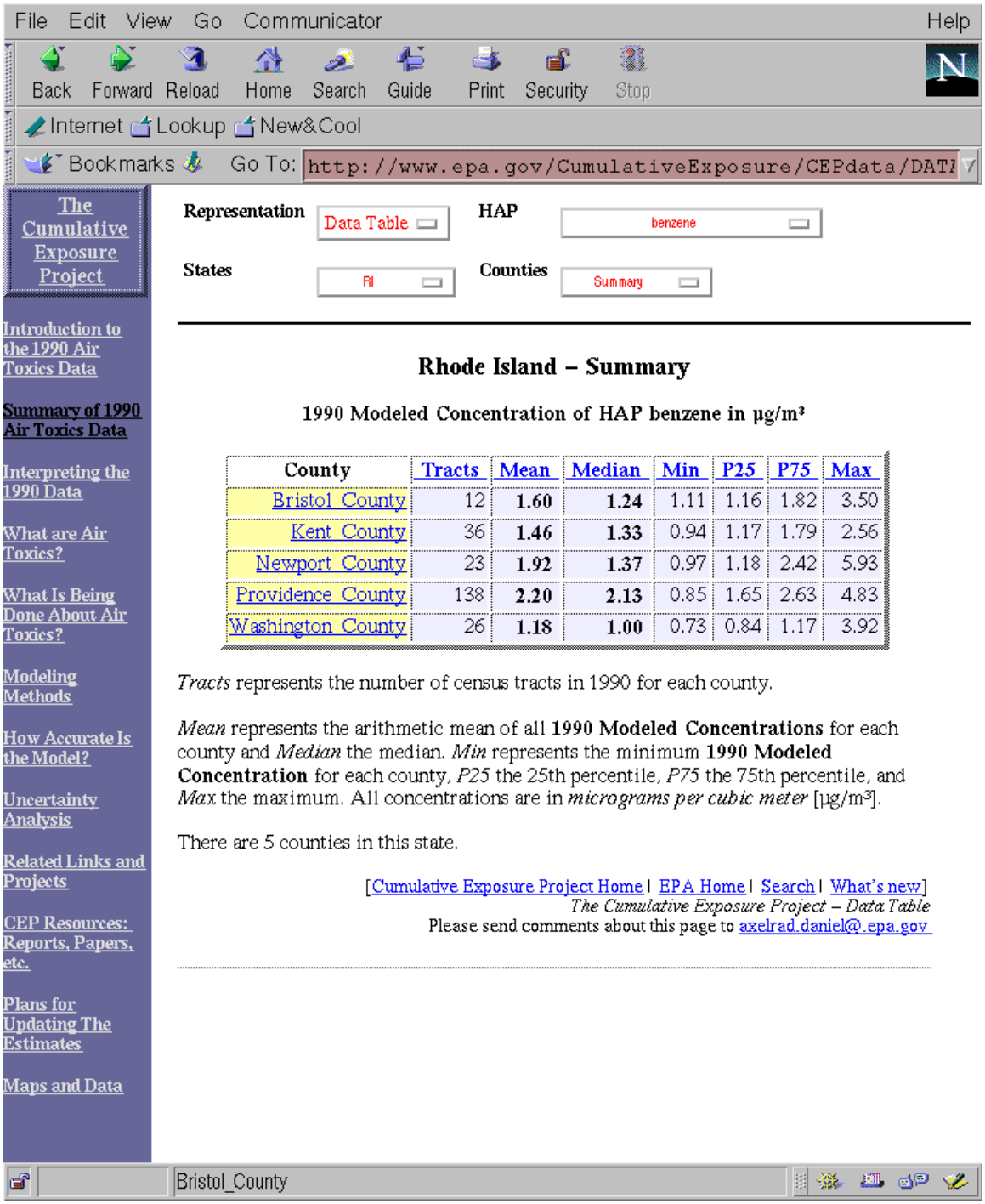


Figure 5: Tabular display at the state level for Rhode Island for all of its 5 counties. Displayed is the modeled 1990 concentration for benzene in micrograms per cubic meter. Summary statistics include number of census tracts, mean, median, minimum, 25th percentile, 75h percentile, and maximum with respect to the underlying census tracts for each county.

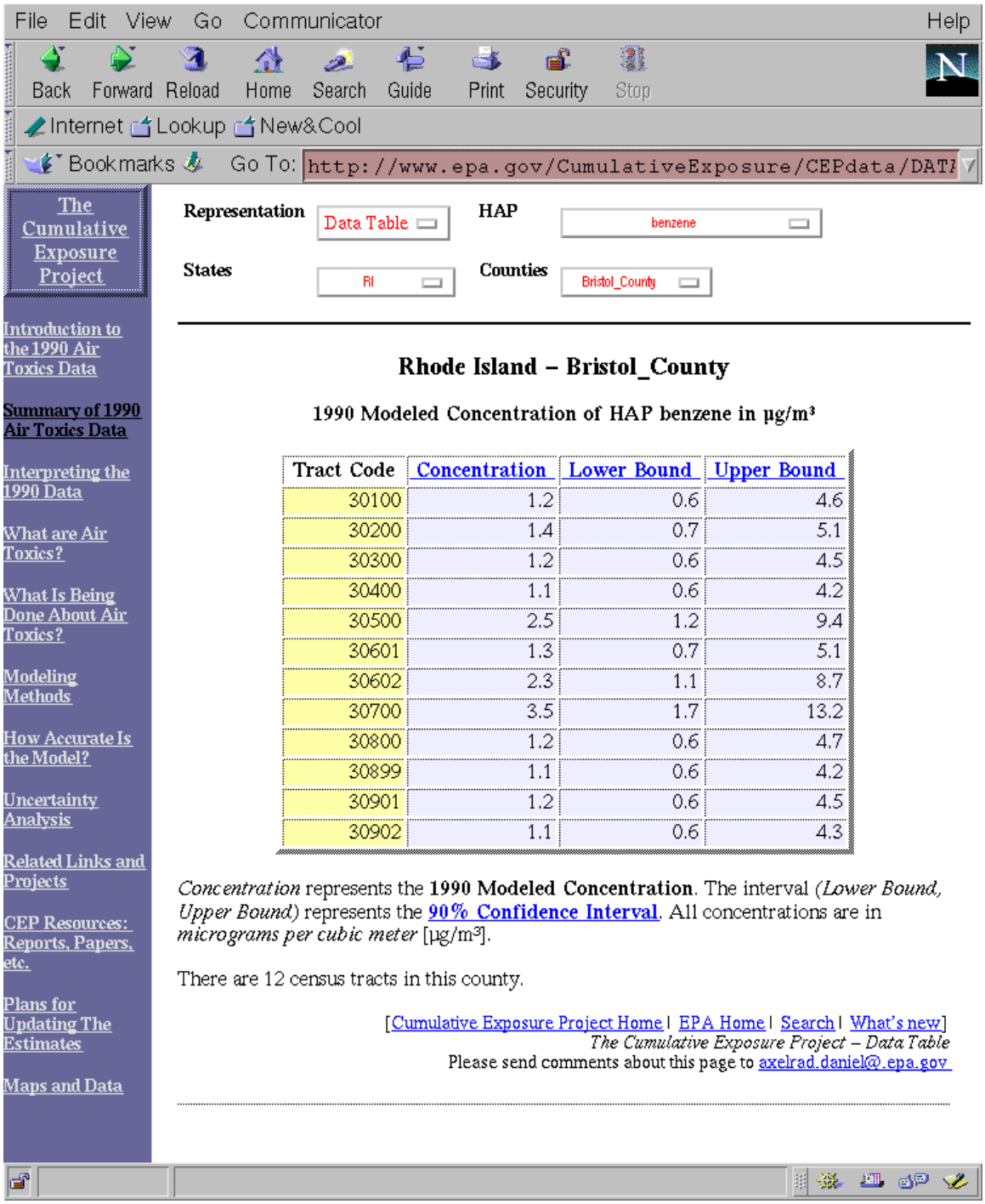


Figure 6: Tabular display at the county level for Rhode Island's Bristol County for all of its 12 census tracts. Displayed is the modeled 1990 concentration for benzene in micrograms per cubic meter. In addition, a 90% confidence interval is displayed.

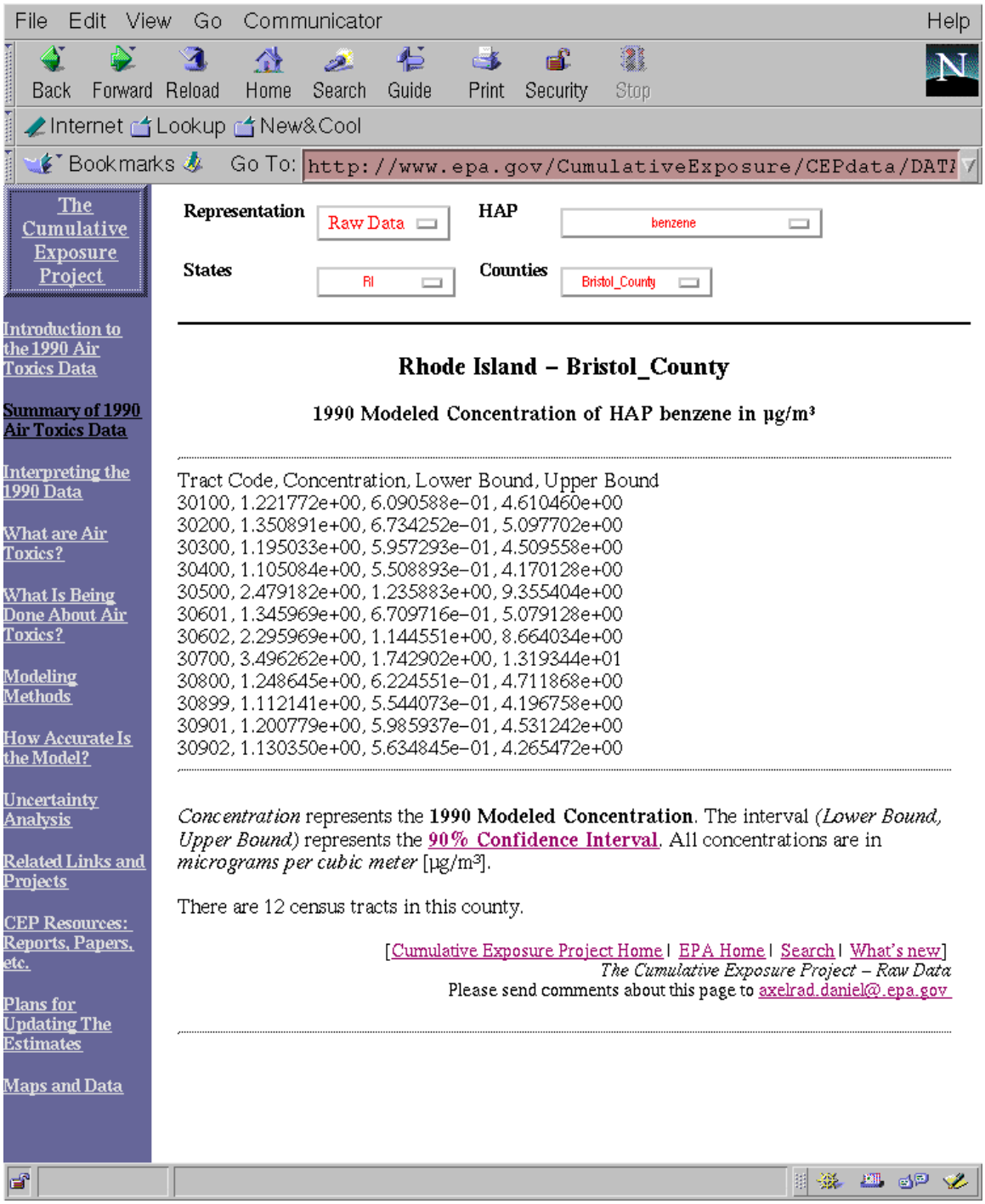


Figure 7: Raw data display at the county level for Rhode Island's Bristol County for all of its 12 census tracts. Displayed is the modeled 1990 concentration for benzene in micrograms per cubic meter. In addition, a 90% confidence interval is displayed.

Developing an NSDI and NBII Clearinghouse Node for the Olympic Peninsula, Washington State

Robert A. Norheim
David L. Peterson
USGS Forest and Rangeland
Ecosystem Science Center
University of Washington Field Station
Box 352100
Seattle, WA 98195-2100
{norheim,wild}@u.washington.edu

Nicholas R. Chrisman
Department of Geography
University of Washington
Box 353550
Seattle, WA 98195-3550
chrisman@u.washington.edu

Teresa Z. Alcock
Olympic Natural Resources Center
P.O. Box 1628
Forks, WA 98331
tza@u.washington.edu

Edward G. Schreiner
USGS Forest and Rangeland
Ecosystem Science Center
Olympic Field Station
600 East Park Avenue
Port Angeles, WA 98362
ed_schreiner@nps.gov

Abstract

We are developing a comprehensive database of metadata for biological and geospatial information for the Olympic Peninsula in Washington state. The metadata are being developed according to FGDC standards, and our metadata collection is included in the NSDI and NBII Clearinghouses. The metadata collection is being delivered to the Olympic Natural Resources Center (ONRC) and is maintained by their geographic information systems lab. This database is designed such that it will be easily accessible and understood by all users and clients who contact ONRC. In addition, we have developed a World Wide Web site for ONRC that will provide access to the database and contain hyperlinks for databases at various agencies and non-governmental organizations. As part of the Clearinghouse work, protocols are being established for maintaining and updating metadata contributed to the database. Training is being provided for database users and metadata contributors to learn how to develop their own metadata and to access the Clearinghouse. The Olympic metadata clearinghouse node will be a single, easy-access point through which land managers, scientists, decision-makers and citizens of the Olympic Peninsula can preview and acquire information on natural resources.

Background

Managing natural resources at large spatial scales, particularly across political land ownership boundaries, is one of the greatest challenges of ecosystem management. Assessing resource distribution, abundance and condition over time in order to make management and policy decisions is often constrained by lack of spatially explicit data, or in some cases by lack of *access* to those data. For any specific biogeographic region, it is valu-

able to have a repository of geo-referenced data on natural resources that can be accessed by both governmental and non-governmental organizations (Goodchild *et al.* 1993, Sample *et al.* 1994, Michener *et al.* 1994, Price *et al.* 1994, Goodchild *et al.* 1996; also see http://biology.usgs.gov/state.partners/description.html).

Inter-organizational cooperation and data sharing are key concepts of the National Biological Information Infrastructure (NBII) and National Spatial Data Infrastructure (NSDI). The value of this approach has been demonstrated in the Sierra Nevada Ecosystem Project (http://ceres.ca.gov/snep/), the Klamath region (Carlson *et al.* 1994; also see www.ced.berkeley.edu/aegis/klam.html>, the Interior Columbia River Basin (http://www.icbemp.gov) and the U.S. Geological Survey Ecosystem Program, focusing on the San Francisco Bay and Delta (http://sfbay.wr.usgs.gov), South Florida (http://sflwww.er.usgs.gov/), Chesapeake Bay (http://www.chesapeakebay.net/bayprogram), Greater Yellowstone region, Platte River and the Mojave Desert, in which coalitions of agencies are pooling natural resources data for large bioregions.

The NSDI and NBII Clearinghouses, sponsored by the Federal Geographic Data Committee (FGDC) and USGS-Biological Resources Division (BRD), are decentralized systems of servers located on the Internet, which contain field-level descriptions of available digital spatial data. This descriptive information, known as metadata, is collected in a standard format to facilitate query and consistent presentation across multiple participating sites. Clearinghouses use readily available Web technology for the end-user (client) side and use the ANSI standard Z39.50 for the query, search, and presentation of search results to the Web client (Nebert 1996).

The Olympic Peninsula is an unusually diverse region which ranges from temperate rainforest to the rainshadow "banana belt" (Henderson *et al.* 1989, Buckingham *et al.* 1995), from alpine glaciers to the ocean. It also has a diversity of ownership, with significant acreage managed by the federal government, state agencies, Native American tribes, and timber companies (Figure 1). Objectives for management of natural resources in this region vary by agency and landowner, yet many physical and biological connections clearly transcend political boundaries (Peterson *et al.* 1997). The current focus on ecosystem management and regional planning makes it imperative that resource data be examined at large spatial scales. Analyses of vegetation patterns on the Olympic Peninsula (Turner *et al.* 1996) and in the western Cascade Range of Oregon (Spies *et al.* 1994) demonstrate the value of using diverse data sources across political boundaries to assess spatial patterns and managerial options. Furthermore, there are important linkages (e.g. anadramous fish) between the terrestrial ecosystems of the Peninsula and the surrounding marine environment (Cederholm *et al* 1989, Bilby *et al.*1996, Stouder *et al.* 1997).

The Olympic Peninsula was one of the centers of conflict during the Pacific Northwest timber crisis of the late 1980s and early 1990s (Dietrich 1992, Yaffee 1994, Durbin 1996). During this time, declining employment in the commercial timber industry due to increased automation and reduced timber supply coincided with the listing of the northern spotted owl (*Strix occidentalis* subsp. *caurina*) as an endangered species and the resultant closure of federal lands to logging. Political and social conflicts arose because of dis-

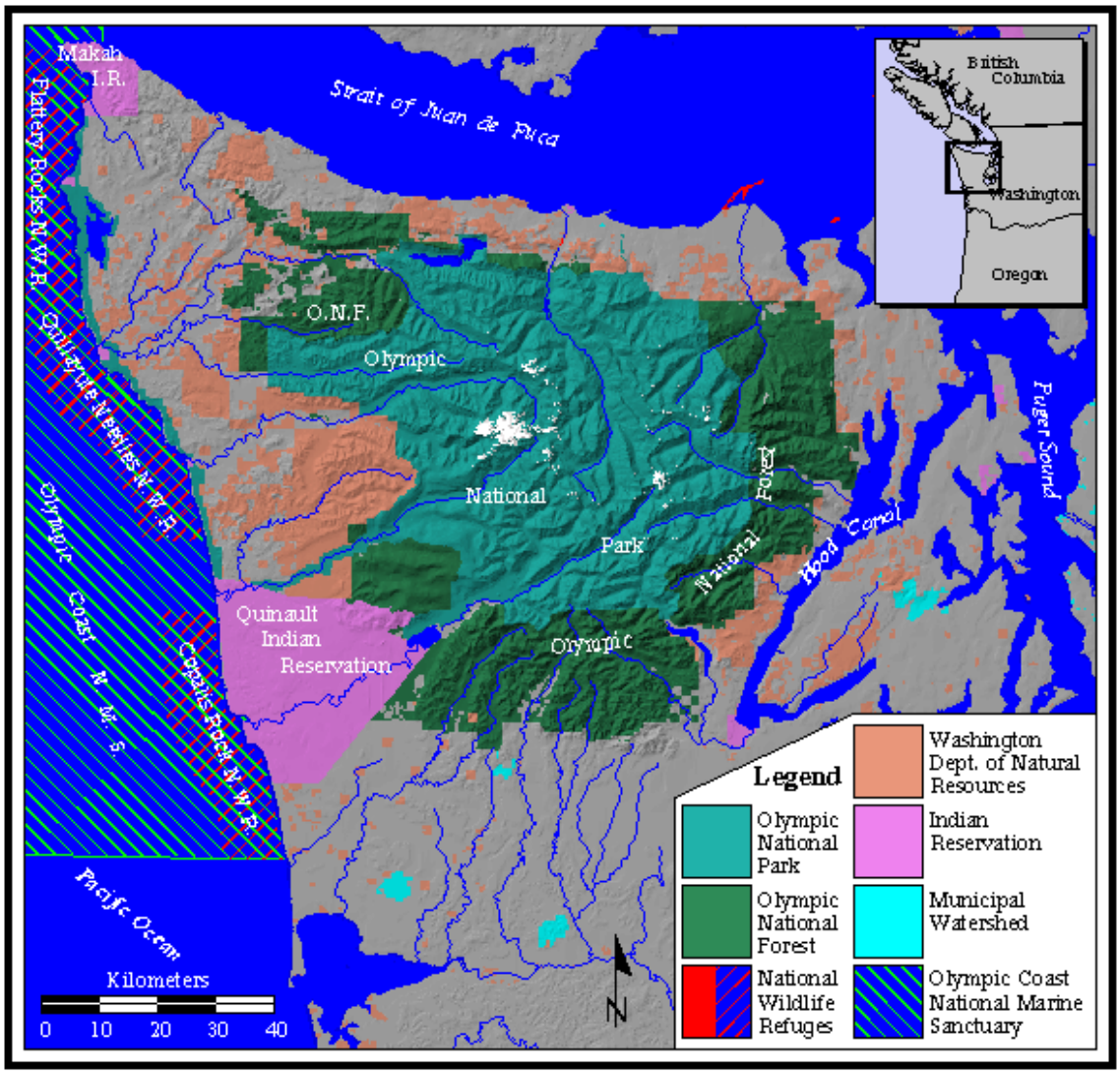


Figure 1 Public land ownership on the Olympic Peninsula

agreements about the use of natural resources on public lands, with the owl and conservationists being blamed by many people for the loss of timber industry jobs. Out of this conflict, the Washington state legislature created the Olympic Natural Resources Center (ONRC), whose mission is "to foster and support the research and education necessary to provide sound scientific information on which to base sustainable forest and marine industries, and at the same time sustain ecological health."

Because of the diversity of ownership interests, there has been no single repository or access point for natural resource data on the Olympic Peninsula. ONRC, recognizing the integrative capabilities of geospatial information and GIS technology, sees these as important tools for accomplishing its mission. Therefore, ONRC is committed to coordinate a regional node of NSDI and NBII. We are now collecting metadata for natural resource data for the Peninsula so that the Center can advance its role of encouraging and facilitating forest and marine resources research and education.

Establishing relationships

As we began this work, we contacted all of the major government agencies which manage land or resources on the Olympic Peninsula, including the Olympic National Park, Olympic National Forest, Olympic Coast National Marine Sanctuary, Washington Department of Natural Resources, several tribes, several counties and regional planning authorities, and also large private landowners (*i.e.* timber companies). We also contacted other agencies that were likely to have geospatial data for the region (such as the U.S. Environmental Protection Agency and state Department of Ecology), as well as non-profit organizations (e.g. the Wilderness Society), and local colleges. Most organizations expressed great enthusiasm and were pleased to have assistance in developing metadata, a task that is required for many of them but which had not yet been accomplished due to lack of funding or time. While some private companies expressed a willingness to participate, several declined because of concerns about the proprietary nature of their data. Some academic researchers also declined to participate due to perceived lack of time and the lack of a university mandate to compile metadata. Despite these negative responses, we quickly ascertained that there were potentially up to 200 organizations and individual researchers who had data sets relevant to the Olympic Peninsula.

We decided to perform a "triage" on these potential contributors. While we welcome all organizations with natural resource data for the Olympic Peninsula to contribute metadata to the ONRC Clearinghouse, we are particularly interested in forestry, biological, and coastal spatial data, because these data are particularly relevant for the missions of ONRC. Our initial efforts have focused on Olympic National Park and Olympic National Forest, which make up the core of the peninsula (over 40% of the land area). We have also worked to get a diversity of organizations represented, and currently have metadata contributed by federal, state, county, tribal, private timberland, non-profit and academic cooperators. Several contributors have indicated that they would participate only if they also saw their peers participating, so it was important to get this cross-section to encourage others to participate.

Another important participant is the Olympic Coast National Marine Sanctuary (OCNMS). Like ONRC, OCNMS also recognizes the importance of the terrestrial environment to the quality of the waters that they oversee, and view geospatial data as a useful approach to charting those linkages. Thus, OCNMS has a keen interest in the success of our project.

Agencies that we have contacted have had a range of exposure to metadata. Federal agencies are generally aware of the Federal Geographic Data Committee (FGDC) metadata standards and the NSDI and NBII, and realize that they are under a mandate to produce metadata for their data sets (Clinton 1994). However, they also consider the requirement to be an ‘unfunded mandate’, and have made little progress in documenting their data sets given more pressing responsibilities. State agencies were also generally cognizant of the FGDC standards, due to the prominent role of the Washington Geographic Information Council (WAGIC), a state coordinating body, but also have made little progress in actually creating metadata. Within other organizations, awareness of metadata was more mixed. Frequently we have had to take on the role of NSDI advocate,

emphasizing that thorough documentation of data sets is important should a data analyst or data manager leave the organization. We also emphasized that on-line delivery of actual data sets, properly documented with metadata, should reduce the drain on an organization's resources in the course of fulfilling data requests.

We have also made a significant effort to document data sets for the Olympic Peninsula compiled by University of Washington researchers under federal funding. This research is subject to the same mandate as federal agencies, because both data and metadata are required to be accessible to the public.

Collecting Metadata

As part of our outreach effort, we have deployed a staff of four 'metadata specialists', who will actually compile metadata for any organization that wants assistance. Many agencies have told us that they wish to produce their own metadata, and that they will share it with the ONRC clearinghouse when the task is completed. They have been happy to receive our advice on metadata tools. Fortunately, several agencies (including Olympic National Forest) have agreed to let us produce metadata for their data collections for them. This has turned into a win-win situation, because the agencies get their unfunded mandates to compile metadata fulfilled and ONRC receives an important collection of metadata.

We initially used a variety of tools to collect metadata. Many metadata contributors seemed concerned about the complexity and length of the 199-element metadata standard (FGDC 1994), and wished to use only a minimal subset of metadata elements. Both WAGIC and the FGDC had defined subsets of 41 basic elements plus 105 working elements (many of which are geospatial data projection system choices). We steered many of these clients to the web-based Metadata Entry System (MES) tool, which handles only 45 elements. However, once the clients became more familiar with and less intimidated by metadata, they quickly wanted to be able to add more metadata elements than MES allowed. This forced us to start almost from scratch using another tool, because it is difficult to import metadata from one tool to another. Thus, a hard lesson we have learned is that one must choose metadata tools carefully from the start of a metadata project.

We have developed NBII-compliant metadata to the greatest extent possible (including the taxonomy elements). We now use the MetaMaker software (version 2.21) developed by the USGS-Biological Resources Division Environmental Management Technical Center for the NBII (http://www.emtc.er.usgs.gov/http_data/emtc_spatial/applications/nbiimker.html), because most other metadata tools do not include the NBII extensions. We have generally been successful with MetaMaker, although we developed specific protocols to follow when developing or revising metadata in order to circumvent limitations of the software (including several "bugs" discovered by our technicians). We have also found that it is useful to have a copy editor carefully review metadata, because the compilation process is sufficiently tedious that even the most careful worker can produce errors.

Challenges of Metadata Compilation and Distribution

As we continue to communicate with current and potential partners, several challenges of metadata compilation and distribution arise. For example, even when we are compiling metadata for an agency, they will still view metadata as a low priority compared to other resource management tasks. This can be frustrating when trying to develop a schedule for metadata compilation and review. The final review is particularly important because it must be completed before the metadata are posted on the clearinghouse. In some cases, we have had to tell the contributor that the metadata will be posted after a period of 30 days regardless of whether they have given us any feedback.

Another issue is that many potential contributors have concerns about sensitive data, especially endangered species data, private asset data, and Native American cultural site data. Contributors express concern about revealing sensitive information through public metadata will jeopardize their efforts to protect private assets, endangered wildlife and preserved cultural resources. Once the public finds out about these sensitive analyses and data, the requests for that data will come, forcing agencies and organizations to develop very selective policies in limiting access and use of those sensitive datasets.

Some potential contributors, especially those with experience in the debates surrounding endangered species management, express concern about the potential litigious use of data. This fear was alleviated in some cases with the ability to state indemnity in the "Distribution Liability" element of metadata records. Therefore, industrial timber land owners have chosen to provide metadata only about a couple of datasets that they already make publicly available, such as their recreation sites data. Native American Tribes have chosen not to share metadata for their culturally sensitive data and the Olympic National Forest has chosen to share metadata on endangered species, but respond to requests for actual data on a case-by-case basis.

In addressing these kinds of challenges, our main selling point has been that the FGDC and NBII content standards specifically incorporate protections for contributors. Some potential contributors still do not feel adequately protected. For example, the Stand Management Cooperative at the University of Washington College of Forest Resources, which uses data on private assets and does not wish to share this sensitive information with the public in any way.

Another interesting difficulty we have encountered is with one major data set, the National Science Foundation's Long-Term Ecological Research (LTER) program. This major potential contributor already has its own internal metadata format. Unfortunately most of these metadata are surprisingly not compliant with FGDC standards, and the program personnel we have contacted expressed little interest in modifying the metadata to meet these standards.

As demonstrated by the lack of resources of agencies and organizations to create compliant metadata, our small team can only encourage organizations like LTER to begin the process of conversion and eventual integration for a safe yet rich metadata environment, setting the stage for eventual data sharing. Our philosophical approach to our contributors has been to the use of FGDC and NBII content standards such that metadata can indeed

provide a safe and effective method for encouraging the inclination to organize and eventually share expensive geospatial and biological information.

The FGDC metadata content standard is, of course, not the final statement on the subject. We have found it complex to produce compliant metadata from the sources available to us. Some fields that are mandatory are much less important to our target users than other "optional" fields.

Relationship to state-wide efforts

We are working closely with WAGIC. Rather than actually maintaining a node to be connected to the NSDI and NBII ourselves, our metadata collection resides on the Washington State Node, and is listed as a "sub-node" with the NSDI. Under an agreement with WAGIC, the University of Washington Libraries has actual responsibility for the computer on which the node resides. Searchers accessing the NSDI will find Olympic Peninsula information whether they select the Washington State node or the Olympic Peninsula node on the NSDI search form.

WAGIC also provides one-day metadata creation training workshops, in which participants bring in a data set to be documented and are exposed to a variety of tools for metadata creation. WAGIC is sponsoring a number of these workshops on the Olympic Peninsula in a cooperative effort to increase the amount of metadata for both the Peninsula and the state. We hope that this relationship will be a model for other regional cooperative efforts in Washington state and beyond.

Expected Outcome and Potential Applications

In the course of this project, we have worked to establish capabilities that endure after the project ends, so that ONRC will be able to maintain the clearinghouse indefinitely. Organizations will be provided protocols for updating their metadata; information on date of acquisition and frequency of updates will help users assess data quality and temporal resolution.

The clearinghouse is providing ONRC with a critical component of its geographic information system (GIS) program. Development of an attractive, user-friendly Web site allows scientists, resource managers, educators and citizens to access the best available information on natural resources of the Olympic Peninsula. On the Web site, we have included links to both metadata and data, so that users can browse available metadata, rather than just finding information via the search capabilities of the clearinghouse. Anyone with a computer and Internet connection (now available in most public libraries) can now access the ONRC Clearinghouse. The clearinghouse will save a substantial amount of work for ONRC staff and provide the ONRC GIS staff with an essential tool for organizing and disseminating information.

The Clearinghouse will facilitate the role of ONRC in synthesizing and distributing information on natural resources. It will help to integrate a wide range of data sources and allow scientists and resource managers to analyze landscape characteristics or bio-

logical resources at large spatial and temporal scales (into the future, or retrospectively with older data sets) and across political boundaries. We anticipate that the clearinghouse can be used as a focus for communication and cooperation among agencies, institutions and residents of the Olympic Peninsula, thereby leading to stronger relationships among the many Peninsula organizations to support digital geographic data coordination and hopefully to provide a common ground for decision-making.

In addition, the Clearinghouse will facilitate the analysis of natural resource data at large spatial scales for the Olympic Peninsula. For example, landscape patterns can be assessed with respect to vegetation type, forest stand characteristics, geological resources, hydrological resources, and potential management impacts. Temporal patterns can be quantified using LANDSAT data, aerial photos, older maps and other retrospective techniques, thereby facilitating analysis of changes in wildlife, forest management practices and economic activity on the Peninsula. The clearinghouse can be used to project future changes as well, by developing various scenarios for land-use change, resource management and natural disturbance (e.g. fire). Finally, it may be possible to use multiple data sources to rectify data-layer classifications at political boundaries, thereby resolving inconsistencies and potential conflicts across different land ownerships.

Acknowledgements

This work has been funded by grants from the Olympic Natural Resources Center, the USGS Biological Resources Division, and the Federal Geographic Data Committee's Cooperative Agreements Program. The authors wish to thank our metadata specialists, Mark Bourgeois, Alan Carter-Mortimer, Ella Elman, and David Jeschke, for their assistance with this paper and for their hard work in the trenches.

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THE FAIRFAX LINE

by David Lee Ingram

The need for a survey of the "Fairfax Line" originates with a land grant by Charles II in 1649. This paper will discuss several areas to include the original land grant, the rulings of the Privy Council, the actual survey, the present day significance of the survey, and the efforts by Surveyors Historical Society to relocate the beginning point during their "Rendezvous '99". The author is a Land Surveyor engaged in private practice in Virginia. He has a specific interest in the Fairfax Line because it forms the North line of his home county.

I. INTRODUCTION

The early development of the Colonies that became the United States of America depended on a great many different acts of man, including surveying the land. From small parcels of land to surveys that helped delineate large grants and political boundaries, the colonial surveyors were in the midst of many great steps of progress enjoyed by our forefathers.

One of the more interesting surveys involved the survey of the "Fairfax Line" to establish the limits of the Northern Neck land grant in Virginia. This land grant of approximately 5,000,000 acres was a "political payoff" by a King to his friends and has lead to all sorts of intrigue down to this very day. Politics continued to play a role through the mid 1700's as we shall discover shortly, the Courts of England and the United States had things to say about the matter, and a great deal of the early development of this country took place therein.

As a general point of reference for orientation, the Fairfax line runs from a point about 3/4 of the way up the East slope of the Blue Ridge Mountain approximately 30 miles North of Charlottesville, Virginia, in a North-Westerly direction to a tri-county corner point in West Virginia near where the Western panhandle of Maryland dips down into West Virginia.

II. THE "FAIRFAX" GRANT

The genesis of the Fairfax Grant is in the year 1649 when the King of England, Charles II, is in exile in France with several of his loyal followers. As a reward to two Culpeper brothers and five other loyal friends, the King issued a grant for a "porcon of Virginia .... bounded by and within the heads of the Rivers Rappahannock and Patawomecke.....". Initially this was pretty much a useless piece of paper since the King was in exile, but Charles II was restored to the throne in 1660 and the grant was recorded and a "proprietary" created in the new world. It is worth noting here that this was taking place within 50 years of the first permanent settlement in the new world and the territory that was ultimately encompassed therein had not been explored and was not known.

Ultimately the 7 original shares ended up being held by the 2nd Lord Culpeper and his only child, a daughter named Catherine, married Thomas 5th Lord Fairfax. Thus the Northern Neck Grant is commonly referred to as the "Fairfax Grant" and the survey that is the focus of this paper is referred to as the survey of the "Fairfax Line". It does pay to marry wealth.

III. THE NEED FOR A SURVEY

Early in the history of this grant, knowing that the boundaries were between two Rivers was adequate for the owners and citizens of the new world, but eventually questions started to arise as to just what were the limits of this massive grant. When one is down near the Chesapeake Bay it is very obvious whether you are between the two Rivers, but as you move upstream every branch of a river raises a question as to which side the boundary follows. There were early opinions that the grant should only go upstream as far as the "falls" past which ships could not sail. Other opinions were suggesting various major forks of the rivers, and, of course, the owners wanted to go all the way to the headspring of each river. But which headspring? If you have ever examined the beginnings of a stream in a mountainous area you are well aware of the fact that many small springs flow together to eventually form a stream which eventually matures into a river.

Even after an agreement has been reached as to just how far upriver to go, then the question arises as to how to connect those two limiting points. Is it a straight line? Do you attempt to follow various natural boundaries? Just where do you go?

These various questions offered many different options and the parties on different sides of the issue obviously had conflicting interests and goals. Enter politics and the courts.

IV. THE EXPLORATION SURVEYS OF THE 1730's

During the early 1700's many different interests came into play regarding the extent of the lands of Lord Fairfax. Obviously Lord Fairfax wanted as much as he could get. The Governor of Virginia wanted him to have as little as possible. At various times Lord Fairfax was astute enough a politician to recognize that if he pushed the issue he would lose. So he delayed. Eventually the matter was going to come to a head and those people that would make the final decision needed some good solid information as to what the fuss was about. Several different things took place in the 1730's that eventually helped resolve the situation.

The County Surveyors of the counties that adjoined the Potomac and Rappahannock Rivers were ordered to survey, map, and provided field notes for those portions of the rivers in their counties.

In 1736 three different survey expeditions were organized with all three having representatives of the Colony of Virginia and Lord Fairfax. One group was to explore and map the Potomac to its head. A second group was to explore and map the North Branch of the Rappahannock and the final group was to explore and map the South Branch (Rapidan River) of the Rappahannock. All three groups consisted of surveyors and commissioners and their works were completed.

The work of the three groups and the county surveyors lead to the preparation of a map of the Northern Neck in the years 1736 and 1737. The map shows the courses of both rivers and of particular interest makes note of Latitudes across the map. The ability and knowledge to relatively easily get an accurate Latitude certainly contributed to the accuracy of the map. The one thing the map does not show, however, is a boundary line across the end of the grant. This was still to be decided.

V. THE PRIVY COUNCIL RULING

Lord Fairfax was wise enough to know that he did not want a decision made by the Colonial Government because he would surly lose. He played the game of politics very shrewdly and managed to get the case transferred to the Privy Council in London. After managing to chose his time very carefully, when he thought he had a sympathetic Council, Lord Fairfax finally got a ruling that was very much to his liking. The Council ruled that has grant would go from the headspring of the Conway River (on the Rapidan, on the Rappahannock) to the headspring of the Potomac River. This indeed turned out to be a very fortuitous ruling for Lord Fairfax because it gave him maximum land area that he could develop and prosper from until after the Revolution.

VI. THE SURVEY OF THE "FAIRFAX LINE"

After the Privy Council had issued its final ruling, the next step was to survey the line connecting the head springs of the two rivers. In 1746 a survey crew was formed consisting of four surveyors, Col. Peter Jefferson (father of Thomas) and Robert Brooke (son of Robert Brooke who was one of the 1736 surveyors exploring the Potomac) representing the Colony and Benjamin Winslow along with Thomas Lewis representing Lord Fairfax. There were also a number of "Gentlemen Commissioners" representing both sides of the matter, and a number of helpers with various other duties. Altogether, the party ultimately consisted of approximately 40 men along with their pack animals, supplies, etc.

We know quite a bit about the actual survey because Thomas Lewis kept a journal during the survey that was part survey notes and part diary. The journal surfaced in a Supreme Court case (Maryland v. the State of West Virginia) and was held by an attorney by the name of George E. Price of Charleston, WV, and later returned to D. S. Lewis. In 1925 a renowned Virginia Historian by the name of John Wayland transcribed the journal, had it typeset, and printed. In the course of this printing project, Wayland used the same spacing and page content as the original and made the following footnote comments:

> "The little book in which Lewis wrote with a quill pen, and which easily went in his pocket is 3.5 inches from side to side and 5 5/8 inches from top to bottom .....In spite of his poor spelling, Thomas Lewis was a man of culture and wide information. He had a literary taste, and was probably one of the best mathematicians in the colony..... It is possible that Lewis wrote this journal day by day on the expedition, sometimes making his entries at night by the flickering light of the camp fire. In

> view of all the conditions we cannot help admiring his straight lines and good penmanship, and we are much disposed to excuse his bad spelling..... We are reminded here again of the fact that only a small part of the story of our country, in the conquest and settlement of the wilderness, has ever been told, and now only a small part can ever be written. It is such documents as Lewis's journal that now and then give to our imaginations the wings of exalted visions."

The journal itself tells a lot about the work and life of the expedition and the characters of the people therein. The next few entries of this paper will be quotes or paraphrases from the journal to give an idea of just what the situation was with the expedition. First, however, a comment or two. It seems that surveyors are not a lot different 250 years later. There appears to be something in the nature of the beast. Another observation is his constant reference to the commissioners as "gentlemen commissioners". One can almost hear the sarcasm in his pen. Since these men might be considered lawyers today, again not much has changed.

Now to the journal: Lewis sets out from home near Staunton, Virginia on Wednesday, September 10, 1746, to go to Capt. Downs to await the commissioners and the other members of the expedition. On Thursday he took sick and had to put up for the day and then proceeded on to Capt. Downs the next day. The others were arriving at the same time and he speaks of spending the day very agreeably. On Sunday they solicited a preacher to preach them a sermon before they departed and they were bid farewell. After getting things ready on Monday and Tuesday, they retired to their camp on Tuesday and spent there hours with a great deal of pleasure and merriment. He was taken ill in the night with violent vomiting. On Wednesday they continued packing and spent the evening "in our usual manner". They were alarmed with a quarrel in Capt. Downs lane amongst a crowd of drunken people. The rails and stakes of Capt Downs fence supplied the want of cudgels and they were applied with tolerable good success.

On Friday, September 19, 1746, they departed Capt Downs and set off to commence their assigned task. They called at Kirkley's and regaled themselves with some very good cider. They were obliged to camp before arriving at the head spring of the Conway. On Saturday the 20th, the mountains made such a dismal appearance that John Thomas took sick and returned home. One gets the idea that just getting to the beginning point of the survey was a major task and it probably was.

The next task was to find the proper beginning spring that had been identified in the 1736 survey. They spent time measuring up several different forks well up in the Blue Ridge Mountain attempting to compare their survey to the prior survey. They eventually satisfied themselves that they had the correct spring and found some trees that probably had been marked ten years earlier. It appears that they really weren't fully satisfied that they had the proper location, but finally the "gentlemen commissioners" agreed upon a starting point. One of the things that may have lead to difficulty is to understand a spring in the Blue Ridge Mountains. Typically they are not a "gusher" high up in the mountain, but rather a "seep" and considering that this is mid-September it is very possible that there was little if any water flowing.

Finally, on September 25, they began to run a trial line to connect the headspring of the Rappahannock River with the headspring of the Potomac River. They probably had a copy of the 1736 & 1737 map, but it would serve only as a general guide for their direction and distance. They made their best estimate as to a direction and proceeded to run on a course of N 41 2/3 W off into the Western wilderness of Virginia. They would occasionally run into settlers on the Western frontier, but for all intents and purposes they were on their own for an indeterminate period of time. They were beginning about 3/4 of the way up the east slope of the Blue Ridge and had to then cross the Blue Ridge and several other major valleys and mountains as they would soon find out.

After crossing the Blue Ridge and coming to the Shenandoah River in Page Valley, they met up with the rest of their baggage which had been sent there from Capt Downs rather than bring it all over the Blue Ridge with them. They encamped at Loungs and dined and regaled themselves with several black jacks of punch. Through out the journal it is obvious that they never passed up the opportunity to have a good drink. Surveyors never change.

On Sunday the 28th at night they took a variation of the needle and found it to be 1.5 degrees West. The four surveyors that were on this journey were probably some of the best the colony had to offer and the survey demonstrates the ability to run a long straight line on a North-West heading, a knowledge of astronomy, and many other aspects of surveying. At this point it was determined that it was not practical to take the horses and baggage over Peaked Mountain so the surveyors struck off on their own while the baggage and "gentlemen commissioners" went around. The day was exceedingly hot, the mountain steep, and they were almost overcome and ready to faint for want of water. They did, however, make it up, across, and down the Mountain in one day all the while running a survey line and measuring the distance. Quite a feat.

By Friday, October 3rd, the survey had gone 31 miles and Lewis notes that several of the horses had like been killed tumbling over rocks and precipices and "ourselves often in the outmost danger this tirable place was called Purgatory". By this time they had crossed the Shenandoah Valley and were headed up into the Allegheny Mountains. Many parts of the Allegheny's are still very rough and rugged to this day and one can only imagine the hardships being faced at this time. Once again the surveyors separated from the main body with a couple of horses while the commissioners and the main part of the baggage went to find an easier way. On Sunday the 5th the surveyors could not "lyeby" because their horses were starving and they only had provisions for the day. Necessity made them press forward.

Here Lewis laments the great many hardships being endured by the surveyors and the horses and observes that they were very lucky that no one broke a shin. At one river crossing they decided to leave their work and go downstream to see if they could discover some inhabitants where they might get some provisions. They only discovered one family of "poor Dutch people" from whom they could have no supply. Fortunately Mr. Brooke returned from having found the main party and brought with him enough supplies so that the surveyors could proceed on. On Tuesday the 7th they ascended a very steep mountain and passed mile 46 on their survey. Again the horses were having problems traversing the terrain and there was no water for them.

On Wednesday the 8th they broke the glass of their compass. On Wednesday the 9th they encamped opposite Coburns. They "went to see Coburn who with his wife & miller a Bucksom lass Repay'd the Visite in the evening we spent very merily". This was the last settlement they expected to encounter and laid over on the 10th to be supplied with fresh provisions, shoe the horses, and do laundry.

The survey continued on October 11 with additional reports of rugged terrain, rough going, success at hunting a bear and three cubs for food, swamps, etc. etc. This was an extremely difficult journey and the men had to know that they would have to do it all again. Never the less, they pressed on because they had a task to finish.

On Friday the 17th they started being concerned that possibly they had missed the head of the Potomac and had gone too far. Streams were running in the wrong direction (towards the Mississippi) and I'm sure frustration was catching up. They pressed on, however, and on the 21st one of the men who had been on the 1736 journey reported after an exploration that he had found the headspring of the Potomac from ten years prior with the previously marked trees plainly visible. The rest of that day and the next the surveyors ran a "traverse" from where they were at to the headspring so that they could calculate their error. That night they dined on a loin of roasted venison and drank to his Majesty's health. After performing their calculations, they discovered that they were only off line by about 4 miles - quite amazing considering that they had run approximately 76 miles on an educated guess. With this information they were able to calculate a corrected bearing for the return trip and on Thursday, October 23rd, they left the head of the Potomac to return to the place of beginning.

The return trip was no less difficult, but Lewis's notes seem to not be as "down" as some of the earlier comments. In fact, on the return trip they had to do additional work of blazing trees and otherwise marking the line the entire way. They also took a couple of occasions to measure up valleys to make a "tie line" to confirm that their return line was running true.

In spite of the hardships encountered, there were however moments of levity. Lewis does note that they stopped at Coburns again on the return trip on the 28th where they dined and had wrestling matches. On Thursday the 30th they concluded the evening with "merriment" since it was his Majesty's birthday, they drank to his health, and discharged 9 guns.

Another moment of levity occurred on the 29th when they held a trial for Fumfire the farrier. Several days earlier he had deserted the group, but reappeared to them on the 28th. At the trial it was determined that his punishment would be to wear a bell about his neck for a week. This was probably a serious event and the thought of wearing a bell was indeed a punishment not to be enjoyed, but one can imagine that the trial might have had a good laugh or two as well. One does get the impression throughout the journal that even though the men were serious about a difficult task, they did know how to have a good time. Once again, a sign that surveyors never change.

Finally, on November 13th, 1746, the survey arrived back at the point of beginning and found that they had "fell about 100 yard below the place of our beginning". They proceeded to mark a rock "FX" and marked and scribed a number of trees. Thus in a little over one and one-half months, this group of intrepid surveyors surveyed over 160 miles of line (76 mile out, 76 miles back, plus tie lines) and managed to end up within 100 yards of their beginning point using a compass and chain. An amazing accomplishment!!!!

They broke camp the next day and all of the men started to proceed home. Jefferson, Winslow, and Lewis set off for Capt Downs and then went to spend an evening with Mr. Madison - one assumes he means the family of James Madison. The leaders of the colony of Virginia, including the surveyors, were a apparently tight knit group and took every opportunity to stay in touch with each other.

VII. THE MAP OF THE SURVEY

The surveyors had one more task to perform and that was to prepare a map of the Fairfax lands. They had agreed that they would meet at Jefferson's on January 1, so Lewis continues in his journal that he departed home on December 30. The schedule did not hold and Lewis did not arrive until January 3rd, but not to worry as Brooke and Winslow had not arrived. After waiting several days and getting tired of the wait, Lewis set off to find the other two. He arrived at Brooke's on the 15th of January to find him preparing for his journey in a few days. They finally ran across Winslow and agreed that they should leave for Jefferson's on the 21st - only three weeks late for a meeting to draw maps. They finally arrived at Jefferson's on the 23rd of January and began work on the 24th. They soon discovered that they needed additional paper and supplies that needed to be ordered from Williamsburg. They continued their work until February 21, 1747, at which time they had completed seven plans of the Northern Neck each about three foot square. He does not say what happened to the seven plans, but one of the original copies is located in the "State Paper Office" in London. In 1860 a hand drawn copy of the original map was made and returned to Virginia where it now resides.

Lewis concludes his journal by saying "Having finished a troublesome and difficult affair wherein I have first and last spent 127 days and cash 3 pounds: 3 shillings". He does not say what, if any, his compensation for the job was.

VIII. PRESENT DAY SIGNIFICANCE OF THE "FAIRFAX LINE"

The Fairfax Line surveyed in 1746 still has significance to this day. In numerous places it represents private property lines, for a significant portion of its length it represents County lines, and for several miles it represents the state boundary between Virginia and West Virginia.

The North-West end theoretically should represent a corner of the State of Maryland, but because of the way the head of the Potomac River bends around it is only a county corner of three West Virginia counties. This issue was only resolved when the Supreme Court ruled against Maryland in determining that Maryland would only go up the Potomac far enough to meet a point where a

North line from the Fairfax stone would cross the branch of the Potomac. Without the ruling, the boundary of Maryland did not make any real sense and was indeterminate.

IX . SURVEYORS HISTORICAL SOCIETY PROJECT

The North-West end of the Fairfax line is known to this very day and is easily accessible. In addition to being a tri-county corner, there is a lane that leads right up to a small park and picnic area with appropriate plaques and other identification.

The South-East end of the Fairfax line is just the opposite. To my knowledge it has not been used or identified since 1746. It is located in a very remote, wooded area in what is now Shenandoah National Park and can not be located other than in a general sense. The Surveyors Historical Society will be gathering in Luray, Virginia, September 23 - 25, 1999, to hold “Rendezvous ‘99" for the purpose of reestablishing the beginning point and, with a little luck, perhaps even find the initials “FX” that the surveyors of 1746 chiseled in the rock. Throughout 1999 additional research will take place, known points along the line will be GPS’ed , and at the rendezvous we will attempt to follow in the footsteps of the surveyors from 1746. You are invited to join us.

(Portions of this paper originally appeared in the September 1997 issue of POB Magazine, pp. 24 –27)

EMERGENCY MEASURES

Bradley Freeman, PLS

ABSTRACT

In January 1997 Triad Associates, a civil engineering, planning and land surveying firm from Kirkland, Washington was contacted by AGRA Earth and Environmental Inc., a geotechnical consulting firm from Kirkland, Washington to help with assessment and analysis of a landslide in Des Moines, Washington. AGRA wanted Triad Associates to develop a means to monitor the movement of the landslide.

After an initial site visit, Triad Associates chose to use GPS (Global Positioning System) surveying techniques over traditional surveying methods. We determined that the size of the site (approximately 40 acres), as well as the existing vegetation and other obstacles to line-of-site surveying necessitated the use of GPS technology. By using GPS technology, we were able to simultaneously observe the control stations and the monitoring points. Furthermore, we determined that only GPS could provide the level of accuracy (2-cm horizontal) the geologists needed to determine how to stop the slide and keep the affected homes safe.

This paper will discuss the methodology that was used to conduct the GPS monitoring survey, the problems that we encountered while performing the monitoring survey and preparation of the reports to be used by AGRA Earth and Environmental, Inc.

INTRODUCTION

In January 1997, residents in the Woodmont area of Des Moines, Washington discovered they were on the move. In fact, their properties were shifting straight toward Puget Sound due to a large, slow-moving landslide affecting eight homes in the suburban waterfront city just south of Seattle. Initial investigations indicated that the landslide movement was approximately 1/8" to ¼" per day.

Since then, the landslide has slowed to a standstill, largely due to the efforts of local geotechnical consultants. However, their work depended on critical survey data. In order to monitor movement of the landslide, Triad Associates, a 100-person civil engineering and land surveying firm with offices in Kirkland, Washington, was subcontracted to establish survey points at strategic locations across the site.

LIVES AT STAKE

The City of Des Moines acted quickly when they received reports regarding damage to private property and utilities in the Woodmont area. The Seattle area had experienced many landslides in early 1997, including one in which a family was killed when their Bainbridge Island home plunged into Puget Sound. Although none of the affected property in Des Moines was publicly owned, the City didn't hesitate to call a local geotechnical engineering firm to investigate the landslide because of the risks involved.

The landslide itself was nothing new; the geotechnical consultants consider it ancient, since the event has been going on for hundreds of years. The head of the landslide in its current state is marked by a curved escarpment approximately 40 feet high and located west of Marine View Drive South and east of 8th Avenue South. The approximate limits of the landslide are the Puget Sound on the west, Marine View Drive S. on the east, S. 272nd Street right of way to the north and S. 276th Street to the south. The crown of the main scarp of the landslide is approximately 350 feet in length with the flanks extending 800 feet at the toe of the slope at the shoreline of Puget Sound. The distance from the crown to the toe of the landslide is approximately 700 feet.

The City of Des Moines contracted AGRA Earth & Environmental geotechnical consultants in Kirkland, Washington, to join their team in late January, 1997 to assist the City with preliminary assessment, data compilation, and monitoring and analysis of the Woodmont landslide. Information obtained through discussions with local residents and city and county personnel suggests that the landslide began moving in early January 1997. The cause of the movement was attributed to heavy snow and subsequent rain that fell during the last week of December 1996 and the first week of January 1997.

AGRA's preliminary evaluation suggests the landslide movement is apparently being caused by excess groundwater pressure acting on the slip surface at depth. Control of surface water run-off was a major concern in minimizing the movement. Simultaneously with AGRA's activities, the residents being impacted by the landslide contracted Golder Associates, Inc. of Redmond, Washington, to evaluate causes of the landslide and propose recommendations to slow or halt the movement. Golder used drilling and field mapping to determine causes of the landslide movement.

METER BY METER

To monitor movement, AGRA subcontracted Triad Associates. The survey data provided by our firm would allow the City of Des Moines to determine if the movement was of a degree sufficient to

cause personal injury to any residents. When Triad Associates' Greg Juneau, PLS, Director of Surveying, Lonnie Carlson, PLS, Construction Surveying Coordinator and I as Specialized Survey Services Manager visited the site, it became obvious that traditional surveying methods would be difficult to use. We determined that the size of the site - approximately 40 acres - as well as the existing vegetation and other obstacles to line-of-sight surveying would probable necessitated the use of GPS technology. Furthermore, we determined, only GPS would provide the level of accuracy (2-cm horizontal) the geologists needed to determine how to stop the slide and keep these homes safe. Also by setting this project up this way, Triad would be able to reobserve the monitoring points at future dates.

After the initial site visit Triad Associates presented to the City of Des Moines a proposal, consisting of three alternatives, to complete the necessary survey work. The first alternative proposed using traditional survey techniques to establish three baselines across the landslide area. Our concern with this method of monitoring was that we might not be able to "anchor" the ends of the baselines to stable areas outside the landslide area. The results of this would be that we would be able to detect movement within a baseline but we could not determine if the whole baseline was moving with the landslide or not. A second alternative was proposed that used the traditional baselines but the end and middle points within the baselines would be checked using GPS survey techniques therefore any overall movement of the entire baselines could be determined.

The third alternative proposed to the City of Des Moines was to do the survey monitoring using just GPS survey techniques. Monitoring points would be established throughout the landslide area with the base stations for the GPS receivers established well outside the known limits of the landslide. This would assure us that we would be detecting the movement of the monitoring points. Along with the monitoring points Triad Associates recommended to the City of Des Moines that ten crack gauges be established near the lower monitoring area in order to quickly see and determine the amount of movement in the area.

There were significant differences in the cost between using traditional survey techniques and GPS survey techniques. We determine that there was about $2,100 difference in cost to setup the project between the two survey techniques and about $400.00 difference in cost during each subsequent monitoring session as shown below.

	Baseline	GPS
Setup Costs	$6,800	$4,700
Cost per Monitoring Session (Baselines Only)	$1,200	
Cost Per Monitoring Session) (Baselines +GPS)	$3,200	$2,800

Based on these cost savings and the accuracy of the readings we believed we could achieve using GPS survey techniques; AGRA and the City of Des Moines decided to proceed ahead with using the GPS technology to monitor the movement of the landslide.

SURVEY PROCEDURES

First, established control points on stable ground as near to the site as possible were needed. The Survey Branch of the King County Department of Public Works had several points in the area on which they had established NAD 83/91 coordinates using GPS technology. Two of these control points were in the right area, approximately one-half mile east of the landslide site; however, they were located in the centerline of a busy arterial street and could not be occupied for several hours continuously. Two secondary control points were subsequently established on quiet side streets. However when we occupied the control points with GPS equipment we could not leave them unattended and therefore we needed a three-person crew to perform the monitoring surveys. If we would have place the control points in a secured area such as a fenced yard or a stable roof with access we would not have needed to use a three person crew.

Twenty monitoring points were established in the landslide area. The points themselves were ½-inch diameter rebar, 30 inches long which were driven into the ground and capped. A major challenge was positioning them so that they would give the geotechnical consultants the information they needed while maintaining the ability to obtain clean signals from the GPS satellites.

Leica GPS System 300 equipment was used during the field sessions. Two GPS receivers were set up over the secondary control points and a third receiver was placed in a backpack to be carried to the various monitoring points. The raw field data was then downloaded into a computer and processed using Leica's Static Kinematic or SKI computer software.

Measurements were obtained on three different days during the last week of January and the first week of February 1997, and were taken every other week until April 1997. Monitoring was then restarted in the fall of 1997. All total between February 1997 and June 1998 fourteen sets of survey data were provided to AGRA.

Greg and I decided early in the process to determine Washington North Zone State Plane Coordinates on each of the monitoring points instead of using a local coordinate system. This would ensure that the work could be easily duplicated years from now, if necessary. The modular features of the SKI software enabled the determination of these coordinates with minimal effort.

We also determined that state plane grid coordinates needed to be converted to ground coordinates because ground movement was the item of primary interest. A computer utility program was

obtained from the Washington State Department of Transportation to convert state plane grid coordinates to ground coordinates. The program used elevation and latitude parameters that we determined for the project area to calculate the combined factor to be applied to each state plane coordinate. The program than applied the combined factor and produced a report showing the ground coordinates of each monitoring point.

We determined the orthometric elevation of each of the monitoring points using the GPS measurements. Fortunately, the two King County survey points used for initial control had established orthometric elevations. From this information, a geoid separation was determined at each of the secondary control points and then averaged to determine the geoid separation for the monitoring points. This averaged geoid separation was then applied to the ellipsoid height determined using the GPS data to derive the orthometric elevation of each of the monitoring points.

A monitoring report in a spreadsheet format was developed to show the movement of each point relative to the original position of each point and also relative to the prior location of the point. The final ground coordinates were entered into the spreadsheet for each of the monitoring points and the relative movements were calculated. The elevation of each of the monitoring points was also shown in the report.

CORRECTIVE MEASURES

Through the utilization of GPS survey techniques Triad Associates was able to promptly obtain survey information and provide it to AGRA in a timely and cost effective manner. Concurrently, the City of Des Moines provided the survey data to Golder to aid them in determining site conditions. Crews and equipment subcontracted to Golder were being used to obtain subsurface conditions and making repairs to stabilize the landslide.

From the drill hole information, it was determined that the slip surface of the landslide was occurring at depths ranging between 50 to 70 feet. The slip surface was occurring at the contact between a very stiff and hard clay and silt. A contractor working for Golder installed water pumps in some of the drill holes, converting them to dewatering wells. Also, an interceptor drain was installed in the central area of the landslide to collect surface run-off. Both dewatering measures helped reduce the groundwater pressure along the slip surface and slowed the movement of the landslide. The monitoring data collected in April 1997 showed that the movement within the landslide had indeed stopped due to the dewatering efforts.

CONCLUSION

Based on the results and the amount of data collected and turned over to AGRA we believe that the utilization of the GPS survey techniques was a key to the success of this project.

As with any large project we did experience problems. The biggest problem was the ability to receive "clean" signals from the GPS satellites. A lot of the survey monitoring points were located in areas of heavy vegetation. When we began the project in the winter of 1997 many of the bushes and trees did not have any leaves. By the spring of 1997 these bushes and trees "leafed out" which created signal acquisition problems. We also had one monitoring point that located right next to a tree with a power line located a few feet above the point. At this point we constantly fought to gather enough "clean" data from which to compute the baseline vectors.

The results of the signal acquisition problems were that we had some variations in the direction and magnitude of the movement of some of the monitoring points from session to session. The greatest variation occurred in the elevations or heights of the monitoring points, which seems to be fairly typical for GPS determined heights. In spite of the occasional anomalous information AGRA was able to utilize the data given to them to adequately depict downslope movement of the landslide as it occurred.

Triad Associates believes that the project was a success. They proved to themselves, and the geotechnical consultants involved with the project, that GPS technology will provide accurate and timely survey data on a fairly large site with lots of environmental constraints. The monitoring of the Woodmont Landslide required the use of GPS technology to be able to simultaneously observe the control stations, which were located on stable ground, and the monitoring points, within the landslide area. This allowed the overall movement of the landslide to be determined. Also by setting this project up this way, Triad will be able to reobserve the monitoring points at any future date desired by the City of Des Moines.

ACKNOWLEDGMENT

The author wishes to thank Greg Juneau of Triad Associates and Henry Brenniman of AGRA Earth and Environmental for their assistance with the preparation of this article. The author also acknowledges the City of Des Moines for their permission to produce this paper.

Portions of this paper originally appeared in the September 1997 issue of POB Magazine, pp. 24 –27.

Bradley Freeman is a Project Surveyor and Specialized Survey Services Manager at Triad Associates, Kirkland, Washington. He is a licensed Land Surveyor in the States of Washington, Idaho, Montana and Alaska.

Evaluating FGCS Guidelines for Establishing GPS-Derived Ellipsoid Heights

Daniel J. Martin
Geodetic Program Supervisor
Geodetic Survey Section
Vermont Agency of Transportation
133 State Street
Montpelier, Vermont 05633
Phone: (802) 828-2813 Fax: (802) 828-2437
dmartin@aot.state.vt.us

Abstract

The Global Positioning System (GPS) has proven to accurately produce three-dimensional coordinates. This has typically been achieved by following strict guidelines designed for the task. There is no question that desired accuracies can be achieved when following these guidelines. However, the guidelines can be restrictive, and expensive to follow. This paper illustrates the ellipsoid height accuracies achieved when following the Federal Geodetic Control Subcommittee (FGCS) guidelines for establishing GPS-derived ellipsoid heights (Standard 2 cm and 5 cm) version 4.3. It also investigates areas of the guidelines where procedural changes suggest no significant loss of accuracies and improves field efficiency.

This paper discusses the project scope and the guideline procedures used; then evaluates the results, and suggestion as to how the guidelines can be improved.

Introduction

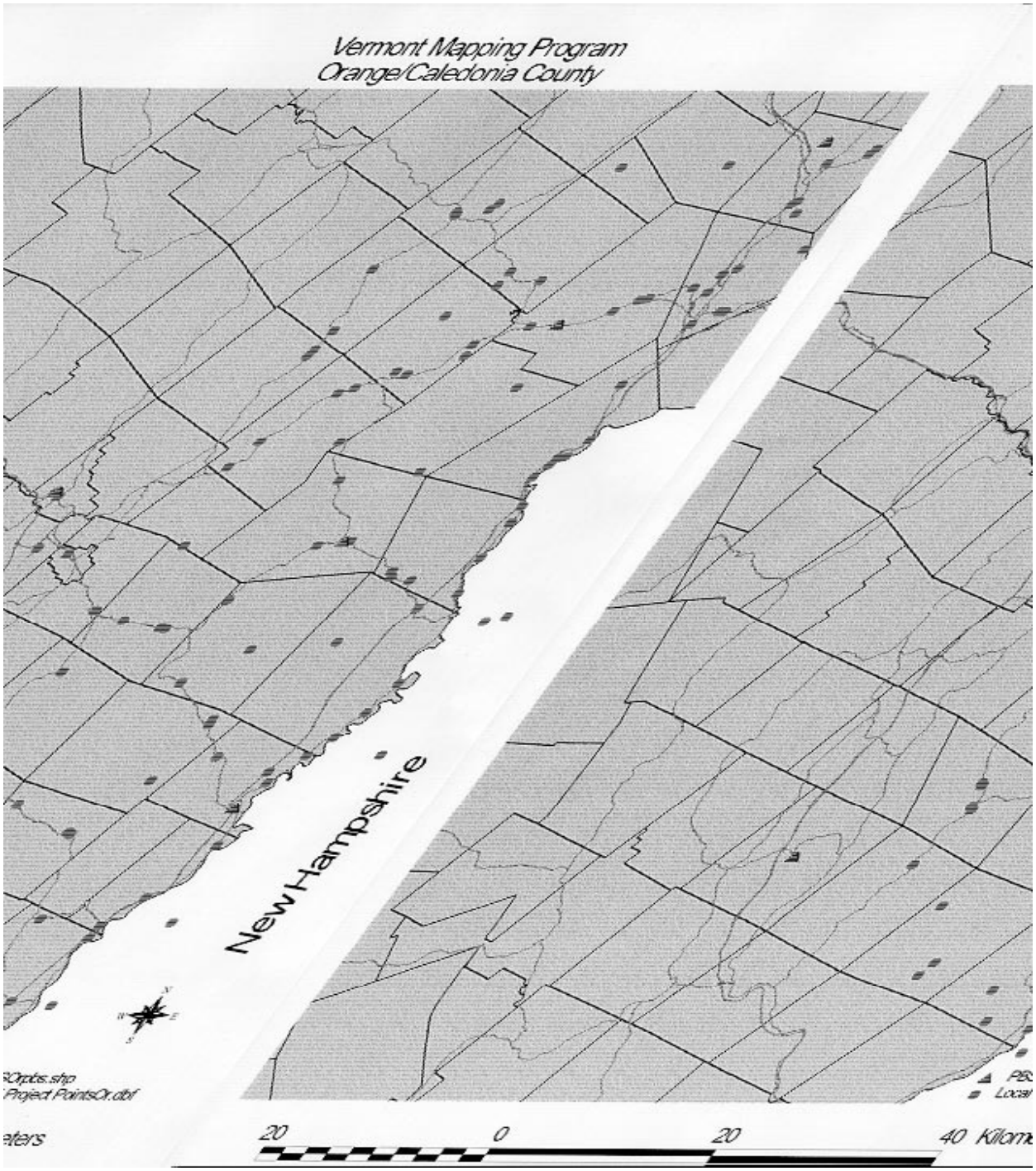
Since 1994, the Vermont Agency of Transportation Geodetic Survey (VTGS) has conducted GPS first-order horizontal control surveys and computed GPS-derived orthometric heights to support the Vermont Department of Taxes' digital orthophotography program. The accuracy of the heights produced between 1994 and 1996 typically ranged from three to six centimeters. In 1997, the demonstrated accuracy of the GPS-derived orthometric heights improved to 17 mm 2-sigma for one project (Martin 1998a, 1998b).

In 1997, the FGCS released Guidelines For Establishing GPS-Derived Ellipsoid heights (Standard 2 cm and 5 cm) version 4.3. (Zilkoski et al. 1997) The Vermont Agency of Transportation with the assistance of Vermont National Geodetic Survey Advisor, applied for and received a grant from the Federal Geographic Data Committee to evaluate these guidelines.

Project Scope

In the spring of 1998, the VTGS conducted ground control for the Vermont Orthophotography Program in the Orange and Caledonia County areas of Vermont. The project would establish 116 first-order horizontal control stations and provide GPS-derived orthometric heights where needed.

Figure 1 is a map of the project area. (Figure 1)



Project Planning

The ellipsoid height guidelines call for establishing Primary Base Stations (PBS) at a 40 km spacing. These stations once established are the basis of ellipsoid heights for the project. The PBS are to be established by conducting five hour sessions on three different days from a minimum of three Primary Control Stations (PCS) which are to be High Accuracy Reference Network (HARN) stations or Continuously Operating Reference Stations (CORS). Vermont has a HARN in place which has a nominal spacing of 30 km. However, due to the age of Vermont's HARN (observed in 1992), the HARN ellipsoid heights are of poor quality, and were not suitable to be used as PCS. Therefore, it was necessary to use stations outside of Vermont as the PCS.

It was decided to use the CORS Network as the PCS. Stations **Brunswick 1** (BRU1) and **Portsmouth 2** (POR2) located in Brunswick, Maine and Portsmouth, New Hampshire respectively, were chosen along with Vermont's CORS, **Vermont Capital CORS** (VCAP). The baseline lengths from POR2 and BRU1 to the project area are approximately 180 km. HARN stations **LYNPORT** (LYNP), **MS 23 RESET** (MS23), **BRADFORD I 91** (BRAD), AND **VTC HARN** (VTCH) along with a new station **U V G** (0UVG) were selected as the PBS.

Instrumentation

The survey used four Trimble 4000SSE dual frequency full wave length receivers and one Trimble 4000SSI duel frequency full wave length receiver, in conjunction with the Trimble compact L1/L2 with ground plane antenna. Dual frequency full wave length receivers are required for baselines longer than 10 km, and is the preferred receiver for all observations (Zilkoski et al. 1997). These observations were taken with 2 meter fixed height tripods, which are required for the 2-cm standard.

Temperature, pressure, and relative humidity observations were taken during the PBS survey using psychrometers and digital barometers.

Observation Scheme

A schedule was developed for the PBS observations. Five hour sessions were scheduled for three days, with the start time staggered each day to use different satellite geometry. The three sessions were scheduled from 6:00 a.m. to 11:00 a.m.; 11:00 a.m. to 4:00 p.m.; and 2:00 p.m. to 7:00 p.m..

A schedule was then developed for the local network stations. The guidelines require that for each adjacent station pair observed, a reobservation must take place 27 to 33 hours later. This was accomplished by scheduling six 45 minute sessions a day with 45 minute move times between sessions. This scenario allowed for sessions A, B, C, D, E, F on day one. Then on day two, sessions A, B, and C were reobservations of sessions D, E, and F from day one. This way three reobservation sessions and three new sessions were conducted each day.

Project Statistics

The project contained the following

- Three Primary Control Stations (CORS)
- Six Primary Base Stations (one was VCAP and was also used as Primary Control)

- 111 Local Network Stations

Total Number of Stations = 116

- 351 Independent Vectors

Total Number of Vectors = 351

- Three Phase one (Primary Base Station) Observation Days
- 16 Phase two (Local Network Station) Observation Days

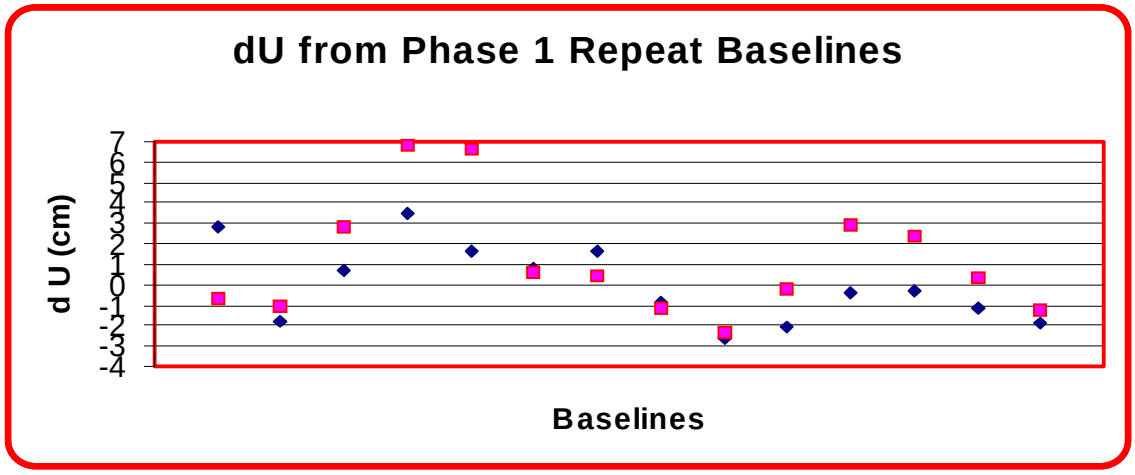
Total Number of Observation Days = 19

Base Line Processing and Adjustments

Phase one

The PBS five hour observations were processed with the precise ephemeris using Trimble's GPSurvey™ software. The advanced controls were set to use the Hopfield atmospheric model with a two-hour zenith delay and standard meteorological data for tropospheric modeling. Repeat base lines were compared using the NGS software DELTAHI (Ed Carlson Personal Communication). A summary of the repeat baselines are contained in Figure 2. The baselines were then adjusted using Trimble's TRIMNET™ v2.3 adjustment software.

The raw data was also submitted to NGS where it was processed using the NGS software OMNI



(Mader et al. 1991), and adjusted with the NGS ADJUST v4.1 software (National Geodetic survey 1997). Table 1 contains the final coordinate differences as computed by NGS and VTGS.

 Station	_Lat (m) VTGS-NGS	_Lon (m) VTGS-NGS	_h (m) VTGS-NGS
LYNP	0.0072	-0.0048	0.023
MS23	0.0078	-0.0055	0.008
0UVG	0.0048	-0.0050	0.000
BRAD	0.0033	-0.0065	-0.013
VTCH	0.0063	-0.0044	-0.016

Table 1

Figure 3 shows a map of the ellipsoid height differences from Table 2. Though the differences are not large, they are larger than we had hoped. On examination of the differences in figure 3, it can be suggested that a tilt exists south to north. Figure 4 shows a plot of the ellipsoid height differences south to north. The horizontal scale represents the cumulative change of latitude in meters.

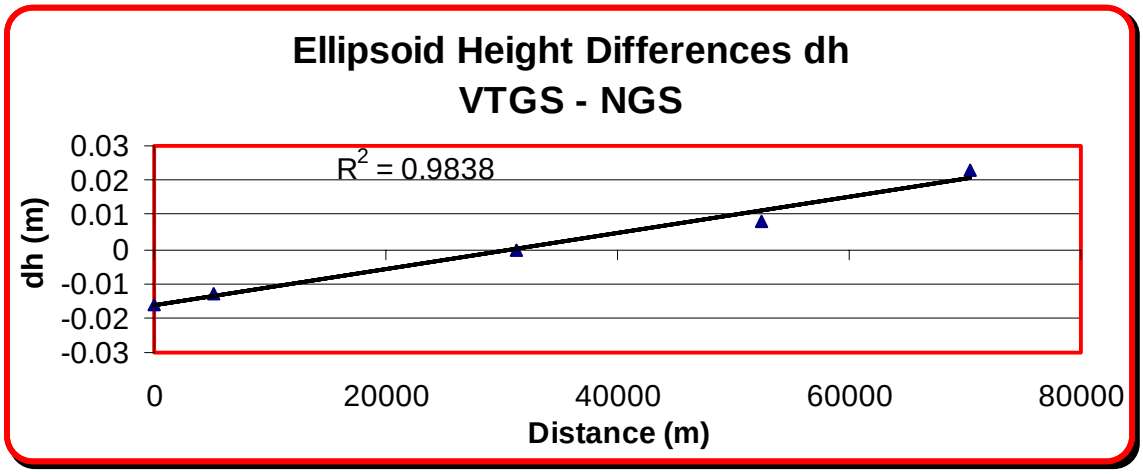
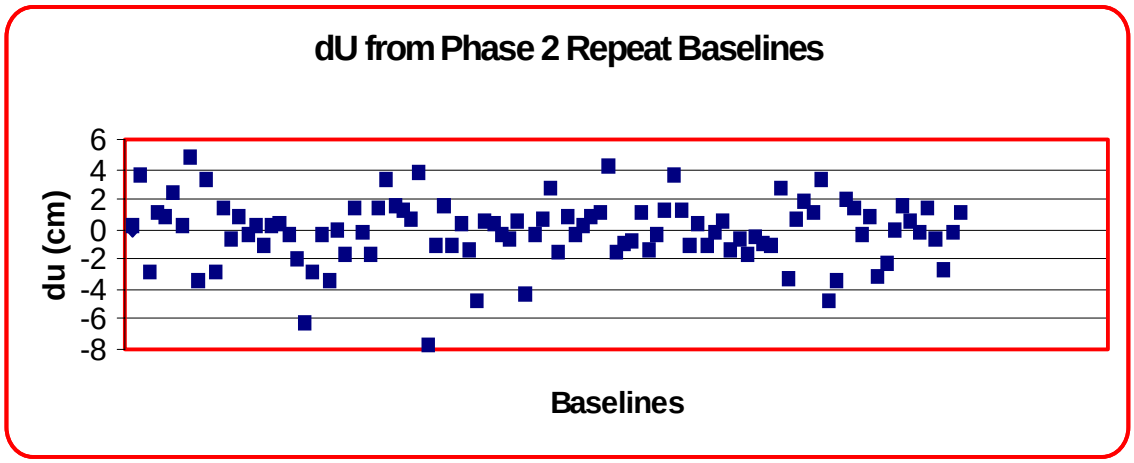


Figure 4 shows that the differences in the computed ellipsoid heights contain a linear trend, and is a function of the latitude. A linear regression was computed and is represented in Figure 4 by the solid line.

Phase two

The observations were processed with the precise ephemeris using Trimble's GPSurvey™ software. The advanced controls were set to use the Hopfield atmospheric model with no zenith delay and standard meteorological data for tropospheric modeling. Repeat base lines were compared using the NGS software DELTAHI (Ed Carlson Personal Communication). A summary of the repeat baselines is contained in Figure 5. The baselines were then adjusted using Trimble's TRIMNET™ v2.3 adjustment software.

In order to meet the 2-cm standard, ellipsoid height differences from repeat baselines must agree to 2 cm or better. Forty-five (approximately 25 percent) of the repeat baselines fail at the 2-cm level. In order to meet the 2-cm standard, those baselines would need to be reobserved. Reobserving



these lines would have added two to three additional days of observations to the project. Do to time

constraints no reobservations were made.

The ellipsoid height differences from the repeat baselines were used to compute a ellipsoid height accuracy at a 95% confidence. The ellipsoid heights test 3.94 cm at 95%. If the 45 problem vectors are rejected, the ellipsoid heights test 1.96 cm at 95%.

Orthometric Height Computations

The GPS vectors were reformatted in the NGS Blue Book format and network adjustments were computed using the NGS ADJUST software. Orthometric heights were computed in conjunction with the GEOID96 Geoid model (National Geodetic Survey 1996).

Forty-one of the stations in phase two were National Geodetic Reference System (NGRS) vertical control stations. These stations provided known orthometric heights with which to compare the GPS-derived orthometric heights.

Orthometric heights were first computed by means of the equation

$$H_{88}=h-N_{96}$$

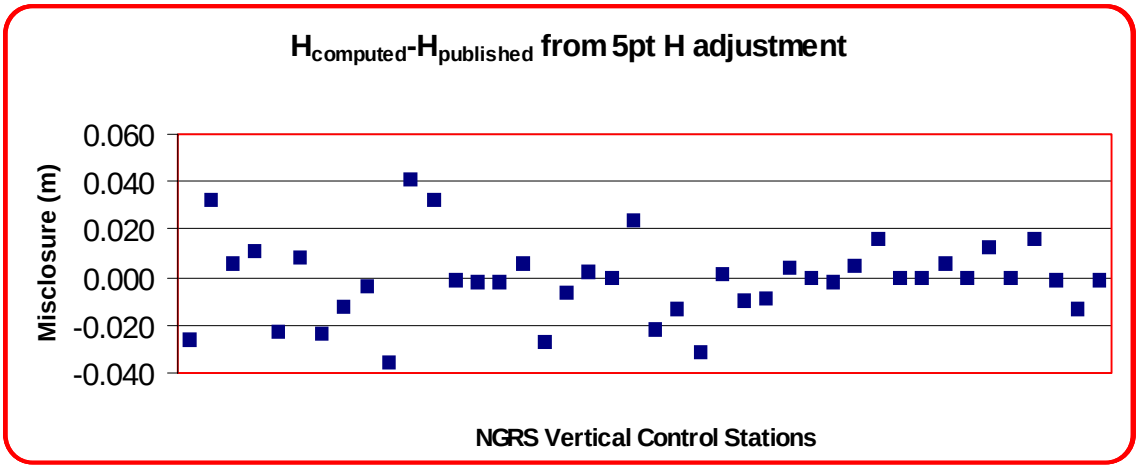
where N_{96} is the Geoid96 geoid height, h is the adjusted ellipsoid height and H_{88} is the orthometric height in the North American Vertical Datum of 1988 (NAVD88). This is generally the least accurate method of computing GPS-derived orthometric heights as it relies only on a geoid model and the adjusted ellipsoid height.

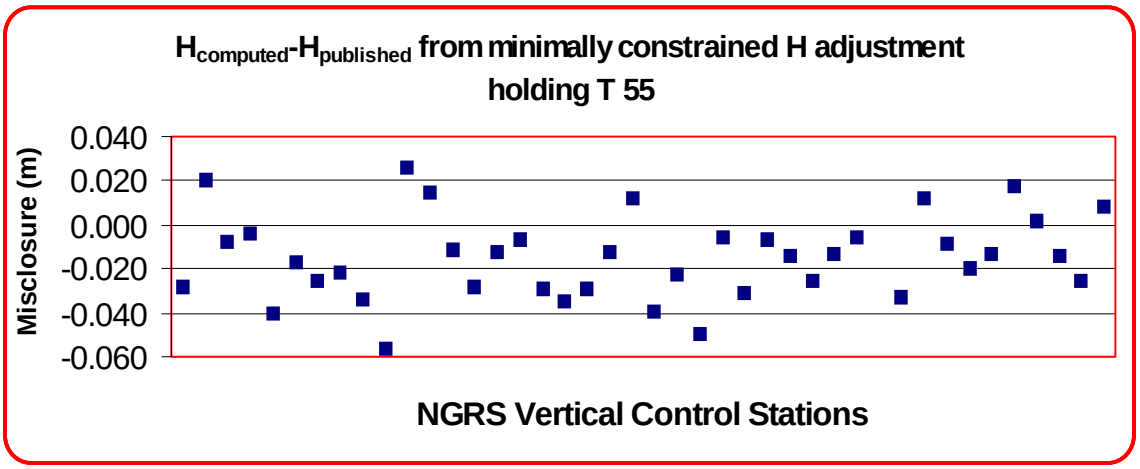
Orthometric heights were then computed by constraining the orthometric height of a known NGRS vertical control station in a minimally constrained least-squares adjustment.

Finally, a partially constrained vertical adjustment was performed by constraining five of the NGRS vertical control stations in a least-squares adjustment. One station was constrained as near the four corners of the project as possible and one near the middle. Doing this effectively removes any tilt or bias in the geoid model.

Figures 6, 7 and 8 show the orthometric misclosures between the computed and published orthometric heights for the NGRS vertical control stations.

The misclosures from these calculations were used to compute vertical accuracies at the 95 percent level. All data was included in the accuracy computations, including those vectors which yielded ellipsoid height residuals greater than 2 cm. The results of the accuracy tests shown in Table 2 indicate that there is not a significant difference in the quality of the orthometric height between the constrained ellipsoidal adjustment and the partially constrained orthometric adjustment. This would indicate that the GEOID96 geoid model is well fitted to the project area, i.e. the shape of the model is correct. However, it is clear from looking at figures 6 and 7 that a bias exists, which is causing the majority of the residuals to be negative.





Orthometric Accuracy Tests (95% confidence level)		
Adjusted Ellipsoid Heights and GEOID96	Minimally Constrained Orthometric Adjustment	Partially Constrained Vertical Adjustment (5 Point)
3.4 cm	3.7 cm	3.2 cm

Table 2

Summary

Approximately 25%of the ellipsoid height differences produced in this project did not meet the 2-cm standard. As mentioned earlier, the vectors that did not meet the standard were not reobserved. Had the vectors been reobserved, we are confident that the standard would have been met. This can be said since the only criteria for meeting the standard is that the repeat baseline ellipsoid height differences be less than 2 cm. This could have been achieved if reobservations were made.

Guideline Evaluation

On the surface, the guidelines were a little difficult to follow. However, after discussions with NGS personnel, we got a better sense of what we needed to do.

The actual accuracy determination is the simplest part of the standard. It requires only that all repeat baselines agree to 2 cm or better. As standards go, this is easy to prove.

However, this only gives a level of certainty between stations which have direct observations between them. The guidelines offer no means of assessing cumulative error determination between stations not directly connected. For example, if station 1 is connected to station 2, and station 2 is connected to station 3, a determination could be made about the relative height accuracy between stations 1 and 2 or between stations 2 and 3, but no determination could be made between stations 1 and 3.

One possible solution is that the standard could incorporate criteria for vertical loop closures. This would allow accuracies to be computed throughout a series of stations similar to differential leveling computations.

The guidelines do make the statement "Final coordinates and their quality assessment shall be determined by using least-squares adjustment software and by analysis of repeated vectors and free-adjustment residuals and loop misclosures (most loops consisting of repeated vectors).". However, the guidelines do not indicate what the acceptable range would be for loop misclosures or free-adjustment residuals.

Horizontal and Orthometric Standards

Another area where the guidelines fall short is that they only incorporate standards for ellipsoid heights. This is a problem because most survey projects by nature are horizontal. If the ellipsoid height guidelines alone are followed, the survey may not meet first order horizontal standards.

The ellipsoid height guidelines state that they will be expanded in the future to incorporate the establishment of GPS-derived orthometric heights.

If horizontal coordinates were produced with GPS and orthometric heights were produced with differential leveling, as in the past, then it would be necessary to have two sets of specifications. Since it is frequently the intent to establish horizontal and vertical positions simultaneously with GPS, it would be most practical to have a single standard which addressed both the horizontal and vertical components in common terminology.

A single guideline should be developed that supports Latitude, Longitude, Ellipsoid Height, and GPS-derived Orthometric Height.

Observations

One potential problem with the guidelines is that they require repeat observations to be taken 27 to 33 hours later. This is to reduce the risk of introducing a bias from using the same set of satellites with the same geometry and to provide different atmospheric conditions. While this theory is sound, in practice this means that the repeat observation is not always taken under ideal conditions, i.e. fewer number of satellites and poor satellite geometry. Often, this can result in the repeat observation yielding less than desirable results, possibly with larger errors than would be introduced by observing under the same conditions.

When we planned the observations, we found that we had to adopt a generic schedule, i.e. the same every day. This was the only convenient way to schedule the repeat baselines so that the observations would meet the criteria. Unfortunately this did not allow any flexibility and was generally inefficient.

Future Research and Conclusion

The ellipsoid height guidelines, though somewhat difficult to navigate, will yield the intended results. This evaluation also suggests that these guidelines will lend themselves to producing GPS-derived orthometric heights that approach the same accuracies with the inclusion of proper specifications.

One error source which we believe is pertinent to ellipsoid heights is the proximity of thunderstorms during observations. Though not documented here, VTGS has experienced poor height results when a thunderstorm has been in the area. Others GPS users that we have spoken with have also experienced this phenomenon. This could be pointed out in the guidelines as a "heads-up" while additional research is being conducted.

It has been the experience of VTGS that it is possible to achieve the same level of vertical accuracy by following the existing specifications for a first order horizontal control survey (FGCC 1988) with only minor modifications (Martin 1998a, 1998b). We believe this is possible due to the increased level of redundancy in a well designed network.

This survey incorporated a "network approach". Since all baselines were repeated, we believe that we can remove most of the repeat baselines with the exception of some key vectors (perimeter), and duplicate the results demonstrated in this paper. Our next approach will be to replace repeat baseline computations with vertical loop misclosures and least-squares free-adjustment residuals to test the ellipsoid height differences. This approach could significantly reduce field work while maintaining the desired accuracy.

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A Geodemographic Population Diffusion Model for Ohio

Joseph Szakas
ICPSR

Barry Bennett
Robert Schmidley
Edward Simmons
Ohio Department of Development

ABSTRACT

This paper sets forth a model for predicting the geographic distribution of population in the state of Ohio. Where population growth and development will occur in the future is of vital importance to state and local officials as well as private sector decision-makers in anticipating and planning for change.

The goal of the model is to provide a rational method for spatially diffusing statewide and county-by-county population projections to smaller units of geography. It combines demographic and geospatial factors to identify those areas where population increases are likely to occur as well as the regions that are unlikely to experience growth. The model is implemented in a computer geographic information system and employs both vector and cell-based (raster) methods. It incorporates factors such as population and housing unit density, commuting patterns, sewer access, proximity to interstate highways and urban centers as well as available land. Results of the application of the diffusion model to population projections for Ohio are presented. Potential limitations of the model are also considered.

INTRODUCTION

Growth or decline of population over time has a profound impact on society and its use of land. An increasing number of people in an area can strain local government's ability to deliver services such as education, fire and safety. New schools may have to be planned and built to accommodate a greater number of students. Police and fire departments may have to expand their staffs to patrol and protect an increasing number of residential structures. In a similar way, loss of population can result in pressure to close schools. More people and houses can overburden existing water and waste treatment systems and clog a region's road network with traffic.

Private sector decision-makers also have a keen interest in population change. Locating a new business or expanding an existing facility must take into account the people who will provide labor or purchase its goods and services. A decreasing population can lead to reduced real estate and land values and affect an area's merchants. Population changes dramatically influence the natural environment. More people generally means more buildings so that open space, farms or woodlands may be

converted to residential, commercial or industrial land uses. It is clear that anticipating and planning for population change is an important issue to a broad base of public and private decision-makers.

An attempt has been made to model the potential for population change across the state of Ohio in a geographic information system (GIS). The model focuses on the allocation of county level population projections through 2015 to smaller units of geography within counties. It is based on population projections created by the Ohio Department of Development, demographic data from the 1990 Census, and census tract level geography. While only an initial exploration, it is hoped this effort will establish the groundwork for a more robust method for predicting where people will concentrate in the future.

FACTORS RELATED TO POPULATION CHANGE

Demographic information used in this analysis was obtained from the 1990 Census of Population and Housing, Summary Tape File 3 (Census of Population and Housing, 1990). This source reflects sample data obtained from the "long form" questionnaire used in the 1990 Census. It is an extremely rich portrait of our society and a case could be made that virtually every data field should be used in population modeling. Of course, a small subset of the fields was chosen for the GIS model presented here.

The specific characteristics chosen include the following items.

Persons	Number of persons
Employment Status	Number civilian employed and unemployed
Family Income	Average (computed) family income
Housing Units	Number of housing units
Water Source	Percent public system or private company
Sewage Disposal	Percent public sewer
Year Built	Median year (housing unit) structure built
Value	Median value (owner occupied) housing units

(Since the model was implemented at the tract level, Summary Level 140 data were used.)

Certainly the density of population and housing (number of people divided by tract area) strongly influences how much growth can occur. Heavily settled areas are not likely to experience much increase in the number of residences and, therefore, people. In addition, the extent to which a region is served by water and sewer systems (as opposed to wells and septic tanks) limits how much development can occur. Urbanization requires extensive sewer and water systems. The year housing units are built may provide an indication of likely settlement. Urban zones with older housing are less likely candidates for development than suburban fringes with newer housing. (Of course, the ripest areas for new development may be sparsely settled far fringes with relatively old residences!)

Economic health of a region is another key factor influencing population change. Presumably, regions with strong economies are magnets for people while depressed places tend to decline as folks seek their fortunes elsewhere. Family income, employment, and housing values were selected in an attempt to reflect economic well being.

Of course, many environmental factors come in to play in determining how much population a patch of real estate can sustain. Soils, geology, drainage, and terrain influence where settlement will occur. For example, the ease with which sewer lines can be extended is highly dependent on the terrain. While data for such variables were not used in this analysis, proxies for other geographic variables were included in the modeling process. Approximations for the following factors were developed from other data:

Available land	Amount of land available for development
Urban	Proximity to urban center
Highway	Within interstate highway corridor

In general, development will cluster where there is available land close to existing population centers and within reach of adequate highways.

DIFFUSION MODEL

The model involves a two-stage process. In the first part, a growth potential map for the state was derived. This map consists of polygons to which growth potential rankings were assigned as follows:

4	High potential for population growth
3	Moderate potential for population growth
2	Low potential for population growth
1	Virtually no potential for population growth

A fifth rank of zero was applied for those areas (tracts in Lake Erie, for example) that will not sustain people.

In the second stage of the model, the various county level population projections (He, 1997) through the year 2015 were allocated to polygons in the growth potential map. The allocation of an increase or decline in countywide population (from 1990 to 2015) was made in rough accordance to the following factors:

60% increase	Potential rank 4
25% increase	Potential rank 3
10% increase	Potential rank 2
5% increase	Potential rank 1
60% decrease	Potential rank 1

25% decrease	Potential rank 2
10% decrease	Potential rank 3
5% decrease	Potential rank 4

(Some adjustment to these allocation factors was necessary. The reason is a county might not have sufficient area in one or more of the potential ranks over which to apportion the change.)

The development of the growth potential map was accomplished by creating and combining several map layers using the demographic and geographic variables identified above. Map layers for the following factors were generated.

Urban core areas	Census tracts with population density of 1,000 persons per square mile, buffered to 5 miles
Economic health	Ranking of census tracts by unemployment rate, average family income, and housing unit value
Available land	Buffer general road network line map to 1,000 feet
Highway proximity	Buffer interstate highway map to 5 miles
Demographic growth factors	Ranking of census tracts by population and housing unit density and age of housing stock
Water/sewer access	Census tracts with 85% or more sewer access buffered to 3 miles, other tracts ranked by both water system and sewer access

These individual maps were then combined to form the overall growth potential map.

RESULTS

The final population potential map is somewhat visually unappealing as it retains many artifacts from the census tract polygons from which it was created. It also reflects a rather unrealistic spidery network of "unavailable land" in rural areas. A more suitable model for available land could probably be obtained from a more complete road map (Census Tiger/Line data). Block group level analysis (as opposed to tracts) might provide a smoother more appealing potential map. The overall impression of growth potential in Ohio is, nevertheless, somewhat reasonable. Substantial growth may be expected along the corridor stretching from Cleveland southeast to Columbus as well as north from Cincinnati to Dayton.

The map of Miami County illustrates the results of the actual population allocation from the second stage of the model. This map also reflects some of the unappealing artifacts of the growth

potential map. However, it provides a not-too-unreasonable picture of the percentage change in the county's population through 2015.

LIMITATIONS OF THE MODEL: FUTURE WORK

The shortcomings of the model considered in this paper are significant and point the way to future refinements that can be proposed.

A limitation of the model is that it does not take into account social and political factors on population change. It is difficult to identify, quantify and map the institutional parameters, such as restrictive zoning or lack of zoning that can affect the course of future development. Another weakness is the failure of this analysis to reflect spatial interactions among the various geographic units. As an example, large, economically strong metropolitan areas will have a more far-reaching and profound influence on their regions than smaller densely settled places in rural counties. A more complex model is needed to account for such differing interactions.

No attempt was made to consider the sampling errors inherent in the Census data and their affect on model outcome. While this problem would be exacerbated with block group level analysis, the higher spatial resolution might enhance results. Block groups will in general be less inhomogeneous than census tracts.

Information from the 1980 Census could be compared to 1990 data to isolate the factors most highly correlated to population change. Variables for the present model were chosen intuitively and combined in a rather naive manner. Several of the factors may have been correlated and thus over counted. A more rigorous approach would identify and differentiate the significant causal factors and model them with more fidelity. In a few years, once the 2000 Census is completed, it will be possible to take advantage of the distinct GIS snapshots of population for the years, 1980, 1990 and 2000.

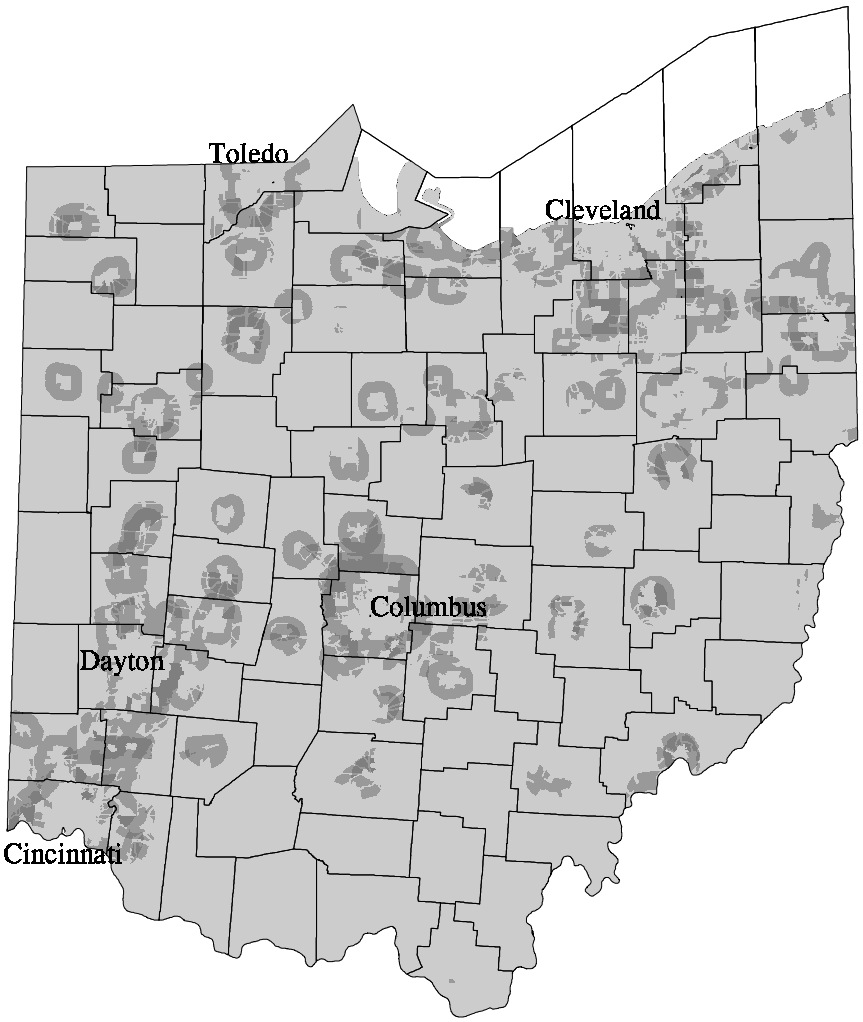
The authors readily acknowledge the results presented here are only a first attempt at establishing a rational method predicting future population distributions. However, they hope to undertake some of the enhancements noted in order to develop a more appropriate model. While predicting population for a small town or township twenty years out may be out of the question, it should be possible to obtain a generalized notion of where the state's population will be in the future.

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Census of Population and Housing, 1990: Summary Tape File 3 on CD-ROM (Ohio) [machine-readable data files]/prepared by the Bureau of the Census.--Washington: The Bureau [producer and distributor], 1992.

He, 1990. Population Projections Ohio and Counties by Age and Sex 1990 to 2015. Ohio Department of Development, P.0. Box 1001, Columbus, Ohio 43266-1001.

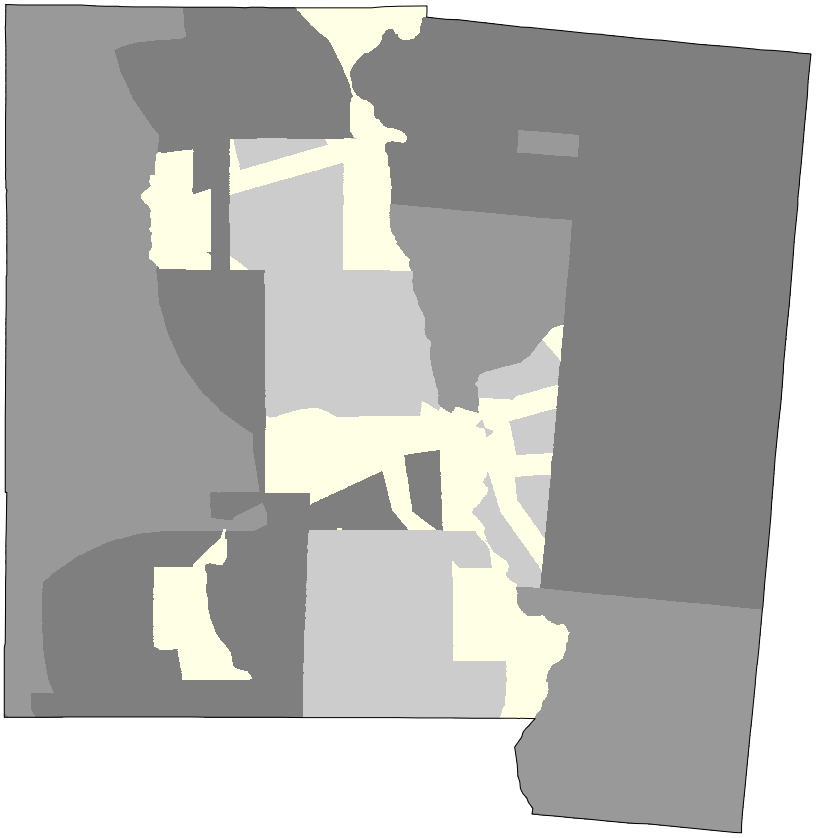
Ohio Population Potential
1990 - 2015



Areas of high potential (dark) are likely to experience substantial population growth while areas of low potential (light) will experience little or no increase in population.

Population Growth Potential
- High potential
- Moderate potential
- Low potential
- Virtually no potential
- Void tract

Predicted Population Change
Miami County, Ohio
1990 - 2015



Map shows percentage change
in population predicted by model.
Overall county increase is 16,018.

Percent Change
- 30% or more
- 15.00% - 29.99%
- 7.50% - 14.99%
- 0.00% - 7.49%

The California Spatial Reference Center:
Partnership for a Sustainable Spatial Reference System

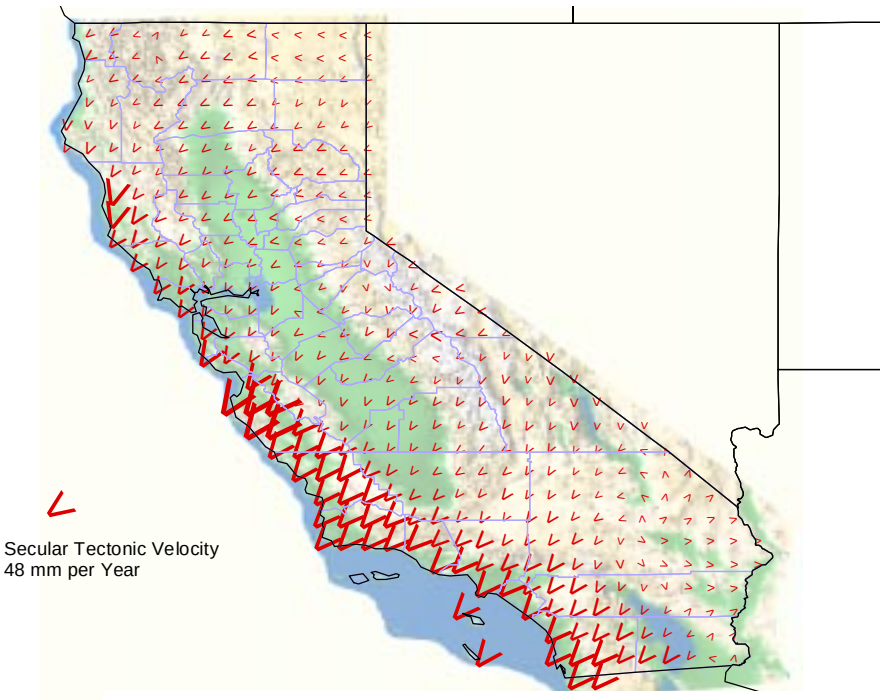
Gregory A. Helmer
Robert Bein, William Frost & Associates
Irvine, California
Email: gahelmer@rbf.com

Introduction

Perhaps nowhere else can one find the diversity and magnitude of challenges confronting the implementation and maintenance of a geodetic reference network than in the state of California. Crustal motion in the form of tectonics, local and regional seismic activity, volcanic expansion, and land subsidence makes any geodetic control campaign temporal at best. Difficult terrain in the form of vast deserts, remote mountains and offshore areas introduce questions as to the reliability of the geoid model, and compound the difficulty of water vapor modeling in the troposphere. These unique challenges are issues within the most populous state in the nation -- a region that on its own is the seventh largest economy in the world. Decreasing federal programs frustrate the problem and place increasing responsibility for solutions at the local level. The demand for high-precision and reliable spatial data has never been greater. Critical decisions effecting health and safety, from land use and environmental regulation to emergency response efforts, are increasingly dependent upon accurate knowledge of spatial relationships. The California Spatial Reference Center (Center) has been formed as a partnership with federal, state and local agencies, business and educational institutions to provide for a state-of-the-art spatial reference framework for the state of California. The Center is based at the University of California San Diego's Scripps Institution of Oceanography. Building upon the infrastructure of Continuous Operating Reference Stations (CORS), data processing and data management at the Scripps Institute for Geophysics and Planetary Physics, the Center is dedicated to providing the necessary geodetic services to ensure accurate and consistent spatial referencing of information within California.

GEO-DYNAMICS AND THE SPATIAL REFERENCE SYSTEM

Episodic movement resulting from significant earthquakes is only a portion of the impact from crustal motion on spatial data. The continuous secular component of the movement constantly erodes away at the integrity of the spatial reference system. The High Precision Geodetic Network (HPGN), established in 1991, has been the State's primary horizontal reference system, and is very quickly approaching its practical life expectancy. In the nearly eight years since its completion, the 238 stations of the California HPGN have experienced an average displacement of 17 cm relative to the North American plate (mean = 16.7 cm, sigma = 15.8 cm). While the majority of this displacement is



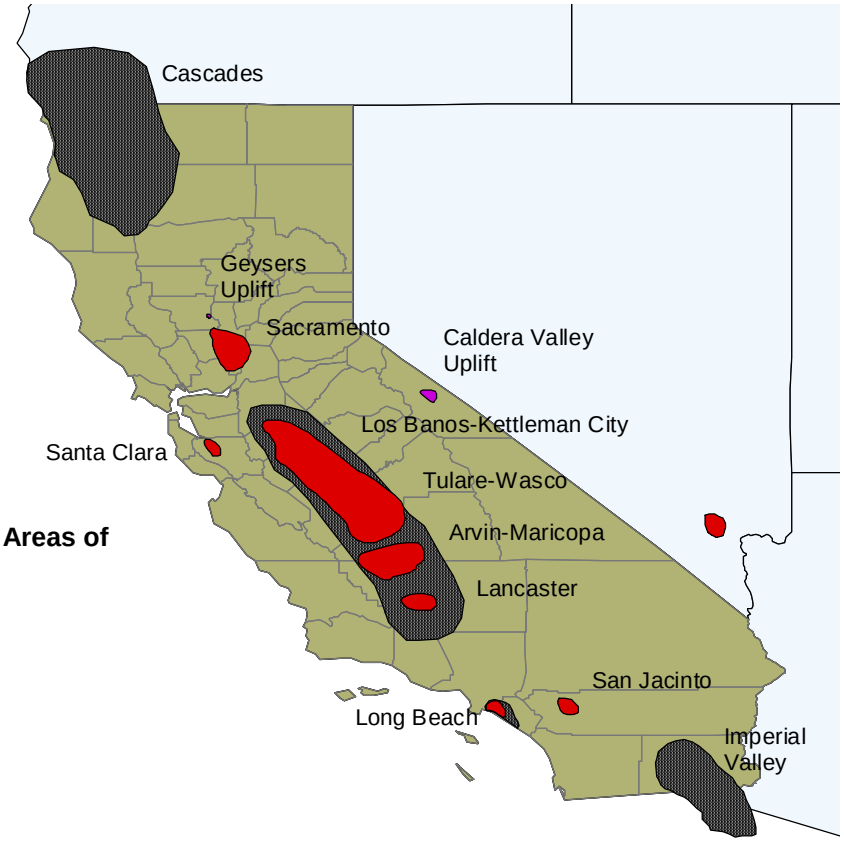
Horizontal Velocity Field

uniform when looked at within any particular local area, relative distortions exist and are becoming more significant with time. Even excluding the Landers and Northridge earthquakes, average relative distortions between the HPGN stations at epoch 1999.0 are estimated at 1.10 ppm. This exceeds the original error budget for the network and places the B-Order accuracy in question. Re-observation of the Federal Base Network in California was conducted in May of 1998. This campaign included 66 stations completed by NGS and 75 stations by other cooperative efforts. Also included was data from the 74 CORS in operation at the time. Cartography and GIS invest tremendous effort developing systems which, to derive the maximum return, must be relatable to a common reference frame over time. This is impossible with the current system at the levels of accuracy available from modern positioning technology. Clearly provisions must be made to accommodate crustal motion in our geodetic reference system if we wish to participate in the advances in GPS and GIS technologies.

Secular movement is virtually constant over any historical time period. The crustal blocks move past each other continually but friction along the fault restricts that movement. This action causes the crust to bend under the force of the accumulated strain. As strain increases, the west side of the fault bends southerly and the east side bends northerly. In theory this bending approximates a tangent function. At some point the force overcomes the friction releasing the strain in the form of an earthquake. The elasticity of the crust allows it to return to its zero-strain shape with an offset occurring along the fracture zone. Dislocation theory has been verified using historical geodetic measurements and geologic investigations along fracture zones. Precise GPS observations before and after the Landers event revealed displacements of over 2 meters near the epicenter. Offsets in stream courses found imbedded in geologic strata show regular 300 to 700 year displacements of 2 to 20 meters in the Salton Trough area.

VERTICAL DEFORMATION

Vertical deformation is found in many areas of North America. Some of the numerous causes include fluid withdrawal, post glacial uplift, seismic deformation, sedimentary loading and magma intrusion. While vertical deformation is not unusual, California is again unique having regions of deformation from at least three different causes. Subsidence resulting from fluid withdrawal is found throughout large regions of the Sacramento, San Joaquin and Imperial Valleys. Seismic deformation is found to varying degrees



Known and Suspected Areas of Vertical Deformation

throughout the San Andreas fracture zone and uplift due to magma intrusion is known to occur in the Mammoth Lakes area. Vertical deformation does not lend its self to simple modeling. The forces influencing vertical deformation are neither time dependent nor predictable. The amount of ground water removed, or replaced, in any given area can vary dramatically, and the removal may not produce the same magnitude of subsidence throughout the effected area. Seismic and volcanic activity, as we are so acutely aware, is essentially unpredictable. Forty-two centimeters of uplift spread over a 700 Km^2 region were experienced following the Northridge Earthquake. Further complicating the situation is the lack of studies to monitor vertical deformation on a regional scale. The high cost of precise leveling has limited the ability to investigate vertical deformation. What is clear is that California represents a unique challenge in the definition of a vertical reference system do to the described vertical deformation.

ECONOMIC IMPACT

In addition to the destruction of geodetic control, crustal motion problems and the incoherent spatial reference system impose significant losses that could be mitigated through better management. Subsidence is a problem with very direct cost implications and is clearly poised to get worse. The California Department of Water Resources estimates aquifers in the State are over drafted by 10 million acre-feet annually — an increase of 75% over the last five years. The National Research Council conservatively estimated the annual costs due to increased flooding and structural damage from subsidence in the US to be in excess of $125 million. FEMA has estimated that a two-foot difference in elevation for a typical California residence translates into a difference of more than $18,000 in flood insurance premiums over the life of the loan. The security of the region's water supply is precariously dependent upon the precise elevations across the State. Aqueducts in Southern California traverse some of the most active geophysical areas in the nation, and the Sacramento-San Joaquin Delta is not only an area of known subsidence, but is an extremely sensitive political and environmental issue. From precision agriculture and intelligent vehicle highways to earthquake monitoring and emergency response management, a multitude of advanced applications are being implemented which require accurate spatial integrity to be fully realized. The Center proposes to employ the same network of continuous operating GPS reference stations (CORS), used to monitor the movement of the earth's crust during the Landers and Northridge earthquakes, to provide a spatial reference data impervious the dynamics of the region. The program is designed to support a wide range of positioning and mapping applications.

ORGANIZATION

The Center is a non-profit organization affiliated with Scripps Institution of Oceanography of the University of California – San Diego. It has been formed under a constitution and by-laws, which includes nine stated objectives.

1. Maintain a high-accuracy horizontal and vertical reference network, based on GPS and related technology, and consisting of in-ground monuments, and Continuously Operating Reference Stations (CORS) and geodetic models necessary to meet the spatial positioning needs within California for surveying, engineering, geomatics, scientific (e.g., crustal motion studies), geographic information systems, and other activities dependent upon reliable spatial referencing.

2. Designate this network, through legislation, as the "California Spatial Reference System" (CSRS) and specify in such legislation that the CSRS is California's mandated spatial reference system.

3. Process and adjust GPS observations and provide the National Geodetic Survey (NGS) with appropriate electronic data files and related information for inclusion into the NSRS, all in accordance with approved NGS procedures.

4. Distribute, by electronic means, accurate time-dependent geodetic coordinates, velocities, GPS observational data, and supplemental station information (historical coordinates, seismic data, etc.).

5. Maintain an archive of selected modern geodetic data and related metadata.

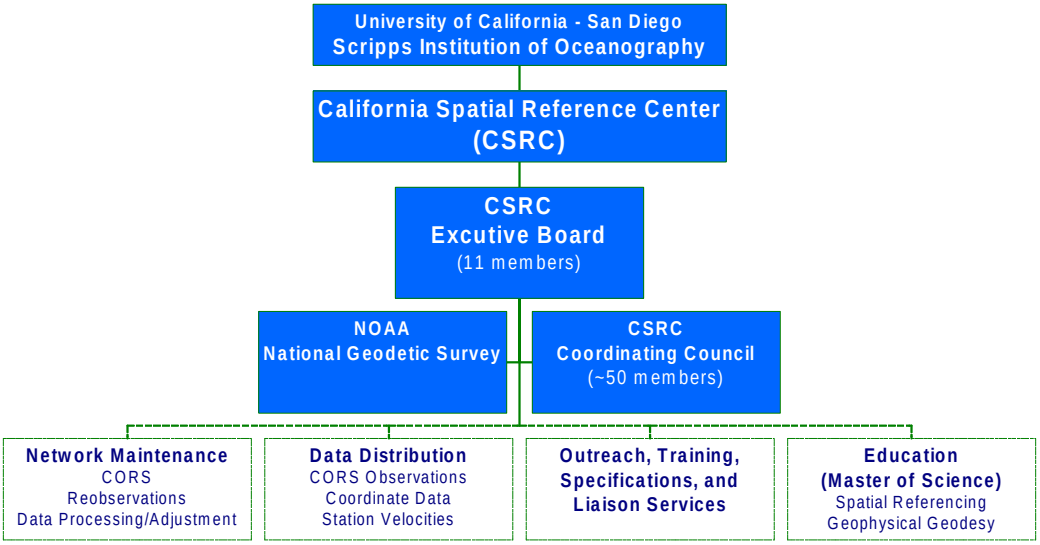
6. Develop specifications and guidelines for GPS surveys, not otherwise provided at the national level.

7. Educate users and potential users on (a) spatial referencing issues; (b) availability and application of the Center's data, products, and other information; and (c) services available from the Center.

8. Foster cooperation and partnerships among other agencies, research entities, and organizations involved in spatial referencing issues.

9. In cooperation with the University of California - San Diego, develop a Master of Science degree program in spatial referencing and geophysical geodesy.

An Executive Board and a Coordinating Council govern the Center. Membership represents a broad range of professionals from the public and private sector, educational institutions, and various spatial data users.



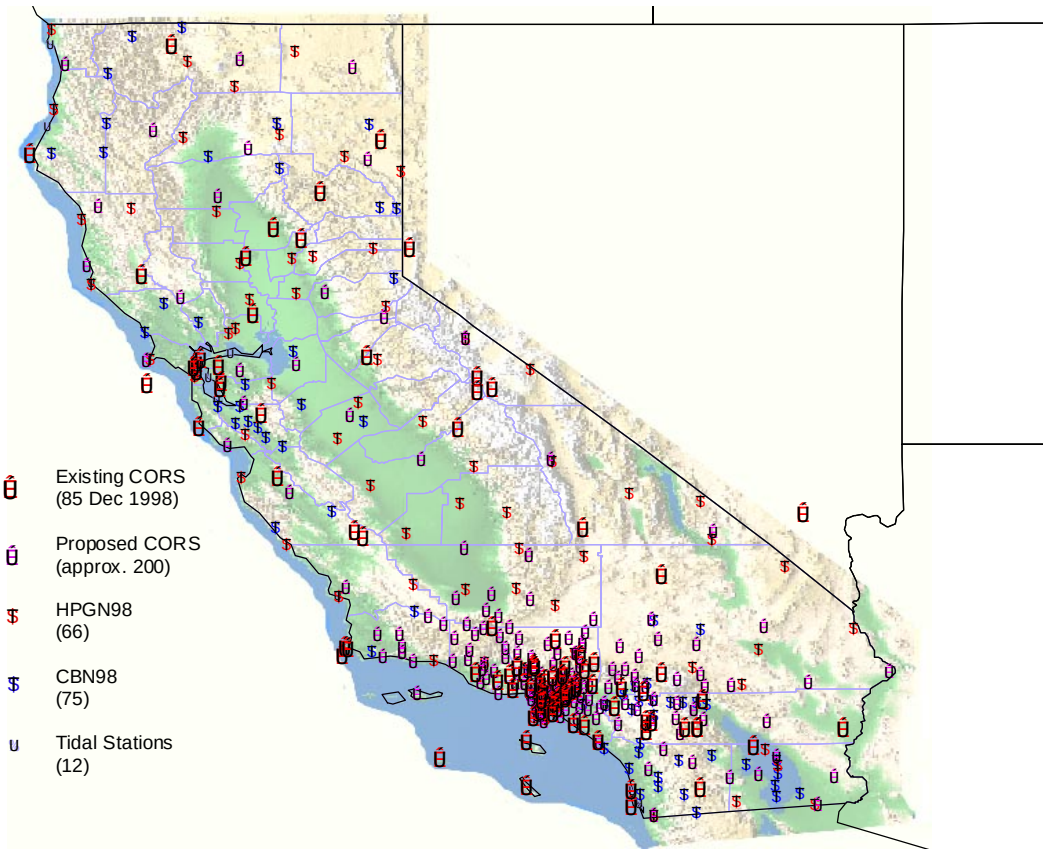
GEODETIC NETWORK

The Center proposes to provide one accurate, consistent spatial reference network, throughout the State of California, that will fulfill the many diverse spatial referencing needs of California's users. The network design includes the following basic concepts:

- The network shall be part of the National Spatial Reference System (NSRS).
- The network shall conform to the datum, accuracy classifications, guidelines and methodologies accepted by the National Geodetic Survey (NGS).
- The basis of the network shall be derived from Continuously Operating Reference Stations (CORS) of NGS and the International GPS Service (IGS).
- The network shall be established, maintained, and monitored through GPS geodetic survey procedures conforming to the standards and specifications of NGS.
- Each network station shall have, at the minimum, values for horizontal position, ellipsoidal and orthometic heights, horizontal velocities, and a stated accuracy standard for each value.
- The geodetic values for the network stations shall be kept current or supplemental information shall be provided to enable the user to determine the current geodetic values. An exception is vertical geodetic values for stations within identified subsidence areas.
- The network design shall be based on providing network stations and geodetic data for GPS positioning techniques -- both horizontal and vertical.
- In general, accuracy standards and station spacing shall conform to the provisions of these Bylaws. Exceptions are permitted upon approval of the Executive Board.
- The objective of the Center's geodetic network is to provide a statewide "basic" (framework) spatial reference system; it is not intended to provide specific project-related control monuments.

The Center's geodetic network will consist of four components -- CORS, Primary Stations, and Secondary Stations, and the geodetic models used in application of the system. The accuracy standards, station spacings, and the Center's involvement for each component is expected to include the following:

Component	**Standard (cm)**		**Station Spacing** (km)		**Center Involvement**		
	Horiz.	Ortho.	Nominal	Minimum	**Establish**	**Maintain**	**Monitor**
CORS	1	2	80	7	Yes	Yes	Yes
Primary	1	2	40	7	Yes	Yes	Yes
Secondary	1	2	20	7	No	No	Yes



Geodetic Models

As part of the reference network, the Center will cooperate with geodetic research organizations to develop, approve and distribute the geodetic models necessary for use of the CSRS network and data. These models are expected to include the following:

a. Transformation Parameters: Seven parameters (i.e. delta [X, Y, Z] rotation [X, Y, Z] and Scale) for transformation from the current International Terrestrial Reference Frame (ITRF) datum to the current CSRS datum.

b. Horizontal Velocity: Secular and episodic crustal motion data compiled into a computer readable format for coordinate conversion between epoch dates.

c. Geoid Heights: The most current geoid model approved by NGS.

d. GPS Orbits: Precise post-fit ephemeris for the GPS constellation.

CONCLUSION

It is clear that California represents a particularly unique challenge to spatial data science. The California Spatial Reference Center proposes to meet these challenges with a state-of-the-art GPS monitoring network, advanced data processing and analysis capability, and electronic data distribution techniques. To insure the benefits of this system are available into the future, the Center will maintain close coordination with the earth science researchers at Scripps, NGS and other institutions, and will work with the University to develop the education programs for the spatial data experts of the future.

The procedures formulated and implemented by the California Spatial Reference Center will further national and international initiatives, and the applications supported by the system offer tremendous opportunities for the local region.

COAST GUIDE

MERIDIAN RETRACEMENT

1858 – 1998

Submitted by:

Richard T. Sterling

14 Breckenridge Drive

Hoquiam, WA 98550

360-533-4735

December 7, 1998

COAST GUIDE MERIDIAN RETRACEMENT 1858 – 1998

The activities of many professional surveying societies include perpetuation of historical surveying monuments or boundaries. The Land Surveyors Association of Washington Historical Society has been active in this work since 1985. The commemoration of the 140th anniversary of the Coast Guide Meridian is an example of cooperative projects that professional and other organizations generate to provide ties between the past and the future.

LOCATION AND EARLY HISTORY

Grays Harbor is located along the Pacific Ocean coast of Washington State. It is a large natural inlet draining the Chehalis River Basin; a large portion of Southwest Washington has rainfall of 60 to over 100 inches annually. Grays Harbor is detailed on U.S. Coast & Geodetic Service (NOOA) Sheet No. 6195, Washington State 1:100,000 maps and U.S. Geological Survey 1:24,000 quadrangles. It is closest to Latitude 47° North and Longitude 124° West. A 1791 map based on Spanish exploration between Baja California and the Aleutian Chain shows an inlet north of 47 degrees North which has been historically assigned to Point Grenville but was named the Bay of Martyrs for the party of sailors lost in 1775 as they attempted to obtain water, firewood and possibly a mast. (Voyage of the Sonora...1775 Journal, Ye Galleon Press, Fairfield, WA 1987)

Following development of chronometers, coastwise sailing developed into widespread navigation of the open ocean and the voyages of Captains Cook and Vancouver are well documented. It remained for a commercial voyage from New England by Captain Robert Gray in 1792 to discover the mouth of the Columbia River and insure the claim of the United States to the Pacific Northwest. He also located Grays Harbor but upon sailing into it met with hostile natives. He named it Bulfinch Harbor after a business partner; Vancouver called it Whidbey Harbour but the crew of the Lady Washington named it after their commander, who first carried our flag around the globe. Trading and shipping into Puget Sound and the Columbia River evolved into British settlement. One of the recorded visits to Grays Harbor was overland from Fort Vancouver by plant explorer, Scotsman David Douglas in 1824-25. Expansionist policies of Andrew Jackson eventually produced the U.S. Exploring Expedition (ref. Leroy Middleton - Evergreen State Surveyor, Fall 1997, pp. 15-19) under Lt. Charles Wilkes, a very capable surveyor but much maligned commander of a joint surveying and scientific expedition (1838 – 1842). Their maps of Puget Sound and the Columbia River

guided the General Land Office in choosing a point of origin for the Willamette Meridian in 1851 (ref. C. Albert White – Initial Points of the Rectangular Survey System, PLS of Colo., 1996, pp. 179 - 184)

A Wilkes party under the command of Midshipman Henry Eld traveled to Grays Harbor from Puget Sound via the Black and Chehalis Rivers. Their map of the Lower Chehalis River and Grays Harbor contains names of the party, most prominently Brackenridge Bluff, named after another Scot horticulturist, William. D. Brackenridge and Stearns Bluff, named after a capable U.S. Marine Sergeant, Simeon Stearns. (ref. David B. Tyler - U.S. Exploring Expedition, Amer. Philosophical Soc. 1968, p. 98, p. 208, p. 287)

GOVERNMENT LAND OFFICE SURVEYS AND SETTLERS

A flood tide of emigrants from the Eastern U.S., fueled by interest in Oregon and then the California Gold Rush, provided impetus to form the Department of the Interior in 1849 and move the General Land Office from the Treasury Department where it had been since 1812. Not one principal meridian had been initiated since 1833. Land surveys were required for settlement of the country under rules using the technology that had developed.

Instructions of the GLO Commissioner dated 1855, Diagram "A" (ref. C. Albert White – A History of the Rectangular Survey System, G.P.O. 1982) illustrates the theoretical progression of work controlled by standard parallels and true meridians to correct for convergency of meridional range lines which totaled 0.77 chains per township at 45 1/2^{O} North latitude. The plan had to be revised due to the influx of settlers into the Oregon Territory after 1836 resulting in scattered claims and the Donation Land Claim Statute in 1850. The Willamette Meridian and Standard Parallels at 24 mile intervals progressed north but the meridian encountered Puget Sound and a Puget Sound Guide Meridian was employed to proceed to the 49th Parallel. Unfortunately, Meridians and Parallels encountered "mountainous" topography which slowed and caused difficulties unenvisioned by surveyors of gentler topography in the East.

As settlers arrived in groups they banded together to form communities strengthened by the experiences of the emigrant trail. Communities such as the Grand Mound – Upper Chehalis Valley provided the GLO survey contractors and crew to perform the surveys of the public domain. A list of donation land claims by counties, (ref. Charlotte Shackleford, Building A State; Sec. XVI, pp. 403 – 452, 1939) has many

names which occur in the field note records. The record is further enhanced by a diary. (ref. – Patterson F. Luark, Unpublished Manuscript of Handwritten Diary 1843 - 1868, Hoquiam Public Library) Luark's donation land claim, Cert. No. 104, Pts. Secs. 24, 25, T 15 N, R 3 W was on Ford's Prairie. (Centralia). 1865 (A. H. Johnson) and 1994 (WA DNR) maps compare how place names have changed. After the 4th Standard Parallel survey by Jared Hurd to Grays Harbor in 1855, (GLO Notes Vol. 8, p. 2) plans for a Coast Guide Meridian to the 8th Standard Parallel were made using the Wilkes Map. The crossing from Stearns Bluff to Brackenridge Bluff scaling about four miles was not beyond the ability of available equipment. Deputy GLO surveyor Alleck C. Smith, an attorney from Illinois, obtained Contract No. 35 for the Coast Guide Meridian between Ranges 10 and 11 West from the 4th Standard Parallel to the 5th Standard Parallel. The field notes do not contain any specifications except those of the standard 1855 instructions which called for Burt's Solar Compass or equivalent on Meridional Lines.

In 1854 William S. Burt printed the first edition of a manual for the adjustment and use of the solar compass after it had been in use for 20 years. Smith's party included his father-in-law, Anson G. Henry – a GLO Deputy surveyor from Illinois by way of Marion County, Oregon who had a contract to survey in Twps. 19 and 20 North, Ranges 10, 11 and 12 West and eventually the 5th Standard Parallel to the ocean. Three generations of the Henry family made a large contribution to surveying – The Henry Homestead at the mouth of Grays Harbor served as a magnet to fortune hunters when plans for a gold mining venture developed. Captain, also Doctor Henry was a long time political ally of Abraham Lincoln and had aspirations to become Indian agent but settled for appointment as Surveyor General of Washington Territory in 1861. He had gold fever which caused survey lines to wander in the vicinity of streams.

Smith's party commenced their survey on May 8, 1858 and there is an entry in the Luark Diary ... "1858 – May – Sat – 8 – spent the day with Doctor Henry and Alex C. Smith who were commencing a job of surveying the first part of which was to run a meridian line from 4 to 5 Standard Parallell (sic) and between R 10 – 11 W." Luark was familiar with the 4th Standard Parallel because he had used it to make a preliminary survey of a territorial road from the Upper Chehalis Valley to the Grays Harbor in February, 1858. His diary contains references to GLO Deputy Surveyors Gardner, David and Charles Byles, T. F. Berry, T. F. Whitworth, Henry Stearns, and the first coast survey vessels on Grays Harbor.

The survey of the Coast Guide Meridian (GLO Notes Vol. 9, pg. 1 and Plat) reached the south shore of Grays Harbor at 67.50 chains where Smith established a meander corner and with favorable low tides proceeded on the sandy tide flat (photo and map) to lay out a base line for triangulation.....the notes do not mention double chaining but show two sets of chainmen. Smith's 25 chain baseline includes two angles to a projected flagged point over one channel in the tide flat. He then proceeded to chain across a second flat to the main channel where another flag was established; then with a mile long base line established from a point projected to the north side of the tide flat, he again measured two angles back to the second flag. Using the means of the observations he calculated the total distance across the water at 327.67 chains. NOTE: There is a computational error of 0.63 chains on the distance to the first flag which clearly involve using the cotangent of 19 degrees not 19 1/2 degrees. (ref. (G. H. Co. Forestry Dept. Map 1:1000')

The survey proceeded 24 miles as instructed, then Smith and Henry commenced to survey portions of Twps. 17 through 19 North, Ranges 11 and 12 West toward the ocean in 1858 under separate contracts. In 1859 Smith surveyed in portions of Twps. 17 and 18 North, Ranges 7 through 10 West, closing on the Guide Meridian. Henry surveyed in Twp. 20 North, Range 11 West and the 5th Standard Parallel North west to the ocean. In 1861 Smith surveyed the original Quinault Reservation boundary without ties to the 5th Standard Parallel North. (ref. Jerry L. Olson and Others...The First Land Surveyors of Washington, LSAW Historical Committee.) Smith returned to the practice of law served as County Prosecutor (Luark) but moved to Lewiston in 1863 and served as an Associate Justice of the Idaho Territory Supreme Court (Circuit Judge). (Idaho Supreme Court Records, Boise, ID) His estate was probated in Kelso, Cowlitz County, Washington in 1871.

Meanwhile other surveys were contracted in 1858 to close on the 4th Standard Parallel North and Coast Guide Meridian by David F. Byles, north from Willapa Harbor in Twps. 14 through 16 North, Rges. 10 through 12 West and by Henry N. Stearns, east from Satsop in Twp. 17 North, Rges. 7 through 11 West. Both men were experienced surveyors beginning in 1855 and 1857. (Olson, Op. Cit.)

DAVID F. AND CHARLES N. BYLES

The family of David F. Byles (1832 – 1897) and Charles N. Byles (1844 – 1897) emigrated in 1852 (Isabel Rosmond Collins – A Brief History of the Byles Family and Medcalf Family, 1983) to the Upper Chehalis Valley. David began surveying in the Rochester - Oakville area in 1855 (GLO Contract No. 9). An entry in the Luark Diary "1855 – Oct – Sat – 20 - ...D. F. Byles and Mr. Smith ate dinner with us on their way in from surveying." The Luark diary is replete with contacts with the family of Rev. Charles Byles (1809 – 1869) and his brother, James Biles.

When Luark moved to Pt. Chehalis, David Byles survey in Twps. 16 North, Ranges 10, 11 and 12 is mentioned... "1858 – August – Thurs – 19 - Spent half the day with D. F. Byles surveying...ran through to the ocean on line 20 rods north of where I intend to build" and "I and J. M. (son) went to Ocean Beach and moved 2 posts on township line making new witnesses at one - - - - - - - of D. F. Byles Dept. Surveyor."

Luark made reference to the early attempts of the Coast Survey to map the fickle entrance to Grays Harbor in 1862 (NOAA Microfilm) and later saw steamships replace his sail and row boat. He served as Probate Judge, Postmaster, U. S. Commissioner, and was appointed as physician to the Quinault Indian Agency where he was affectionately known as "Le Pome" (apple) Doctor." He eventually homesteaded a site more suited for farming near Montesano where he and his brother Michael's descendents reside to this day. A nephew, Elijah Luark Wade was a GLO Deputy Surveyor in the 1890's.

Luark describes Henry N. Stearns (Donation Land Claim Cert. No. 87, S 36, T 14 N, R 3, Claquato)... "1858 – January – Thurs – 7 ... School taught by Mr. Stearns on Mound Prairie;" "1858 – May – Fri – 21 ... Started home about 11 a.m....had a nice sail up bay against tide in river canoe...Had Mr. Stearns and 2 of his hands (surveying party) with us to - - - - - - - - - -. Landed at Maces after dark" and on Oct 18 describes a visit by Thos. James and Stearns followed by a trip to Cape Shoalwater Lighthouse. Stearns later surveyed extensively in the Chehalis area as GLO Deputy until 1875. He was elected as Lewis County Surveyor but was primarily an educator and served a superintendent of schools for several terms.

Charles Newton Byles worked for and with David F. Byles as Deputy GLO Surveyor from 1869 to 1875. He was educated at the Territorial University, Seattle and Business School, Portland; taught school, and owned a farm, part of which he platted

into Montesano. David had earlier platted the first townsite at Cosmopolis but later moved to Elma where he platted the first townsite plat and farmed along the road to Centralia. The brothers were involved in surveying half a million acres of public domain, Indian reservation, donation land claims and townsites in 30 townships. In 1871 they extended the 5th Standard parallel north to the east some 20 miles from Smith's survey. Both men were active in their communities all of their lives. The imposing Nineteenth Century C. N. Byles mansion stands across from the Grays Harbor County Courthouse.

C. N. Byles' grandson, Frederick Rosmond Byles, was educated as a logging engineer, later was a partner with his brothers, Robert and John in a cedar lumbermill and is retired in Forks. On a visit there, I inquired about the Byles family surveys. He produced the compass which C. N. Byles had used and related me how it had been passed via C. N. Byles' children. A daughter, Martha married his father, Fred Rosmond in 1906. The Rosmond family homesteading on Grays Harbor was serialized. (Ocean Observer, Ocean Shores Washington, 1990- 91) Fred worked for the Land Commissioner in Olympia for four years and was Grays Harbor Auditor from 1906 to 1910. A photograph of the compass, manufactured by Richard Patten and Sons, Baltimore, and letter to Silvio Bedini, Smithsonian Institute Historian, resulted in an article in Professional Surveyor, 1987.

DEVELOPMENT

After the Civil War, interest in access via steamship resulted in hydrographic surveys and the development of sawmills in the 1880 – 1890 period brought railroad surveys to the area. The south shore of Grays Harbor was developed with speculatively platted townsites. (Southharbor in Section 31, T 17 N, R 10 W and Ocosta in Section 9, 10, 11 and 16 of T 16 N, R 11 W in 1890) The 1892 United Railroads of Washington survey did not find the meander corner but tied to the 1/4 S 36 and 31 on the Guide Meridian. (State Land Comm., Olympia.)

1931 oyster tideland surveys by H. S. Shorey, PLS 669 and subsequent survey in the plat of Southarbor by W. L. Lovejoy, C. E. in 1932 fail to show the meander corner and surveys by the Northern Pacific only tie to the original standard corner on the 4th Standard Parallel North. The County Engineer (Job 137, 1931) and State Route 105 survey maps (Land Comm.) tie to this point. Timber cruisers have perpetuated the point physically. (W. H. Peebles, Cruise Book 42, page 32; PLSO Olympia) A 1952 survey by Robert D. Powell, PLS 4303 in Section 31 originated on the 4th Standard Parallel

North and re covered evidence at the 1/4 S 31 and 36 which was corroborated by Washington Dept. of Game surveys by Don Pitzler, PLS 5526, W. E. Berglund, PLS 5658, and Dan G. Williams, PLS 7202. Timber sales by the Department of Natural Resources have relied on the corners.

LSAW HISTORICAL SOCIETY RETRACEMENT ON 140TH ANNIVERSARY

As early as 1900 USC & GS hydrographic and topographic surveys of Grays Harbor have used a point at Stearns Bluff to tie between the ocean and the port development in Hoquiam and Aberdeen. Recent Corps of Engineer GPS control through the Harbor has upgraded the accuracy of the survey network, primarily using new stations with satellite visibility. When the State Department of Transportation Geographic Services Survey Branch was contacted about a retracement of the Coast Guide Meridian which would cover the area between Highways 105 (Aberdeen – Westport) and Highway 109 (Hoquiam – Ocean Shores), they expressed interest in working with the LSAW Historical Society. Five new GPS stations plus USC & GS RM "Stearns 2 No 1 1939" were used to control the south end of the meridian using static one hour minimum observation periods for vectors over 10 kilometers and fast static technique with 10 - 15 minute observations for vectors under 10 kilometers.

The north side of Grays Harbor was homesteaded after 1860 and developed by the timber industry after 1880. The speculative townsite of Grays Harbor City contained land on both sides of the Guide Meridian and the 1900 Northern Pacific Railroad Survey did not find a meander corner but tied to the 1/4 S of Sections 1 and 6. The corner has been accepted by the County Engineer since 1890. (A. M. Holman/J. O. Stearns County Roads) 1998 reconnaissance with ties to recent Rayonier Inc. and highway surveys indicated that two existing GPS stations and four new stations were needed to control the resurvey on the north side of Grays Harbor. In addition, USC & GS "Brack 1909" was brushed out for observations.

To obtain a stronger triangulation system net, simultaneous observations were made at Westport, Central Park, Bowerman Airfield and Grass Creek on May 18, 1998 using eight Trimble GPS stations operated by Geographic Services employees under the supervision of Kurt Iverson with the help of fifteen LSAW members and local surveyors and family or friends. (See Appendix) Ties to the GLO corners were made with a Topcon 303 and the station at Grass Creek re-occupied when the GPS observation at "Brack 1909" was performed. After processing the data the ties were

calculated and a single proportion applied to calculate the position of the lost M Cs (diagram).

In June, Denny De Meyer and Dick Sterling met in Forks at the home of Frederick Rosmond and video-interviewed him with the compass set up on the tripod. It is in excellent condition and the needle very responsive. Following the interview the instrument and tripod were forwarded to the Chehalis Valley Historical Museum in Montesano where it is prominently displayed.

PERPETUATION AND REMONUMENTATION

In early August the initial point, a 1 1/4 inch iron pipe, was perpetuated with a 36" diameter by 24" deep reinforced concrete monument with a Berntsen 3" brass and appropriate BLM marking; together with a bronze plaque dedicating the monument by the cooperators and a second bronze plaque with the state plane coordinates and latitude and longitude. At present no vertical data is available. Two Reference Monuments and one reference tree witness the monument.

On August 8, 1998, a ceremony was held at the monument with representatives of the Byles family, Frank Byles of Olympia, his son, Brian and grandson of Seattle, including their spouses and friends, in addition to LSAW members and local surveyors and foresters. Dan G. Williams of Hoquiam identified what we had called a blaze on an old snag as his cruiser mark, a "crowfoot" chopped there in 1952. Photos and reminiscences familiar to surveyor gatherings followed. The Byles compass had been borrowed from the museum for the occasion with maps and photos of the project displayed.

The proportioned position of the southerly MC fell on the tidelands below the railroad embankment in rubble from rip-rap rock and a witness corner was established on the south edge of the abandoned railroad ballast with two spruce reference trees. The attempt to establish the northerly MC developed problems because the calculated point fell at the foot of the railroad cut in the bluff. This position was not established and a new position calculated using the 0.63 chain conflict in the notes which was cited previously. The 1900 field note survey of the Northern Pacific Railway tied to the 1/4 S 1 & 6 and that distance is a reasonable fit. The 1890 plat distances of Grubac's addn. to Grays Harbor Vol. 1, pg. 211 of plats do not agree with the distance between 1/4 S 1 & 6 and Section Corners 1, 12, 6 and 7 found by Rayonier Inc. No bearing tree evidence was found at the section corner in a 1950 survey by George Lonngren and Dan

Williams. It has been accepted but no evidence can be found at this time to support its position.

An error might have occurred in “measurement” up the bluff involving over 130 feet vertical in 115.5 feet (GLO 1.75 chains) horizontal; about 40 degrees or 90% slope). The use of a 33-foot Gunters chain is difficult to nearly impossible and it is challenging to measure such slopes with a modern 200-foot steel tape by letting the chain down from the top. The record distance 1.75 chains may have been measured trigonometrically with a solar compass telescopic attachment using a rod as described in Burt’s manual. At the point calculated from the 1998 re-survey, it is close to a vertical cut and the point would not be visible from the meanders on the tide flat.

Analysis of the data is not conclusive enough to establish the north MC at record distance. A point on line at the proportionate distance indicated by the “bust” in the field note record is to be established as a witness corner to the MC. This will be in a place that is more practical in relation to the meander lines which were run on the gravelly, sandy beaches of lower Grays Harbor at low tide. A search for the monument established by R. W. Glaeser PLS 15024; Record of Survey Vol. 15, pp. 82 – 85 was unsuccessful due to the abrasion from storms and driftwood logs.

No attempt has been made to monument the north end of the Coast Guide Meridian. A survey by Grays Harbor County, ROS v. 65, p. 17 recovered ITT Rayonier corners and perpetuated the position with a standard pipe and aluminum cap. Since 1858 the GLO surveys of Smith, Henry, D. F. Byles and Stearns have been found by GLO contractors; D. F. and C. N. Byles in 1871 and Gilbert Ward in 1883. County and State road surveys have tied to corners along the line. An additional plat in 1927 in Sec. 18, T 18 N, R 10 W (Bayview tracts) adjoins the meridian. Numerous surveys for timber cruising and harvesting over the years prior to the 1972 Survey Recording Act have information of value. Walter H. Peebles timber cruise books 1921 – 1934 call for 24 GLO corners with bearing trees. ITT Rayonier and other company files have been furnished to provide records along this line. Since 1972, nine records of survey of 8 1/2 miles plus 5 1/2 miles of the 1998 survey contain information. A feature that occurs through the entire record is the excess in distance in comparison with the record GLO distance. The excess occurs at the beginning of the line and is a feature with rare exception in most of Smith’s and Henry’s surveys in 1858 and 1859.

The Land Surveyors of Washington Historical Society wishes to express its gratitude to all of the persons who generously gave their time and support for this adventure.

LEGAL ISSUES PERTAINING TO GIS DATA POSITION ACCURACY

Jerry R. Broadus

INTRODUCTION AND CONCERNS

The law is a conservative force that adapts only grudgingly to new technologies. The rapid spread of geographic information technology is anything but conservative. At present very few legal constraints to this spread exist. The best the law can do is attempt to pigeon-hole new concepts into existing precedent, which is always an inefficient process and can result in painful and inappropriate responses to problems. In this article I will address some of the concerns about data position accuracy, the lack of standards in this area, how the law will likely address injuries when they occur, and provide some suggestions for future action.

The July 1996 issue of *Cartography and Geographic Information Systems* includes an excellent research paper entitled "Research Priorities for Geographic Information Science" attributed to The University Consortium for Geographic Information Science (UCGIS). One of the topics of that paper is the need to "lessen the likelihood that geographic information technologies will be misused, or their products misinterpreted, or inappropriate decisions be made based on their products." UCGIS (1996) P. 118. The first concern addressing that and other issues is the need to develop better tools for data integration based on good, quality research. Quoting from the authors:

"Adding to the complexity of the integration task is the existence of two very different types of accuracy. A map or image can capture the relative positions of features with great accuracy; but their absolute positions depend on how successfully the map or image is registered to an earth frame, most notably the system of latitude and longitude. For example, we can know the distance from one mountain peak to another precisely, but have very poor information on their latitude and longitude positions. This difference becomes crucial when two data sets have to be combined-- unless both have high levels of absolute positional accuracy, there will be significant errors of misregistration. Such errors often occur when databases are updated with information that is apparently more accurate." UCGIS P. 118. This concern over different levels of accuracy, and how to handle them in integrated data sets, seems to be one of the most commonly perceived problems in the move to computerize data under a geographic framework. You will see it brought up often in the discussions that follow.

Traditional mapping standards are dependent upon scale, which raises another conceptual difficulty when working with computerized mapping. One is always tempted to display maps on a computer screen at whatever "zoom factor" seems convenient, yet the intended publication scale necessarily drives the budgeting of time spent in accurate gathering of geographic details. "Changing the scale of data without first understanding the effects of such action can result in the representation of processes or pattern that are different from those intended. The spatial scaling problem presents one of the major impediments, both conceptually and methodologically, to advancing all of the sciences that use geographic information... UCGIS P. 122. The authors suggest that crucial scales should be identified at which data content, structures, and patterns begin to change.

In summarizing the accuracy concerns, the authors refer to problems inherent in GIS data

caused by uncertainty in data attributes, location, and spatial dependence. "Unfortunately, geographic data are often used, analyzed, and presented with the assumption that they are free from uncertainty. The beguiling attractiveness, the high aesthetic quality of cartographic products from GIS, and the analytical capability of GIS, further contribute to an at times undue credibility of these products. Undeserved and inappropriate acceptance of the accuracy of these data is often not warranted, however, for the reasons discussed above. Error-laden data, used without consideration of their intrinsic uncertainty, have a high probability of leading to inappropriate decisions." UCGIS P. 126.

As a practicing surveyor, I am often confronted with concerns expressed by others who practice in the GIS arena. The most common complaints I have heard involve: 1) the mixing of "inaccurate" with "accurate" data and the resulting uncertainty of the result: 2) the lack of or the misunderstanding of a meaningful metadata giving relative accuracies of each measurement in a GIS; 3) a fear that the public will be harmed by a misunderstanding and misuse of GIS parcel line positions; 4) a fear that a surveyor's decisions concerning property lines will be forced (by a regulatory requirement for some permit desired by the surveyor's client) to yield to conflicting decisions from a GIS mapper of parcel lines; and 5) a concern that a misunderstanding of the nuances (in the public land states) of the federal surveying system on the part of mappers untrained in surveying will impact property rights. Many of these concerns are refinements of the points made by the UCGIS authors.

As these comments and the following discussions illustrate, accuracy in the locational attributes of GIS data is not a single, easily handled concept. One aspect is the difference between relative and absolute positional accuracy of points described above. Another is the relation of topographic data to a meaningful local framework, most often parcel lines, which involves the accuracy of the parcel lines as well as the topography. There is, of course the need to accurately identify a mapped object (a precise location of a power pole means little if the object is entered into the database as a telephone pole) and, often more problematical, there is the issue of locating the edges of objects that are difficult to precisely identify in a spatial sense (such as the edge of a wetland). Errors in the handling of any of these concepts can lead to a misunderstanding and misuse of the GIS data, and eventually to harm.

If someone suffers damage from such harm, the law will attempt to assign responsibility. This is a messy process, full of concerns for doing justice and advancing perceived social policies, rather than a scientific weighing of the "facts" within the framework of the "law" to reach a conclusion. Two systems exist to attempt to prevent future harm in these sorts of areas. The tort system makes a responsible party pay a monetary award to the victim, thereby encouraging care in the future. The licensing or regulatory system attempts to insure that people practicing in the field possess certain minimal levels of competency, and work under a code of ethics and responsibility enforced by the possibility of losing one's license, thus generally weeding out the careless before they cause harm. Both systems have the potential of profoundly affecting future activities in GIS data collection, although as yet there are no clear trends. How the law will affect GIS mapping in the future is largely conjecture, and that alone is problematical. This is clearly an area where surveyors and other mappers engaged in GIS data collection need to address standards and legislated solutions today, proposing solutions to future problems before the courts begin wielding the hammer of liability.

STANDARDS

Both the tort and the regulatory systems of preventing harm from the dissemination of erroneous data on standards. The tort system accepts as its base line the "ordinary, prudent person performing the particular kind of work in a prudent manner", which is far easier to measure against some written standard. The courts have tended to adopt written standards when they find them, even if not perfectly suited to the business at hand (see *e.g. Bell v. Jones*, 523 A. 2d 982 [DC App. 1986] wherein the court held that the American Congress on Surveying and Mapping Standard Detail Requirements for American Land Title Surveys created a nationwide standard of care for surveyors, even when working on mortgage inspection surveys). Licensing and regulatory laws also lean toward the use of written standards because they are perceived to give clear guidelines against which a person's right to keep his or her license can be measured. Unfortunately, GIS mapping, especially with GPS receivers, has been advancing at a rate that has far outstripped the ability of most standards committee to keep up.

Traditional mapping standards (Office of Management and Budget United States National Map Accuracy Standards, American Society of Photogrammetry and Remote Sensing Accuracy Standards for Large Scale Maps, United States Army Corps of Engineers Topographical Mapping Standards) are scale based, which suffers in the variable scale environment of computer mapping. Data collection standards traditionally have been based on a ratio of one to a denominator computed by dividing the distance between survey points by the propagated standard deviation of distance between survey points obtained from a least squares adjustment, classified as first, second, or third order work. FGCS 1995 The USACE 1994 standards for GIS mapping, for instance, specified Second Order Class I or II field surveying coupled with a mapping tolerance that could vary from one to 100 feet depending on the scale of the map. USACE 1994.

Modern control survey standards (in the United States and Canada) are based on the ability of the survey to duplicate already established control values. Both local accuracy and network accuracy standards are accommodated. Local accuracy is the uncertainty at the 95% confidence level in the coordinates of a point relative to other directly connected, adjacent control points. Network accuracy represents the uncertainty at the 95% confidence level in the coordinates of a point relative to the geodetic datum. FGCS 1995 and CSAS 1997. Thus each point can have two accuracy statements, meeting the concerns of "relative" and "absolute" accuracy described by the UCGIS. The US system classifies various accuracy statements into "ranges". Range VII (0.050-0.100 meters) is recommended for first- level densification of control networks for accurate GIS mapping, while Range XI (1.000-2.000 meters) is recommended for historical and social resource GIS mapping. Some state agencies have adopted very detailed standards for GPS control surveys, in some instances specifying practically every detail of the field survey (acceptable GDOP, period of observation sessions required, number of satellites tracked, etc.) WDOT 1994.

After the control survey, however, standards for the mapping of individual GIS data points are generally left up to the agency controlling the project. A review of just one recent edition of *Surveying and Land Information Systems* reveals articles describing several different approaches to GIS data collection methods: One describes the employment of one single frequency GPS receiver related to known points in a local datum, using a permanent continuously operating base station. Han 1996. This method requires accurately measured local coordinates on the two known points, but does

not depend on accurate network coordinates. Another, describing mapping in the Netherlands, addresses precision of acquisition of coordinates of discreet points as well as what it calls "identification precision", "The corner of a building, for example, can be pointed out much more precisely than the middle of a ditch". Salzmann 1996. In this scheme, the mappers produced a large scale map using GPS and photogrammetry, and added certain (high liability) underground utilities by relating locations to building corners. For their purposes, the mappers opted for high local accuracy over network accuracy. An Israeli project is described, using GPS controlled aerial base maps, wherein ground detail surveys were made by taping or by locating total station control points by resection from building corners. Doytsher 1996. One article addresses parcel boundaries, describing a computerized fitting of separate surveyors' subdivision maps into a larger control scheme by adjusting distortions in pieces at individual block levels. Christensen 1996.

The point, of course, is that the seeming explosion in GIS mapping is a recent enough phenomenum that mappers are still experimenting and developing there own standards as they work through their projects. A small city may not feel it needs to meet high network accuracy standards, hoping that any registration problems will be satisfied through its own local efforts. An agency may or may not use its maps for certain high liability information (such as locations of buried utilities) and will adjust its methods to meet its own comfort levels. While some of the articles mentioned above described a preference for locating utilities from nearby building corners, based apparently on a perception of certainty and ease of use, some city mappers I have spoken with feel that any locations that depend on structures for reference points are subject to misinterpretation and problems of changing conditions. I have discussed underground utility locations with one experienced GPS mapper who uses GPS to coordinate potholes dug down to the utilities and then runs through the same points with a total station, not because of a mistrust of either system but because his comfort level demands checks using two independent methodologies.

What, then, is a reasonable, prudent level of mapping accuracy? You will not, at this point, find a court case to tell you the bottom line, and I submit that you don't want one. It is perfectly reasonable to develop a mapping system to meet the budgetary and needs constraints of your agency or project. Of course, if different levels of mapping accuracy are mixed together into one database even within one agency there is a potential that the overall usefulness of the data will be degraded, but from a liability standpoint the potential danger comes from communicating the data to outside users in ways that misrepresent the value of the information.

Much GIS mapping is driven by state, federal, or local government. Because the initial costs of an accurate GIS are high, there will probably always be a need to recover the taxpayer's investment. Our democratic society demands that most government owned data be available to the public (ostensibly to keep the regulators honest, but there is great practical value in public access to agency mapping data). The public will not be privy to the individual budget and needs constraints of the mapping project that collected the data. Thus there is always the possibility that an agency will sell something that the public will not understand, and will then use to make unfortunate decisions. Of course, a judicious use of metadata and disclaimers will potentially affect the outcome if a lawsuit arises, but tort law still demands some basic attempts on the part of the data provider to understand and accommodate the real needs of the data consumer. Licensing laws can potentially affect the relation between the provider and consumer as well.

TORT LAW PRINCIPLES

Before discussing torts involving data dissemination, a preliminary consideration is worth looking at-- that being certain levels of governmental immunity. Presumably, much of the GIS data given or sold to the public will come from a governmental agency. At one time, governments essentially did not have to worry about repercussions if the data was incorrect, because they were immune from liability. Across most of the United States, however, this common law immunity has disappeared. Also, in those areas where some form of immunity still exists, it typically will not protect the individual employee who provides the data nor will it protect a private contractor who might have provided the data to the agency in the first place.

In general, a municipal corporation may be sued for tort the same as a natural person or a private corporation. McQuillin, 1993 §53.03. Some states have passed statutes exempting cities or counties from liability for certain acts, including negligently misrepresenting information. Interestingly, in some of those states the municipality is liable if it carries insurance. McQuillin, 1993 §53.02.30. Statutory immunity is the exception rather than the rule.

Common law immunity still exists in a few states for "governmental" functions as opposed to "proprietary" functions. The distinction is usually stated in terms of acts carried out for the good of the public as opposed to municipal profit. McQuillin, 1993 §53.02.10. Another line of common law immunity exists in many states for public duty versus "special relationship". Essentially, a municipality is not liable for damages that occur because of the way it handles duties owed to the general public. The most common examples involve police (although the police have a duty to protect the public safety, you have no claim against the city if the police fail in your particular case and you are injured) and building permit officials (although the building officials have a duty to enforce the building code, you have no claim against the city if the building you are in collapses because it was in reality not up to code). "Special relationship" refers to the instances that do create liability in the city, because its employees have undertaken some task that directly affects an individual person. In the mapping arena, suppose that the city zoning map is incorrect, and you pick up a copy as a member of the general public. Relying on the map, you make a decision concerning a piece of property and suffer some economic damage because the map later proves to be wrong. The city is not generally liable. On the other hand, if you specifically ask a person at the zoning counter if you can build something on a particular parcel, and the counter person gives you the wrong answer because the map is incorrect, then the city is generally liable because of the "special relationship" created by your direct request in reference to a specific project. Any of the following factors can create a special relationship: direct contact between municipal agents and the person; an assumption by the municipality, through promises or actions, of a duty to act on the person's behalf; knowledge by the municipal agent that inaction could lead to harm; the person's justifiable reliance on the municipal agent; occurrence of the injury while under the direct control of the municipal agent; or municipal action that increases the risk of harm. McQuillin, 1993 §53.04.25.

Thus there are many instances where the municipality can give out inaccurate information and still not be liable. Still, however, liability will attach (unless there is a specific state statute granting immunity) when a city employee gives out inaccurate information in response to a direct request.

Further, there is always the possibility of liability for the employee or for the person, if a contracter, who gathered the information in the first place.

What is your duty when you disseminate information to another-- either to the public or to a client? The common law assigns different levels of responsibility depending on the risk of physical harm:

Negligent Misrepresentation Involving Risk of Physical Harm

(1) One who negligently gives false information to another is subject to liability for physical harm caused by action taken by the other in reasonable reliance upon such information, where such harm results

(a) to the other, or

(b) to such third persons as the actor should expect to be put in peril by the action taken.

(2) Such negligence may consist of failure to exercise reasonable care

(a) in ascertaining the accuracy of the information, or

(b) in the manner in which it is communicated. (Restatement of Torts, 2d. §311).

This principle puts a rather high level of liability for giving out inaccurate information if there is a likelihood of physical harm. It makes no difference if the information is gratuitous or not. You are expected to inquire into the accuracy of the data: "Where the actor furnishes information upon which he knows or should realize that the security of others depends, he is required to exercise the care of a reasonable man under the circumstances to ascertain the facts, and the judgment of a reasonable man in determining whether, in the light of the discovered facts, the information is accurate. His negligence may consist of failure to make proper inspection or inquiry, or of failure after proper inquiry to recognize that the information given is not accurate." Furthermore, the actor must communicate the information in a way meaningful to the recipient: "It is, therefore, not enough that the actor has correctly ascertained the facts on which his information is to be based and has exercised reasonable competence in judging the effect of such facts. He must also exercise reasonable care to bring to the understanding of the recipient of the information the knowledge which he has so acquired." (Restatement of Torts, 2d. §311 comments d. and e.).

This liability can affect any level of the chain of creating and distributing maps that others could be expected to rely upon, where there is any risk of physical harm. In short, if your company is mapping some utility (as part of a contract with a city) which, if mislocated could create a physically dangerous situation, you are charged with locating it accurately. This charge is not relieved merely because you understand that the city will not be using the map to put itself in harms way-- that is even though you understand you are mapping the utility merely for some inventory rather than construction purpose, you are still charged with gathering accurate information. The charge comes from the danger that mislocation could create, not from the understood use intended by the client. The same can be said for the city employee putting the information into the database, who is (perhaps through his or her supervisor) expected to make a reasonable, and independent, effort to verify the accuracy of the data. The same goes for the person handing out the map. If all this sounds like strict liability, it practically is. The law is especially harsh in this arena because of the risk of physical injury.

How accurate must the location be? The law simply says "reasonably accurate", but of course that involves understanding what is reasonable considering the level of danger involved. While it might seem reasonable on the ground to derive the location of a gas line imprecisely because that is all the mapping budget allows, a jury would most likely find that a reasonable mapper would have exercised more care because of the danger. Is there any balance in this situation? Yes, in that the mapper is only liable if the user of the map took some action "in reasonable reliance" on the map.

In far the majority of cases where danger of physical harm is involved, of course, state laws have been written to provide alternate safeguards, such as requirements that underground pipes be located by utilities prior to starting any excavation in the area. In those instances, one can argue that reliance on a GIS product is unreasonable. The same can be said for reliance on any map that has disclaimers clearly printed on its face. It is not at all clear, however, that information in a metadata layer would provide any protection, at least for the last entity in the chain that provides the map to the consumer. The law requires that each actor in the chain exercise reasonable care in communicating the information. Thus the field mapper might understand one level of accuracy to be desired, and might communicate his or her adherence to that level and thus be protected because the client is, say, an agency that wrote the specifications in the first place. But then if the agency puts the only information concerning the accuracy of the product in a metadata layer, and then gives a copy of the product to a member of the public who would not be expected to understand metadata without some explanation, then the agency's providing of the metadata without explaining them at the consumer's level of understanding would clearly be inadequate. That is, because of the danger of the situation the provider of information is charged with tailoring his or her communication of the information so that the consumer can understand it. If any of this comes to a court case, all of the balancing of reasonable accuracy, reasonable communication, and reasonable reliance would be handled by a jury that might well be swayed more by the severity of the damage than the actual reasonableness of any action in the chain.

An example, brought to my attention recently, involved a city that was developing GIS maps and had spent a great deal of its budget mapping parcel and street right of way lines, but had merely added the fire hydrant (and certain other utility) locations from old office cards without field checking them. The GIS manager became concerned that his map might show hydrants in blocks where they no longer existed. Would the city be liable if the fire department responded to a fire call and lost too much time locating a suitable hydrant? Traditionally, this sort of problem would be handled by the state's view of governmental function (or public duty) immunity. The city can't put out all its fires and the state's courts might just say it has no duty to do so. The potential liability, of course, comes from the argument that the city negligently updated its records; that in creating a new record of the city utility locations it should have, above all, updated its inaccurate hydrant records because of the potential for danger. Liability would not flow to the fire department but it might to the GIS department. Whether this would in any instance create liability would most likely depend on a whole litany of factors concerning how the city dealt with its GIS-- was the fire department led to believe it had an accurate map, was the map used to approve building permits-- that sort of thing. The fear, and its a legitimate one, is what would happen after a team of lawyers picked apart the GIS process and laid it out before a jury.

By far the most likely form of injury from reliance on an innacurate map is economic rather than physical. The law treats these injuries differently:

Information Negligently Supplied for the Guidance of Others

(1) One who, in the course of his business, profession or employment, or in any other transaction in which he has a pecuniary interest, supplies false information for the guidance of others in their business transactions, is subject to liability for pecuniary loss caused to them by their justifiable reliance upon the information, if he fails to exercise reasonable care or competence in obtaining or communicating the information.

(2) Except as stated in Subsection (3), the liability stated in Subsection (1) is limited to loss suffered

(a) by the person or one of a limited group of persons for whose benefit and guidance he intends to supply the information or knows that the recipient intends to supply it; and

(b) through reliance upon it in a transaction that he intends the information to influence or knows that the recipient so intends or in a substantially similar transaction.

(3) The liability of one who is under a public duty to give the information extends to loss suffered by any of the class of persons for whose benefit the duty is created, in any of the transactions in which it is intended to protect them. (Restatement of Torts, 2d. §522).

Liability for pecuniary losses is a bit more limited than liability for physical injury. For the private contractor, who is supplying data to a municipality for a GIS, the law understands that the contractor can only be expected to use the level of care commensurate with his or her understanding of the magnitude of loss likely from the use he knows the recipient is going make of the data: "[T]he duty of care to be observed in supplying information for use in commercial transactions implies an undertaking to observe a relative standard, which may be defined only in terms of the use to which the information will by put, weighed against the magnitude and probability of loss that might attend that use if the information proves to be incorrect. A user of commercial information cannot reasonably expect its maker to have undertaken to satisfy this obligation unless the terms of the obligation were known to him." (Restatement of Torts, 2d. §522 comment b.). In short, if you are supplying information to a municipality you are expected to exercise the level of care necessary to make the data accurate enough for whatever task the municipality actually tells you it is going to undertake with the information, and no more. Generally, if the municipality gives you specifications for the contract, you can assume that you only need to meet those specifications. If, on the other hand, you are asked to help the municipality plan the project, you would be wise to ask the municipality to decide what it wants the information for and to tell you-- then you would be expected to design the data collection so as to meet a level of accuracy sufficient for those needs.

The municipality has a broader duty if it gives the information to the public. This is not to say that any information which the public can get because of general freedom of information requirements must be as accurate as any member of the public might want. Rather, if the municipality uses the GIS

for some specific purpose-- say to issue building permits-- then the city is responsible for seeing to it that the data is accurate enough for that purpose, and this duty runs to anyone who asks for such a permit. For example:

A city had a GIS that showed wetlands related to parcel lines. A property owner requested a subdivision permit on a lot and was told that he would have to perform a wetlands analysis because the GIS map disclosed a wetland on a portion of his property. The owner disputed the map's accuracy, but the city stood firm. The owner specifically went to the affected portion of the property and told the city that the wetland was obviously several hundred feet away. The city still required him to do the analysis. The wetlands biologist hired by the owner agreed with the owner, and the city proceeded to process the application. The owner then filed a claim against the city for the cost of the wetlands analysis, and prevailed. This example actually happened, but there is no court citation for it because the city, on the advice of its attorney, settled the claim out of court. This is a classic example of liability for these sorts of maps, for the following reasons:

First of all, there was a special relationship with the owner, created by his request for a permit and his repeated statements that the GIS map was inaccurate. There was no state law exempting the city from liability in these instances. The city employee had all the power, and could have gone out on the ground, in answer to the owner's concern, and determined for himself that the map was wrong. Instead, the employee placed undue reliance on the map, indicating a lack of understanding of the level of accuracy that went in to the map in the first place. Finally, no matter how accurately the city mapping department prepared its base or utility maps, it very likely would have let this sort of inaccuracy get by. The wetland was entirely on private property, its edge requires special expertise to identify, the mapping is difficult and expensive, and wetlands move, disappear, and reappear anyway. The mapping department made a decision as to how to map the wetlands (based on budget and practicality) and may not have even known what its level of accuracy as to this feature really was at any particular point in time. Thus the lesson the city learned in this case was that its employees must understand the weaknesses of all layers of its mapping and have some alternative procedures in place for dealing with those layers where a legitimate challenge is likely.

The pecuniary damage arena is one where metadata becomes more important (than the physical damage arena, where metadata is little help) and its use gets more complex. Metadata must be treated like fine print in a contract. Remember that the law expects the information supplier to meet some standard, but its a standard relative to the use of the information weighed against the magnitude of likely harm. Some standards have been proposed, for example by the National Society for Professional Surveyors (a section of the American Congress on Surveying and Mapping) and by some state surveying societies, that essentially address the issue by not defining good practice but by allowing the standards to be met by merely stating the level of accuracy in the metadata. How useful is that? If you are giving the information to a sophisticated developer who is making advance plans relative to a site, then merely giving him the metadata may be good enough-- the developer should read the fine print. If, however, as in the example above a property owner is requesting a permit and is actually refused, then obviously the metadata (if its existed) should have been used by the city employee. Metadata is fine but it is not a shield.

LICENSING ISSUES

Licensing the data collector is a different legal approach. The theory is that since tort liability of municipalities is traditionally very low (because of immunities or the complex nature of lawsuits) perhaps the public can be better protected by limiting the number of people who collect data for GIS databases to those who are trained and who face the possibility of losing their licenses if they perform below par. A few states have specifically addressed base mapping in their surveyor licensing statutes:

Here is an example from New Jersey:

The term "practice of land surveying" within the meaning and intent of this chapter shall mean any service or work the adequate performance of which involves the application of special knowledge of the principles of mathematics, the related physical and applied sciences, and the relevant requirements of law to the act of measuring and locating distances, directions, elevations, natural and man-made topographical features in the air, on the surface of the earth, and on the beds of bodies of water for the purpose of determining areas and volumes, and for the establishing of horizontal and vertical control as it relates to construction stake-out, for the monumentation of property boundaries and for the platting and layout of lands and subdivisions thereof and for the preparation and perpetuation of maps, record plats, field notes, records and property descriptions in manual and computer coded form that represent these surveys. The practice of land surveying shall include the establishment and maintenance of the base mapping and related control for land information systems that are developed from the above referenced definition of land surveying. For the purposes of this subsection, "land information systems" means any computer coded spatial database designed for multi-purpose public use developed from or based on property boundaries. N.J. §45:8-28 (e).

The New Jersey approach is unusual, and gives evidence of a considered approach to balancing general GIS mapping with traditional property surveying. Parcel lines on GIS maps are probably not of any more concern to property owners than are the already common tax parcel boundaries on assessor's maps, unless they are used to directly affect the extent of someone's property rights. In that eventuality the New Jersey approach would kick in. A much more common law would be one similar to the following, from Washington:

(9) Practice of land surveying: The term "practice of land surveying" within the meaning and intent of this chapter, shall mean assuming responsible charge of the surveying of land for the establishment of corners, lines, boundaries, and monuments, the laying out and subdivision of land, the defining and locating of corners, lines, boundaries and monuments of land after they have been established, the survey of land areas for the purpose of determining the topography thereof, the making of topographical delineations and the preparing of maps and accurate records thereof, when the proper performance of such services requires technical knowledge and skill. RCW 18.43.020 (9).

Some other common state laws will be similar to the Washington law but might specifically mention "mapping" or preparation of maps. One can easily interpret such laws to cover the type of data collection normally done for GIS databases. The argument in favor of such an interpretation (that the public is best served if such mapping is done under the supervision of a person who is trained, has passed an exam, and faces administrative punishment for poor work) essentially presupposes that inaccurate GIS data will be misused to the public detriment (otherwise, a licensing requirement becomes merely a way to limit access to the profession). Perhaps, if more instances of such misuse

occur, especially with no liability because of the cloak of governmental immunity, then we will see more states taking active approaches similar to the one in New Jersey.

The most daunting practical problem of applying the licensing approach is the above mentioned lack of clear standards. A professional license is a property right that cannot be taken away without due process of law, and to most boards that means that before a license can be revoked there must be clear standards that any licensee always can turn to in order to determine a base level of practice. As the punishment is not based on actual or potential damage, imprecise terms such as "reasonably accurate" are simply not clear enough. Also, prosecution of alleged unlicensed professional practice usually involves pursuing someone for a misdemeanor-- and thus is not high on the list of most prosecuting attorneys.

One approach, which seems to be a sort of backwards justification for inaction, is to state that GIS mapping is not surveying unless it rises to the qualifiers listed in the various licensing statutes. For instance, mappers in Washington might not be making "maps and accurate records thereof, when the proper performance of such services requires technical knowledge and skill." In New Jersey, the triggering language might be "the application of special knowledge of the principles of mathematics, the related physical and applied sciences, and the relevant requirements of law to the act of measuring." In short, if you want an accurate or scientifically sound map, you must use a land surveyor, but not otherwise. Many land surveyors and state boards would probably say that as far as it goes, that statement is true; unfortunately it skirts the issue of whether GIS maps should be accurate for protection of the public. At present, in most states the threat of applying the licensing statutes to GIS mapping merely lurks in the background.

CERTAIN PROPERTY LAW ISSUES

An aspect of GIS mapping that continually raises discussion among land surveyors is the potential for misuse of GIS parcel lines or coordinates in the area of property line determinations. One thing that surveyors learn, and relearn over and over again, is that property boundaries are based on history rather than science. For example, in Washington we have many examples of property lines based on imaginary harbor lines "drawn" over the water. These often came about due to an intense effort by the state to map harbor lines using state plane coordinates in the 1960's and 1970's. Plats were filed showing one, two, or at most three monuments with NAD 27 state plane coordinates, and then the harbor lines were defined with state plane coordinates at the angle points (no bearings or distances, usually). Under classic property law, this scheme means that the monuments reference the harbor lines, rather than the coordinates. That is, it would be a mistake to run all the coordinates on the map through a transformation to NAD 83 coordinates to perform a modern survey. Rather, modern coordinates must be measured to the monuments, and then angles and distances must be computed from the coordinates on the map to derive modern coordinates on the harbor lines. Thus, if the monuments have been destroyed before the modern measurements are made, all the traditional evidentiary tools to replace the monuments in there original positions must be employed before resorting to coordinates to begin the survey.

Other examples abound. One can imagine a city wherein the populace monuments all their lot corners and builds walls around their individual kingdoms. Then the city might survey coordinates

at the property corners. Suppose, due to plate tectonics, the entire town moves out from under the coordinates. Traditionally the property lines would have moved with the plates, the loss being felt by the soul who lives at the edge. The coordinates, however, now mean nothing except in a relative sense. The point, of course, is that property line determination is dependent on much more than measurement. History, use, common sense, and law mean much more than lines on any map. Property law is based on common expectations and actions, rather than mathematical theory. Furthermore, property surveyors generally care very little about network accuracy, because all of the factors that affect property lines are local. The local nature of property measurements may be extreme; often it is improper to extend property line control surveys beyond a block (except to force a tie to a larger network) and sometimes the control is dependent on two or three lots.

All this means that any attempt adjust, fit, or "rubber sheet" GIS parcel lines into a larger network will create parcels that almost certainly will not match property lines. To the extent that such parcel lines are limited to general planning tools, this form of "inaccuracy" has little if any effect. The worry which is expressed by surveyors is that an administrator might require the surveyor to fit his or her locally accurate survey into a general, predetermined network that changes the local accuracy (for instance to qualify for a land use permit). Although potentially a legitimate fear, this concern can be met by education as to what a property survey really is.

The need to recognize that parcel lines are not property lines does not mean that parcels are inappropriate aid for modernizing land information. A graphical land information system is beneficial even if based on imprecisely mapped tax parcel lines. Efficiency of record keeping and retrieval, increased effectiveness of administration, and equitable dissemination of ownership information can all be facilitated by a computerized database. Tulloch, 1996.

CONCLUSION

Legal issues pertaining to GIS data collection are evolving much more slowly than the technology. This is probably good in that it indicates that there have been few injuries from data collection, at least that have made it to the courts. At present, GIS data have been treated mostly as available to the public on a freedom of information basis. Unfortunately, potential liability and legal issues might change that scenario. If more care in gathering or communicating the information to the public becomes a requirement to avoid tort or licensing liabilities, costs of data acquisition and maintenance will rise. Associated issues are the specter of public private competition, copyright, and ownership of the data. Why, reasonably, would a property owner hire a private company to perform a topography survey if he or she felt she could get the same service for less cost from a GIS map. Why, indeed, would a private mapper contract to provide maps to a city if he or she felt the city would then distribute the maps and data in competition? These issues are all expected to eventually affect the price of data availability from GIS. Gerhard, 1996.

Probably the biggest danger that can come from continued rapid development of GIS mapping in its present "style" is that some injury could occur, which then leads a court to impose some inappropriate standards. The industry would be better served to begin now discussing and preparing written standards of practice. Legislation needs to address the licensing issues. Most

survey licensing laws probably do apply to GIS mappers, perhaps they shouldn't. In any case, licensing laws clearly need to be updated in most states, and standards of practice need to be revised to address GIS mapping. The biggest "action" recommendation that can come out of a study such as this is that survey and mapping societies, boards of registration, and legislators need to begin formulating policy choices in this area before courts impose some solution that hinders the process.

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LIMITING THE USE OF DEFECTIVE PUBLIC LAND SURVEY MARKERS

Dexter M. Brinker, PE-PLS
P.O. Box 8142
Pueblo, Colorado 81008-8142
(719) 546-9838

Abstract

The United States Public Land Survey System was devised as an aid for transferring land from the public domain to private ownership. While the system, (including several revisions to the original format), has many excellent features, it also contains several weaknesses which have caused numerous boundary disputes over the years since its inception in 1785. One major difficulty is that a single improperly restored section corner or quarter corner may jeopardize all subdivisionsal boundaries in a two to four square mile area.

A new Colorado Law makes it possible, in many cases, to limit the extent of damage caused by a defective corner marker to the immediate vicinity where property rights have been perfected through use or adjudication, and then to use an original or properly restored corner (instead of the defective one) to control future aliquot surveys outside the limited problem area, thus putting a much higher percent of future boundaries in the "right place" and averting future conflicts.

This paper will explain the proposed use of the new Colorado Law and will also review many of the strong points and shortcomings of the Public Land Survey System.

Background

The design of the Public Land Survey System is attributed largely to Thomas Jefferson, a many of many skills and much insight. He probably had some help, but if so, these helpers are hard to identify. Regardless of the authorship, the system provided a fairly straightforward way of identifying parcels of land for transfer from federal to private ownership. The first application of the method was in eastern Ohio (a project called the "Seven Ranges") and grew quickly, especially after the United States purchased the Louisiana Territory in 1803.

Strong Points of the System

1. Transfer of title to land required "prior survey", that is, the locations of parcels were identified on the ground (at least in a general way) before deeds were written.

2. Monuments were set on the ground marking the essential control points established by these "prior surveys" and official plats of the surveys were made available to the public.

3. The monuments were to be considered as "absolute and controlling", even if their locations were at variance with plat dimensions, so long as they could reasonably be determined to be in their original, undisturbed positions.

4. Standard procedures were established for dividing the large surveyed parcels into smaller parts, probably with the goal of creating practical sized family farms. These procedures were published, but were revised several times as experience was gained; hence early surveys must be carefully researched to determine the correct method for conducting a retracement survey.

Major Weaknesses of the System

While the method as implemented is really quite simple, it has been severely abused, especially in areas where unskilled persons have been allowed to practice land surveying. Other problems have arisen because of defects in the system itself.

1. No provision was made to maintain the original monuments. It may have been assumed that the states, through an office such as County Surveyor or Registrar of Deeds, would attend to this predictable problem. However, in most states there was inadequate funding for this critical service, and in some states no funding at all. As a result, these essential monuments disappeared at an alarming rate.

2. There was no system of recording (in a public office) newer, and presumably more precise, dimensions between accepted monuments (either original or properly restored), and the source of the new information.

3. No dimensional or positional tolerance was specified for subsequent surveys. In the later, and most common, method of section subdivision, this lack of a survey standard or tolerance led to innumerable conflicts concerning the "proper" location of the center of section (center quarter corner) and lesser aliquot markers.

4. The rules for the subdivision of sections were unclear as to whether all subdivisionsal lines had to be surveyed on the ground. This problem was solved (at least for Colorado surveyors) by a law allowing mathematical solutions based on complete section boundary control.

5. Although procedures were published for restoring lost and obliterated section corners and quarter corners, no clear rule was established to limit the use of improperly restored or erroneously monumented corners.

Confusion Created by Defective Corners

Consider the hypothetical case of four fairly regular sections in a rural but rapidly developing area. In the original Public Land Survey of the region (vintage 1890), the corner common to these four sections was set considerable "out of line". This may have happened as a result of difficult terrain, dense vegetation, surveyor blunder, or other causes. Some, but not necessarily all, of the discrepancy was reflected in the official notes and plat. Regardless of the cause of this misalignment, the shape of the sections, as defined by the original approved survey,

cannot be changed.

However, in 1902 some well meaning road builders built north-south and east-west "section-line" roads to serve the area. They did not know about the original section corner stone, which was obscured by dirt and weeds, and therefore used adjacent quarter section evidence to control their alignment. Hence the road intersection, and the "apparent" section corner, ended up far from the stone.

In 1920 a local stake pounder (surveyor) was retained to lay out a site for a country store at one corner of the intersection. The land owner allowed thirty feet off of two sides of his land for county road rights-of-way. To this "surveyor", the road intersection looked good enough for a section corner, so he didn't bother looking for conflicting evidence that would only complicate his job. He probably didn't read the original government survey notes. In fact, he may not even have known that such notes existed, much less where to find them! However, he did "perfect" his survey by driving a Model T Ford axle into the center of the intersection to perpetuate "his" section corner.

In 1990 the area around the axle began to grow and a new-generation surveyor was hired to "break-down" (more properly to "subdivide") one of the sections. He read the original notes and found the stone together with several acceptable pieces of collateral evidence. The question is, "What monument should this surveyor have used as the section corner?" State law tends to support the existing roads (and axle) as boundary markers for the country store tract (because of use and acquiescence), but what about all the other aliquot boundaries in the four sections which are still tied to the stone and still to be surveyed? Obviously, if the axle is used, the subdivision of the four sections will be entirely different from the results if the stone is used.

Now, after ninety years of uncertainty in Colorado's boundary law, the surveyor can use both! Specifically, Colorado Revised Statutes 38-44-113 states that:

> The establishment of a boundary corner through acquiescence confirmed by a court of competent jurisdiction, or by written agreement pursuant to section 38-44-112, shall not alter the location or validity of any existing or properly restored public land survey monument in the vicinity. Such existing or properly restored public land survey monument may be used to control future land surveys in the region when such surveys are not related to the boundary corner established by acquiescence or agreement.

The word "may" is extremely important, since the procedure is permissive, not mandatory. Every case will be different, depending on how many boundaries in the affected area have been perfected through use, monumentation or adjudication.

Summary

While adjudication will still be needed in many cases, surveyors, lawyers and judges in Colorado now have a better guideline for resolving boundary conflicts which result from defective Public Land Survey monument restoration. It is essential to recognize that this law does not create new corners or boundaries. It simply allows two or more monuments in a localized area to be used for entirely different purposes in order to maximize the protection of all property rights in the affected region.

Line Segment Reliability

Larry E. Stanfel
Management Science & Statistics
University of Alabama
Tuscaloosa, AL 35487-0226
LSTANFEL@CBA.UA.EDU

Abstract

The paper presents a geometric model of a problem proposed earlier in this journal. It proceeds to derive exact integral expressions for the expected value of the area of the figure formed by the line segment joining two measured points with the segment joining two exact points. Finally, the question of exact solution is approached by means of a computer system for symbolic integration.

Review

In their paper (Stanfel and Stanfel 1993) the authors introduced the problem of measuring the disparity between a line segment considered to join two intended points, (x_1, y_1), (x_2, y_2) and a second segment joining $(x_1 + a_1, y_1 + b_1)$ and $(x_2 + a_2, y_2 + b_2)$. The setting of the problem was that the first pair of points had coordinates specified by some measurement process, whereas measurement error had caused the second pair of points to be those actually in use. A possible realization was a measured pair of corner points of a survey boundary vs. a true pair.

In order to standardize a measure of difference, it was assumed that the corresponding end points of the two segments lay on vertical lines through the respective points, as in Fig. 1 and Fig. 2.

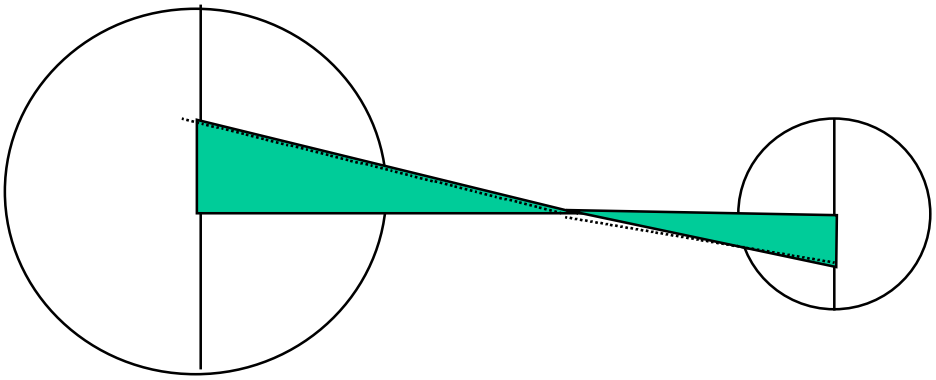


Figure 1. Restricted model with two triangles

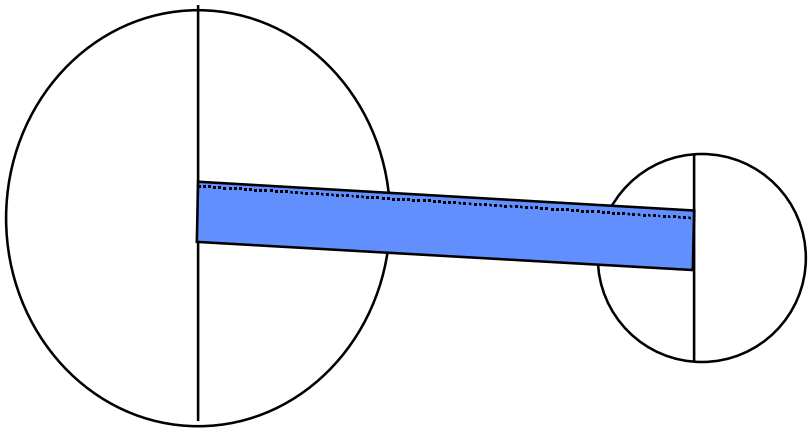


Figure 2. Restricted model with a quadrilateral

There, the circles represent the regions of error in which the respective true points actually lie. The shaded regions are the areas formed by the two segments, and it was suggested that the area is a reasonable measure of the difference between the two versions of boundary segment.

If one assumed some probability distributions of measurement error over the two circles, then, he could proceed to express the expected value of that difference area. (Stanfel and Stanfel 1993) assumed uniform distributions of errors inside the circles and found a simple expression for the average area of the error figure.

Notice that standardizing the end point configurations amounted to assuming $a_1 = 0$, or, equivalently that the second segment had been extrapolated or truncated on each end so as to have its end points lying on verticals through the circles' centers.

Of course, standardizing the length of the second segment changed the correct geometry, and in a second paper (Stanfel and Stanfel 1994), the authors attempted to generalize the model. In this version, the end points of the second segment were not modified, and the goal was again to compute the average area of the error figure after assuming a uniform distribution of the probability of the position an end point in a circle. As previously, an exact expression for the value was obtained.

However in (Easa 1995) an err or in the geometric model of (Stanfel and Stanfel 1994) was shown, so that the expression found was incorrect. Easa conjectured that the quadrangle integral would probably lack an exact solution, anyway, and proceeded to advocate calculating the area by simulation.

What eluded all the authors up to this point was that the geometric cases enumerated in (Stanfel and Stanfel 1994) were not exhaustive, that another pair of conditions, beyond the trapezoid and the two triangle cases presented, are possible. Furthermore, the new conditions challenge the definition of area as taken in all three papers.

The present contribution obtains all the geometric cases for error figures formed by the two segments and discusses the difficulty in applying the original area definitions caused by the new cases. After making a reasonable definition to resolve that disparity it proceeds to obtain correct expressions for the various terms of the total area, and provides evidence on the question of exact integrability of them.

Problem Description

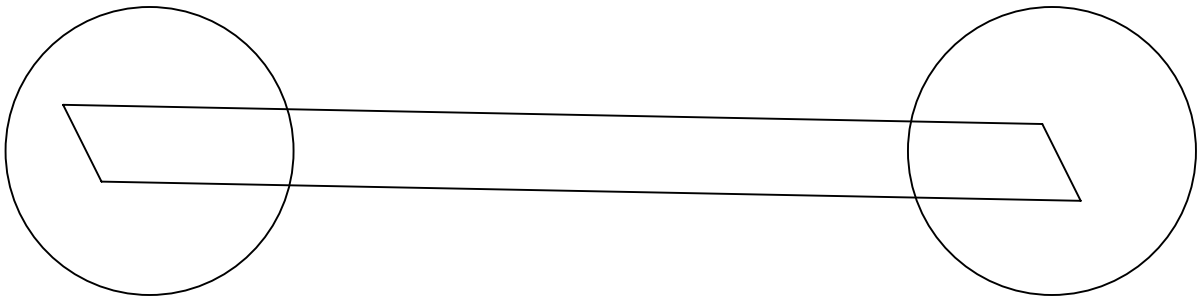
As before we describe two circles and the various points of interest.

There is a circle of radius r with center at (0, 0) = O_1 and inside it is a Random point with coordinates (a_1, b_1), $a_1^2 + b_1^2 \leq r^2$. A second circle of radius s has center (x_2, 0) = O_2. X_2 is a constant, and it is probably large relative to the a's and b's. Inside this circle is a random point, (x_2 + a_2, b_2), $a_2^2 + b_2^2 \leq s^2$.

The interpretation of interest is that the segment joining O_1 and O_2 is the supposed boundary segment; r and s are radii of error associated with the point location process; (a_1, b_1) and (x_2 + a_2, b_2) are the end points of the second segment. Thus, it was reasonable to comment that x2 is large relative to the a's and b's and also, relative to r and s.

A set of simplified sketches will illustrate all the possibilities for the figure formed by the two segments. The segment joining the circle's centers is drawn as a solid line, the segment connecting the random points, as a dashed line. The segments to close the figures in a reasonable way for calculating area are also solid.

If bot random points lie on the same side of the solid segment, the figure is a quadrilateral, as in Fig. 3.





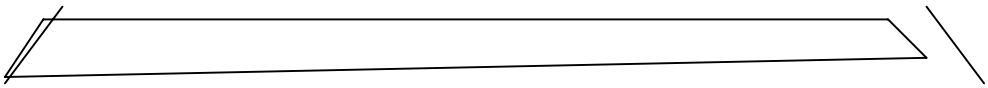


Figure 3. The quadrilateral case.

If the random points lie on opposite sides of the solid segment (i.e., the line of circle centers), the difference figure is two triangles, as in Fig. 4.

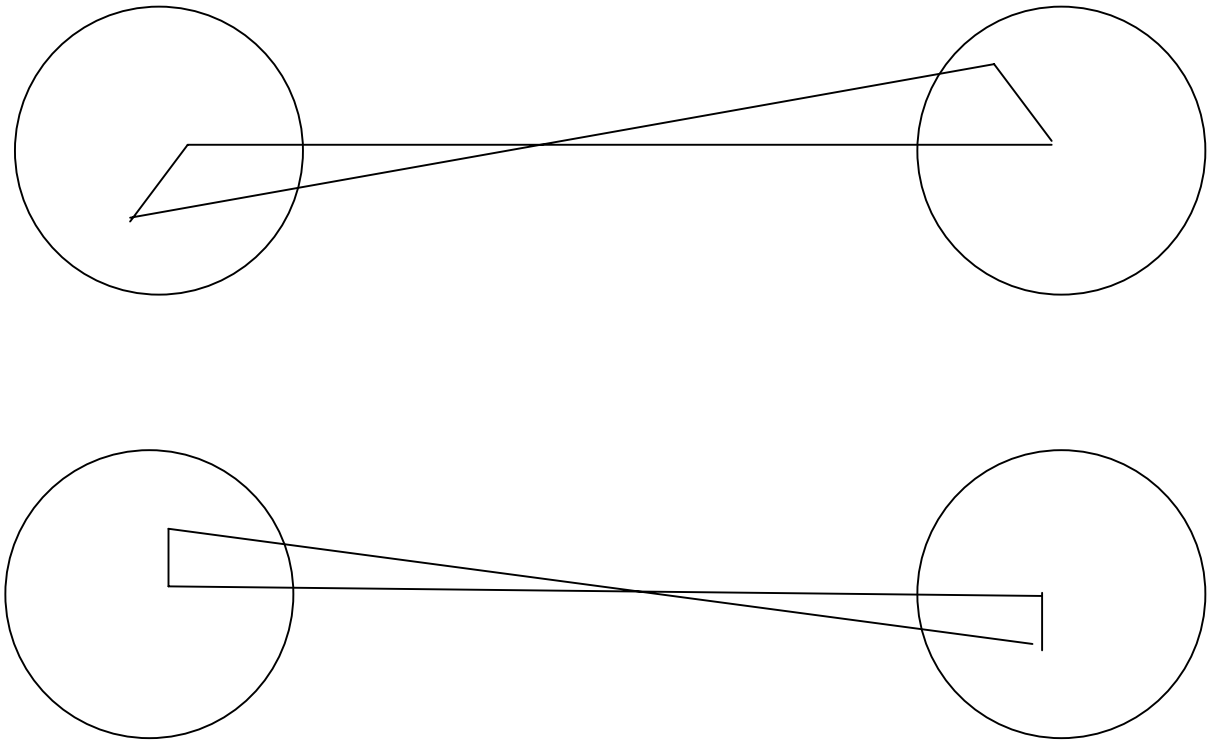


Figure 4. Cases of two triangles

But it is exactly here that previous analyses did not go far enough, because the segments' point of intersection need not be between the centers.

Labeling the centers O_1, O_2; the random points A, B; and the point of intersection of the segments, I we may realize the conditions in Fig. 5.

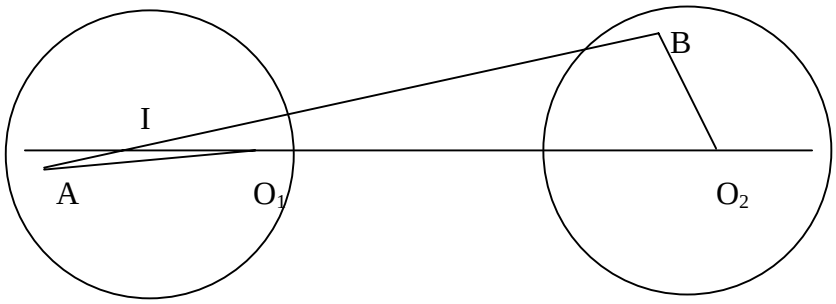


Figure 5. Another case of the opposite side of geometry.

Here is a case where the definition of area applicable in Fig. 4 is not necessarily as reasonable. As before, the random points are connected by solid segments to their respective centers, and AIO_1 is the correct figure to use in Fig. 4 under the same labeling convention. However, it is not obvious in Fig. 5 that the total area of interest is $AIO_1 + IBO_2$; the AIO_1 part is no longer the same relationship as it was. Further investigation as to what makes sense to practitioners regarding areas of difference or discrepancy is required here, but this investigation proceeds with $AIO_1 + IBO_2$ because it is dictated by the convention in Fig. 4, and one has some consistency that way.

Similarly, Fig. 6 shows the intersection, still not between the centers, to occur inside the right-hand circle.

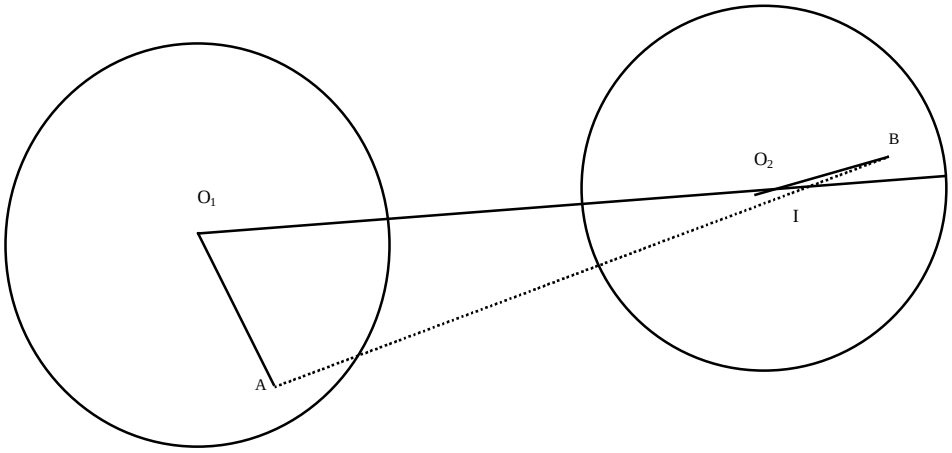


Figure 6. Opposite side anomaly in the right hand circle.

Here, we would use O_1AI and O_2BI for area.

Both Fig's. 5 and 6 correspond to $b_1 \le O,\ b_2 \ge O$. While we do not include drawings it is apparent that there are similarly $I < O_1$ and $I > O_2$ cases when $b_1 \ge O,\ b_2 \le O$.

The immediate task is to decide under what values of the random variables each of the conditions will occur and then to obtain expressions for the error figure in each contingency.

To begin, the line joining the two centers has the parametric representation

$\lambda(O,O)+(1-\lambda)(x_2,O)=$

$((1-\lambda)x_2,O)$ for real values of λ (1)

while that through

(a_1, b_1) and $(x_2 + a_2, b_2)$ may be expressed

$z\ (a_1, b_1) + (1\text{-}z)\ (x_2 + a_2, b_2)$ for real z (2)

From (1) and (2) one finds the point of intersection, when there is an intersection, to be

$$I=\left(\frac{b_1x_2+b_1a_2-b_2a_1}{b_1-b_2},O\right)=(x,O) \tag{3}$$

Since all the areas of interest will be viewed in terms of triangles, a formula for the area of a triangle is necessary. For a triangle with vertices (u_1, v_1), (u_2, v_2), (u_3, v_3) oriented as in Fig. 7.

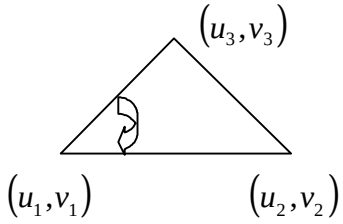


Figure 7. Point positions for validity of the area formula.

The area is (Gellert, et al 1975)

$$\tfrac{1}{2}\left[u_1v_2 - u_1v_3 + u_2v_3 - u_2v_1 + u_3v_1 - u_3v_2\right] \qquad (4)$$

Application of (4) to arrangements where the (u_1, v_1), (u_3, v_3) segment is not in a positive rotation from the other one will produce the negative of the correct area.

Since each of the area figures in values generally four variables a_1, b_1, a_2, b_2, which may vary over certain ranges, the expressions for area are four-dimensional integrals. It is necessary to obtain the expressions for each area and, importantly, establish the limits of integration. The cases, identified by the shapes and intersection type, are treated below.

Quadrilateral

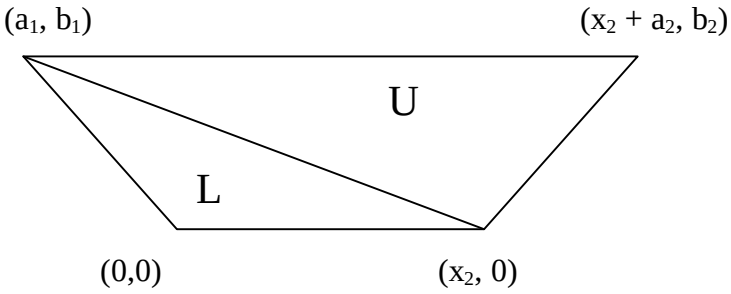


Figure 8. The area of a quadrilateral.

Using (4) with $(u_1, v_1) = (0,0)$, $(u_2, v_2) = (x_2, 0)$, $(u_3, v_3) = (a_1, b_1)$, the lower triangle, L, has area

$$\frac{1}{2}\left(b_1x_2\right) \qquad (5)$$

For U, take $(u_1, v_1) = (a_1, b_1)$, $(u_2, v_2) = (x_2, 0)$, $(u_3, v_3) = (x_2 + a_2, b_2)$ and determine U's area to be

$$\tfrac{1}{2}\left(-a_1b_2 + x_2b_2 + b_1a_2\right) \qquad (6)$$

The quadrilateral's area is then

$$Q = \frac{1}{2}\left[b_1 x_2 - a_1 b_2 + x_2 b_2 + b_1 a_2\right] \tag{7}$$

and the reader is reminded that x_2 is a constant.

Fig. 8 was drawn with both $b_1, b_2 \le 0$ but notice that a quadrilateral also occurs for $b_1, b_2 \le 0$. Notice, also, that a_1, a_2 are unrestricted as to sign, so that the appropriate limits are

$$\begin{aligned} &b_1, b_2 \ge 0 \\ &-r \le a_1 \le r \\ &-s \le a_2 \le s \\ &0 \le b_1 \le \sqrt{r^2 - a_1^2} \\ &0 \le b_2 \le \sqrt{s^2 - a_2^2} \end{aligned} \tag{8}$$

and also

$$\begin{aligned} &b_1, b_2 \le 0 \\ &-r \le a_1 \le r \\ &-s \le a_2 \le s \\ &-\sqrt{r^2 - a_1^2} \le b_1 \le 0 \\ &-\sqrt{s^2 - a_2^2} \le b_2 \le 0 \end{aligned} \tag{9}$$

In the case of (9) the appropriate area expression is just the negative of (7), which we anticipate from the fact that the signs of the b's are reversed between the two cases.

Two Triangles. $I < O_1$.

This is the Fig. 5 case, and

$$\frac{b_1 x_2 + b_1 a_2 - b_2 a_1}{b_1 - b_2} < 0 \tag{10}$$

Here, $b_1 \le 0, b_2 \ge 0$, so $b_1 - b_2 < 0$.

From (10), consequently,

$$b_1\left(x_2 + a_2\right) > b_2 a_1 \tag{11}$$

and since $x_2 + a_2 > 0$, this case cannot occur when $a_1 > 0$.

So suppose $a_1 < 0$, and that (11), the necessary condition, is satisfied.

Then $\frac{b_1}{a_1}\left(x_2 + a_2\right) < b_2$

or

$$b_1 > \frac{a_1 b_2}{x_2 + a_2} \tag{12}$$

Since, for Fig. 5, it was decided that $AIO_1 + IBO_2$ would be used, the area of interest is

$$\tfrac{1}{2}\left[b_1 x_2 + b_1 a_2 - b_2 a_1 + b_2 x_2\right] \tag{13}$$

We have

$$\begin{aligned} &-r \le b_1 < 0 \\ &0 \le b_2 \le \sqrt{s^2 - a_2^2} \\ &-s \le a_2 \le s \\ &-\sqrt{r^2 - b_1^2} \le a_1 < \frac{b_1\left(x_2 + a_2\right)}{b_2} \end{aligned} \tag{14}$$

Two Triangles. $I > O_2$

This is the situation in Fig. 6 and is analyzed similarly to the $I < O_1$ case. By assumption

$$\frac{b_1 x_2 + b_1 a_2 - b_2 a_1}{b_1 - b_2} > x_2 \tag{15}$$

and b_1-b_2 is still < 0, with $b_1 \le 0$, $b_2 \ge 0$.

Then (15) gives

$$b_1 a_2 < b_2 (a_1 - x_2) \tag{16}$$

Again, $b_1 \leq 0$, $b_2 \geq 0$, and since a_1 is small relative to the segment length x_2, $a_1 - x_2 < 0$. (16), then, cannot be realized with $a_2 < 0$.

Assuming that $a_2 > 0$, the other necessary condition for (16) is

$$b_2 < \frac{a_2 b_1}{a_1 - x_2} \tag{17}$$

The area in the case, $O_1AI + O_2BI$ in Fig. 6 is

$$\tfrac{1}{2}\left[-b_1 x_2 - b_1 a_2 + b_2 a_1 - b_2 x_2\right] \tag{18}$$

where

$$\begin{aligned} &-\sqrt{r^2 - a_1^2} \leq b_1 \leq 0 \\ &s \geq b_2 \geq 0 \\ &-r \leq a_1 \leq r \\ &\sqrt{s^2 - b_2^2} \geq a_2 > \frac{b_2 (a_1 - x_2)}{b_1} \end{aligned} \tag{19}$$

Notice also from Fig. 5 that if A and B is each made to fall on the opposite side of the line through O_1, O_2, it is possible to realize $I < O_1$; and from Fig. 6 that if each of A, B is made to fall on opposite sides of the line through O_1, O_2, it is possible to find $I > O_2$.

We omit these expressions and their defining limits and later will refer to expressions <u>similar</u> to (13), (18) and to limits <u>similar to</u> (14), (19). It will be clear then that this is not detrimental to the conclusions.

<u>Two Triangles</u>. $O_1 \leq I \leq O_2$

Consider the case of the upper figure in Fig. 4. For the left triangle, let $(u_1, v_1) = (a_1, b_1), (u_2, v_2) = (x, 0), (u_3, v_3) = (0, 0)$ and find area = $\frac{1}{2}(-b, x)$ by using (4).

For the right hand triangle, with $(u_1, v_1) = (x, 0)$, $(u_2, v_2) = (x_2, 0)$, $(u_3, v_3) = (x_2 + a_2, b_2)$, we get area $\frac{1}{2}[-xb_2 + b_2 x_2]$, so in this case the total area is $\frac{1}{2}[-(b_1 + b_2)x + b_2 x_2]$, or

$$\frac{1}{2}\left[(-b_1 - b_2)\left(\frac{b_1 x_2 + b_1 a_2 - b_2 a_1}{b_1 - b_2}\right) + b_2 x_2\right] \tag{20}$$

Notice that the lower figure in Fig. 4 gives area

$$\frac{1}{2}[(b_1 + b_2)x - b_2 x_2] \tag{21}$$

which, in lieu of the reversal of the signs of b_1, b_2 between the two cases, is what could be predicted.

We next include the limits for the case corresponding to the area expression (20); that is, for the upper case of Fig. 4, where $b_1 \le 0$, $b_2 \ge 0$. In words, we just enumerate the variable possibilities that exclude the Fig. 5 and Fig. 6 cases for these respective ranges of b_1, b_2. The four subcases are

$$\begin{aligned} &-r \le a_1 < 0 \\ &-\sqrt{r^2 - a_1^2} \le b_1 < \frac{a_1 b_2}{x_2 + a_2} \\ &-\sqrt{s^2 - b_2^2} \le a_2 < 0 \\ &s \ge b_2 \ge 0 \end{aligned} \tag{22}$$

$$
\begin{aligned}
&-r \le a_1 < 0 \\
&-\sqrt{r^2 - a_1^2} \le b_1 < \frac{a_1 b_2}{x_2 + a_2} \\
&s \ge a_2 > 0 \\
&\sqrt{s^2 - a_2^2} \ge b_2 > \frac{a_2 b_1}{a_1 - x_2}
\end{aligned}
\tag{23}
$$

$$
\begin{aligned}
&r \ge a_1 > 0 \\
&-\sqrt{r^2 - a_1^2} \le b_1 < 0 \\
&s \le a_2 < 0 \\
&\sqrt{s^2 - a_2^2} \ge b_2 > 0
\end{aligned}
\tag{24}
$$

$$
\begin{aligned}
&\sqrt{s^2 - a_2^2} \ge b_2 > \frac{a_2 b_1}{a_1 - x_2} \\
&s \ge a_2 > 0 \\
&r \ge a_1 > 0 \\
&-\sqrt{r^2 - a_1^2} b_1 < 0
\end{aligned}
\tag{25}
$$

Without damage to the conclusions we may also simply refer to the expression (21) subject to limits similar to (22), (23), (24), (25).

What is of interest is the expected value of the area of the figure that results when the random points (a_1, b_1), $(x_2 + a_2, b_2)$ occur. Previously (Stanfel and Stanfel 1993), (Stanfel and Stanfel 1994) it was assumed that (a_1, b_1), and $(x_2 + a_2, b_2)$ were independent and each described by a uniform distribution over its corresponding circle.

In general we write $\mathrm{p}(a_1, b_1, a_2, b_2)$ for the density function and abbreviate this to p in the expressions below.

Then the thing to compute is

p [expression (7)] integrated over (8)

+ p [negative of (7)] integrated over (9)

+ p [expression (13)] integrated over (14)

+ p [expression (18)] integrated over (19)

+ p [similar to (13)] integrated over similar to (14)

+ p [similar to (18)] integrated over similar to (19)

+ p [expression (20)] integrated over (21), (22), (23), and (24)

+ p [expression (21)] integrated over similar to (21), (22), (23), and (24).

Results

The first purpose of the analysis here is to get a correct expression of the quantity of interest, the expected area. Aside from the omission of exact formulations for a couple of cases quite similar to others and which a reorder could fill in by looking at those derived in detail, this has been accomplished.

The question that motivated the work, however, is whether or not the consequent problem has an exact solution; that is, whether or not these integrations can be done exactly.

As pointed out in [Easa, 1995] it is common that multiple integrals cannot be solved exactly and that approximations are the only resort. And as shown in [Stanfel, 1996], the state of this approximation art is much advanced. Still, anyone prefers an exact result if it exists not only on the grounds of accuracy, sensitivity analyses, and the like, but also for the convenience of a formula in lieu of the necessity of a computer run in order to get an answer.

Our integrands are not so complicated - in fact, generally very simple - but the limits complicate the successive integrands, and the expressions are daunting for hand solution.

But another state of the art that has advanced rapidly is that of symbolic mathematics done by a computer. Thus, for a test, just one of our subproblems was subjected to the Mathcad system, a product of Mathsoft Corporation.

As expression (20) is the most complicated of the integrands, it was selected, and since (22) is as involved as any of the limit sets for that problem, that was the integral selected.

Expression (26), then, is the integral input to the symbolic integration system, which was running on a Pentium processor PC.

$$\int_{-\sqrt{r^2-a_1^2}}^{a_1 b_2/(x_2+a_2)} \int_{-\sqrt{s^2-b_2^2}}^{0} \int_{0}^{s} \int_{-r}^{0} [\text{expression (20)}]\, da_1 db_2 da_2 db_1 \qquad (26)$$

The system ground away on this problem for twenty-four hours without solving it. Although the system failed to declare the integral could not be done exactly, the length of time expended unsuccessfully was strong evidence of that condition, and the author feels that to be the case. Therefore it appears that numerical techniques offer the only possibilities and that one must be content with approximations to the average area.

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The Local Table Map — A GIS Aggregate Cell Construct

Micha Pazner, Associate Professor
Department of Geography
The University of Western Ontario
e-mail: pazner@julian.uwo.ca

ABSTRACT. This paper is about GIS *Aggregate Pixel Constructs* with a particular focus on one example — the *Local Table Map,* where embedded within each multi-cell block is a sub-array of cells representing a micro 'table'. Local table maps allow the GIS user to map the interrelation of several variables, on a single image, for many (e.g. millions of) locations — at each and every location. Furthermore, an LTM can be overlayed on conventional point, line, area, or surface maps. The aggregate cell model, discussed in the paper, approaches the classic cartographic map overlay problem in a non-standard and unique way. By changing cell resolution, multi-cell constructs are produced which are then manipulated to achieve a micro-spatial-neighborhood based overlay. Various types of LTM multivariate data modeling applications are presented. Local Table Maps can have substantial information content and aesthetic appeal. The paper discusses information representation advantages of aggregate constructs over the single-cell image processing paradigm. Current image processing programs have brought single cell image processing to the desktop. The case is made for the support of multi-cell pattern creation and processing capabilities in future generations of cartographic and other image processing software.

"Note the effectiveness and elegance of *small spots of intense saturated color* for carrying information —a design secret of classical cartography and, for that matter, of traffic lights." (Tufte, 1990)

Introduction

This paper describes the development of an *aggregate pixel construct* and discusses its multivariate data modeling applications for researching bodies of information. The broad problem is envisioning large multivariate datasets in the context of what has been referred to as “the interface of the display with the user” by NATO IST-05 (Visualization for Massive Military Datasets, 1997). One specific facet of this problem is the challenge of creating multivariate cartographic overlays. For instance, how can the six variables represented

as small multiple maps in *Figure 1,* be overlayed on top of the relief and hydrology (two additional variables) base map — without obliterating any information?

The three-traffic-light-colors scheme is well known in cartography, as the quote at the beginning of this paper attests. This scheme may also be found in graphs and diagrams, as well as in tables. A good example is the comparative summary table shown in *Appendix A* (from Steinitz C., D. Olson, et al. 1994). The table is an extensive summary of information based on millions of mapped locations. Can this scheme be implemented in an intensive mode, at each of those locations? The Local Table Map, discussed below, is an attempt to apply this type of representation at the micro level as well as at the macro level — for spatial and other applications.

The Local Table Map

Local Table Maps (LTM) are a specific example of an aggregate cell construct. Local Tables Maps are image-based representations created using a multi-cell block. In the example used here and presented in *Figures 2 and 3,* embeddded in each 4x4 =16 cell block are six cell tiles representing a micro 'table' that holds a value for one of six variables. Each variable cell can have only one of the following *Traffic Light* hues: Red, Yellow, Green. Gray is used to turn a variable's traffic light "Off" in a visual sense. Keeping the data values for each variable in separate numeric ranges lets the user vary the numeric thresholds for each variable's hues separately, thus achieving a form of data brushing (MacEachren et al. 1994). This means that each variable 'traffic light' can be tuned differently (e.g.: at what numeric value does it turn to Red, etc.), and that this tuning can be easily adjusted. The alerting Red hue is generally used to flag extreme or high values. Alternatively, normalizing the variables using a standard numeric scale (such as 0 - 100) allows the user to co-vary the thresholds simultaneously for all normalized variables.

Local Table Maps allow us to map and record the interrelation of several carto/graphic variables, on a single image, at each location, for a very large number of locations (e.g. millions of pixels), with minimal loss of information, and fairly economically *(Figures 3, 4, and 5)*. Local Table Maps can be overlayed on conventional map and or graph elements such as points, lines, areas, or surfaces. *Figure 5* shows how six variables organized as an LTM are superimposed on a pseudo-shaded-relief (surface) map that also has (areas of) water bodies. Therefore, a hybrid Map-Table construct is further mixed in with additional graphic elements, allowing for a more contextual, information rich, and visually diverse multivariate information display. The study area, shown in *Figure 5,* is in the Campbell Hills, District of Mackenzie, NWT (Canada). Appendix B is an aerial photograph showing the area in Figure 5 that is delineated between the two lines.

Aggregate Cell Constructs

The case for cell aggregate constructs is made based on the severe limitations to using the single pixel — which can only represent information using a single graphic element: tone or color. Moving away from the single cell to cell aggregates enables the deployment of visual variables such as: color, length, width, orientation, shape, size, pattern and texture. These visual variables or graphical elements can be designed and implemented for optimal visualization impact (Buttenfield, 1993) (Bertin, 1983). For example, a 4x4 = 16 cells aggregate construct was used to design the Local Table block shown in *Figure 2*. Other examples of aggregate cell constructs include the Multivariate Tile Pattern Map (Pazner, 1997) and GIS Icon Maps (Pazner and Lafreniere, 1997) (Lafreniere et al., 1996). Research is currently underway to experiment with the notion of organizing the traffic light variable scheme used in an LTM in patterns based on a Road Junctions metaphor. The multivariate traffic-light junction metaphor may result in what could be termed Variable Cell Junction Maps.

Aggregate pixel designs have a special ability to take into account the classic elements of photointerpretation. Therefore the aggregate-cell-map maker has control over designing elements of interpretation (Avery and Berlin, 1992). We are not only able to interpret a given external representation, but are also better able to design, generate, and manipulate the external representation. Tufte (1990) stresses in his book: "Escaping (this) flatland is the essential task of envisioning information..." Should we not seek to break out of the imprisoning (single) cell representation in our quest to escape flatland? Current image processing software have brought single cell image processing to the desktop. The case is made here for the need and utility of multi-cell pattern creation and processing capabilities in future generations of image processing software. This would allow users to make and read images containing multi-celled constructs.

The work presented in this paper uses a raster (image based) geographic information system (GIS) to create and manipulate the aggregate cell patterns (Tomlin, 1990) (Pazner, 1995). The procedure for creating multi-cell constructs is based on a unique spatial/neighborhood overlay approach. In these image-maps, the basic assumption of each cell representing a single location, is relaxed to permit multiple cells per original location. Every cell is transformed in the local table map into a patterned matrix of output cells—the multi-cell block *(Figure 3)*. The key step in the procedure is a GIS model that generates an internal block addressing system. The addressing system is created using spatial, arithmetic and logic operations. The whole procedure, including the internal addressing model, is stored as a script ("macro") in the software used (see the *Software* endnote).

A New Overlay Method?

The aggregate cell model is unique in that it approaches the map overlay problem in a non-standard way. Here, the overlay is achieved by changing (increasing — i.e. making more fine) cell resolution, and then making use of the new small multi-cell constructs. In effect, this is a 'pseudo spatial neighborhood based overlay' in which micro graphical tables, icon elements, and conventional themes (0D-3D) can be placed *(Figure 3)* at each original location — after it has been transformed to a pseudo-neighborhood. This type of overlay method can be seen as a micro-juxstaposition based overlay, or a 'sub block' (of cells) overlay where an aggregate (multi cell) construct is first created, and then advantage is taken of its sub-structure. As opposed to conventional overlay methods which operate at the standard cell resolution of the map, this overlay technique is achieved by working at two levels: with cell aggregates (a seemingly higher level), but that are at a sub-original cell-resolution (i.e. at a lower level). As a result this technique has both an aggregate and a sub-cell flavor. Yet another way to look at the method is as user-controlled cell-block cartographic half-toning. Where the half-toning pattern is used to create point-symbol or iconic elements inside a cell block.

Use

How can aggregate cell constructs, such as local table maps, be used? The interpretation of aggregate cell maps involves explicit comparison of the inter-relation of variables mapped deliberately for co-examination. The resultant maps reveal macro interaction patterns *(in Figures 5 and 6)*, while retaining in a zoomed-in micro view *(Figure 3)*, full and easily visualized hue and numeric information for every variable at each location. These maps are viewed at a continuum of display scales from micro *(Figure 3)*, through intermediate meso scales *(e.g.: Figure 4)* to macro views *(Figure 5)*. Consequently, aggregate cell constructs can be used effectively for co-visualizing multivariate interrelation. In addition, when viewed in electronic form, it is possible to use screen tools provided by the software, such as a 'numeric magnifying glass tool' that allows the user to see the underlying data values.

Local table maps represent an interesting cross between a modern pixel-based image and a map composed of cartographic ‘table symbols’ with a legend key. The potential exists to reap the advantages of interpreting image data while reading simple cartographic table symbols. In reading a local table map, there is an emphasis on comparing between the variables that are depicted on the map. This is an exercise in interpreting isolated-and-joined elements that is different from the interpretation of normal imagery. The LTM presented in *Figures 3 — 6* show eight to nine variables, out of which six are in LTM form. The six micro tabulated variables are: elevation, steepness, orientation, drainage, impervious, and soil *(see Figures 1 and 2)*. In addition, areas of hydrology and vegetation *(in Figure 6)* are depicted as conventional themes, and the terrain is shown using shaded relief *(Figures 5 and 6)*.

Extending the Use from Maps to Graphs

Aggregate cell constructs, even when made using a GIS, are applicable not only to maps, but also to graphs. Graphing applications come under the rubric of spatialization using GIS. The term *spatialization* refers to the spatial representation and processing of non-spatial information such as numeric arrays, tables, graphs, text, etc. The term *GRASP* (GRaph and Array Spatial Processing) was coined and is used by Pazner for raster GIS based spatialization applications. Selected references on spatialization and GRASP include: (Kuhn, 1996) (Malczewski et al., 1997), (Pazner, 1994), (Scott and Pazner, 1992) and (Skupin and Buttenfield, 1997). Examples of Spatialization/GRASP work include:

- Multiple Comparative Overlays (by Anderson et al. , 1998)
- Graphic Timetables (Scott and Pazner, 1992)
- GIS modeling of Color Model Spaces (Pazner, 1998)
- Line Graphs: Transect Profiles (Ripley and Pazner, 1998)
- Scatterplot Modeling (Pazner and Liu, 1992)

Applicability

Possible applications for Local Table Maps include work that involves learning about information: coincidence, context, conditions, comparison, interrelation, interaction, anomalies, change detection, classification, hypotheses generation, explanation, causality, and location. One type of application that appears particularly suitable is exploratory data visualization (EDV), where the *data* can take the form of: variables, criteria, indices, gradients, error, uncertainty, etc. The data may be spatially distributed or spatialized (ie. non spatial data).

With proper variable selection, graphic design, and colorization Local Table Maps that have substantial information content and aesthetic appeal can be created. Local Table Maps represent an interesting table/map (in image format) hybrid information representation whose implementation can additionally mix in conventional, 3D, and icon pattern map constructs. It is planned for instance, to experiment with LTM in conjunction with translucent overlay methods, in order to further increase the dimensionality of the map, i.e.: the number of variables that can be carried by a single layer.

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Software

The software used is ThinkSpace's raster GIS for desktop computers (PC and Macintosh) — *MF Works* (Formerly Map•Factory — reviewed in Limp, 1996). Most, if not all of the capabilities needed to create aggregate cell constructs exist in other similar systems such as Idrisi (Clark Labs, Clark University, MA).

Acknowledgements

Research in aggregate cell constructs and spatialization was actively supported and funded in 1996-1997 by the National Center for Geographic Information and Analysis (NCGIA) under the Directorship of Dr. M. Goodchild. NATO's IST-05 N/X Network of Experts on Visualization for Massive Military Datasets has provided an ongoing framework for conducting this research. This study is being conducted in collaboration with the Graduate School of Design at Harvard Unversity. Special thanks to Lori Tesolin, a student research assistant. The two research thrusts — aggregate cell constructs and spatialization — represent the continuing evolution of research areas the author has been active in for the last decade of this millenium.

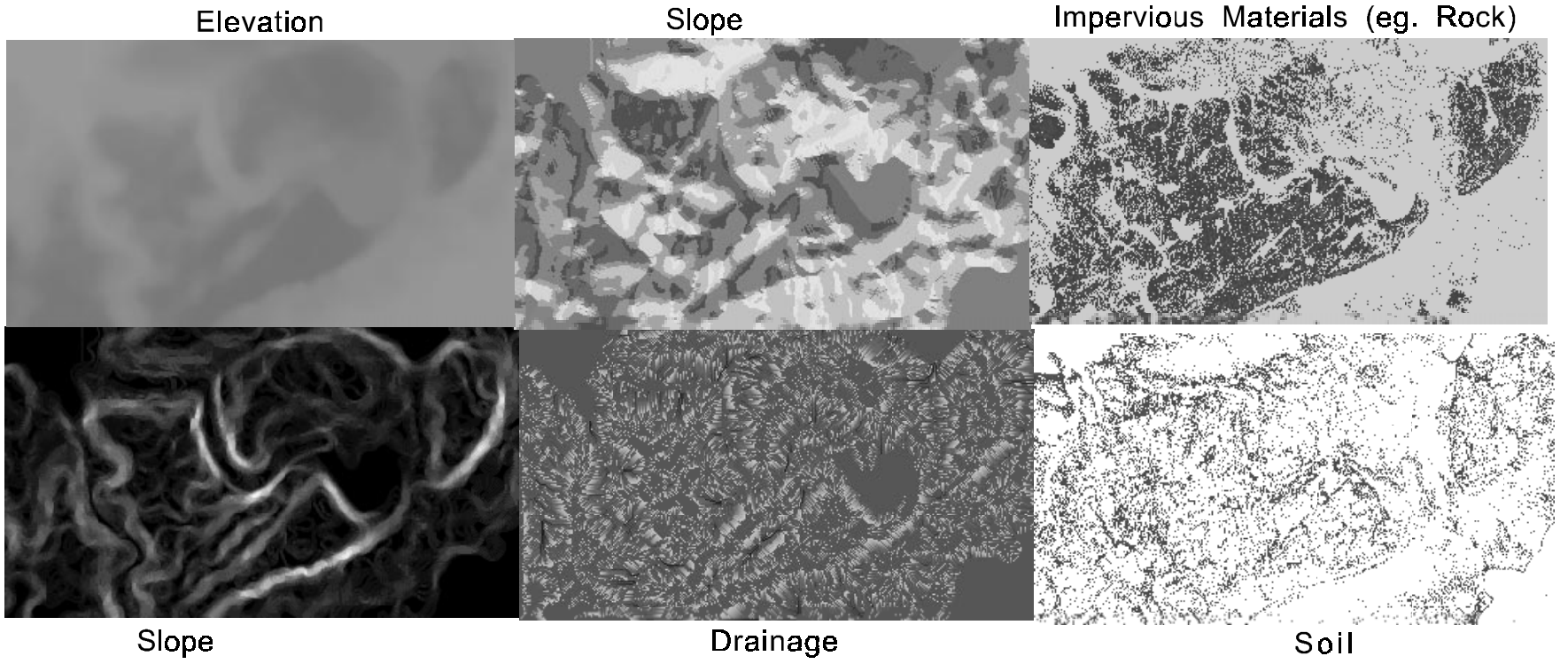
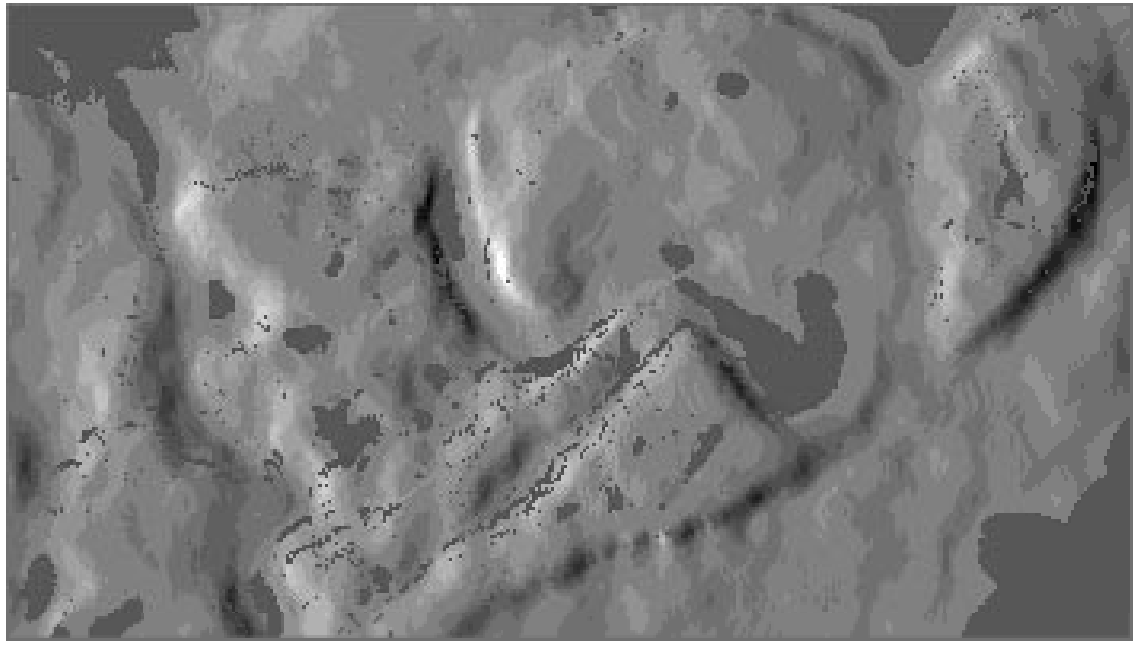


Figure 1

The spatial distributions of the six variables. The maps are colorized individually. The additional (enlarged) map shows the terrain and hydrology from satellite data.



Shaded Relief, Water Bodies, and Wet Areas from Satellite

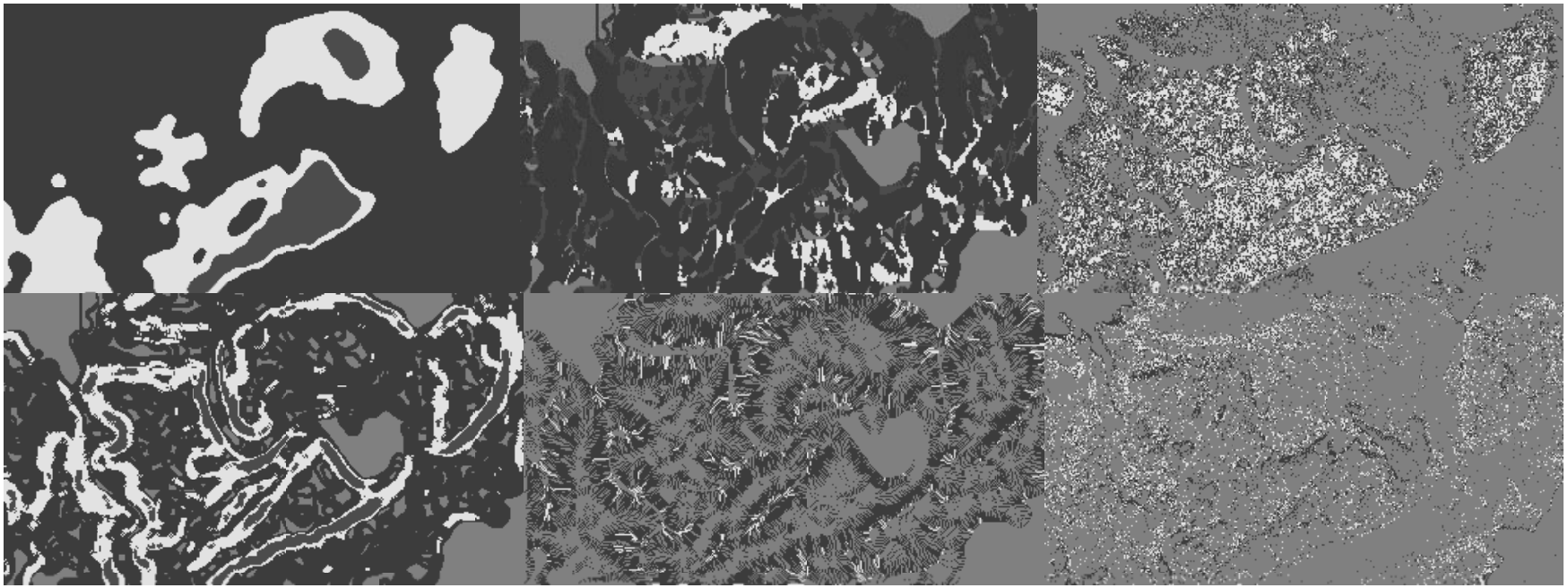


Figure 2

The spatial distributions of the six variables colorized as they appear in the Local Table Map

Elevation
- High Elevation
- Intermediate Elevation
- Low Elevation

Orientation
- North
- South
- NE, East, SE, SW, West, NW
- Horizontal

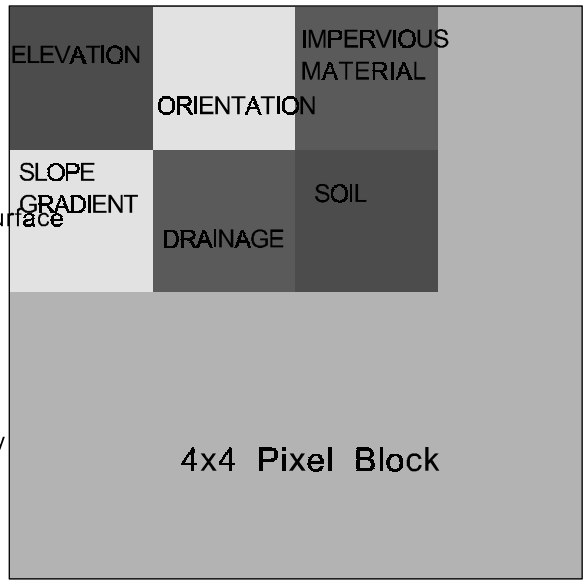
Impervious Material
- Highly Reflective Surface
- Intermediate Reflective Surface
- Low Reflective Surface
- Non Impervious

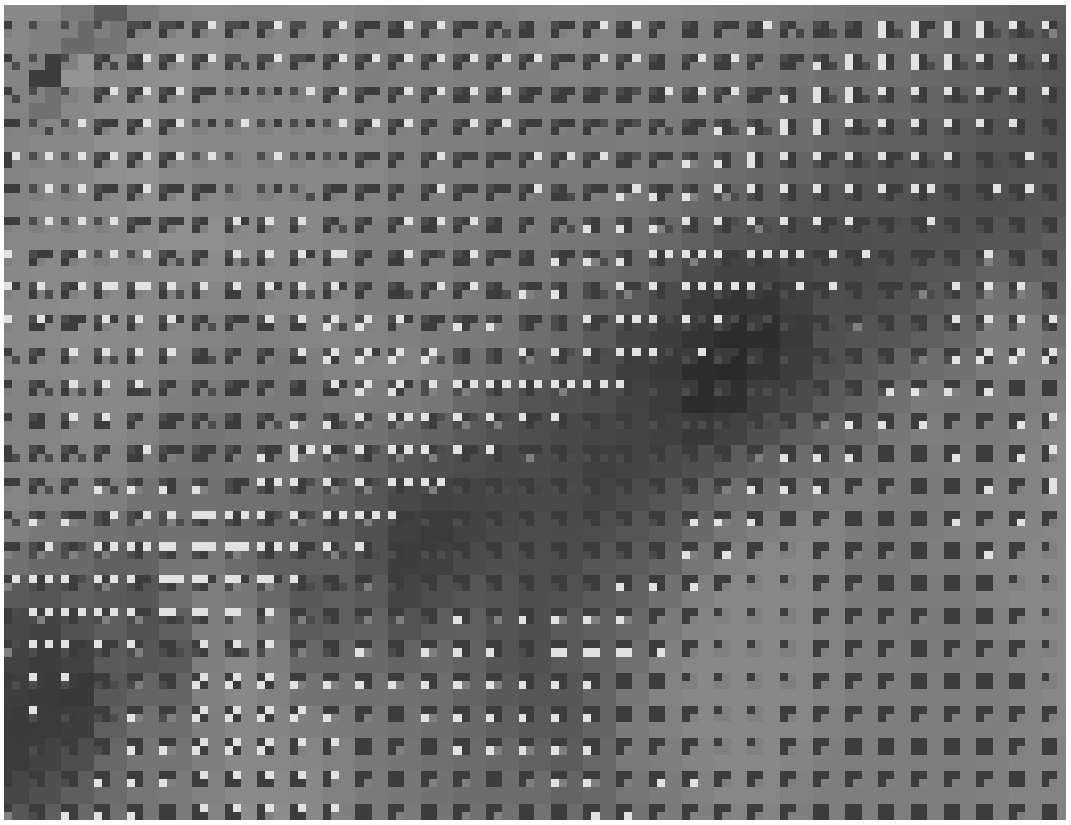
Slope Gradient
- High Steepness
- Intermediate Steepness
- Low Steepness
- Flat

Drainage
- High Drainage
- Intermediate Drainage
- Low Drainage
- Negligable Drainage

Soil
- High Soil Certainty
- Intermediate Soil Certainty
- Low Soil Certainty
- Non Soil

Local Table



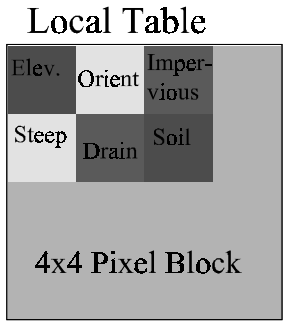


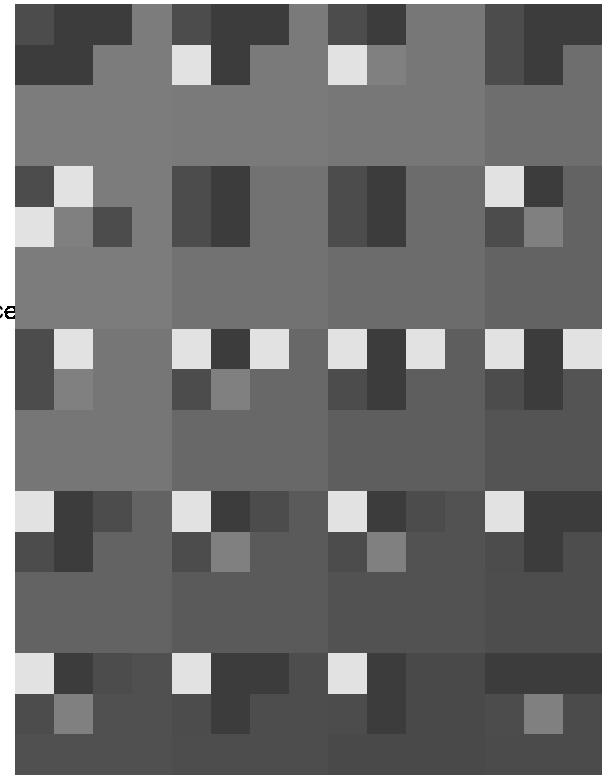
Local Tables Maps are created using a 4x4 (16 cells) block. In this example, within each block there is a six-celled table that holds a value for one of the variables: Elevation, Slope Gradient, Orientation, Drainage, Impervious Material, and Soil Coverage. The different variables are given values of high, intermediate or low and assigned their respective "Traffic Light" colours. The alerting Red is generally used to flag extreme or high values.

Figure 3

Local Table Map

'micro scale'





Elevation

- High Elevation
- Intermediate Elevation
- Low Elevation

Orientation

- North
- South
- NE, East, SE, SW, West, NW
- Horizontal

Impervious Material

- Highly Reflective Surface
- Intermediate Reflective Surface
- Low Reflective Surface
- Non Impervious

Slope Gradient

- High Steepness
- Intermediate Steepness
- Low Steepness
- Flat

Drainage

- High Drainage
- Intermediate Drainage
- Low Drainage
- Negligable Drainage

Soil

- High Soil Certainty
- Intermediate Soil Certainty
- Low Soil Certainty
- Non Soil

Figure 4: Local Table Map - a 'meso scale'

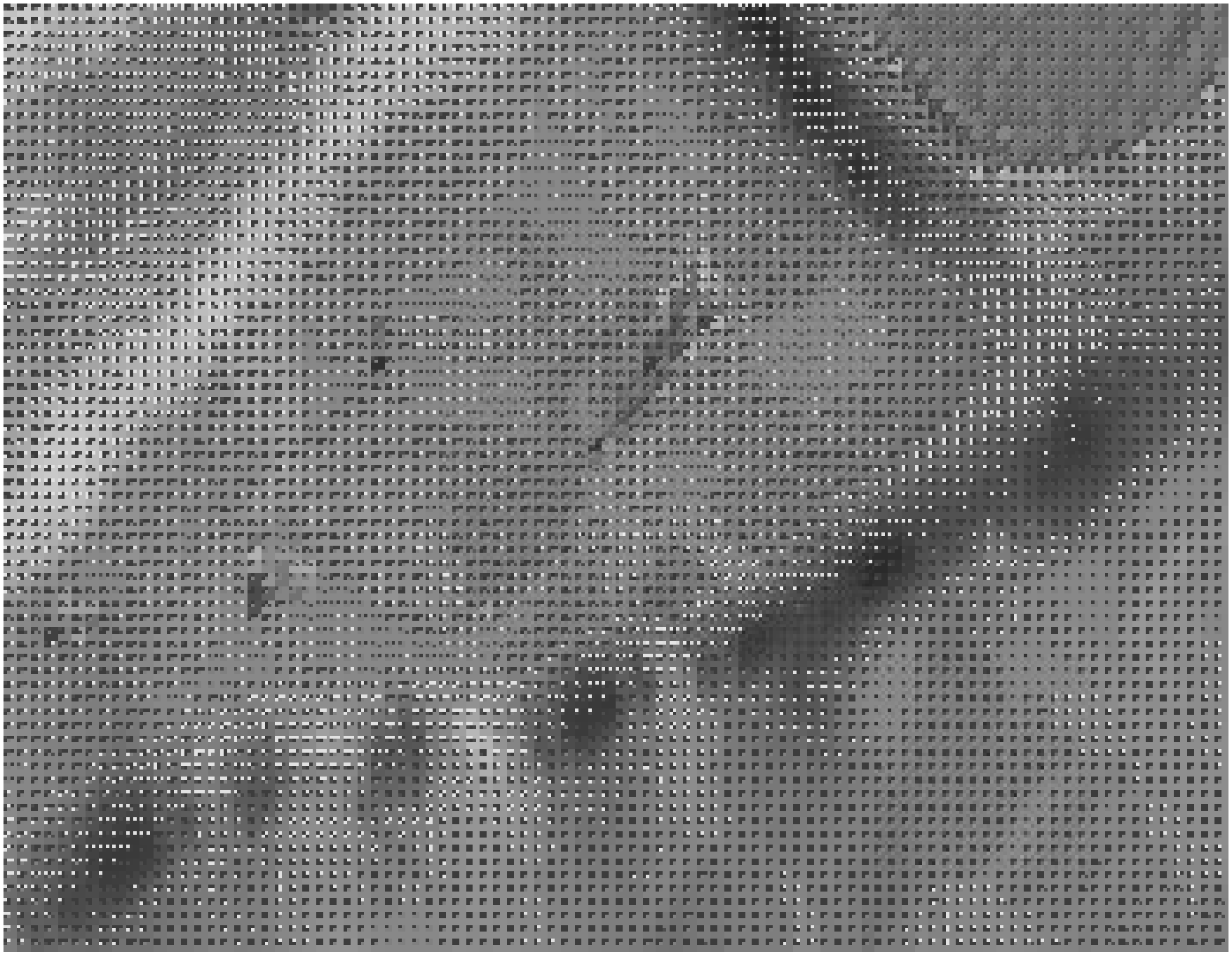


Figure 5: Local Table Map

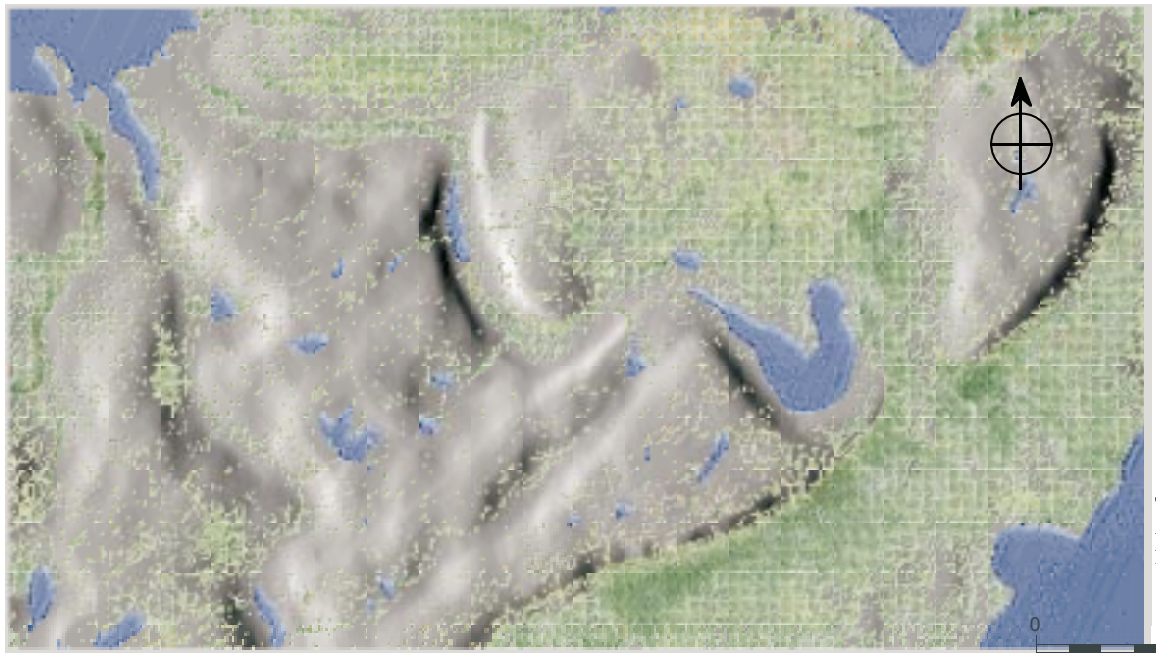


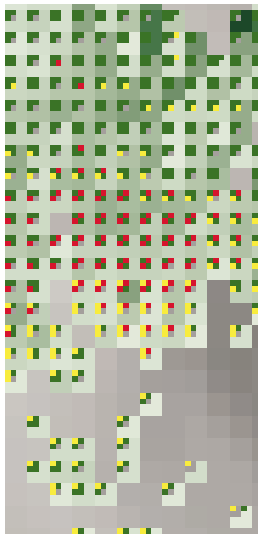
0 1,000 2,000 Metres

1:30,000

The area that is delineated between the two lines represents the view from an oblique air photo taken from the NorthEast.

Figure 6: Local Table Map – Limited to Vegetated Areas





The North Arrow also points to the location of the detailed section above.

0 1,000 2,000 Metres

1:36,500

Elevation
- High Elevation
- Intermediate Elevation
- Low Elevation

Orientation
- North
- South
- NE, East, SE, SW, West, NW
- Horizontal

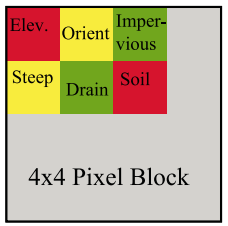
Impervious Material
- Highly Reflective Surface
- Intermediate Reflective Surface
- Low Reflective Surface
- Non Impervious

Slope Gradient
- High Steepness
- Intermediate Steepness
- Low Steepness
- Flat

Drainage
- High Drainage
- Intermediate Drainage
- Low Drainage
- Negligable Drainage

Soil
- High Soil Certainty
- Intermediate Soil Certainty
- Low Soil Certainty
- Non Soil

Local Table



Appendix A

A Table exemplifying the three-traffic-light-colors scheme (from Steinitz C., D. Olson, et al. 1994). The table is an extensive summary of information based on millions of mapped locations. Can this scheme be implemented in an intensive mode, at each of those locations? The Local Table Map attempts to apply a similar representation at the micro level of the individual locations or data observations.

	Existing	Plan Trend	Buildout	Township	Southern	Spine	Park
Geologic Landscape							
Surface Water Quality	●	●	●●	○	○	○	●●
Water Recharge Areas	●	●	●●	●	○	●	●
Agricultural Soils	●	●	●●	●	●	●	●
Biologic Landscape							
Biodiversity	●●	●	●●	○	●	○	●●
Bear Habitat	●●	●●	●●	○	●	●	●●
Special Natural Areas	●	●●	●●	●	●	●	●●
Visual Landscape							
Scenic Elements	●●	●●	●●	○	●	●	●●
View Quality	●●	●●	●●	●	●	●	●●
Demographics							
Population Capability	97,000 ●	600,000 ●●	400,000 ●	300,000 ●	350,000 ●	350,000 ●	300,000 ●
Economics ($millions)							
Cost of Public Action	20	20	20	1,733	1,854	1,996	2,072
Annual Cost If 20 yr/4% Bond	0 ●	1 ●	1 ●	104 ●	120 ●	106 ●	135 ●●
Politics							
Private Roles	●●	●	●	●	●	●	●●
Township Roles	●●	●●	●●	●	●	●	●●
County Roles	○	●	●	○	●	●●	●●

Legend: ●● Most Positive; ● Positive; ○ Neutral; ● Negative; ●● Most Negative

108. A COUNTY LEVEL COMPARATIVE SUMMARY

Source: Steinitz C., D. Olson, E. Bilda, J. S. Ellis, Y-Y Hung, E. Katz, P. Meijerink, A. W. Shearer, H. R. Smith, A Sternberg (1994) *Alternative Futures of Monroe County, Pennsylvania,* Harvard University Graduate School of Design, MA, 89p.

Appendix B



Aerial view of the study area – What the 'real thing' portrayed by a Local Table Map looks like (in winter).

RECRUITMENT - A WAY OF LIFE

James K. Crossfield
Coordinator, Geomatics Engineering
CSU, Fresno
Fresno, CA 93740-0094
(559) 278-4827
james_crossfield@csufresno.edu

ABSTRACT

A Geomatics Engineering student recruitment initiative has been implemented at CSU, Fresno. Factors leading to recent enrollment declines are listed. Three critical initiatives: changing the program name, developing recruitment literature and initiation of a large scholarship program for new students were implemented with strong support by the Geomatics Engineering Advisory council. Continued program strengths are itemized. Recruitment activity during the 1997-1998 academic year is described and summarized in detail. Approximately two separate recruitment activities are necessary for each new student generated.

INTRODUCTION

Geomatics Engineers manage the global spatial infrastructure. This process includes coordinating the acquisition, manipulation, analysis, use and provision of information about the relative locations of objects on the surface of the earth. Professional responsibilities include but are not limited to: imaging systems, aerial photogrammetry, industrial alignment, environmental monitoring, geographic information systems, earthquake monitoring and prediction, property boundary creation and relocation, remote sensing, applications software development, transportation alignment, mapping systems and geodesy.

The faculty and students in the CSU, Fresno Geomatics Engineering Program have created a unique educational enterprise. Organized in 1971 as the first four year surveying degree program in the nation, CSU, Fresno has produced more than 500 graduates. Many have gone on to become industry leaders within California and across the nation. Donn Liddle (1988) prepared digital imagery of the damaged MIR space station last year in his job as imaging specialist for NASA in Houston. Steven Frank (1989), Associate Professor of Geomatics at New Mexico State University, recently organized the 16th Annual North American Geomatics Teachers Conference in Las Cruces, New Mexico. Barbara Littell (1993) lead an effort by the United States Geological Survey to correctly position the South Pole in January of 1996. Leonard Gabrielson (1995) relocated the monument marking the center of California and appeared in the weekly Television show titled “California Gold”. Several graduates own or manage their own consulting firms. Most graduates have become licensed professionals. Collectively, these graduates manage the Global Spatial Infrastructure.

While the program continues to produce quality graduates, enrollment began a steady decline starting in about 1993. Several factors contributed to this decline.

1. The supply of graduates from high school dwindled. This supply is expected to increase substantially after the year 2000. A traditional program could reasonably expect a large enrollment boost from this source. Historically, however, surveying enrollment at Fresno State has been dominated by non-traditional students. The average graduate is 27 years old when he or she leaves the program. A major recruitment campaign to the high schools could increase our share of traditional students.

2. Annual fees doubled in the CSU system between 1992 and 1997. Recent legislative efforts to roll back some of this increase as well as a period of time without major increases may mitigate this problem..

3. A Caltrans summer job program was suspended for two years in 1995 and 1996. This program had provided guaranteed summer employment opportunities for program students since 1988 and was reinstituted during the Summer of 1997. Since the number of student positions is expected to increase over the next few years, we can now legitimately advertise guaranteed summer jobs to potential students.

4. A prohibitive fee assessed to students seeking their second bachelors degree ($150 per semester hour) was imposed about five years ago, severely limiting our enrollment from students in this category. This fee was rescinded about 2 years ago. Since a substantial number (5-10) of our students have historically sought a second degree, we expect enrollment form second degree students to increase over time.

5. Recently, the general societal perception of engineering and technology (with the exception of computers) has not been very good. Engineering enrollments nationwide have suffered during the last six years as the end of the cold war minimized the need for defense based initiatives. Engineering enrollment declines at CSU, Fresno have contributed to a Geomatics Engineering enrollment decline.

A continuing demand for educated professionals and a simultaneously declining enrollment was leading to a serious supply problem. Clearly, action was necessary to maintain adequate long term enrollment so that industry and societal needs could be met.

THE RESPONSE

The CSU, Fresno Geomatics Engineering Advisory Council played a key role in setting the agenda for this change. Council members include: Ken Meme, Larry Fenske, Axel Hoffman, Hans Haselbach, Alan Mikuni, Paul Enneking, Dennis Meyer, Patricia Caldwell, Howard Brunner and Earl Cross. Three initiatives, essential to the viability of any serious recruiting effort, received vital Council input and support. A fourth essential initiative was

due to enhanced recruitment efforts within the School of Engineering and Computer Science.

First came the radical decision to change the program name to Geomatics Engineering. This was unanimously recommended by the Council in 1995. The Council was motivated by the need to increase enrollment and the desire to insure that graduates were recognized throughout the Pacific Rim and the world. Six graduates already have Geomatics Engineering degrees as of August 1998. They are: Grant Dorman, Tony Hunt, Marcelino Rodriquez, Donnie Andrade, Noe Hernandez and Jim Koo. All course prefixes changed to GME in the 1998-1999 academic catalog. The prefix "GE" could not be used because of the obvious correlation (unfortunately) with General Education.

Second, the Advisory Council urged that appropriate recruiting literature be developed to support new recruiting efforts. Advisory Council contributions (both ideas and funds) helped produce a very nice full-color folder and poster. Other organizations including PG&E, East Bay Chapter of CLSA, and the student chapters of SPSA and ACSM supported this effort as well. Early in the 1997 Fall semester, 1000 folders and 200 posters went to high school counselors and math instructors across California and Nevada.

The third major response to the recruitment problem was collection of prospective student addresses on cards at recruiting activities. Use of the cards was instituted by the School of Engineering and Computer science starting in the 1996-1997 academic year. The names and addresses of all respondents expressing interest in Geomatics Engineering were entered into a mailing list of Geomatics prospects. As an added bonus, many names were added to the School prospects list as well, suggesting some longer term benefits due to possible later migration into the Geomatics Engineering major later by some of them.

The fourth initiative dealt with solving the problem of attracting top quality high school students directly into Geomatics Engineering. The Freshman scholarship that had been offered since 1986 provided $500 per semester for a year but this had generated little excitement in recent years. The new Geomatics Engineering Honors Scholars Program, promises to generate much more enthusiasm. A $2,000 per year scholarship is provided to qualified new Freshmen for up to two years. A service commitment and high academic standards are critical parts of the program. The Council and industry have already contributed to this program. Applications have been received and several recipients will be chosen for the 1999-2000 academic year.

ADDITIONAL FACTORS

The described critical components; name change, recruitment literature, prospects mailing list and large scholarship offerings are the cornerstones of a successful recruitment effort. Yet several additional factors continue to make Geomatics Engineering study at CSU, Fresno an attractive choice for prospective students. These include:

1. A strong external scholarship support program generates approximately $50,000 each year for program students. Geomatics students typically receive ten times more scholarship support than the average Fresno State student.

2. Hiring of graduates remains strong. All graduates are employed. Starting salaries are excellent, and salary increases average $3,200 per year.

3. Students have exceptional opportunities to interact and link with the geomatics profession at all levels. Local, regional and national conference attendance by Fresno Geomatics Engineering students remains strong.

4. Our Geomatics students continue to win national recognition at record levels. Since 1984, Fresno Geomatics students have collected more national scholarships and Projects of the Year awards than students at any other university in the nation.

5. Several Geomatics students are employed each year on a variety of faculty initiated research projects. This directly links the latest developments in the discipline to many of our better students. These ideas are routinely incorporated into the regular classes.

6. Geomatics students continue to publish an excellent news magazine (Foresight!) twice each year. Sent to over 6000 professionals nationwide, this publication provides vital linkage between the program and the profession. Newsletters are now routinely incorporated into our Geomatics Web Page.

RECRUITMENT ACTIVITY

The color folders and posters highlighting Geomatics Engineering at CSU, Fresno were sent to 1000 high school counselors and 200 math instructors throughout California and Nevada during early September, 1997. The response was significant. School representatives began calling as soon as the mailings reached them. Scheduling visits all over the state was challenging, especially when trying to combine several visits to more than one school in a day. Double visits were accomplished four times and one rare triple visit was managed. From Westminster (south of Los Angeles) to Anderson (Near Redding), from the California coast to Fernley Nevada (east of Reno) recruiting visits were made from September 1997 through May 1998. More than twenty-five visits to high schools were conducted. Graduate Kurt Lehnhardt helped at Moorpark High School, where 350 students in 11 classes throughout six periods were addressed in one day. Current students and graduates (current professionals) assisted on approximately 25% of the visits.

Additional recruiting visits were also conducted to Fresno City College, Reedley College, Bakersfield College, Merced College, Yuba College and Fresno college night at the local convention center. College of Engineering and Computer Science information cards were distributed at each of these events and returned by students interested in Engineering in

general and Geomatics Engineering at CSU, Fresno in particular. As of December 1998 more than 450 prospective Geomatics Engineering students have been identified by these cards. More are added at each recruiting activity.

The typical recruitment visit to a high school class was well organized. The steps include:

1. Introduce yourself
2. Hand out a career guidance quiz and have them start to fill it out. See Appendix A.
3. Since they don't understand all the terms, interrupt to show several overheads describing Geomatics Engineering. Answer any questions that arise.
4. Now take a few minutes to provide specific details about several different aspects of the discipline. A short applicable video clip or computerized presentation helps here.
5. Have them finish (and/or revise) the career guidance quiz.
6. Give several page information packets (including hype sheets, brochures, newsletters, etc.) to all who score appropriately on the quiz. Also give this material to all others who express interest.
7. Now talk about guaranteed summer work in Geomatics Engineering, super scholarship opportunities for new Freshmen, professional and career opportunities and conferences, etc.
8. Finally, when interest is at the peak, distribute the interest address cards and urge them to check Geomatics Engineering.
9. Collect the address cards and insure they get added to the data base. Follow up with regular communications.

The information cards were used to generate a Geomatics Engineering prospects mailing list. Monthly mailings to these students have generated a positive link in their minds to Geomatics Engineering at CSU, Fresno. These mailings have included a welcome letter, the Geomatics Perspectives Newsletter, the Foresight! magazine, announcements about student meetings, open house, our annual conference and other notices and communications. We have included high school and junior college counselors and math instructors in these mailings as well. The goal is to portray a dynamic functional program that the students would want to join.

The result of 35 recruiting events at 31 sites in two states over a nine month period (September 1997 - May 1998) has been studied in detail. A total of 1,610 potential students were addressed, approached, cajoled and lobbied about a Geomatics Engineering career and the educational program offered at CSU, Fresno. More than 375 prospective students indicated some degree of interest in our major.

Nine visits were to career centers where interested students would gather for a special presentation. Thirty-three class presentations were made at fifteen different sites directly to math, science and CAD classes. Finally eleven career fair type activities were visited at 10 different locations. The results of the visits were interesting. Success was measured as the percent of contacted students that checked Geomatics Engineering on an interest card. Exactly 33.3% (158 of 474) of contacts made at career fairs became Geomatics

Engineering prospects. Almost 28.4% (38 of 134) of contacts made at high school career centers and nearly 19% (190 of 992) of classroom contacts became Geomatics Engineering prospects. A further breakdown indicated that 20.4% (120 of 587) in nineteen math classes, 16.3% (33 of 203) in eight physics classes, 1.4% (1 of 65) in two geology classes, 33.3% (6 of 18) in one English class and 25% (30 of 119) in five CAD/Drafting classes became GME prospects. The hottest prospects were the 91 of 460 precalc or calculus students and the physics students since it is assumed that they have the math skills and intellect to make it through the program. The Cad/Drafting students are the next most promising category since they typically have computer skills.

ASSESSMENT

The results are in for the Fall 1998 semester. A total of thirty-nine potential new students were expected after all applicants, potential applicants and interview records were correlated. Twenty-five of them actually began the semester by attending Geomatics Engineering classes at CSU, Fresno. They include: Tony Alvarado, Joyce Anson, Alex Nanpei, Brian Bowers, Matt Morrow, Tim Lackey, Nathan Duffy, Mark Carter, Casey Lowry, Jeremy Nickel, Jim Buongiorno, Shawn Dyer, Zach von Felton, Guadalupe Contreras, Adam McFadden, Gary Babcock, Jason Beck, Jeffrey Beck, Oscar Carreon, Daraxay Phengdara, Richard Ward, Nick Minas, Juma Al-Nuaimi, Obaid Saeed Al-Kaabi and Jonathan Franz.

The fourteen no shows are aggravating to say the least. One, a high school valedictorian from Tulare went to Cal Poly San Luis Obispo to study Environmental Engineering. One from Blythe decided to attend a local Community College there. One from Fresno went into Construction Management. One young man joined the Navy, two fellows from Saudi Arabia (Aramco) went to Ohio State and two other local persons are preparing to enroll in the Spring 1999 semester. No information is available on the other six. All likely enrollees were sent a welcome letter as soon as they were identified. An example of this letter may be seen in Appendix B.

We expect five new admits in Spring 1999. This number may grow a bit, but as is usually the case, this number just about matches the number of students who became phantoms during the Fall Semester. Fall 1998 phantoms include a student with family problems, one who took a (non geomatics and low-paying) job, and two or three others who dropped out for unknown reasons.

The reasons why the 25 new students entered our program are many and varied. About Fifteen came as a direct result of the recruitment effort last year. Another five came at least partly due to that effort last year, and the other five came for no apparent reason related to the recruitment effort. Fall 1998 recruitment efforts have added another 100 names to the prospects mailing list. We hope to see a significant number of these students in the coming years.

One conclusion is clear from this admittedly limited data set. Approximately one new Geomatics Engineering student will be directly generated for every two recruiting visits conducted. A substantial number of additional new students will be generated since the program is in the recruiting mode and looking for new students more actively.

RECRUITMENT GOALS

A legitimate recruitment program requires long term effort. It is unrealistic to expect huge enrollment increases in one or two years. A steady improvement can be reasonably expected. A modest 5% improvement in the number of enrolled Geomatics Engineering students by the end of Fall 1998 was envisioned. The table provided below indicates reasonable goals for each of the following five years. These projections are predicated upon the continued validity associated with the assumptions and initiatives indicated in the preceding paragraphs.

CSU, FRESNO GEOMATICS ENGINEERING ENROLLMENT PROJECTIONS

YEAR	MINIMUM	MAXIMUM	TARGET
1997-1998	60	70	60
1998-1999	70	80	70
1999-2000	80	95	85
2000-2001	95	115	100
2001-2002	115	140	125

TABLE 1.

Initial progress has been made towards meeting these objectives. It appears that 1998-1999 goals will be met. The success of the overall recruitment effort in subsequent years will be measured in terms of the actual number of enrolled students. Keeping up the level of intensity has become a primary objective over the next few years. The faculty and students at CSU, Fresno are excited and expect to see a gradual enrollment increase over the next few years as our recruiting efforts become "A WAY OF LIFE".

CONCLUSIONS

A substantial recruitment plan has been implemented at CSU, Fresno based upon four critical innovations. These factors were: changing the program name, developing colorful recruiting literature, developing a prospective student address database and initiating the Geomatics Engineering Honors Scholars program. Assuming that background parameters remain essentially stable, we expect a modest (70%) to significant (100%) enrollment increase throughout the next five years. Continued efforts to increase the prospective enrollment base and sustained efforts to maintain communication with these prospects should yield ongoing success. The long term nature of this challenge, requires constant commitment. As the enrollment grows, we expect that our graduates will continue to lead the Geomatics industry in California and the Nation.

APPENDIX A

SAMPLE GEOMATICS ENGINEERING CAREER GUIDANCE QUIZ

STATEMENT	TRUE	FALSE
I want a well paid, high technology related job after I graduate from college.		
I like computers.		
I like maps.		
I like mathematics.		
I enjoy talking to other people, learning from them, and sharing my ideas and knowledge with them.		
I am a good writer.		
I am interested in well paying, career focused summer jobs while in college		
I would appreciate receiving TEN times more scholarship money than other students.		
I want to protect our constitutional rights concerning private property ownership		
I would like an opportunity to manage some aspect of the global spatial infrastructure.		

TOTALS _____ _____

Tabulate the true and false responses. If you answered seven or more statements TRUE, then you should investigate a career in Geomatics Engineering.

For more information contact:

James K. Crossfield, Ph.D., L.S.
Coordinator, Geomatics Engineering
CSU, Fresno
Fresno, CA 93740-0094
(559) 278-4827
(559) 278-7002 (FAX)
james_crossfield@csufresno.edu

APPENDIX B

SAMPLE WELCOME LETTER TO NEW STUDENTS

Timothy W. Lackey
4925 N. Hills Dr.
Bakersfield, CA 93308

June 30, 1998

Dear Tim,

It is my pleasure to welcome you to the Geomatics Engineering Program at California State University, Fresno. This is an exciting time for you to enroll. Jobs are plentiful. As a graduate from the nation's only Geomatics Engineering degree program you will be prepared to manage the global spatial infrastructure.

Incoming Freshman, without any college credit, should consider the following courses (semester hours in parenthesis) for the Fall 1998 Semester: GME 1 (1), GME 5 (3), GME 15 (3), GME 15L (1), GME 66 (2), Math 75 (4) and Engl 1 (3). If you have transferable college credit, be sure to talk to an advisor during the summer to verify appropriate course selection. I will be available during the second summer orientation/advising day (July 29. If this poses a problem, contact me by phone to resolve any advising questions.

You have made a wise choice. Next summer, you will have your pick of exciting summer work experiences. Additionally, you will be eligible for thousands of dollars worth of scholarship support throughout the year. The 38th Annual Geomatics Engineering Conference in January will be exceptional. I urge you to get involved with it and a host of other exciting Geomatics Engineering activities. Tentatively, we are planing to organize a group hike to Clouds Rest (Higher and East of Half Dome) during the Labor Day weekend. Could be great fun.

If you have any questions please, call, write or E-mail.

Sincerely yours,

James K. Crossfield, Ph.D., L.S.
Coordinator, Geomatics Engineering
CSU, Fresno
Fresno, CA 93740-0094
(559) 278-4827 (W)
(559) 297-1866 (H)
(559) 267-6759 (FAX)
james_crossfield@csufresno.edu

Information Enriched GIS Representation of Map Transect Profiles

Neil Ripley and Micha Pazner
Department of Geography
The University of Western Ontario
London, Ontario, Canada
N6A 5C2
nripley@julian.uwo.ca

Abstract

We have developed a method for constructing, processing and enriching profiles of surface transects in a raster GIS environment. For instance, the transect may be of a road section winding its way through hilly terrain. Once constructed, the profile graph can be further processed by the GIS, as if it were a map. This allows us to develop new derivative pixel-based profile graphs based on data extracted from the original map representation, as well as from a profile graph representation. The results are raster GIS representations of profiles that are augmented by information of analytical or visualization value. These representations offer new image-based alternatives to classic line-based profile graphs. We describe the model used to build the profile representation. This model is unusual in that the raster GIS derivatives have transformed axes that represent graphs. The raster GIS model is presented and cartographic issues are discussed. Planned future developments include adapting the method for other types of axis transformation problems in raster GIS.

Introduction

The use of profiles for the presentation of cartographic information has a long history. The technique appears to have originated during the Middle Ages (Wallis and Robinson, 1987), when cartographers began adding coastal profiles to charts to assist sailors in recognizing sections of coasts. Building on this representation technique, we have developed new methods for the construction and enhancement of profiles. Our methods operate in a raster GIS environment (Tomlin, 1990) and allow us to develop, process and enrich profiles of surface transects. The development of a model that leads to transformed axes products within a GIS environment constitutes the key contribution of this research.

Motivation

A multi-disciplinary team of students at the University of Western Ontario (London, Canada) is constructing a solar powered car named the *SunStang*. The team is scheduled to compete in two races;

the *Sunrayce* from Washington D.C. to Orlando Florida (USA) during June 1999, and the *World Solar Challenge* from Darwin to Adelaide (Australia) during October 1999.

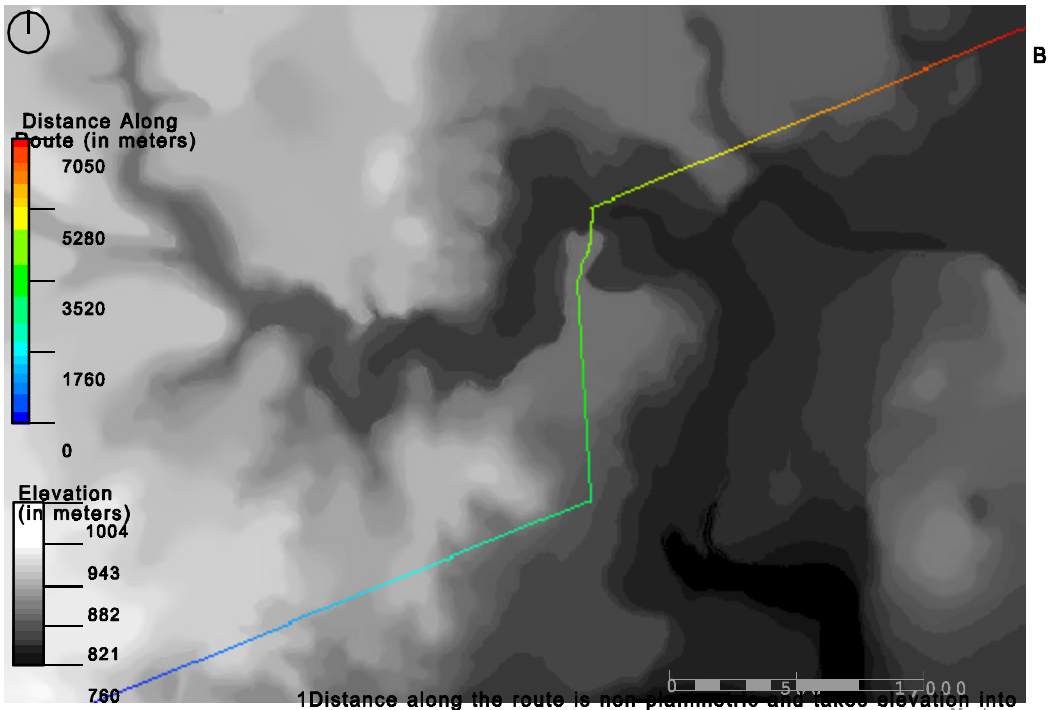
The SunStang team enlisted our assistance in the development of a Spatial Decision Support System (SDSS) (Heywood *et al*, 1995). The SDSS will include a GIS and a Spatial Database (SDB) with information about the route such as grades, lengths and orientations of slopes. Calculations based on SDB information will enable the team to predict the power consumption of the solar vehicle at any point along the race route. The vehicle's rate of energy consumption varies depending on the terrain it encounters (Cassidy and Gurley, 1990). As might be expected, it requires more energy to travel uphill than it does to traverse a flat surface, and it requires less energy to travel downhill.

The team hopes to gain a competitive edge in race strategy planning with detailed information about the route terrain organized within an SDSS framework. The goal is to optimize the vehicle's performance over the race route by analyzing, prior to the race, the terrain it will encounter.

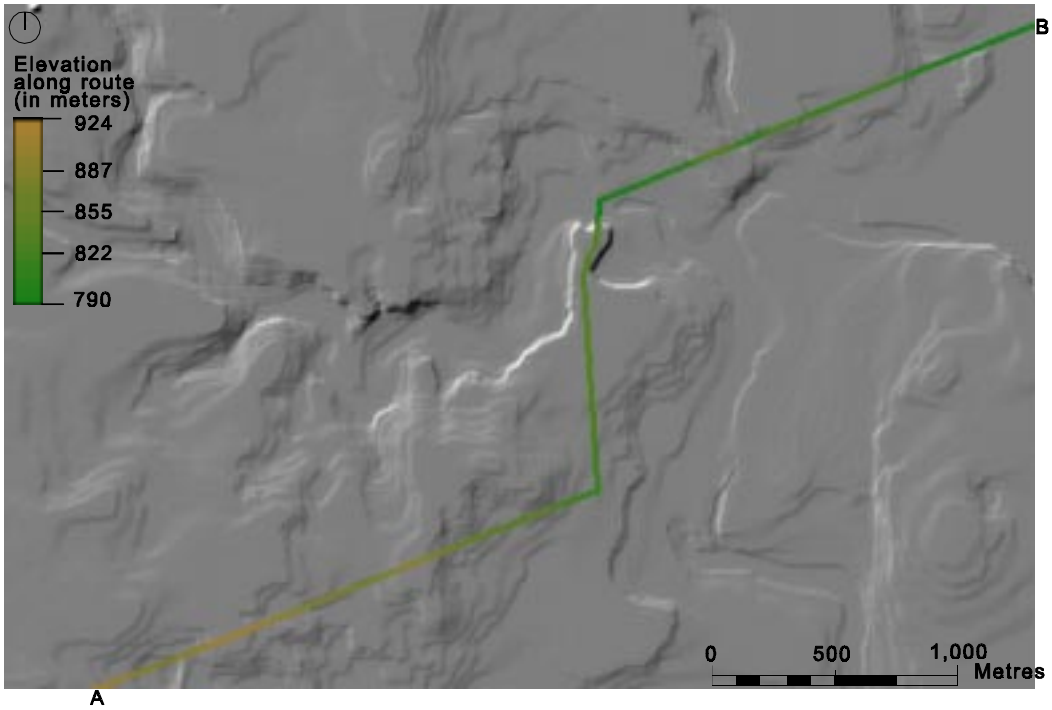
Data and its transformation

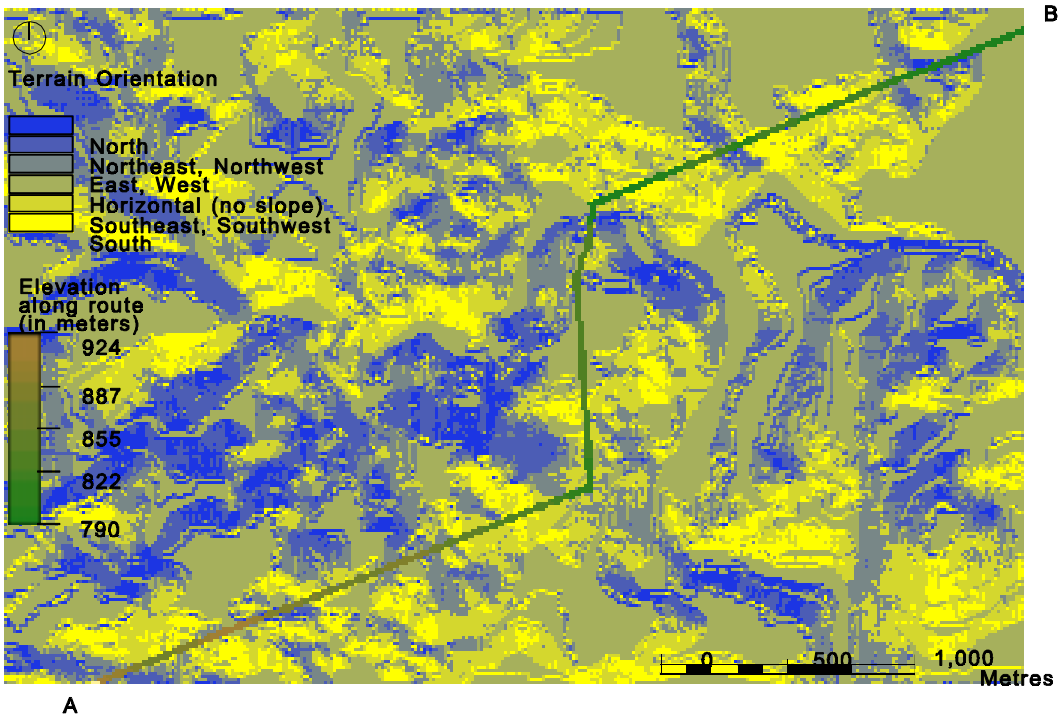
In order to develop the transect profile model, a hypothetical route winding its way through some hilly terrain was selected. The route, shown in *Figure 1*, is in London (Ontario) and runs along a selection of city streets. Elevation varies along this route between 790 meters and 924 meters. The route is 7 050 meters in length from starting-point A to end-point B. The route length measurement takes into account the 'rise over run' distance (Tomlin, 1990). This leads to a slightly longer profile than would result from a planimetric distance measure. However, it more accurately reflects the distance that would be covered by traversing the profile.

The terrain which the route passes through is represented using a digital elevation model (DEM) shown in *Figure 1*. To facilitate visualization we have rendered it as a pseudo-shaded-relief map in *Figure 2*. The DEM was also used to derive a terrain slope orientation map (*Figure 3*).



1Distance along the route is non-planimetric and takes elevation into account (i.e."rise over run" distance).


Elevation
along route
(in meters)
924
887
855
822
790
B
A
0
500
1,000
Metres


B
Terrain Orientation
North
Northeast, Northwest
East, West
Horizontal (no slope)
Southeast, Southwest
South
Elevation along route (in meters)
924
887
855
822
790
0
500
1,000
Metres
A

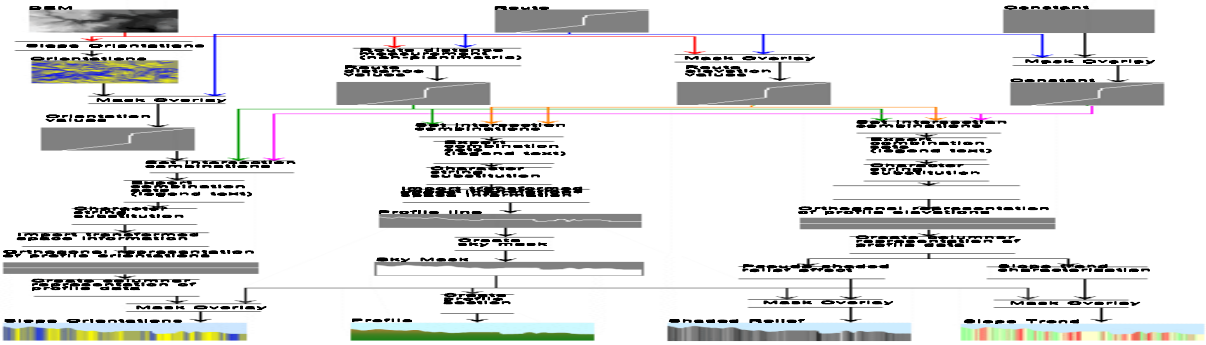
Figures 1 and *3* present some obstacles to interpreting important information about the route. Elevations and terrain slope orientations along the route are difficult to quantify visually when the data is presented this way. These orthogonal representations of data are not as suited to conveying information about transects as profile graphs are. By way of analogy, consider an aerial-photograph of Manhattan. An aerial photo is not able to communicate a sense of the building heights nearly as well as a ground, or side-view-based, panoramic photograph of the skyline profile. In a similar way, a profile graph representation of the route transect reveals different detail about the route than an orthogonal representation.

The development of our profile graph representations makes use of a GIS to generate the new X' and Y' axes. The new X' axis is derived by applying a distance measurement operation which: (i) limits the measurement only to cells that lie on the route; and (ii) takes into account the elevation changes along the route. That is, it explicitly calculates the length of the transect using an 'over the terrain' distance rather than the planimetric distance measure used in more traditional profiles, as described by Monkhouse and Wilkinson (1978). The resulting transect profile X' axis is slightly longer and will more accurately reflect the true distance along the route.

The new Y' axis is a scaled elevation range that provides an exaggerated but visually effective exposition of the route's profile (see transect elevation values in *Figure 2 & 3*). The profile needs to be exaggerated to draw out detail since vertical changes in a topographical surface are practically imperceptible when depicted at the same scale as the horizontal (Muehrcke, 1986).

A description of the Process

The model shown in *Figure 4* is unusual in the world of raster GIS, where the Z value is normally transformed as a function of the X, Y or Z values in one or more layers, and X, Y transformations are limited to re-scaling. In layman's terms, the result of a GIS transformation normally does not result in the creation and transition to a completely new 'space'. In our case, the X and Y values must be transformed into a new and different set of axes. The new



X' axis is the result of the distance measurement along the transect, while the new Y' axis is either held to a constant value (see either of the Orthogonal representation graph icons in *Figure 4*) or is composed of transect elevation data (for example the Profile line graph icon in *Figure 4*). Two different Y' axes are required since the development of a complete Profile graph requires the creation of two separate representations of transect data. The first representation arranges the transect data orthogonally while the second representation develops a profile line graph. A mask is derived from the profile line and is laid over the orthogonal representation to produce the completed Profile graph.

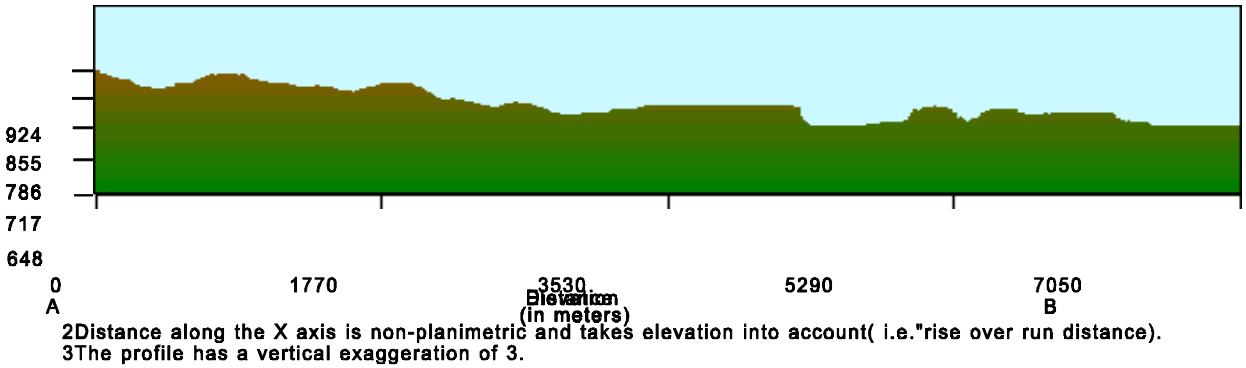
The transformation process begins in 'map space' with a DEM as the base map. A fundamental step at the beginning of the model is a transect distance measurement operation to generate the new X' axis values. Derivatives in 'map space' are developed from operations acting on the DEM data set. Examples of such derivatives include Terrain Slope Orientations and Route Transect Orientations.

The axis transformation begins with the development of the new X' and Y' axes and pairing these with Z' data values, i.e. the generation of the new map intersection combination sets - (X',Y',Z'). These three information variables will be required to create the profile graph representation. For any given point on the route, the X' component is the non-planimetric distance value measured along the route transect. The Y' component is either an elevation value (required to develop the Profile line graph representation) or some constant value (needed for the Orthogonal representations). The Z' value carries the information of interest for that point, i.e.: its slope orientation or elevation. To illustrate this, consider the point 3530 meters along the route transect. If we are developing the profile line graph representation of transect elevation data, then the X' component is 3 530, Y' is 840 (the elevation at that point) and Z' is also 840. Thus the derived map intersection combination set is (3 530, 840, 840). If an orthogonal elevation representation were being developed, the X' component of 3 530 would be paired with an arbitrary constant value such as 50, and an elevation of 840 — giving the combination set (3 530, 50, 840).

The next stage entails exporting the combination sets from the GIS as a text file. This file is arranged in three data columns. The file is composed of either purely numeric data or a combination of alphabetic characters and numeric values. The combination set file is processed using a text editor. Extraneous characters such as commas are stripped from the file and any alphabetic sequences (e.g.: 'North' or 'Southeast' that result from the terrain slope orientation operation) are converted into numeric codes that lend themselves to representations as X', Y', Z' coordinate values.

In the final stage, the transformed space information is imported back into the GIS as an X'Y' coordinate map. At this point, the transect data is now transformed from the initial 'map space' to the new 'profile space'. The basic representation of the transect is the profile graph depicted in *Figure 5*. It shows the route transect elevations much more clearly than was possible on the DEM.

Profiles and their derivatives in a raster GIS environment



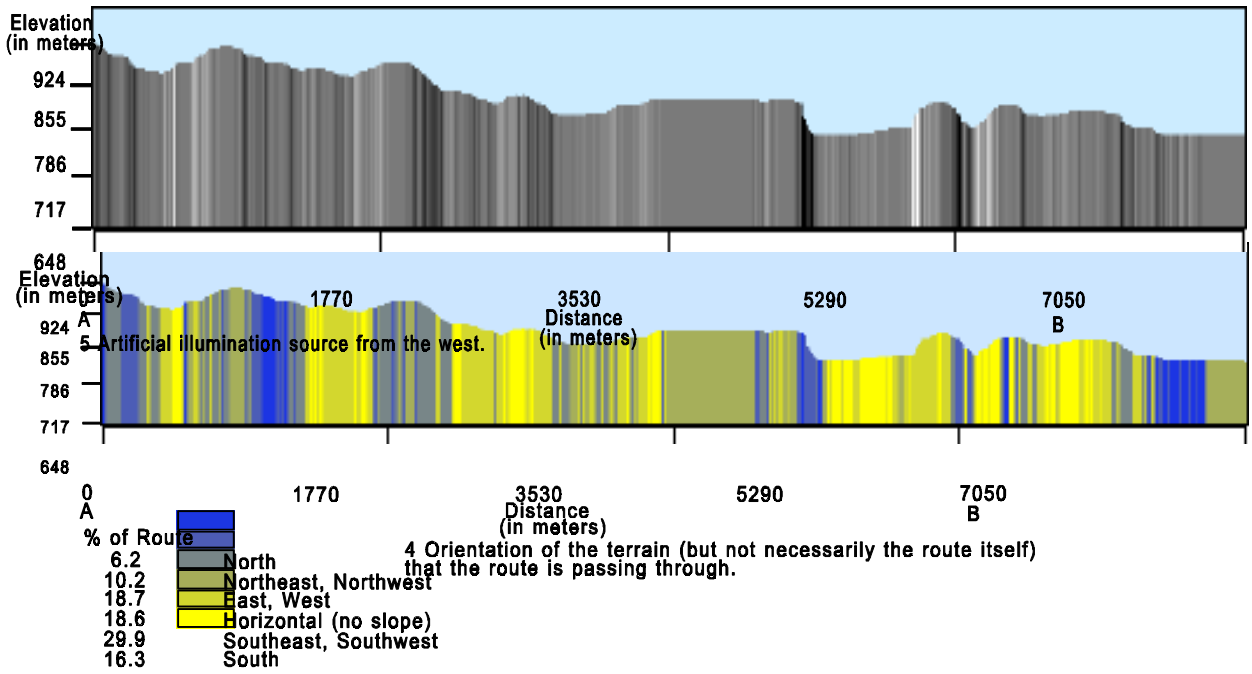
2Distance along the X axis is non-planimetric and takes elevation into account(i.e."rise over run distance).
3The profile has a vertical exaggeration of 3.

The transformation of transect data makes profile graphs that are available for processing as regular GIS data layers - as if they were maps. This allows new derivatives to be created in 'profile space'. The new derivative pixel-based profile graphs can be based on data extracted from the original map representation, as well as from the profile graph representation.

The results are raster GIS representations of profiles that are enriched with added information. A terrain slope orientation profile graph is depicted in *Figure 6*. Here the traditional profile graph showing the elevations along the transect is enhanced with information about the slope orientations of the terrain that the transect cuts through. Accompanying the graph is a legend giving the breakdown of the various

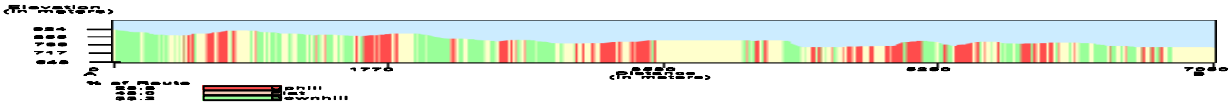
orientations. Analysis of the orientations along the route can improve the estimates of potential solar energy available to the vehicle.

The profile in *Figure 7* is enriched with information about slope trends showing uphill, flat and



downhill sections of the transect. This information can be used to improve the energy consumption estimates of the SunStang.

Profile graphs can also be enhanced by using a pseudo-shaded-relief effect for visualization



purposes (*Figure 8*). This makes it easier to visualize the transect terrain by producing a three dimensional effect. The terrain appears to stand out from the page.

Current and Future Prospects

There are several possible applications for using this technique. The method was originally conceived for creating the SunStang SDSS, where it continues to be an integral part of that effort. It is planned to fully implement transect-based route energy absorption and expenditure models in time for the scheduled races. These models will constitute key features of overall race strategy planning.

The technique could also be extended to non-route transects. Other disciplines such as geology, hydrology, biology and remote sensing often make use of profile graphs. This profile modeling technique makes such graphs easy to produce and, as shown before, we can use the technique to create profiles enriched with extra information. This can be a powerful tool for analysis.

Profile modeling might prove itself quite useful in the development, enrichment and analysis of transect profiles over non geographic surfaces such as multivariate graph surfaces. Or, alternatively, the method could be applied to entirely different types of axis transformation problems using raster GIS. Further research on the technique could lead to transect graphing and analysis of different point, line and area based graphs such as scatterplots, histograms, and bar charts.

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Software

The profile model in this paper was developed in ThinkSpace's Map• Factory raster based GIS.

HAPPY MAPPERS VS MULTIDISCIPLINARY MAPPERS

ROBERT M. STOLLARD
COLORADO SURVEYORS
3165 SOUTH RALEIGH STREET
DENVER, COLORADO 80236-2138

BIOGRAPHICAL SKETCH

Robert M Stollard is a Professional Land Surveyor registered in Colorado, Wyoming, and Kansas. He has an associate title in drafting from the University of Nebraska, at Omaha, and a B.S. in education from the University of Colorado at Denver. He served in the U.S. Navy and was an instructor of surveying in the Navy Reserve Construction Battalion. His experience spans over forty-four years in surveying and engineering. He served as the Professional Land Surveyor for the Jefferson County Mapping Department from 1982 through 1995. He is a member and past president of Professional Land Surveyors of Colorado, a member of ACSM, NSPS, and WFPS.

ABSTRACT

I could not believe it! "This is a bad dream," I said to myself. I pinched myself hard and found I was awake. But it was impossible! I was looking at a parcel map created by the assessors office which was in the fifth year of their new map project, to be used only by the assessors office. There were several parcels designated as "owner unknown." The assessor gave the legal descriptions to the county treasurer to be put on the tax sale list because of nonpayment of taxes. The treasurer was not a mapper, but questioned the large number of parcels in this one area to be added to the tax sale list. A couple of years previous, he and the county commissioners were sued by a person who acquired a Treasurers Deed for a parcel of land that did not exist on the ground. The treasurer sent the parcel legals to be verified by the Jefferson County Mapping Department which was a new modern up to date, state plane coordinate, computerized, county office. I was chosen to check it out. This is the story of two county mapping offices in Jefferson County, Colorado.

INTRODUCTION

Colorado became a State August 1, 1876, one hundred years and one month after The United States of America became an independent country. In 1976, Colorado House Bill No. 1089 was passed and approved April 29, 1976, and took effect on July 1, 1976. It is Section 39-5-103.5 C.R.S. which reads as follows:

39-5-103.5. Maps of parcels of land in the county. (1) Prior to January 1, 1981, each assessor shall prepare and maintain full, accurate, and complete maps showing the parcels of land in his county. The maps shall include a master county index map,

together with applicable township, section, and quarter-section maps depending on density. Guidelines shall be established by the administrator to produce uniformity throughout the state. The guidelines shall include the definition of a parcel, the development of a parcel numbering system, map size, map scale, and suggestions for minimum information to be plotted.

(2) In fulfilling the duty imposed upon him by subsection (1) of this section, the assessor may employ other resources or maps available to him. .....".

JEFFERSON COUNTY MAPPING DEPARTMENT

The Jefferson County Mapping Department (JCMD) was created in 1977, and I was employed on February 1, 1982, as the only licensed surveyor. Of course, this was before Global Positioning System (GPS). My first job was to obtain surveying equipment, a truck, and one surveyor helper, on a very limited budget. The budget was made by non-surveying people that did not know what was needed. We needed a theodolite, distant meter, and the gear necessary to traverse. The Purchasing Department was the real problem. I did not realize how much red tape was involved in obtaining a purchase order. I was asked if a regular light bulb wouldn't work as well as a theodolite. Then I had to have pictures of everything. But the Mapping Department was great. Jefferson County, Colorado was one of six counties in the United States that had a computer driven, state plane coordinate, multiple layer, including an orthophoto layer, mapping program. Jefferson County had triangulation stations (control monuments) placed on mountain tops and other high places every two to three miles apart. My part was to run traverses from one triangulation station, through section and quarter section corners, and close on another triangulation station and compute state plane coordinates for the control layer. The director of the JCMD was Billie Swenson who had high standards to make the best accurate maps possible. She insisted on survey accuracy standards instead of the usual mapping accuracy standards. The book entitled ***NEED FOR A MULTIPURPOSE CADASTRE,*** (1980) was one factor that guided her high standards. The book was the report of a one-year study that began in late 1979 by the Panel on a Multipurpose Cadastre of the Committee on Geodesy in the National Research Council's Assembly of Mathematical and Physical Sciences. Before chapter one, the first four pages of the book is the Executive Summary. The first sentence is italicized. I quote the first two sentences. *"There is a critical need for a better land-information system in the United States to improve land-conveyance procedures, furnish a basis for equitable taxation, and provide much-needed information for resource management and environmental planning.* Land-information systems in the United States can be characterized in general as title and assessment records systems, most of which have under-gone relatively small changes in the last hundred years, and land-planning and-management systems, which have evolved in the last 20 years." Another book by the same committee entitled, ***PROCEDURES AND STANDARDS FOR A MULTIPURPOSE CADASTRE,*** (1983) gave Billie Swenson and the personal of the JCMD a shot in the arm because they gave Case Studies of four areas in the United States, one being **the JEFFERSON COUNTY MAPPING DEPARTMENT.**

Rather than a sewer district, a city water company, an electrical company, the county assessor, and other entities, all making maps for their own use, Billie Swenson's basic idea was to have a combined effort to have a control layer with land corners with state plane coordinates that could serve all mapping agencies. Also, many layers of maps could be combined to serve the best interest of cities, counties, law enforcement agencies, fire districts, utility companies, the general public, and the list could go on and on. The list could include the county assessor.

JEFFERSON COUNTY ASSESSOR

In response to House Bill 1089, the Jefferson County assessor, Mike Sobieski, with the backing of, or perhaps under the direction of the county attorney, stated that until the State provided funding, there would be no new maps. Five other counties fell in line. Jefferson County is a large county with mostly large homes in the foothills and the mountains and a large budget. Gilpin County and Clear Creek County are small counties and lie immediately west of Jefferson County. Gilpin County and Clear Creek County are in the mountains and were gold and silver producing from 1859 to 1940. Both have a large portion of federal lands and small budgets. But both Gilpin and Clear Creek Counties had maps made by M A P INC. of Grand Junction, Colorado, in the 1970s. M A P INC. had land surveyors on their staff, and the maps looked professional. It was embarrassing to me when I was working on the county boundary line and had to match their assessors maps with our assessors maps. Jefferson County assessors maps looked like chicken scratchings compared to their maps.

In the fall of 1982, the year I went to work for the Mapping Department, there was the general election and Judy Pettit was elected Assessor. She had a B.S. in mathematics and a masters degree in public administration. She was an appraiser for the previous assessor, but not a mapper. Jo Kildow, the woman in charge of assessor mapping was from the old school and afraid of computers. The methods of mapping for the last 100 years was good enough for Jo Kildow. And she made it difficult for anyone wanting to get information from "her" maps or records. After Judy Pettit was settled in her office as the new county assessor, Billie Swenson, the director of the JCMD, went to the new assessor and offered her state plane coordinates from the JCMD data base or any other assistance she needed to help the assessor mappers. Judy Pettit said she would check with her mappers but never did get back to Billie Swenson. In the late 1980's, Judy Pettit decided to comply with House Bill No. 1089, and started a new mapping program. However she did not ask for any help from JCMD. Billie Swenson went to Jo Kildow and offered again the state plane coordinates or any other help. Jo Kildow answered in a huff! ***"It has been decided that this is our program and we don't need your coordinates."*** The assessor hired more draftspeople and the new assessor mapping project began. Metes and bounds legal descriptions from land parcels, subdivision plats, and copies of boundary surveys were the tools the assessor mappers used. If they couldn't figure out the parcels, they were told to be **HAPPY**, square it off and get it done.

ASSESSOR MAP 60-02

Assessor Map 60-02 is Township 6 South, Range 70 West, Section 2. In the Southeast Quarter of said Section 2, T6S, R70W of the 6th PM, the parcels in the south half of said Southeast Quarter were tied to the South Quarter Corner, and parcels in the north half of said Southeast Quarter were tied to the Center Quarter Corner. The assessors draftsperson drafted the outside boundaries of the Southeast Quarter with data from their records and plotted all the parcels. That left gaps between the parcels in the north half and the parcels in the south half of the Southeast Quarter. Well what to do? The assessors people would not stoop so low as to ask the JCMD about the problem. But what to do? The people in the Office of the County Assessor in Jefferson County, Colorado, made a great decision. They created parcels in the gap area by adding lines here and there to make the parcels look authentic, and labeled them as "A.D." or "P" which stood for ambiguous description or private (show assessor map 60-02). Jo Kildow told me that if anyone asked about a "P" parcel, they were told to go to the Title company and ask them about the parcel ownership. She didn't want me or anyone else in the JCMD checking her parcels. This is an unjustified procedure that a competent, responsible person/office would not do. Then someone from the assessors office would write out a legal description for these bogus parcels and they were given to the County Treasurer to be put on the tax sale. The treasurer has issued Treasurers Deeds for other bogus parcels and has been sued (see Buchwald Petrol Resources, Inc. v Thomas F McTurk 91-CV-1703, Div 2).

The first sentence of 39-5-103.5 (1) reads as follows: "Prior to January 1, 1981, each assessor shall prepare and maintain full, accurate, and complete maps showing the parcels of land in his county." This did not happened in Jefferson County. A typical case is on the assessors map 60-02 where parcels 17 and 41 were owned by the same party. On the old assessors map, these two parcels were drafted together with the north line of parcel 17 being the same location as the south line of parcel 41. On the ground, in the real world, these two parcels existed as shown on the old map. However, the assessors new map shows two other parcels side by side between parcels 17 and 41. The owner called me and said he had purchased a new home which was contingent on the sale of his present home and property, but the title company would not issue a policy on parcels 17 and 41 because the assessors maps showed a "P" parcel and a " A.D. parcel" between his parcels 17 and 41. He said someone from the assessors office told him to call and ask me, which was the official map, the old one or the new one?????? This is unreal. How would you like this to happen to you, your brother, sister, parents, or children? Who is responsible for the time, money, worry and stress that the taxpayer has to suffer? I have had taxpayers call me and tell of their problems with the assessors office. Another typical case is assessors' maps 60-06 and 60-071,072. The south line of map 60-06 should be the north line of map 60-071,072, but the two maps will not fit together. How many, if any, of the assessor's new maps are really valid? One person said, "The taxpayers money is going down the drain, and the service that should be rendered has turned out to be a detriment to the taxpayers. It is a fleecing of the taxpayers."

MOSS ROCK

It was reported by a private land surveyor, Robert Feroldi, that he believed the subdivision of Moss Rock was not located on the East-West centerline and the North-South centerline of Section 10, T 4 S, R 71 W of the 6th PM as described on the original record plat. I consider him one of the best surveyors in Colorado. His original job was to locate the boundaries of a block of vacant lots, but found a very large house on the property. This is not what a surveyor wants to find and lose sleep over. He tried to find an error in his work but only found more evidence confirming his work. The house belonged to the adjacent property owners, and was surveyed by Cecil Broyles, licensed as Professional Engineer and Professional Land Surveyor. I have been told that he is a good engineer. The JCMD was interested in the true location of the subdivision to place it correctly on the map of Section 10. The JCMD made a survey of said Section 10, including the Northerly and Easterly lines of the Moss Rock subdivision I was the Professional Land Surveyor for the JCMD and performed the survey. There was an abundance of information available in old records and existing land corner monuments of said Section 10. Only the North Quarter Section corner monument was missing, but the position of that monument was tied to the steel tower of the electrical transmission power line. Therefore, there was not a problem in determining the boundaries and the centerlines of Section 10. The original lot corner monuments of the Northerly and Easterly lines of the Moss Rock subdivision were clearly visible but rotated in position. The survey I performed was a DEPENDENT RESURVEY which is a survey that depends on the original monumentation of the area, both of the section corners and quarter corners of the section and the original monuments that was set for the subdivision. Using state plane coordinates the true location of all of these corners were plotted on a transparent velum, and then a velum of the subdivision plat was superimposed over the monuments as found on the ground. Then these vellums were superimposed over the orthophotos. The Public Service Power line, the houses and the roads of the Moss Rock subdivision fit perfectly in place. So I proved that the entire Moss Rock subdivision was rotated in position in Section 10. The Northeast corner of the subdivision was intended to be the center of Section 10, but the original monument of the subdivision is 140 feet away, which is evident in the filing of the Summit Ranch Subdivision that is south of Moss Rock subdivision. A land surveyor has to do his surveying in accordance with statutory law, boundary law, case law and court records. One of the best papers surveyors have to guide us in our work is a paper by Chief Justice Cooley, of the Michigan Supreme Court 1864-1885, entitled ***"The Judicial Functions of Surveyors."*** Surveyors have to follow in the footsteps of the original surveyors even if they placed their monuments in the wrong location**.** ***"The original monuments shall forever remain in place."*** A property owner who relied on the original monuments in 1925 to build his house, cannot be told in 1995 that he has to move his house 140 feet.

Broyles performed an INDEPENDENT RESURVEY in 1969 and layed out lots in Moss Rock subdivision where they intended to be but not were located by the original

surveyors. He stated that all the houses built before 1969 and all the roads were wrong. He never checked Summit Ranch Subdivision. Phillip and Nancy Stevenson bought their property in the Moss Rock subdivision from Richard and Edith Hansen, who had a boundary survey made and went to court to seek to quiet title of their part of the land between the existing north line of the subdivision and the intended north line of the subdivision (see Hansen v. Warren, et al 92CV236 Div 9). The boundary survey by Robert Feroldi clearly shows the facts of the case. It has been my understanding that assessors shall change their maps to agree with court action. It has not happened.

The difference in the assessors map and the dependent resurvey of the JCMD, in the location of the Moss Rock subdivision is that the assessors map shows where the subdivision was intended to be located, and the JCMD survey shows where the subdivision was actually located. An analogy of this situation is the case where a house plan was drawn properly, but the lady of the new house had the contractor change some walls, add another room, and enlarge the garage. Does the appraiser appraise the house plan or the house "as built?" Is the assessor concerned? Is it money out of her pocket to do a professional job? If a taxpayer sues the assessor does she have to pay the county attorneys to defend her? Does she care? One time Judy Pettit had one of her staff ask me for information on using state plane coordinates. I sent them all the necessary information including a one page example with blanks to be filled in, but never asked for any questions. Remember that Judy Pettit had a college degree in mathematics. Her office has never used state plane coordinates.

CONCLUSION

But this bogus mapping of the Jefferson County assessor is not necessary. Other assessors of Colorado have a different set of values. Their goal is not to build a great kingdom beyond the reach of their taxpayers, but to serve their taxpayers. Grand County has a full time County surveyor and helps the assessor mappers when they request his help. One of the assessors on the western slope tried to talk to Judy Pettit, but was given a deaf ear. There is plenty of help in this country to produce a good set of maps that would be a service to the taxpayers of Jefferson County. At the annual convention of the American Congress of Survey and Mapping (ACSM) in April of 1994, I met Sheldon Bluestein, who was at that time, the State Mapping Coordinator of the Idaho State Tax Commission. He gave a paper entitled, "Mapping for the Rest of Them: Turning Today's Parcel Mappers into Tomorrow's GIS Experts." He resigned his position and traveled to Russia to help them in their new system of private property ownership. Idaho has a great mapping education program for assessors which includes a solid background in boundary law and the Public Land Survey System which is the system of Sections, Townships, and Ranges as designed by Thomas Jefferson in 1785. Another paper was given in GIS IN THE ROCKIES in Golden, Colorado, in September of 1994 entitled "Fundamentals of Parcel Mapping" by Gary Moos and Richard Brewster of Colorado Springs Utilities. With one knowledgeable person with the proper attitude, and a mapping education

program like Idaho, the Jefferson County assessor mapping program can turn around and serve the taxpayers.

AFTERMATH

The County Commissioners had a study and report made of county departments by a private company named U. G. C. Consultants. About a year late, and after the company nearly went under, the report was delivered in March of 1994. They recommended that the Mapping Department become the G. I. S. Department, the Assessor take over the mapping of the county, members of the Mapping department that checked subdivision plats in the platting process be transferred to the Planning Department, and that the surveyors of the Mapping Department and the Department of Highways & Transportation be combined in their own department. The above recommendations were carried out except for the surveyors. The surveyors of the Mapping Department were combined with the surveyors of the Department of Highway & Transportation. So the bogus maps of the assessor were declared the official county maps. September 1, 1994, the Jefferson County Mapping Department became Geographic Information Systems. January 3, 1995, Jo Kildow retired from the assessors office. In June of 1995, I wrote a memo to one of the top attorneys of the office of the Jefferson County Attorney, which included all the material in this paper. I discussed the information with her and showed her the maps. Nothing ever happened. Billie Swenson thought she could not do any more for Jefferson County and retired July 31, 1995. I also though I had done my part for the citizens of Jefferson County and retired January 15, 1996 The assessors maps mentioned above, 60-02, 60-06, 60-071,072 have not been corrected as of December 1998. In 1997, all of the maps of the entire county, hand drafted by Judy Pettit's mappers, were photo reduced and copyrighted by a private company. Judy Pettit chose not to run for reelection in November 1998, and will leave office in January 1999. I will close with the following.

"Half-measures always result in loss of time and money. The only way to sort out the confusion in the field of general land records is to proceed with the surveying and evaluation of each individual land parcel in all the communities of the Empire. A good cadastre will constitute a complement of my Code as far as land possession is concerned. The maps must be sufficiently precise and complete so that they could determine the boundaries between individual properties and prevent litigations."

Napoleon Bonaparte, 1807